

SSF Formulations

Scott Forrest

2026-08-25

In this script we demonstrate different approaches to including covariates in an SSF model.

Table of contents

Load required packages	2
Import data and clean	3
Create a step object	4
Plot the data spatially	4
Pick out a single individual	5
Import spatial covariate	6
Select local extent	7
Sample random steps to fit models	9
Different SSF formulations	10
Set RSS min and max values	12
Linear covariate	12
Model formula	12
Fit model	12
Response curve	15
Plot habitat selection	16
Quadratic covariate	17
Model formula	17
Fit model	17
Generate the response curve	19
Response curve	21
Plot habitat selection	22
Smooth term	23
Model formula	23

Fit model	24
Alternatively, generate the response curve with the <code>predict</code> function	25
Plot habitat selection	27
Temporal dynamics	28
Static model for comparison	28
Pull out a response curve from one of the hours	30
Fit harmonic model with quadratic terms	32
Create table of NDVI and hour values	33
Easier to get the curves with GAM <code>predict</code> function	35
Fit model	35
Generate the response curve	36
Harmonic regression	37
Fit model	38
Pull out the coefficient across time	39
Calculate confidence intervals across time and plot	40
Pull out a response curve from one of the hours	42
Manually create the selection surface	44
Create table of NDVI and hour values	44
Fit harmonic model with quadratic terms	46
Manually create the selection surface	48
Create table of NDVI and hour values	49
Pull out a response curve from one of the hours	51
Temporal dynamics with smooth terms	53
Pull out a response curve from one of the hours	55
GAM selection surface	57
Fit 2D GAM smooth	281
Plot the smooth selection surface	282
Pull out a response curve from one of the hours	287
References	289

Load required packages

```
library(tidyverse)
packages <- c("amt", "sf", "terra", "RColorBrewer", "mgcv", "gratia")
walk(packages, require, character.only = T)
```

Import data and clean

```
buffalo_data <- read_csv("data/buffalo.csv")
```

New names:

Rows: 133161 Columns: 11

-- Column specification

```
----- Delimiter: "," chr
(2): node, dates dbl (7): ...1, lat, lon, height, accuracy, heading, speed dtm
(2): timestamp, DateTime
i Use `spec()` to retrieve the full column specification for this data. i
Specify the column types or set `show_col_types = FALSE` to quiet this message.
* `` -> `...1`
```

```
# remove individuals that have poor data quality or less than about 3 months of data.
# The "2014.GPS_COMPACT copy.csv" string is a duplicate of ID 2024, so we exclude it
buffalo_data <- buffalo_data %>% filter(!node %in% c("2014.GPS_COMPACT copy.csv",
                                                    2029, 2043, 2265, 2284, 2346))
```

```
buffalo_data <- buffalo_data %>%
  group_by(node) %>%
  arrange(DateTime, .by_group = T) %>%
  distinct(DateTime, .keep_all = T) %>%
  arrange(node) %>%
  mutate(ID = node)
```

```
buffalo_clean <- buffalo_data[, c(12, 2, 4, 3)]
colnames(buffalo_clean) <- c("id", "time", "lon", "lat")
attr(buffalo_clean$time, "tzone") <- "Australia/Queensland"
buffalo_clean$sex <- "f"
buffalo_clean$life_stage <- "adult"
head(buffalo_clean)
```

A tibble: 6 x 6

	id	time	lon	lat	sex	life_stage
	<chr>	<dtm>	<dbl>	<dbl>	<chr>	<chr>
1	2005	2018-07-25 10:04:02	134.	-12.3	f	adult
2	2005	2018-07-25 11:04:23	134.	-12.3	f	adult
3	2005	2018-07-25 12:04:39	134.	-12.3	f	adult
4	2005	2018-07-25 13:04:17	134.	-12.3	f	adult
5	2005	2018-07-25 14:04:39	134.	-12.3	f	adult
6	2005	2018-07-25 15:04:27	134.	-12.3	f	adult

```
tz(buffalo_clean$time)
```

```
[1] "Australia/Queensland"
```

```
buffalo_ids <- unique(buffalo_clean$id)

write_csv(buffalo_clean, "data/buffalo_clean.csv")
```

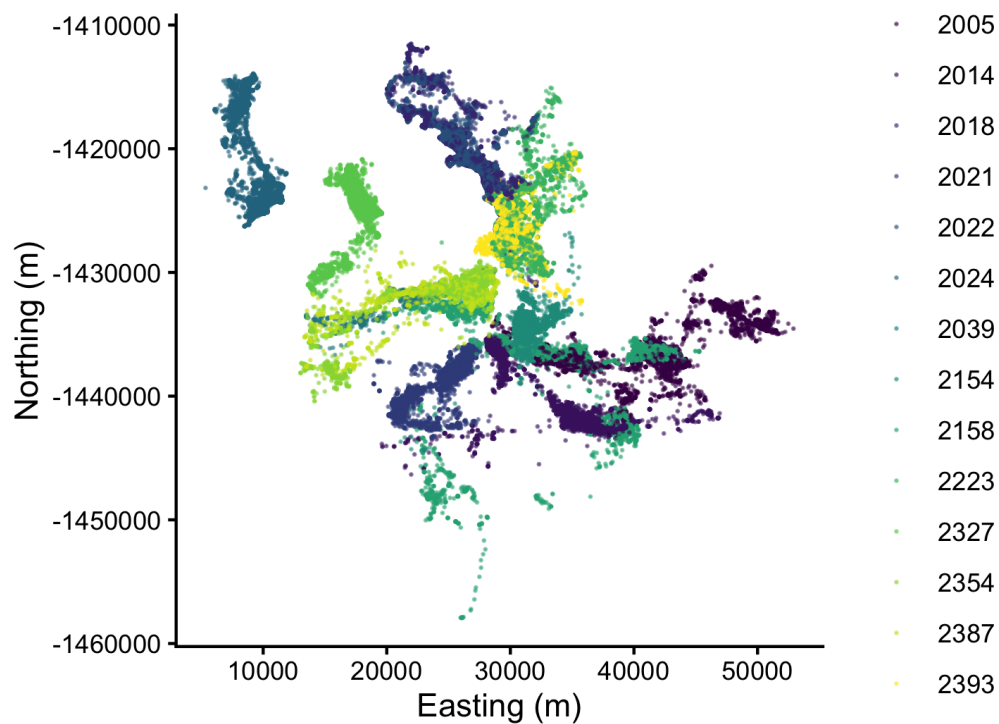
Create a step object

Use the `amt` package to create a trajectory object from the cleaned data.

```
buffalo_all <- buffalo_clean %>% mk_track(id = id,
                                           lon,
                                           lat,
                                           time,
                                           all_cols = T,
                                           crs = 4326) %>%
  transform_coords(crs_to = 3112, crs_from = 4326) # Transformation to GDA94 /
# Geoscience Australia Lambert (https://epsg.io/3112)
```

Plot the data spatially

```
buffalo_all %>%
  ggplot(aes(x = x_, y = y_, colour = id)) +
  geom_point(alpha = 0.5, size = 0.1) +
  coord_fixed() +
  scale_x_continuous("Easting (m)") +
  scale_y_continuous("Northing (m)") +
  scale_colour_viridis_d() +
  theme_classic() +
  theme(legend.position = "right")
```



```
# ggsave("outputs/data_prep/buffalo_djelk_map.png",
#         width = 150, height = 150, units = "mm", dpi = 600)
```

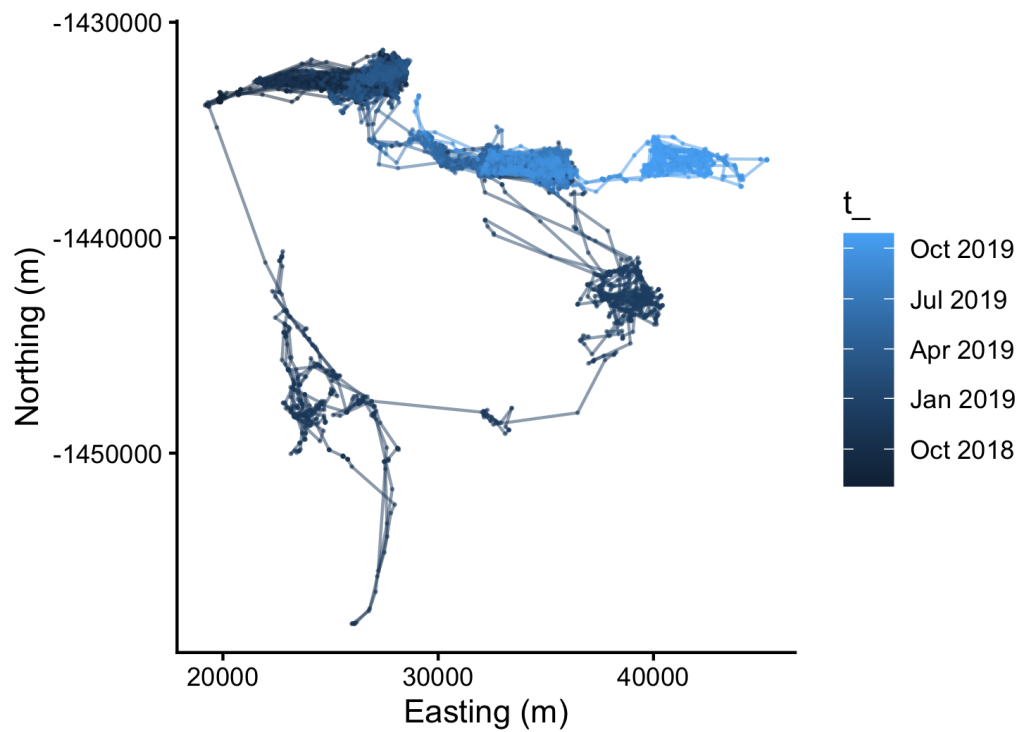
Pick out a single individual

```
which_buffalo <- "2158" # select a single buffalo ID

# create directory to save plots for each buffalo
plot_save_path <- paste0("outputs/buffalo_", which_buffalo)
dir.create(plot_save_path, showWarnings = FALSE)

buffalo_id <- buffalo_all %>% filter(id == which_buffalo)

buffalo_id %>%
  ggplot(aes(x = x_, y = y_, colour = t_)) +
  geom_path(alpha = 0.5) +
  geom_point(alpha = 0.5, size = 0.1) +
  coord_fixed() +
  scale_x_continuous("Easting (m)") +
  scale_y_continuous("Northing (m)") +
  theme_classic()
```

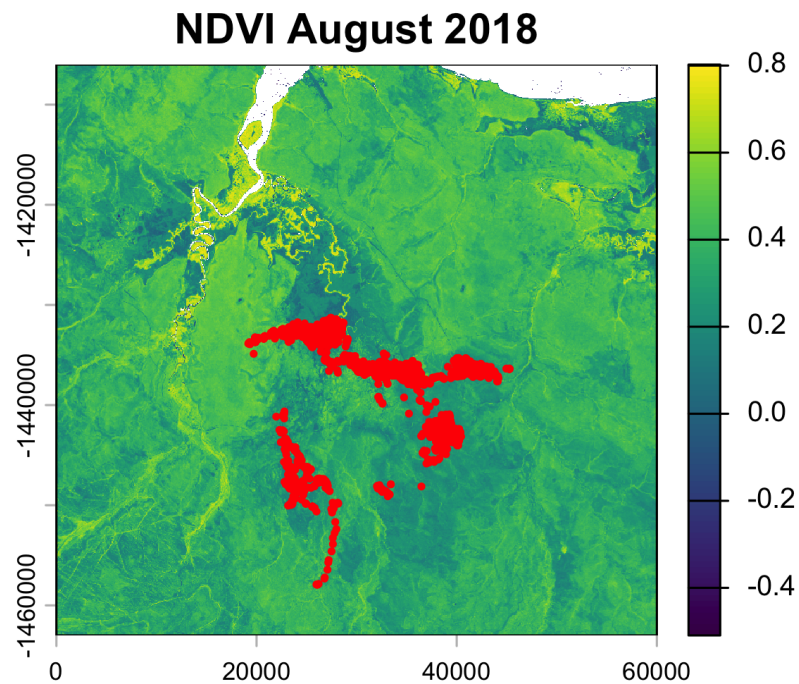


```
ggsave(paste0(plot_save_path, "/buffalo_id_", which_buffalo, "_map.png"),
       width = 150, height = 150, units = "mm", dpi = 600)
```

Import spatial covariate

Although NDVI changes over time, and we have access to monthly layers, we will just select a single month here.

```
ndvi <- rast("mapping/ndvi_aug_2018.tif")
plot(ndvi, main = "NDVI August 2018")
points(buffalo_id$x_, buffalo_id$y_, col = "red", pch = 16, cex = 0.5)
```



```
ndvi
```

```
class      : SpatRaster
size       : 2280, 2400, 1  (nrow, ncol, nlyr)
resolution : 25, 25  (x, y)
extent     : 0, 60000, -1463000, -1406000  (xmin, xmax, ymin, ymax)
coord. ref.: GDA94 / Geoscience Australia Lambert (EPSG:3112)
source     : ndvi_aug_2018.tif
name       : ndvi
min value  : -0.544105
max value  : 0.808655
```

Select local extent

We just want to look at an area centred on a single location, which might be combined with our movement probability surface to generate a next-step probability surface.

We'll centre the extent on an interesting looking part of the landscape, and then buffer it by 1500m (with a buffer of the cell resolution/2 to ensure there is a central cell), which contains most of the step lengths, and therefore most of the movement kernel.

```

cell_resolution <- 25 # in metres

buffalo_single_point <- buffalo_id[7500,] # select one of the locations

buffer <- 1500 + (cell_resolution/2) # buffer in metres

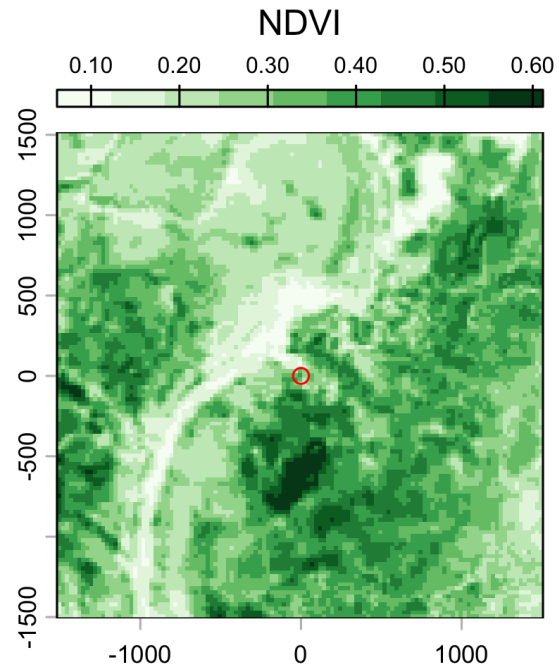
window_extent <- ext(buffalo_single_point$x_ - buffer,
                     buffalo_single_point$x_ + buffer,
                     buffalo_single_point$y_ - buffer,
                     buffalo_single_point$y_ + buffer)

ndvi_window <- crop(ndvi, window_extent)

# set the extent to 0 at the buffalo location
ndvi_window <- terra::shift(ndvi_window,
                           dx = -buffalo_single_point$x_,
                           dy = -buffalo_single_point$y_)

# plot the NDVI layer with the buffalo location as the centre point
plot(ndvi_window,
     # main = paste0("NDVI"),
     col = brewer.pal(9, "Greens"),
     plg=list(x="top", title="NDVI", tics="out", cex=.8))
points(x = 0, y = 0, col = "red")

```

```
# Save the plot
png(filename = paste0(plot_save_path, "/ndvi_local_", which_buffalo, ".png"),
     width = 100, height = 100, units = "mm", res = 600)
plot(ndvi_window,
     # main = paste0("NDVI"),
     col = brewer.pal(9, "Greens"),
     plg=list(x="top", title="NDVI", tics="out", cex=.8))
points(x = 0, y = 0, col = "red")
dev.off()
```

```
quartz_off_screen
2
```

Sample random steps to fit models

```
buffalo_id_steps <- buffalo_id %>%
  steps()

# fitting step length and turning angle distributions to all locations
gamma_dist <- fit_distr(buffalo_id_steps$sl_, "gamma")
vonmises_dist <- fit_distr(buffalo_id_steps$ta_, "vonmises")
```

```
# movement parameters
gamma_dist$params$shape
```

```
[1] 0.4513375
```

```
gamma_dist$params$scale
```

```
[1] 621.3796
```

```
vonmises_dist$params$kappa
```

```
[1] 0.1531606
```

```
vonmises_dist$params$mu
```

```
Circular Data:
```

```
Type = angles
```

```
Units = radians
```

```
Template = none
```

```
Modulo = asis
```

```
Zero = 0
```

```
Rotation = counter
```

```
[1] 0
```

```
# sample random steps
```

```
buffalo_ssf_data <- buffalo_id_steps %>%
```

```
  random_steps(n = 10,
```

```
              sl_dist = gamma_dist,
```

```
              ta_dist = vonmises_dist) %>%
```

```
  extract_covariates(ndvi) %>%
```

```
  mutate(ndvi_sq = ndvi^2,
```

```
         log_sl = log(sl_),
```

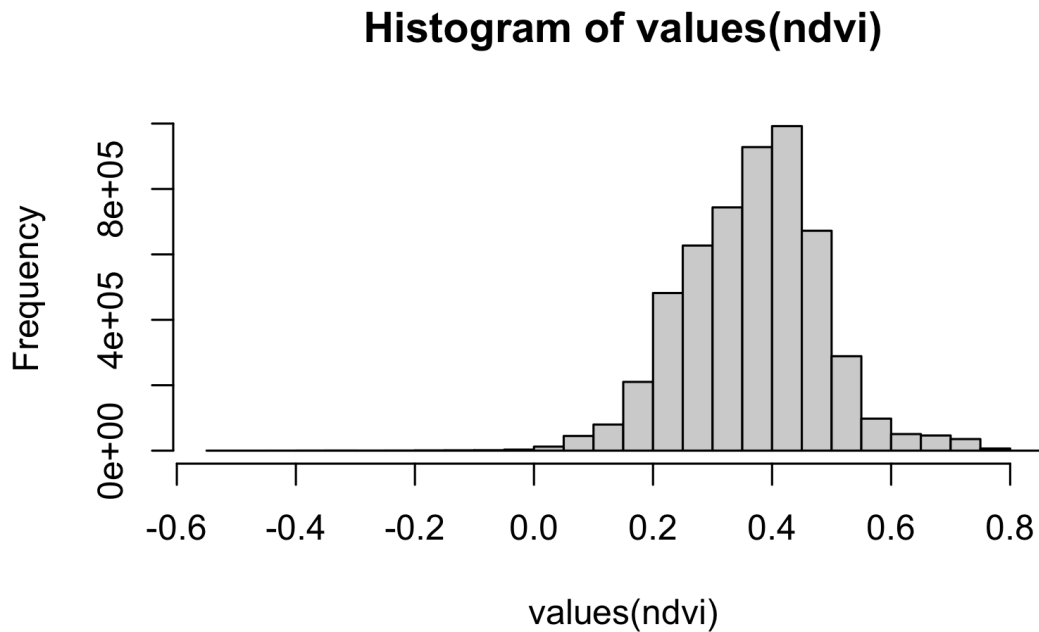
```
         cos_ta = cos(ta_),
```

```
         times = 1) # create a dummy column for times (used in a Cox PH model, but not relevant)
```

Different SSF formulations

Firstly, we want to pull out the range of NDVI values to plot the curves with.

```
# histogram of NDVI values
hist(values(ndvi))
```



```
# 1% and 99% quantiles of NDVI values
ndvi_quantiles <- quantile(values(ndvi), probs = c(0.005, 0.995), na.rm = TRUE)
ndvi_quantiles
```

```
      0.5%      99.5%
0.06121541 0.71819625
```

```
# plot across the range in the local window
ndvi_values <- seq(min(0, ndvi_quantiles[1]),
                  ndvi_quantiles[2],
                  length.out = 100)

pred_data <- data.frame(ndvi = ndvi_values,
                       ndvi_sq = ndvi_values^2,
                       sl_ = 0,
                       log_sl = 0,
                       cos_ta = 0)
```

Set RSS min and max values

This is hard-coded as the maximum rss value across the different formulations, to ensure the plotting extent is consistent across approaches.

```
rss_min <- 0
rss_max <- 2.1
```

Linear covariate

Model formula

$y \sim \text{ndvi}$

Fit model

```
# fit an issf model
ssf_linear <- amt::fit_issf(case_ ~
                           ndvi +
                           sl_ + log_sl + cos_ta +
                           strata(step_id_),
                           data = buffalo_ssf_data)

summary(ssf_linear)
```

Call:

```
coxph(formula = Surv(rep(1, 106678L), case_) ~ ndvi + sl_ + log_sl +
      cos_ta + strata(step_id_), data = data, method = "exact")
```

n= 106678, number of events= 9698

	coef	exp(coef)	se(coef)	z	Pr(> z)	
ndvi	-8.813e-01	4.143e-01	1.443e-01	-6.105	1.03e-09	***
sl_	1.147e-06	1.000e+00	3.174e-05	0.036	0.971	
log_sl	-2.394e-04	9.998e-01	5.688e-03	-0.042	0.966	
cos_ta	-4.305e-03	9.957e-01	1.503e-02	-0.287	0.774	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
ndvi	0.4143	2.414	0.3122	0.5497
sl_	1.0000	1.000	0.9999	1.0001

log_sl	0.9998	1.000	0.9887	1.0110
cos_ta	0.9957	1.004	0.9668	1.0255

Concordance= 0.494 (se = 0.004)
Likelihood ratio test= 38.92 on 4 df, p=7e-08
Wald test = 37.62 on 4 df, p=1e-07
Score (logrank) test = 37.62 on 4 df, p=1e-07

```
AIC(ssf_linear)
```

```
[1] 46478.66
```

```
# fit an ssf gam model with parametric terms
ssf_linear_gam <- mgcv::gam(cbind(times, step_id_) ~
                           ndvi +
                           sl_ + log_sl + cos_ta,
                           data = buffalo_ssf_data,
                           family = cox.ph,
                           weight = case_)

summary(ssf_linear_gam)
```

Family: Cox PH

Link function: identity

Formula:

cbind(times, step_id_) ~ ndvi + sl_ + log_sl + cos_ta

Parametric coefficients:

	Estimate	Std. Error	z value	Pr(> z)
ndvi	-8.813e-01	1.443e-01	-6.105	1.03e-09 ***
sl_	1.147e-06	3.174e-05	0.036	0.971
log_sl	-2.394e-04	5.688e-03	-0.042	0.966
cos_ta	-4.305e-03	1.503e-02	-0.287	0.774

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Deviance explained = -0.0544%

-REML = 23254 Scale est. = 1

n = 106678

```
AIC(ssf_linear_gam)
```

```
[1] 46478.66
```

```
# Get predictions on log-scale with standard errors
predictions <- predict(ssf_linear_gam,
                      newdata = pred_data,
                      type = "link", # log-scale
                      se.fit = TRUE)

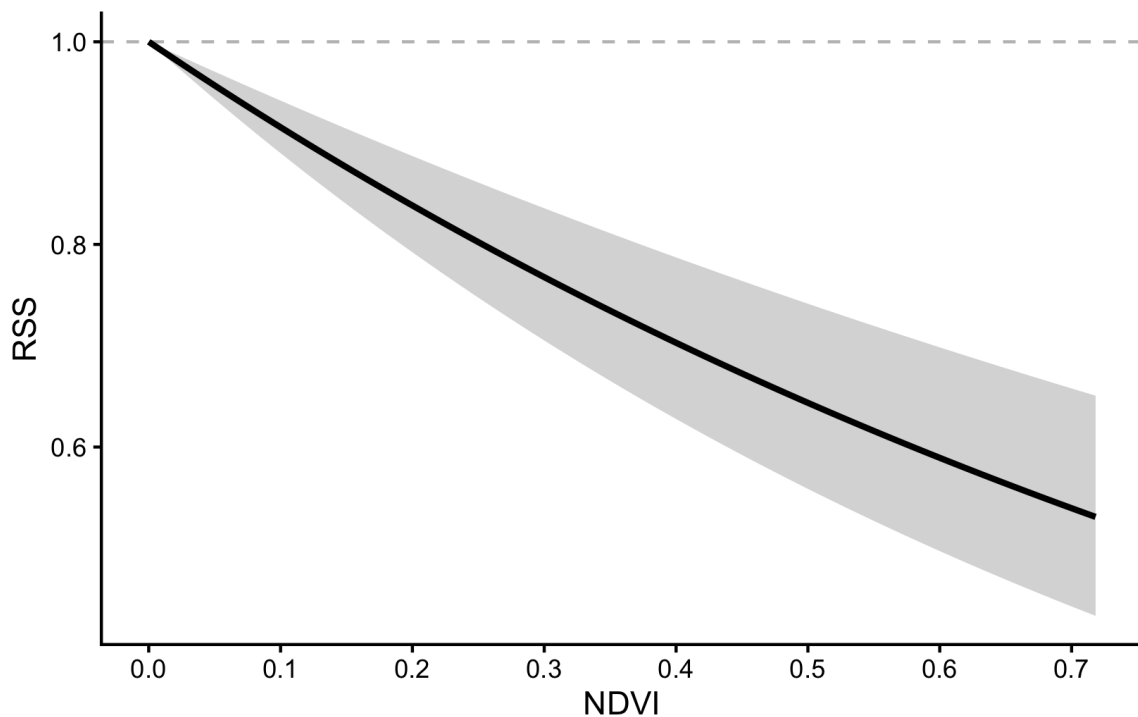
# Extract log RSS and standard errors
log_rss <- predictions$fit
se_log_rss <- predictions$se.fit

# Calculate confidence intervals on log-scale
log_rss_lower <- log_rss - 1.96 * se_log_rss
log_rss_upper <- log_rss + 1.96 * se_log_rss

# Transform to RSS scale
rss <- exp(log_rss)
rss_lower <- exp(log_rss_lower)
rss_upper <- exp(log_rss_upper)

# Create your dataframe
ndvi_coef_df <- data.frame(
  ndvi_values = ndvi_values,
  log_rss = log_rss,
  log_rss_lower = log_rss_lower,
  log_rss_upper = log_rss_upper,
  rss = rss,
  rss_lower = rss_lower,
  rss_upper = rss_upper
)

# plot the response curve
ggplot(data = ndvi_coef_df) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey") +
  geom_ribbon(aes(x = ndvi_values, ymin = rss_lower, ymax = rss_upper),
            fill = "grey70", alpha = 0.5) +
  geom_line(aes(x = ndvi_values, y = rss),
            colour = "black", linewidth = 1) +
  scale_x_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  labs(x = "NDVI", y = "RSS") +
  theme_classic()
```



Response curve

```
# extract the NDVI coefficient and its standard error
ndvi_coef <- ssf_linear$model$coefficients[1]
ndvi_coef_se <- sqrt(diag(ssf_linear$model$var))[1] # SE of coefficient

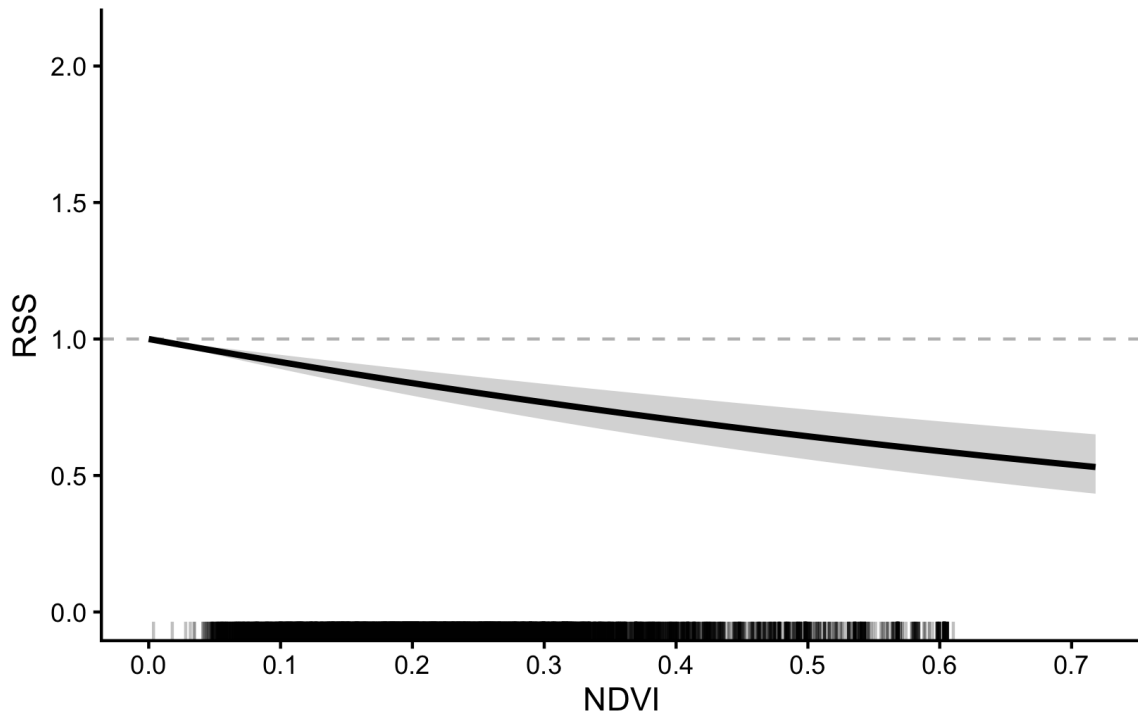
# calculate the RSS for each NDVI value
ndvi_coef_df <- data.frame(
  ndvi_values = ndvi_values,
  log_rss = ndvi_values * ndvi_coef,
  rss = exp(ndvi_values * ndvi_coef),
  se_prediction = abs(ndvi_values) * ndvi_coef_se # SE of prediction at each NDVI value
)

# plot the response curve
ggplot(data = ndvi_coef_df) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey") +
  geom_ribbon(aes(x = ndvi_values, ymin = exp(log_rss - 1.96 * se_prediction), ymax = exp(log_rss + 1.96 * se_prediction)),
    fill = "grey70", alpha = 0.5) +
  geom_line(aes(x = ndvi_values, y = rss),
    colour = "black", linewidth = 1) +
  geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
```

```

aes(x = ndvi), alpha = 0.25, sides="b") +
scale_x_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
scale_y_continuous(limits = c(rss_min, rss_max)) +
labs(x = "NDVI", y = "RSS") +
theme_classic()

```



```

ggsave(paste0(plot_save_path, "/ndvi_linear_response_id", which_buffalo, ".png"),
       width = 120, height = 100, units = "mm", dpi = 600)

```

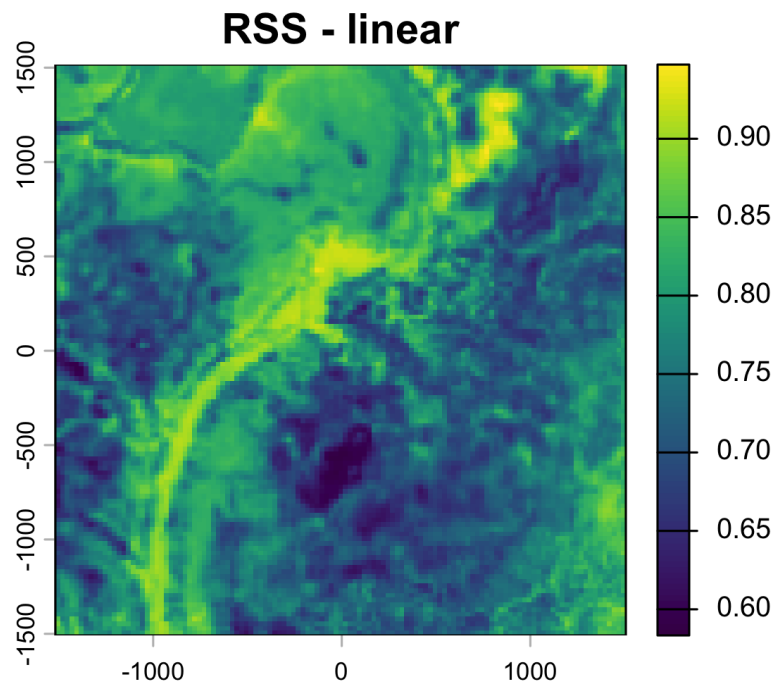
Plot habitat selection

```

# calculate the RSS for the local NDVI values
ndvi_linear <- exp(ndvi_window * ndvi_coef)

# plot the habitat selection (spatial RSS)
plot(ndvi_linear, main = "RSS - linear")

```

```
# Save the plot
png(filename = paste0(plot_save_path, "/rss_linear_", which_buffalo, ".png"),
     width = 100, height = 80, units = "mm", res = 600)
plot(ndvi_linear, main = "RSS - linear", range = c(0.5, 1.75))
dev.off()
```

```
quartz_off_screen
2
```

Quadratic covariate

Model formula

$$y \sim ndvi + ndvi^2$$

Fit model

```
# fit an issf model
ssf_quadratic <- amt::fit_issf(case_ ~
                               ndvi +
```

```

        ndvi_sq +
        sl_ + log_sl + cos_ta +
        strata(step_id_),
        data = buffalo_ssf_data)

summary(ssf_quadratic)

```

Call:

```
coxph(formula = Surv(rep(1, 106678L), case_) ~ ndvi + ndvi_sq +
      sl_ + log_sl + cos_ta + strata(step_id_), data = data, method = "exact")
```

n= 106678, number of events= 9698

	coef	exp(coef)	se(coef)	z	Pr(> z)
ndvi	4.455e+00	8.602e+01	4.973e-01	8.958	<2e-16 ***
ndvi_sq	-8.924e+00	1.332e-04	8.122e-01	-10.988	<2e-16 ***
sl_	-1.610e-05	1.000e+00	3.183e-05	-0.506	0.613
log_sl	3.391e-03	1.003e+00	5.720e-03	0.593	0.553
cos_ta	-4.258e-03	9.958e-01	1.504e-02	-0.283	0.777

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
ndvi	8.602e+01	1.163e-02	3.245e+01	2.280e+02
ndvi_sq	1.332e-04	7.510e+03	2.711e-05	6.542e-04
sl_	1.000e+00	1.000e+00	9.999e-01	1.000e+00
log_sl	1.003e+00	9.966e-01	9.922e-01	1.015e+00
cos_ta	9.958e-01	1.004e+00	9.668e-01	1.026e+00

Concordance= 0.53 (se = 0.004)

Likelihood ratio test= 171 on 5 df, p=<2e-16

Wald test = 142.3 on 5 df, p=<2e-16

Score (logrank) test = 142.2 on 5 df, p=<2e-16

```
AIC(ssf_quadratic)
```

```
[1] 46348.59
```

```

# fit an ssf gam model with parametric terms. -ends up being the same as above,
# but allows us to use the `predict` function to get predictions on the log-scale with stand
# which is a bit more work to do with the `amt` model object.
ssf_quadratic_gam <- mgcv::gam(cbind(times, step_id_) ~
                               ndvi +

```

```

        ndvi_sq +
        sl_ + log_sl + cos_ta,
        data = buffalo_ssf_data,
        family = cox.ph,
        weight = case_)

summary(ssf_quadratic_gam)

```

Family: Cox PH

Link function: identity

Formula:

cbind(times, step_id_) ~ ndvi + ndvi_sq + sl_ + log_sl + cos_ta

Parametric coefficients:

	Estimate	Std. Error	z value	Pr(> z)
ndvi	4.455e+00	4.973e-01	8.958	<2e-16 ***
ndvi_sq	-8.924e+00	8.122e-01	-10.988	<2e-16 ***
sl_	-1.610e-05	3.183e-05	-0.506	0.613
log_sl	3.391e-03	5.720e-03	0.593	0.553
cos_ta	-4.258e-03	1.504e-02	-0.283	0.777

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Deviance explained = -0.0158%

-REML = 23187 Scale est. = 1 n = 106678

```
AIC(ssf_quadratic_gam)
```

```
[1] 46348.59
```

Generate the response curve

```

# Get predictions on log-scale with standard errors
predictions <- predict(ssf_quadratic_gam,
                        newdata = pred_data,
                        type = "link", # log-scale
                        se.fit = TRUE)

```

```

# Extract log RSS and standard errors
log_rss <- predictions$fit
se_log_rss <- predictions$se.fit

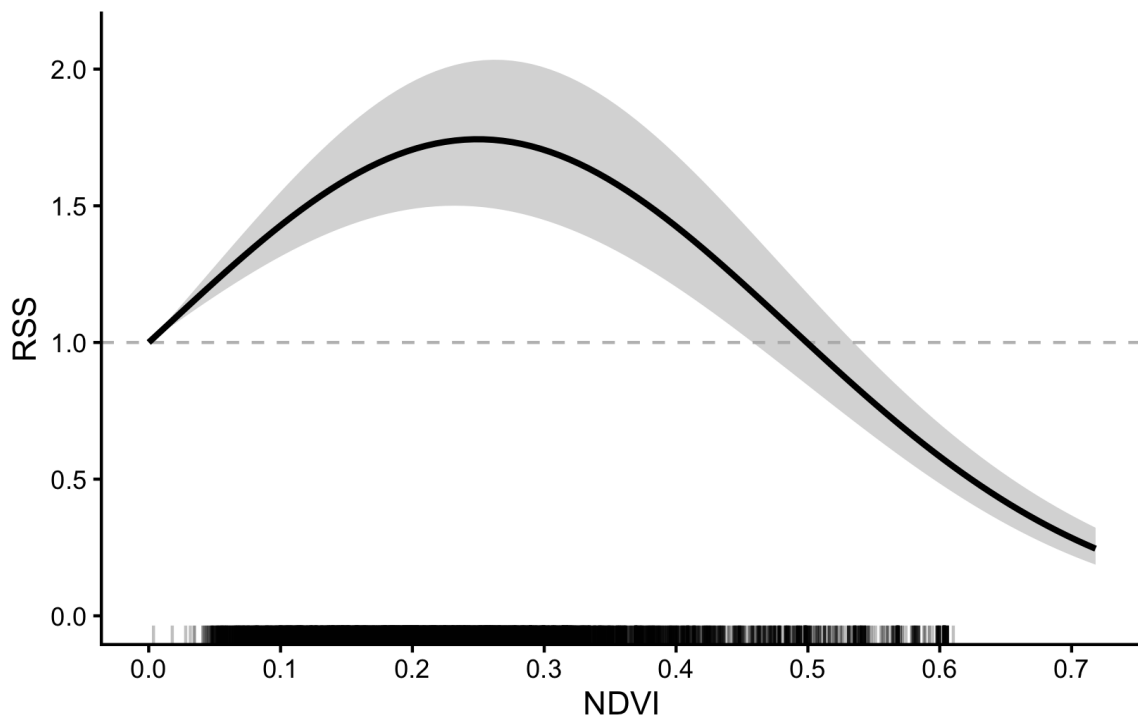
# Calculate confidence intervals on log-scale
log_rss_lower <- log_rss - 1.96 * se_log_rss
log_rss_upper <- log_rss + 1.96 * se_log_rss

# Transform to RSS scale
rss <- exp(log_rss)
rss_lower <- exp(log_rss_lower)
rss_upper <- exp(log_rss_upper)

# Create your dataframe
ndvi_coef_df <- data.frame(
  ndvi_values = ndvi_values,
  log_rss = log_rss,
  log_rss_lower = log_rss_lower,
  log_rss_upper = log_rss_upper,
  rss = rss,
  rss_lower = rss_lower,
  rss_upper = rss_upper
)

# plot the response curve
ggplot(data = ndvi_coef_df) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey") +
  geom_ribbon(aes(x = ndvi_values, ymin = rss_lower, ymax = rss_upper),
    fill = "grey70", alpha = 0.5) +
  geom_line(aes(x = ndvi_values, y = rss),
    colour = "black", linewidth = 1) +
  geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
    aes(x = ndvi), alpha = 0.25, sides="b") +
  scale_x_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  scale_y_continuous(limits = c(rss_min, rss_max)) +
  labs(x = "NDVI", y = "RSS") +
  theme_classic()

```



```
ggsave(paste0(plot_save_path, "/ndvi_quadratic_response_id", which_buffalo, ".png"),
       width = 120, height = 100, units = "mm", dpi = 600)
```

Response curve

```
# extract the NDVI coefficient
ndvi_coef <- ssf_quadratic$model$coefficients[1]
ndvi_coef_sq <- ssf_quadratic$model$coefficients[2]

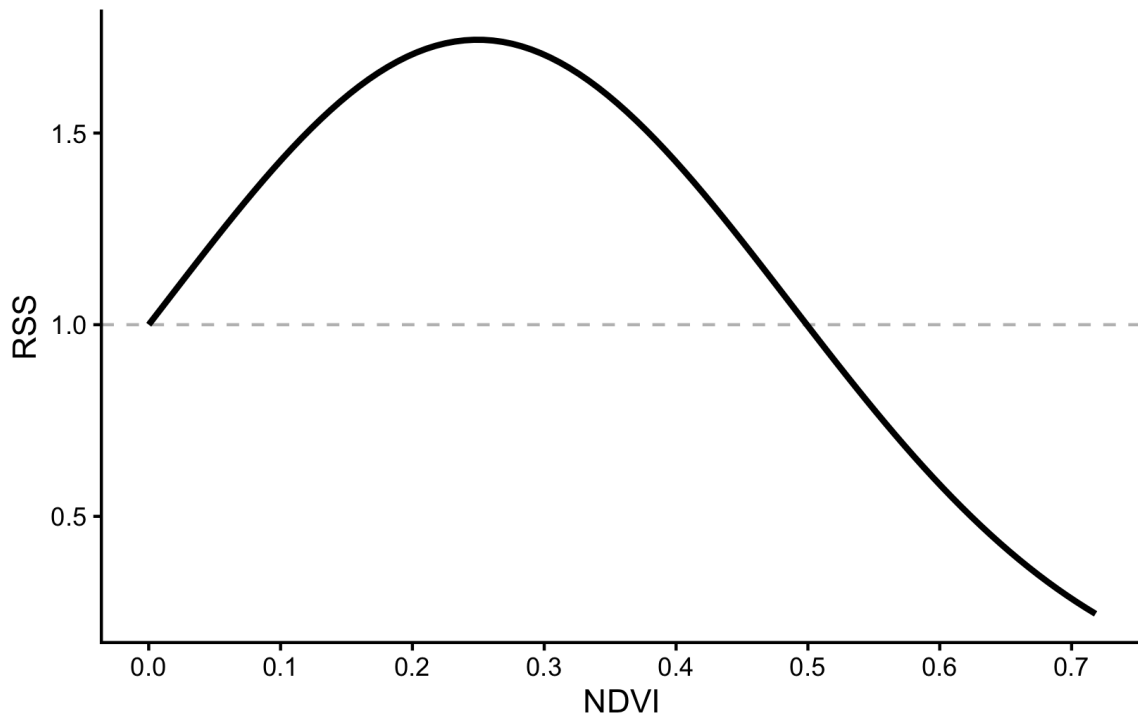
ndvi_se = sqrt(diag(ssf_quadratic$model$var))[1]
ndvi_sq_se = sqrt(diag(ssf_quadratic$model$var))[2]

log_rss = (ndvi_values * ndvi_coef) + (ndvi_values^2 * ndvi_coef_sq)
rss = exp(log_rss)

# calculate the RSS for each NDVI value
ndvi_coef_df <- data.frame(ndvi_values, log_rss, rss)

# plot the response curve
ggplot() +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey") +
```

```
geom_line(data = ndvi_coef_df, aes(x = ndvi_values, y = rss),
          colour = "black", linewidth = 1) +
scale_x_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
labs(x = "NDVI", y = "RSS") +
theme_classic()
```



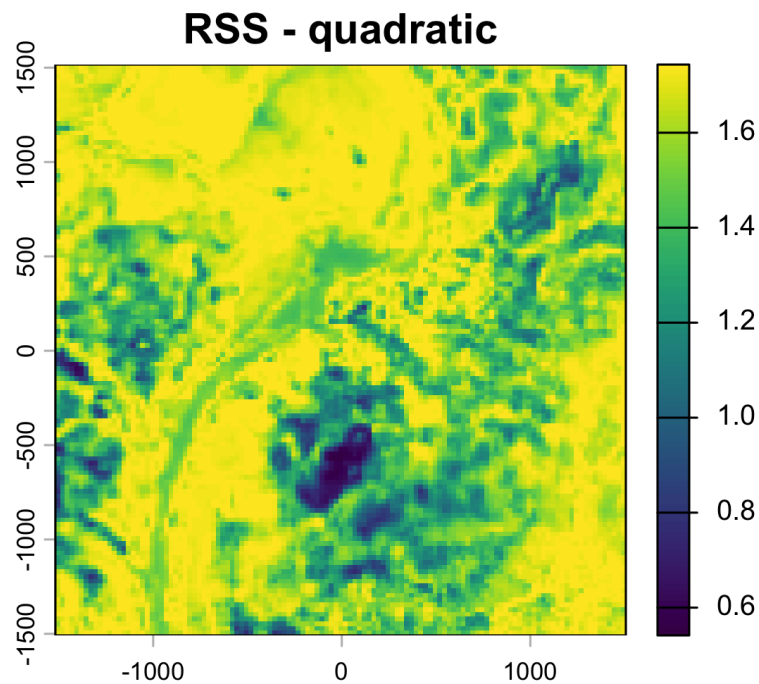
```
# ggsave(paste0(plot_save_path, "/ndvi_quadratic_response_id", which_buffalo, ".png"),
#         width = 100, height = 100, units = "mm", dpi = 600)
```

Plot habitat selection

```
# extract the NDVI coefficient
ndvi_coef <- ssf_quadratic$model$coefficients[1]
ndvi_coef_sq <- ssf_quadratic$model$coefficients[2]

# calculate the RSS for the local NDVI values
ndvi_quadratic <- exp((ndvi_window * ndvi_coef) +
                     (ndvi_window^2 * ndvi_coef_sq))

# plot the habitat selection (spatial RSS)
plot(ndvi_quadratic, main = "RSS - quadratic")
```



```
# Save the plot
png(filename = paste0(plot_save_path, "/rss_quadratic_", which_buffalo, ".png"),
     width = 100, height = 80, units = "mm", res = 600)
plot(ndvi_quadratic, main = "RSS - quadratic", range = c(0.5, 1.75))
dev.off()
```

```
quartz_off_screen
2
```

Smooth term

Fit a GAM to the data, with a smooth term across the range of NDVI values (Klappstein et al. 2024).

Model formula

$$y \sim s(ndvi)$$

Fit model

```
# fit an ssf gam model
ssf_gam <- mgcv::gam(cbind(times, step_id_) ~
  s(ndvi, bs = "cr") +
  # s(ndvi) +
  sl_ + log_sl + cos_ta,
  data = buffalo_ssf_data,
  family = cox.ph,
  weight = case_)

summary(ssf_gam)
```

Family: Cox PH

Link function: identity

Formula:

```
cbind(times, step_id_) ~ s(ndvi, bs = "cr") + sl_ + log_sl +
  cos_ta
```

Parametric coefficients:

	Estimate	Std. Error	z value	Pr(> z)
sl_	-4.585e-06	3.181e-05	-0.144	0.885
log_sl	8.339e-03	5.769e-03	1.445	0.148
cos_ta	-5.143e-03	1.505e-02	-0.342	0.733

Approximate significance of smooth terms:

	edf	Ref.df	Chi.sq	p-value
s(ndvi)	8.448	8.863	386.8	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Deviance explained = 0.397%

-REML = 23088 Scale est. = 1 n = 106678

```
AIC(ssf_gam)
```

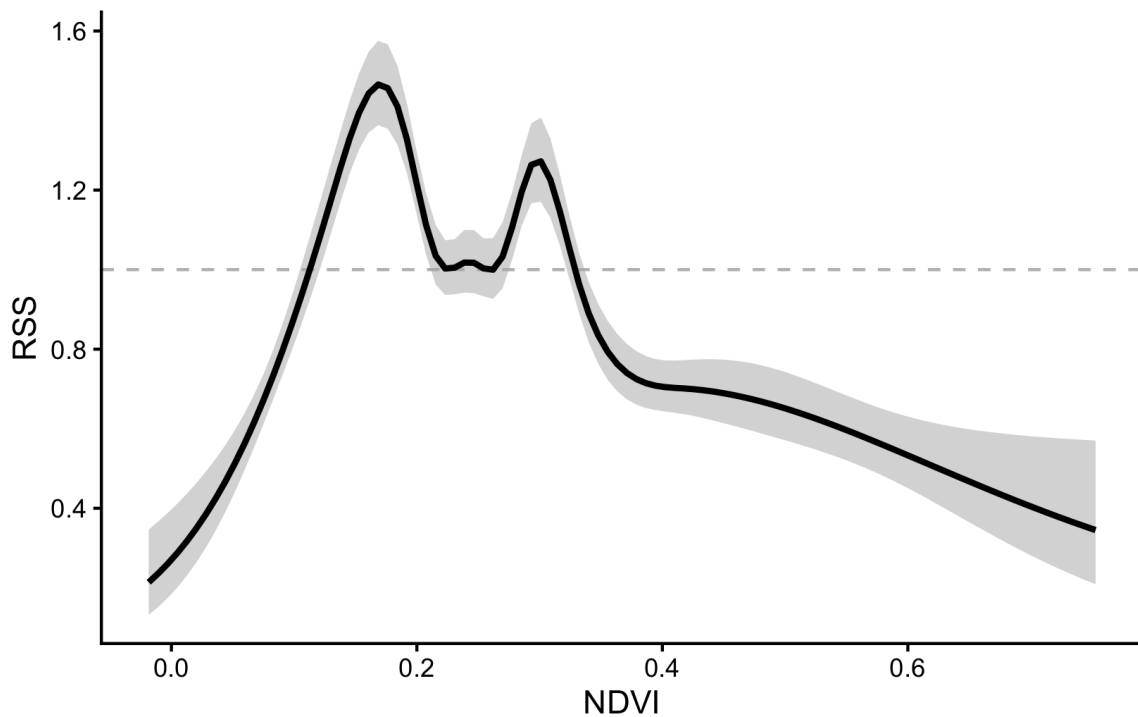
```
[1] 46119.42
```

```
# coef(ssf_gam)[4]
```

```
gam_smooths <- gratia::smooth_estimates(ssf_gam)
```


Using ggplot with the `smooth_estimates` dataframe from `gratia`

```
ggplot(data = gam_smooths) +  
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey") +  
  geom_ribbon(aes(x = ndvi, ymin = exp(.estimate - 1.96 * .se), ymax = exp(.estimate + 1.96 *  
    .se), fill = "grey70", alpha = 0.5) +  
  geom_line(aes(x = ndvi, y = exp(.estimate)),  
    colour = "black", linewidth = 1) +  
  labs(x = "NDVI", y = "RSS") +  
  # scale_x_continuous(limits = c(ndvi_quantiles[1], 0.5)) +  
  # scale_y_continuous(limits = c(0.65, 1.3)) +  
  theme_classic()
```



```
ggsave(paste0(plot_save_path, "/ndvi_smooth_response_id", which_buffalo, ".png"),  
  width = 120, height = 100, units = "mm", dpi = 600)
```

Alternatively, generate the response curve with the `predict` function

```
# Get predictions on log-scale with standard errors  
predictions <- predict(ssf_gam,  
  newdata = pred_data,
```

```

        type = "link", # log-scale
        se.fit = TRUE)

# Extract log RSS and standard errors
log_rss <- predictions$fit
se_log_rss <- predictions$se.fit

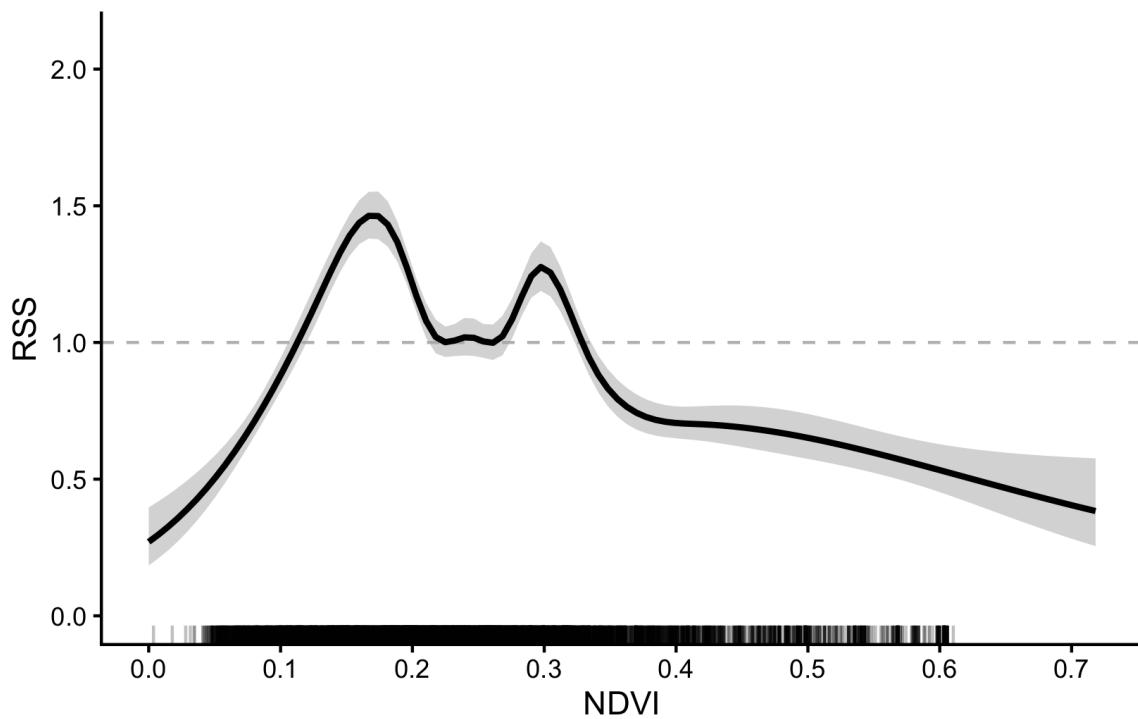
# Calculate confidence intervals on log-scale
log_rss_lower <- log_rss - 1.96 * se_log_rss
log_rss_upper <- log_rss + 1.96 * se_log_rss

# Transform to RSS scale
rss <- exp(log_rss)
rss_lower <- exp(log_rss_lower)
rss_upper <- exp(log_rss_upper)

# Create your dataframe
ndvi_coef_df <- data.frame(
  ndvi_values = ndvi_values,
  log_rss = log_rss,
  log_rss_lower = log_rss_lower,
  log_rss_upper = log_rss_upper,
  rss = rss,
  rss_lower = rss_lower,
  rss_upper = rss_upper
)

# plot the response curve
ggplot(data = ndvi_coef_df) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey") +
  geom_ribbon(aes(x = ndvi_values, ymin = rss_lower, ymax = rss_upper),
    fill = "grey70", alpha = 0.5) +
  geom_line(aes(x = ndvi_values, y = rss),
    colour = "black", linewidth = 1) +
  geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
    aes(x = ndvi), alpha = 0.25, sides="b") +
  scale_x_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  scale_y_continuous(limits = c(rss_min, rss_max)) +
  labs(x = "NDVI", y = "RSS") +
  theme_classic()

```



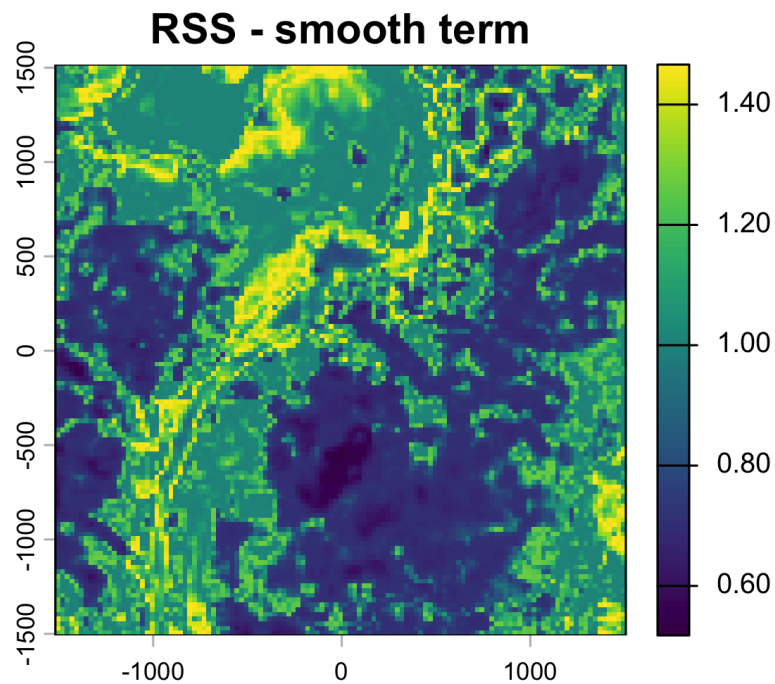
```
ggsave(paste0(plot_save_path, "/ndvi_smooth_response_id", which_buffalo, ".png"),
       width = 120, height = 100, units = "mm", dpi = 600)
```

Plot habitat selection

```
predicted_log_rss <- predict(ssf_gam,
                           newdata = data.frame(ndvi = values(ndvi_window),
                                                  sl_ = 0,
                                                  log_sl = 0,
                                                  cos_ta = 0),,
                           type = "link")

ndvi_gam <- ndvi_window
values(ndvi_gam) <- exp(as.vector(predicted_log_rss))

# Plot the predictions
plot(ndvi_gam, main = "RSS - smooth term")
```



```
# Save the plot
png(filename = paste0(plot_save_path, "/rss_smooth_", which_buffalo, ".png"),
     width = 100, height = 80, units = "mm", res = 600)
plot(ndvi_gam, main = "RSS - smooth", range = c(0.5, 1.75))
dev.off()
```

```
quartz_off_screen
2
```

Temporal dynamics

Static model for comparison

```
ssf_static <- amt::fit_issf(case_ ~
                           ndvi + sl_ + log_sl + cos_ta +
                           strata(step_id_),
                           data = buffalo_ssf_data)

summary(ssf_static)
```

Call:

```
coxph(formula = Surv(rep(1, 106678L), case_) ~ ndvi + sl_ + log_sl +  
      cos_ta + strata(step_id_), data = data, method = "exact")
```

n= 106678, number of events= 9698

	coef	exp(coef)	se(coef)	z	Pr(> z)
ndvi	-8.813e-01	4.143e-01	1.443e-01	-6.105	1.03e-09 ***
sl_	1.147e-06	1.000e+00	3.174e-05	0.036	0.971
log_sl	-2.394e-04	9.998e-01	5.688e-03	-0.042	0.966
cos_ta	-4.305e-03	9.957e-01	1.503e-02	-0.287	0.774

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
ndvi	0.4143	2.414	0.3122	0.5497
sl_	1.0000	1.000	0.9999	1.0001
log_sl	0.9998	1.000	0.9887	1.0110
cos_ta	0.9957	1.004	0.9668	1.0255

Concordance= 0.494 (se = 0.004)

Likelihood ratio test= 38.92 on 4 df, p=7e-08

Wald test = 37.62 on 4 df, p=1e-07

Score (logrank) test = 37.62 on 4 df, p=1e-07

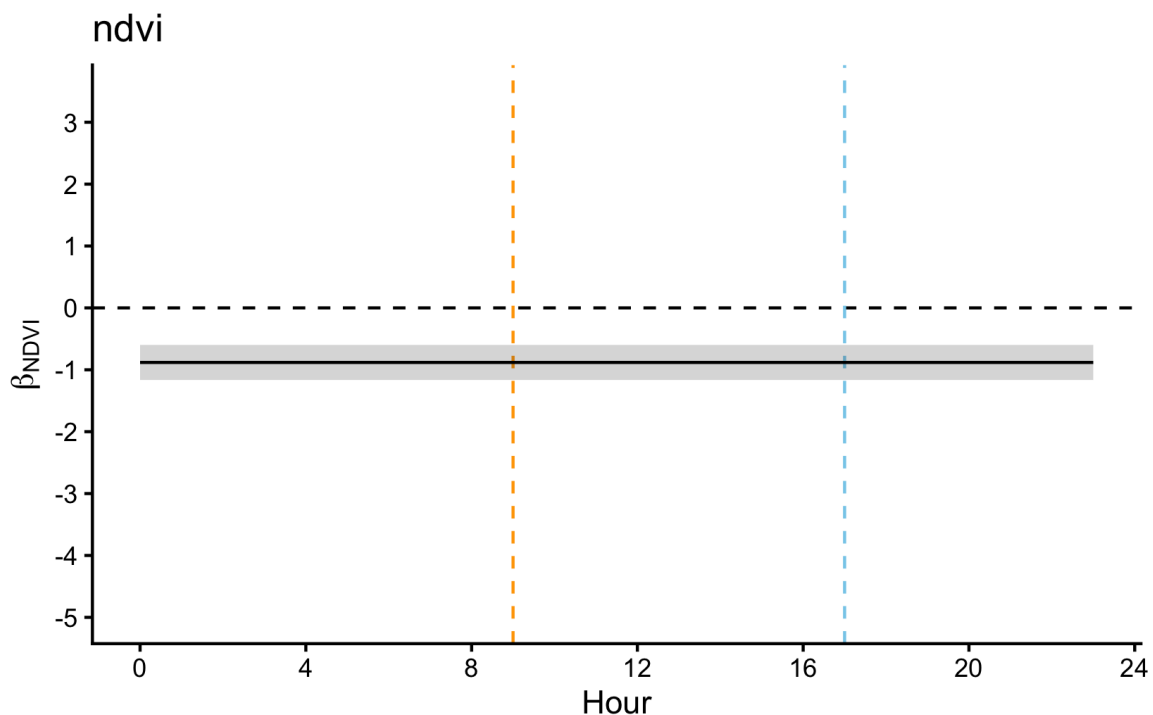
```
AIC(ssf_static)
```

```
[1] 46478.66
```

```
ndvi_coef <- ssf_static$model$coefficients["ndvi"]  
ndvi_se <- sqrt(diag(ssf_static$model$var))[1] # SE of coefficient  
  
ndvi_static_coef_df <- data.frame(hour = seq(0, 23, length.out = 231),  
                                   ndvi_coef = ndvi_coef,  
                                   ndvi_se = ndvi_se,  
                                   ndvi_lo = ndvi_coef - 1.96 * ndvi_se,  
                                   ndvi_hi = ndvi_coef + 1.96 * ndvi_se)
```

Warning in data.frame(hour = seq(0, 23, length.out = 231), ndvi_coef =
ndvi_coef, : row names were found from a short variable and have been discarded

```
# Plot with ribbon
ggplot(ndvi_static_coef_df, aes(x = hour)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "black") +
  geom_vline(xintercept = 9, linetype = "dashed", colour = "orange") +
  geom_vline(xintercept = 17, linetype = "dashed", colour = "skyblue") +
  geom_ribbon(aes(ymin = ndvi_lo, ymax = ndvi_hi), alpha = 0.2) +
  geom_line(aes(y = ndvi_coef)) +
  labs(title = "ndvi", x = "Hour", y = expression(beta[NDVI]), fill = NULL) +
  scale_x_continuous(breaks = seq(0, 24, by = 4)) +
  scale_y_continuous(limits = c(-5, 3.5), breaks = seq(-5, 4, by = 1)) +
  theme_classic()
```



```
ggsave(paste0(plot_save_path, "/ndvi_static_coef_response_id", which_buffalo, ".png"),
        width = 150, height = 80, units = "mm", dpi = 600)
```

Pull out a response curve from one of the hours

```
sample_hours <- c(9, 17)

ndvi_coef_df <- purrr::map_dfr(sample_hours, function(sample_hour) {
```

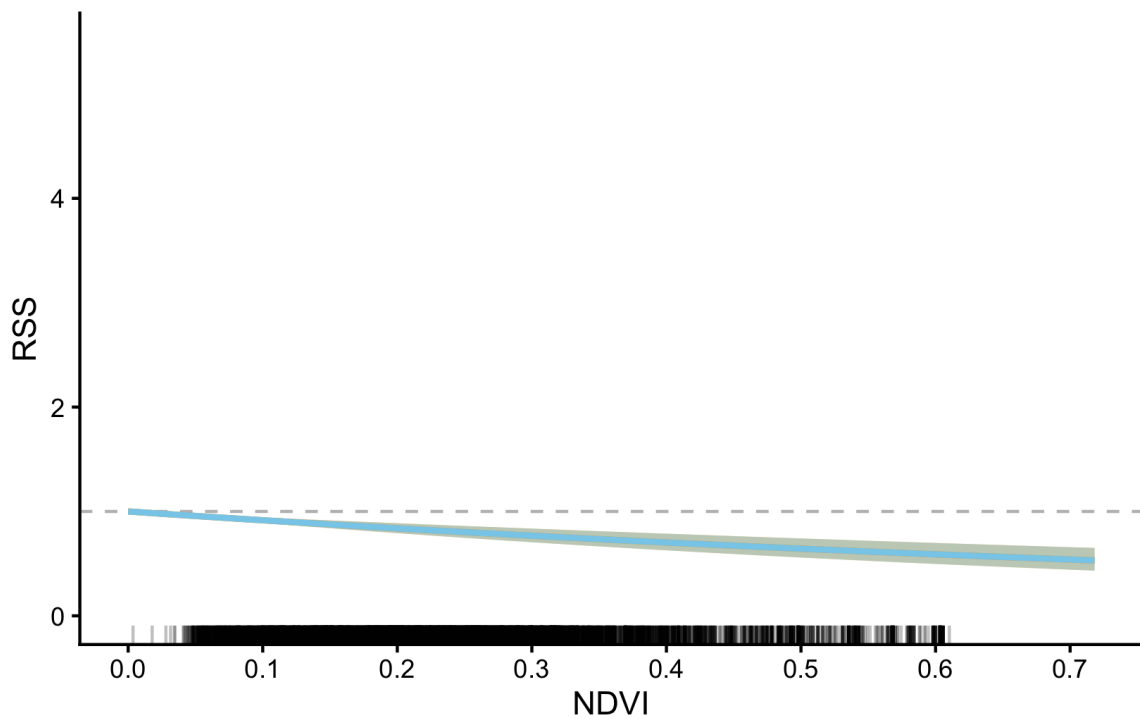
```

# extract the NDVI coefficient and its standard error
ndvi_coef <- ndvi_static_coef_df %>% filter(hour == sample_hour) %>% pull(ndvi_coef)
ndvi_coef_se <- ndvi_static_coef_df %>% filter(hour == sample_hour) %>% pull(ndvi_se)

# calculate the RSS for each NDVI value
data.frame(
  ndvi_values = ndvi_values,
  log_rss = ndvi_values * ndvi_coef,
  rss = exp(ndvi_values * ndvi_coef),
  se_prediction = abs(ndvi_values) * ndvi_coef_se
) %>% mutate(hour = as.factor(sample_hour), .before = ndvi_values)
})

# plot the response curve
ggplot(data = ndvi_coef_df) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey") +
  geom_ribbon(aes(x = ndvi_values,
    ymin = exp(log_rss - 1.96 * se_prediction),
    ymax = exp(log_rss + 1.96 * se_prediction),
    fill = hour),
    alpha = 0.5) +
  geom_line(aes(x = ndvi_values, y = rss, colour = hour),
    linewidth = 1) +
  geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
    aes(x = ndvi), alpha = 0.25, sides="b") +
  scale_colour_manual(values = c("orange", "skyblue")) +
  scale_fill_manual(values = c("orange", "skyblue")) +
  scale_x_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  scale_y_continuous(limits = c(0, 5.5)) +
  labs(x = "NDVI", y = "RSS") +
  theme_classic() +
  theme(legend.position = "none")

```



```
ggsave(paste0(plot_save_path, "/ndvi_static_response_sample_hour_id", which_buffalo, ".png"),
       width = 70, height = 80, units = "mm", dpi = 600)
```

Fit harmonic model with quadratic terms

```
ssf_static_quadratic <- amt::fit_issf(case_ ~
  # linear term
  ndvi +
  # quadratic terms
  ndvi_sq +
  sl_ + log_sl + cos_ta +
  strata(step_id_),
  data = buffalo_ssf_data)

summary(ssf_static_quadratic)
```

Call:

```
coxph(formula = Surv(rep(1, 106678L), case_) ~ ndvi + ndvi_sq +
  sl_ + log_sl + cos_ta + strata(step_id_), data = data, method = "exact")
```

n= 106678, number of events= 9698

	coef	exp(coef)	se(coef)	z	Pr(> z)
ndvi	4.455e+00	8.602e+01	4.973e-01	8.958	<2e-16 ***
ndvi_sq	-8.924e+00	1.332e-04	8.122e-01	-10.988	<2e-16 ***
sl_	-1.610e-05	1.000e+00	3.183e-05	-0.506	0.613
log_sl	3.391e-03	1.003e+00	5.720e-03	0.593	0.553
cos_ta	-4.258e-03	9.958e-01	1.504e-02	-0.283	0.777

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
ndvi	8.602e+01	1.163e-02	3.245e+01	2.280e+02
ndvi_sq	1.332e-04	7.510e+03	2.711e-05	6.542e-04
sl_	1.000e+00	1.000e+00	9.999e-01	1.000e+00
log_sl	1.003e+00	9.966e-01	9.922e-01	1.015e+00
cos_ta	9.958e-01	1.004e+00	9.668e-01	1.026e+00

Concordance= 0.53 (se = 0.004)
Likelihood ratio test= 171 on 5 df, p=<2e-16
Wald test = 142.3 on 5 df, p=<2e-16
Score (logrank) test = 142.2 on 5 df, p=<2e-16

```
AIC(ssf_static_quadratic)
```

```
[1] 46348.59
```

Create table of NDVI and hour values

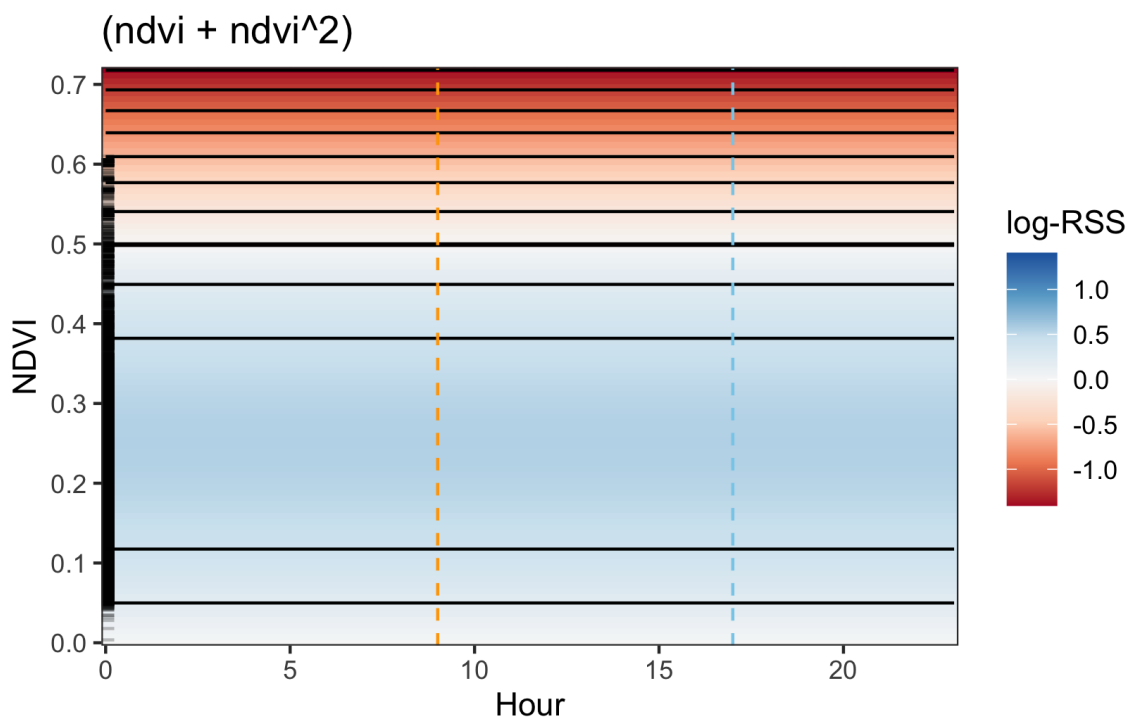
```
ndvi_hour_values <- expand.grid(ndvi = ndvi_values,
                                hour = seq(0, 23, length.out = length(ndvi_values)))

ndvi_lin_coef <- ssf_static_quadratic$model$coefficients["ndvi"]
ndvi_quad_coef <- ssf_static_quadratic$model$coefficients["ndvi_sq"]

ndvi_hour_values <- ndvi_hour_values %>% mutate(
  ndvi_lin_coef = ndvi_lin_coef,
  ndvi_quad_coef = ndvi_quad_coef,
  ssf_quadratic_log_rss = (ndvi * ndvi_lin_coef) + (ndvi^2 * ndvi_quad_coef)
)

max_abs_colour <- max(abs(range(ndvi_hour_values$ssf_quadratic_log_rss)))
```

```
# plot the selection surface
ggplot() +
  geom_raster(data = ndvi_hour_values,
    aes(x = hour, y = ndvi, fill = ssf_quadratic_log_rss)) +
  labs(title = "(ndvi + ndvi^2)", x = "Hour", y = "NDVI", fill = "log-RSS") +
  geom_contour(data = ndvi_hour_values,
    aes(x = hour, y = ndvi, z = ssf_quadratic_log_rss),
    colour = "black") +
  geom_contour(data = ndvi_hour_values,
    aes(x = hour, y = ndvi, z = ssf_quadratic_log_rss),
    breaks = 0, colour = "black", linetype = "solid", linewidth = 0.75) +
  geom_vline(xintercept = 9, linetype = "dashed", colour = "orange") +
  geom_vline(xintercept = 17, linetype = "dashed", colour = "skyblue") +
  scale_fill_distiller(palette = "RdBu", type = "div", direction = 1,
    limits = c(-max_abs_colour, max_abs_colour)) +
  coord_cartesian(expand = FALSE) +
  geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
    aes(y = ndvi), alpha = 0.25, length = unit(0.015, "npc"), sides="l") +
  scale_y_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  theme_bw()
```



```
ggsave(paste0(plot_save_path, "/ndvi_static_quadratic_response_id", which_buffalo, ".png"),
       width = 150, height = 80, units = "mm", dpi = 600)
```

Easier to get the curves with GAM predict function

Fit model

```
# fit an ssf gam model with parametric terms. -ends up being the same as above,
# but allows us to use the `predict` function to get predictions on the log-scale with stand
# which is a bit more work to do with the `amt` model object.
ssf_quadratic_gam <- mgcv::gam(cbind(times, step_id_) ~
                               ndvi +
                               ndvi_sq +
                               sl_ + log_sl + cos_ta,
                               data = buffalo_ssf_data,
                               family = cox.ph,
                               weight = case_)

summary(ssf_quadratic_gam)
```

Family: Cox PH

Link function: identity

Formula:

cbind(times, step_id_) ~ ndvi + ndvi_sq + sl_ + log_sl + cos_ta

Parametric coefficients:

	Estimate	Std. Error	z value	Pr(> z)
ndvi	4.455e+00	4.973e-01	8.958	<2e-16 ***
ndvi_sq	-8.924e+00	8.122e-01	-10.988	<2e-16 ***
sl_	-1.610e-05	3.183e-05	-0.506	0.613
log_sl	3.391e-03	5.720e-03	0.593	0.553
cos_ta	-4.258e-03	1.504e-02	-0.283	0.777

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Deviance explained = -0.0158%

-REML = 23187 Scale est. = 1

n = 106678

```
AIC(ssf_quadratic_gam)
```

```
[1] 46348.59
```

Generate the response curve

```
# Get predictions on log-scale with standard errors
predictions <- predict(ssf_quadratic_gam,
                      newdata = pred_data,
                      type = "link", # log-scale
                      se.fit = TRUE)

# Extract log RSS and standard errors
log_rss <- predictions$fit
se_log_rss <- predictions$se.fit

# Calculate confidence intervals on log-scale
log_rss_lower <- log_rss - 1.96 * se_log_rss
log_rss_upper <- log_rss + 1.96 * se_log_rss

# Transform to RSS scale
rss <- exp(log_rss)
rss_lower <- exp(log_rss_lower)
rss_upper <- exp(log_rss_upper)

# Create your dataframe
ndvi_coef_df <- data.frame(
  ndvi_values = ndvi_values,
  log_rss = log_rss,
  log_rss_lower = log_rss_lower,
  log_rss_upper = log_rss_upper,
  rss = rss,
  rss_lower = rss_lower,
  rss_upper = rss_upper
)

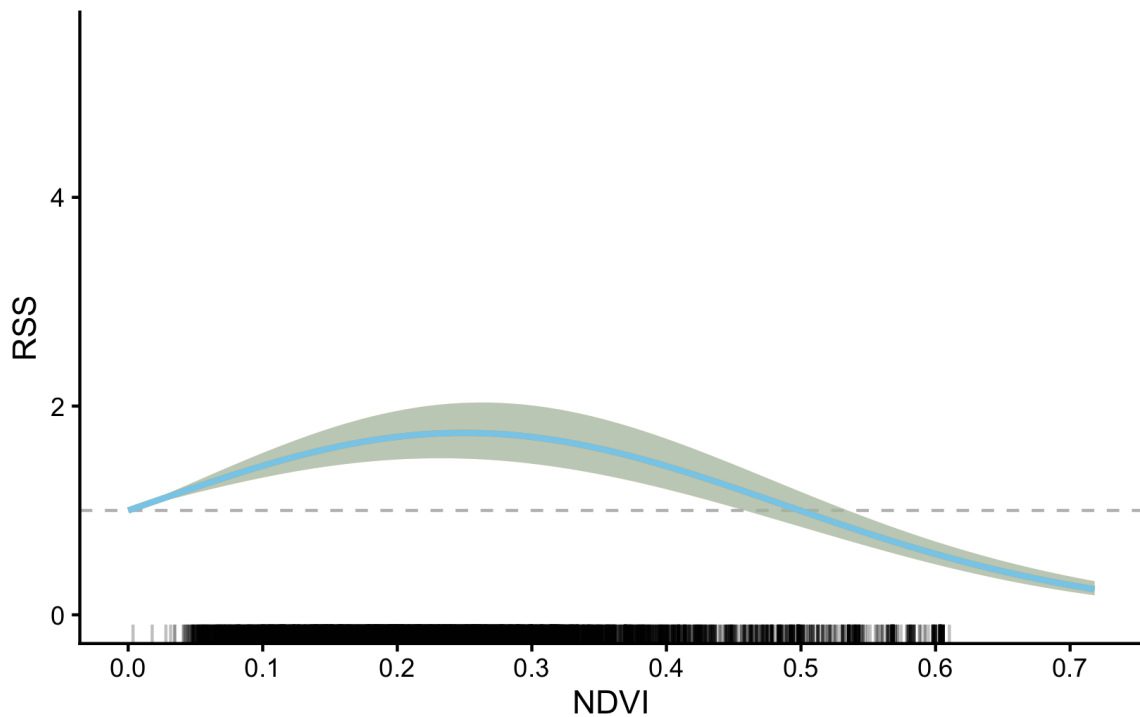
ndvi_coef_df <- rbind(data.frame(hour = 9, ndvi_coef_df), data.frame(hour = 17, ndvi_coef_df))

# plot the response curve
ggplot(data = ndvi_coef_df) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey") +
  geom_ribbon(aes(x = ndvi_values, ymin = rss_lower, ymax = rss_upper, fill = as.factor(hour))
```

```

    alpha = 0.5) +
  geom_line(aes(x = ndvi_values, y = rss, colour = as.factor(hour)),
    linewidth = 1) +
  geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
    aes(x = ndvi), alpha = 0.25, sides="b") +
  scale_colour_manual(values = c("orange", "skyblue")) +
  scale_fill_manual(values = c("orange", "skyblue")) +
  scale_x_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  scale_y_continuous(limits = c(0, 5.5)) +
  labs(x = "NDVI", y = "RSS") +
  theme_classic() +
  theme(legend.position = "none")

```



```

ggsave(paste0(plot_save_path, "/ndvi_static_quadratic_response_sample_hour_id", which_buffal
  width = 70, height = 80, units = "mm", dpi = 600)

```

Harmonic regression

```

buffalo_ssf_data <- buffalo_ssf_data %>% mutate(

```

```

# create hour term
hour = lubridate::hour(t1_),

# harmonic terms
sin_1_hour = sin(2 * pi * hour / 24),
cos_1_hour = cos(2 * pi * hour / 24),
sin_2_hour = sin(4 * pi * hour / 24),
cos_2_hour = cos(4 * pi * hour / 24),

)

```

Fit model

We'll use the `fit_issf` function from `amt`, with manually specified harmonic terms.

```

ssf_harmonics <- amt::fit_issf(case_ ~
                                ndvi +
                                ndvi:sin_1_hour + ndvi:cos_1_hour +
                                ndvi:sin_2_hour + ndvi:cos_2_hour +
                                sl_ + log_sl + cos_ta +
                                strata(step_id_),
                                data = buffalo_ssf_data)

summary(ssf_harmonics)

```

Call:

```

coxph(formula = Surv(rep(1, 106678L), case_) ~ ndvi + ndvi:sin_1_hour +
        ndvi:cos_1_hour + ndvi:sin_2_hour + ndvi:cos_2_hour + sl_ +
        log_sl + cos_ta + strata(step_id_), data = data, method = "exact")

```

n= 106678, number of events= 9698

	coef	exp(coef)	se(coef)	z	Pr(> z)	
ndvi	-9.534e-01	3.854e-01	1.474e-01	-6.467	9.97e-11	***
sl_	-9.277e-06	1.000e+00	3.189e-05	-0.291	0.77113	
log_sl	4.017e-04	1.000e+00	5.699e-03	0.070	0.94381	
cos_ta	-4.982e-03	9.950e-01	1.503e-02	-0.331	0.74036	
ndvi:sin_1_hour	1.195e+00	3.304e+00	2.165e-01	5.520	3.38e-08	***
ndvi:cos_1_hour	-7.601e-01	4.676e-01	1.976e-01	-3.846	0.00012	***
ndvi:sin_2_hour	-1.063e+00	3.453e-01	2.066e-01	-5.148	2.64e-07	***
ndvi:cos_2_hour	8.069e-01	2.241e+00	2.074e-01	3.891	9.97e-05	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
ndvi	0.3854	2.5946	0.2887	0.5145
sl_	1.0000	1.0000	0.9999	1.0001
log_sl	1.0004	0.9996	0.9893	1.0116
cos_ta	0.9950	1.0050	0.9661	1.0248
ndvi:sin_1_hour	3.3036	0.3027	2.1613	5.0495
ndvi:cos_1_hour	0.4676	2.1385	0.3174	0.6889
ndvi:sin_2_hour	0.3453	2.8963	0.2303	0.5176
ndvi:cos_2_hour	2.2410	0.4462	1.4925	3.3648

Concordance= 0.521 (se = 0.004)

Likelihood ratio test= 122.8 on 8 df, p=<2e-16

Wald test = 115.8 on 8 df, p=<2e-16

Score (logrank) test = 115 on 8 df, p=<2e-16

```
AIC(ssf_harmonics)
```

```
[1] 46402.8
```

Pull out the coefficient across time

```
ndvi_harmonic_coef_reconstruction <- function(ssf_harmonic_model, hour) {

  ndvi_coef <- ssf_harmonic_model$model$coefficients["ndvi"]
  ndvi_sin_1_coef <- ssf_harmonic_model$model$coefficients["ndvi:sin_1_hour"]
  ndvi_cos_1_coef <- ssf_harmonic_model$model$coefficients["ndvi:cos_1_hour"]
  ndvi_sin_2_coef <- ssf_harmonic_model$model$coefficients["ndvi:sin_2_hour"]
  ndvi_cos_2_coef <- ssf_harmonic_model$model$coefficients["ndvi:cos_2_hour"]

  sin_1_hour <- sin(2 * pi * hour / 24)
  cos_1_hour <- cos(2 * pi * hour / 24)
  sin_2_hour <- sin(4 * pi * hour / 24)
  cos_2_hour <- cos(4 * pi * hour / 24)

  coef <- as.numeric(ndvi_coef +
    sin_1_hour * ndvi_sin_1_coef +
    cos_1_hour * ndvi_cos_1_coef +
    sin_2_hour * ndvi_sin_2_coef +
    cos_2_hour * ndvi_cos_2_coef)

  return(coef)
}
```

```

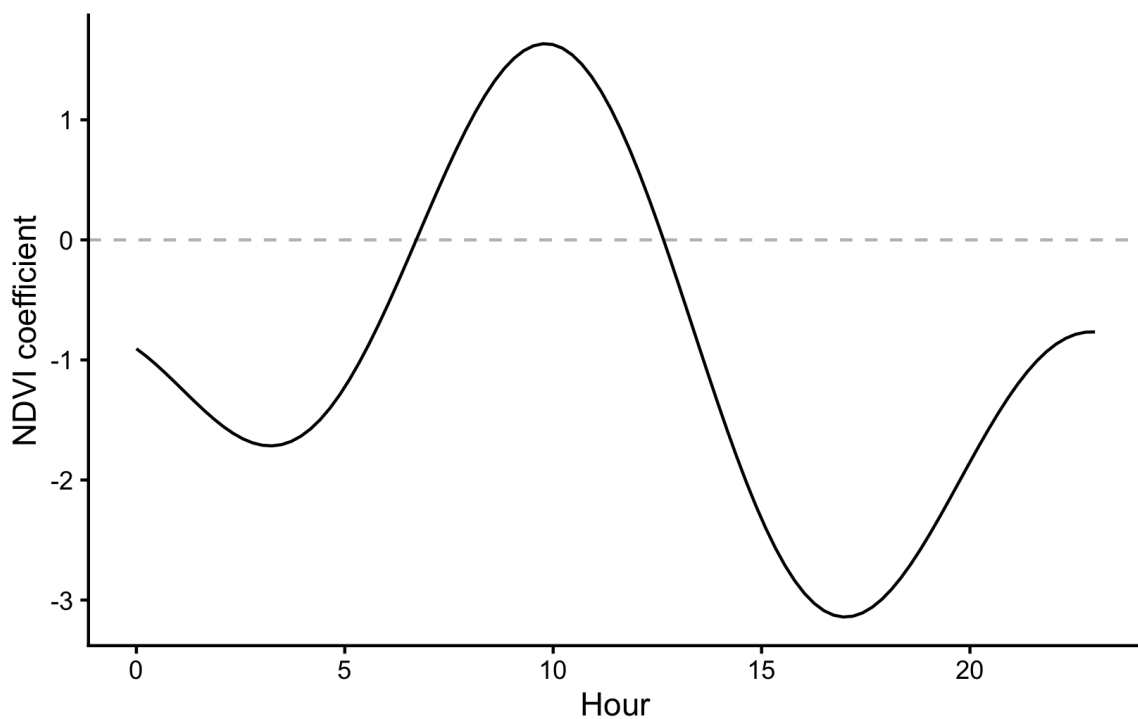
}

ndvi_temporal_coef_df <- data.frame(hour = seq(0, 23, length.out = 100))

ndvi_temporal_coef_df$ndvi_coef <- sapply(ndvi_temporal_coef_df$hour,
                                           ndvi_harmonic_coef_reconstruction,
                                           ssf_harmonic_model = ssf_harmonics)

# plot the response curve
ggplot(data = ndvi_temporal_coef_df) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey") +
  geom_line(aes(x = hour, y = ndvi_coef),
            colour = "black") +
  labs(x = "Hour", y = "NDVI coefficient") +
  theme_classic()

```



Calculate confidence intervals across time and plot

```

sample_hour <- 9 # hour to plot with a vertical dashed line

```



```

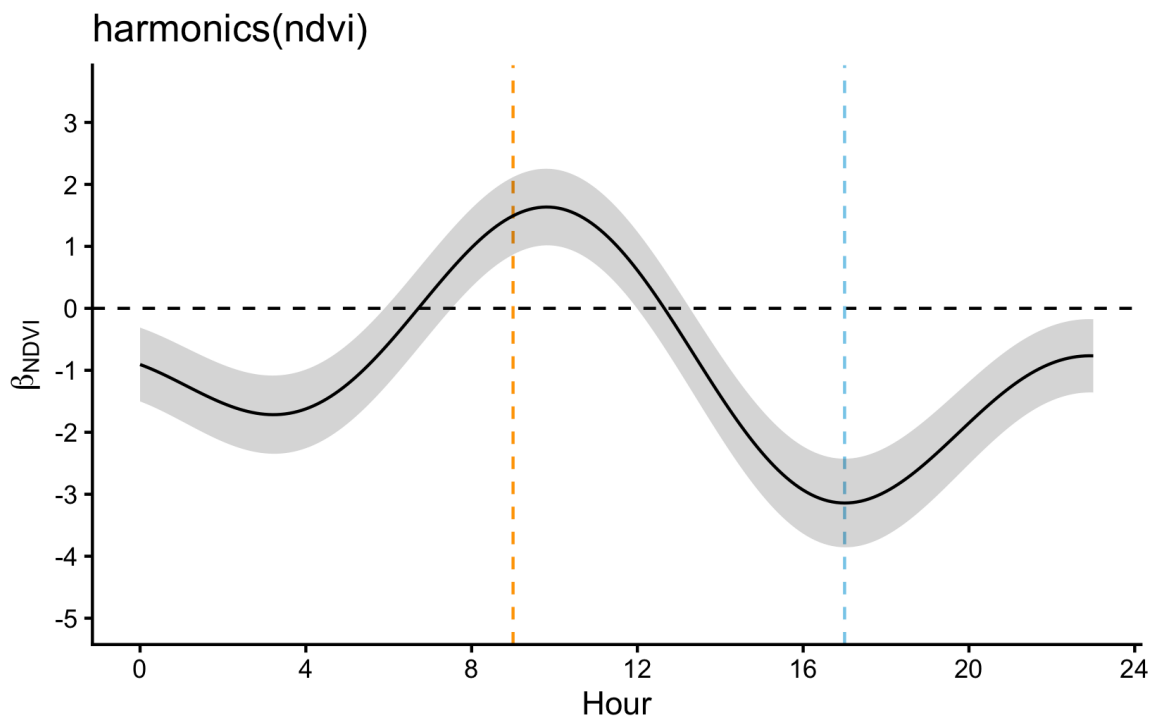
# Coefficient names we need
coef_names <- c("ndvi",
               "ndvi:sin_1_hour", "ndvi:cos_1_hour",
               "ndvi:sin_2_hour", "ndvi:cos_2_hour")

# coefficients and V from the underlying clogit model
beta <- coef(ssf_harmonics$model)[coef_names]
V      <- vcov(ssf_harmonics$model)[coef_names, coef_names, drop = FALSE]

# Build design for each hour
ndvi_temporal_ci <- tibble(hour = seq(0, 23, length.out = 231)) %>%
  mutate(
    sin_1 = sin(2 * pi * hour / 24),
    cos_1 = cos(2 * pi * hour / 24),
    sin_2 = sin(4 * pi * hour / 24),
    cos_2 = cos(4 * pi * hour / 24)
  ) %>%
  rowwise() %>%
  mutate(
    x = list(c(1, sin_1, cos_1, sin_2, cos_2)),
    est = as.numeric(matrix(unlist(x), nrow = 1) %*% beta),
    se  = sqrt(as.numeric(matrix(unlist(x), nrow = 1) %*% V %*% matrix(unlist(x), ncol = 1))),
    lo  = est - 1.96 * se,
    hi  = est + 1.96 * se
  ) %>%
  ungroup() %>%
  select(hour, est, se, lo, hi)

# Plot with ribbon
ggplot(ndvi_temporal_ci, aes(x = hour)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "black") +
  geom_vline(xintercept = 9, linetype = "dashed", colour = "orange") +
  geom_vline(xintercept = 17, linetype = "dashed", colour = "skyblue") +
  geom_ribbon(aes(ymin = lo, ymax = hi), alpha = 0.2) +
  geom_line(aes(y = est)) +
  labs(title = "harmonics(ndvi)", x = "Hour", y = expression(beta[NDVI]), fill = NULL) +
  scale_x_continuous(breaks = seq(0, 24, by = 4)) +
  scale_y_continuous(limits = c(-5, 3.5), breaks = seq(-5, 4, by = 1)) +
  theme_classic()

```



```
ggsave(paste0(plot_save_path, "/ndvi_harmonic_coef_response_id", which_buffalo, ".png"),
       width = 150, height = 80, units = "mm", dpi = 600)
```

Pull out a response curve from one of the hours

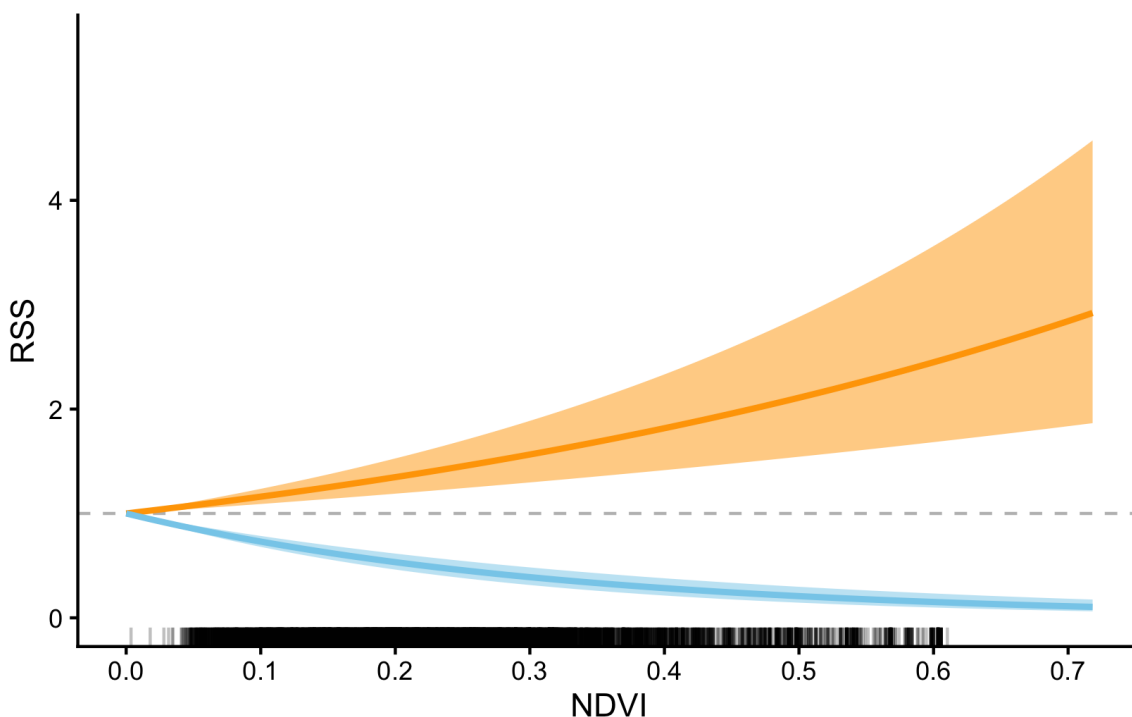
```
sample_hours <- c(9, 17)

ndvi_coef_df <- purrr::map_dfr(sample_hours, function(sample_hour) {

  # extract the NDVI coefficient and its standard error
  ndvi_coef <- ndvi_temporal_ci %>% filter(hour == sample_hour) %>% pull(est)
  ndvi_coef_se <- ndvi_temporal_ci %>% filter(hour == sample_hour) %>% pull(se)

  # calculate the RSS for each NDVI value
  data.frame(
    ndvi_values = ndvi_values,
    log_rss = ndvi_values * ndvi_coef,
    rss = exp(ndvi_values * ndvi_coef),
    se_prediction = abs(ndvi_values) * ndvi_coef_se
  ) %>% mutate(hour = as.factor(sample_hour), .before = ndvi_values)
})
```

```
# plot the response curve
ggplot(data = ndvi_coef_df) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey") +
  geom_ribbon(aes(x = ndvi_values,
                 ymin = exp(log_rss - 1.96 * se_prediction),
                 ymax = exp(log_rss + 1.96 * se_prediction),
                 fill = hour),
            alpha = 0.5) +
  geom_line(aes(x = ndvi_values, y = rss, colour = hour),
            linewidth = 1) +
  geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
           aes(x = ndvi), alpha = 0.25, sides="b") +
  scale_colour_manual(values = c("orange", "skyblue")) +
  scale_fill_manual(values = c("orange", "skyblue")) +
  scale_x_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  scale_y_continuous(limits = c(0, 5.5)) +
  labs(x = "NDVI", y = "RSS") +
  theme_classic() +
  theme(legend.position = "none")
```



```
ggsave(paste0(plot_save_path, "/ndvi_linear_response_sample_hour_id", which_buffalo, ".png")
       width = 70, height = 80, units = "mm", dpi = 600)
```

Manually create the selection surface

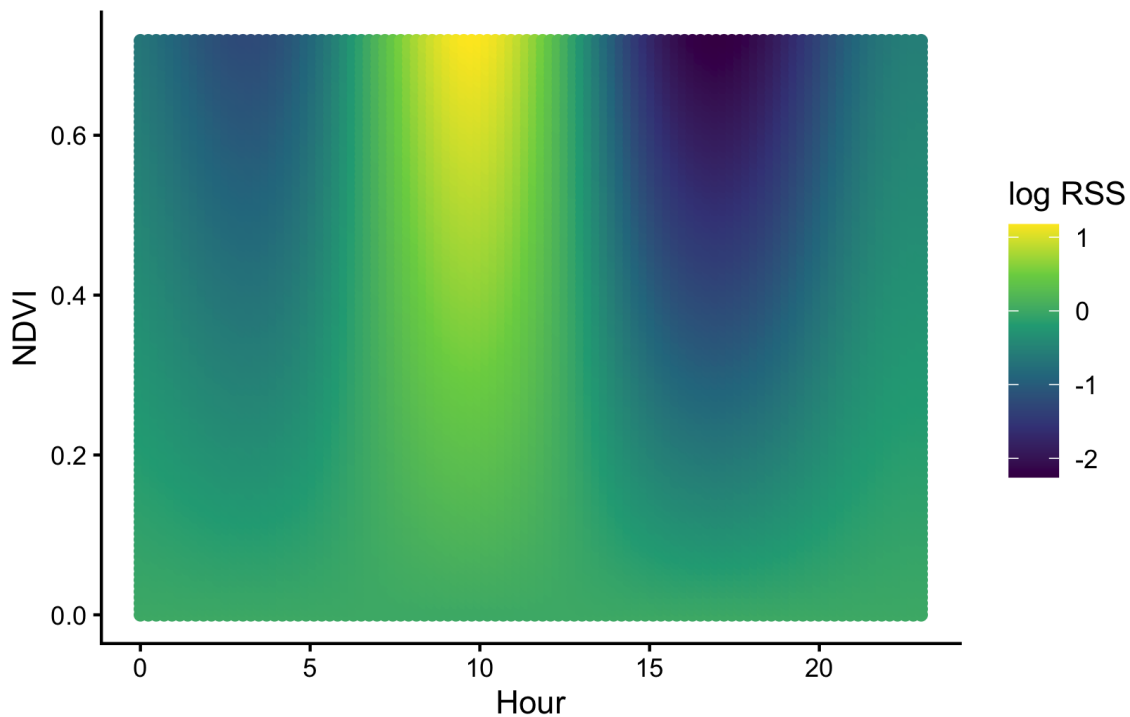
```
ndvi_harmonic_reconstruction <- function(ssf_harmonic_model, ndvi_value, hour) {  
  
  ndvi_coef <- ssf_harmonic_model$model$coefficients["ndvi"]  
  ndvi_sin_1_coef <- ssf_harmonic_model$model$coefficients["ndvi:sin_1_hour"]  
  ndvi_cos_1_coef <- ssf_harmonic_model$model$coefficients["ndvi:cos_1_hour"]  
  ndvi_sin_2_coef <- ssf_harmonic_model$model$coefficients["ndvi:sin_2_hour"]  
  ndvi_cos_2_coef <- ssf_harmonic_model$model$coefficients["ndvi:cos_2_hour"]  
  
  sin_1_hour <- sin(2 * pi * hour / 24)  
  cos_1_hour <- cos(2 * pi * hour / 24)  
  sin_2_hour <- sin(4 * pi * hour / 24)  
  cos_2_hour <- cos(4 * pi * hour / 24)  
  
  log_rss <- as.numeric(ndvi_value * ndvi_coef +  
                        ndvi_value * sin_1_hour * ndvi_sin_1_coef +  
                        ndvi_value * cos_1_hour * ndvi_cos_1_coef +  
                        ndvi_value * sin_2_hour * ndvi_sin_2_coef +  
                        ndvi_value * cos_2_hour * ndvi_cos_2_coef)  
  
  return(log_rss)  
}  
  
# test function  
ndvi_harmonic_reconstruction(ssf_harmonics,  
                             ndvi_values[1],  
                             hour = 12)
```

```
[1] 0
```

Create table of NDVI and hour values

```
ndvi_hour_values <- expand.grid(ndvi = ndvi_values,  
                               hour = seq(0, 23, length.out = length(ndvi_values)))  
  
ndvi_hour_values$ssf_linear_log_rss <- mapply(ndvi_harmonic_reconstruction,  
                                              ssf_harmonic_model = list(ssf_harmonics),  
                                              ndvi_value = ndvi_hour_values$ndvi,  
                                              hour = ndvi_hour_values$hour)
```

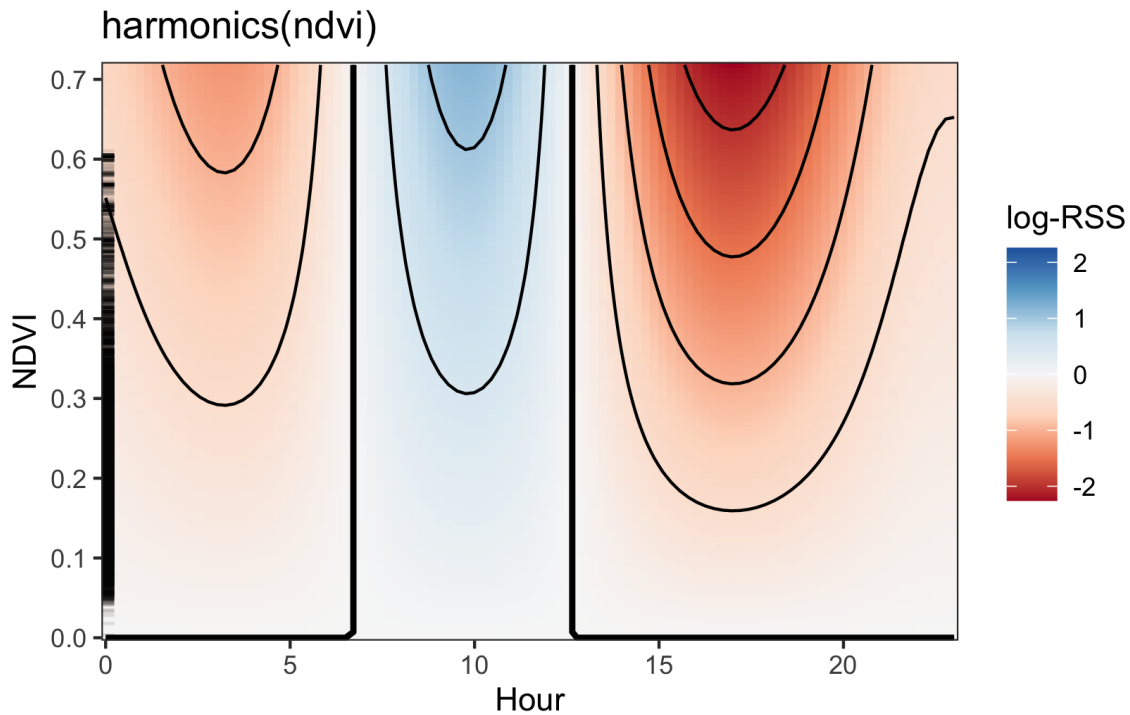
```
# plot the response curve
ggplot(data = ndvi_hour_values) +
  geom_point(aes(x = hour, y = ndvi, colour = ssf_linear_log_rss)) +
  labs(x = "Hour", y = "NDVI", colour = "log RSS") +
  scale_colour_viridis_c() +
  theme_classic()
```



```
max_abs_colour <- max(abs(range(ndvi_hour_values$ssf_linear_log_rss)))

# plot the selection surface
ggplot() +
  geom_raster(data = ndvi_hour_values,
    aes(x = hour, y = ndvi, fill = ssf_linear_log_rss)) +
  labs(title = "harmonics(ndvi)", x = "Hour", y = "NDVI", fill = "log-RSS") +
  geom_contour(data = ndvi_hour_values,
    aes(x = hour, y = ndvi, z = ssf_linear_log_rss),
    colour = "black") +
  geom_contour(data = ndvi_hour_values,
    aes(x = hour, y = ndvi, z = ssf_linear_log_rss),
    breaks = 0, colour = "black", linetype = "solid", linewidth = 1) +
  scale_fill_distiller(palette = "RdBu", type = "div", direction = 1,
    limits = c(-max_abs_colour, max_abs_colour)) + # symmetric around 0
  coord_cartesian(expand = FALSE) +
```

```
geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
         aes(y = ndvi), alpha = 0.1, length = unit(0.015, "npc"), sides="l") +
scale_y_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
theme_bw()
```



```
ggsave(paste0(plot_save_path, "/ndvi_harmonic_response_id", which_buffalo, ".png"),
       width = 150, height = 80, units = "mm", dpi = 600)
```

Fit harmonic model with quadratic terms

```
ssf_harmonics_quadratic <- amt::fit_issf(case_ ~

# linear terms
ndvi +
ndvi:sin_1_hour + ndvi:cos_1_hour +
ndvi:sin_2_hour + ndvi:cos_2_hour +

# quadratic terms
ndvi_sq +
ndvi_sq:sin_1_hour + ndvi_sq:cos_1_hour +
```

```

ndvi_sq:sin_2_hour + ndvi_sq:cos_2_hour +

sl_ + log_sl + cos_ta +

strata(step_id_),
data = buffalo_ssf_data)

summary(ssf_harmonics_quadratic)

```

Call:

```

coxph(formula = Surv(rep(1, 106678L), case_) ~ ndvi + ndvi:sin_1_hour +
  ndvi:cos_1_hour + ndvi:sin_2_hour + ndvi:cos_2_hour + ndvi_sq +
  ndvi_sq:sin_1_hour + ndvi_sq:cos_1_hour + ndvi_sq:sin_2_hour +
  ndvi_sq:cos_2_hour + sl_ + log_sl + cos_ta + strata(step_id_),
  data = data, method = "exact")

```

n= 106678, number of events= 9698

	coef	exp(coef)	se(coef)	z	Pr(> z)
ndvi	4.496e+00	8.970e+01	5.067e-01	8.873	< 2e-16 ***
ndvi_sq	-9.138e+00	1.075e-04	8.400e-01	-10.878	< 2e-16 ***
sl_	-2.582e-05	1.000e+00	3.197e-05	-0.808	0.41923
log_sl	4.076e-03	1.004e+00	5.731e-03	0.711	0.47695
cos_ta	-5.770e-03	9.942e-01	1.505e-02	-0.384	0.70134
ndvi:sin_1_hour	8.621e-01	2.368e+00	7.214e-01	1.195	0.23209
ndvi:cos_1_hour	1.219e+00	3.385e+00	7.034e-01	1.734	0.08299 .
ndvi:sin_2_hour	-2.196e+00	1.113e-01	7.186e-01	-3.056	0.00225 **
ndvi:cos_2_hour	1.471e+00	4.353e+00	7.043e-01	2.088	0.03676 *
ndvi_sq:sin_1_hour	6.008e-01	1.824e+00	1.226e+00	0.490	0.62410
ndvi_sq:cos_1_hour	-3.012e+00	4.919e-02	1.126e+00	-2.675	0.00747 **
ndvi_sq:sin_2_hour	1.688e+00	5.411e+00	1.176e+00	1.436	0.15104
ndvi_sq:cos_2_hour	-5.083e-01	6.015e-01	1.174e+00	-0.433	0.66493

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
ndvi	8.970e+01	1.115e-02	3.322e+01	2.422e+02
ndvi_sq	1.075e-04	9.298e+03	2.073e-05	5.580e-04
sl_	1.000e+00	1.000e+00	9.999e-01	1.000e+00
log_sl	1.004e+00	9.959e-01	9.929e-01	1.015e+00
cos_ta	9.942e-01	1.006e+00	9.654e-01	1.024e+00
ndvi:sin_1_hour	2.368e+00	4.223e-01	5.759e-01	9.738e+00
ndvi:cos_1_hour	3.385e+00	2.954e-01	8.528e-01	1.344e+01
ndvi:sin_2_hour	1.113e-01	8.985e+00	2.722e-02	4.551e-01

```

ndvi:cos_2_hour      4.353e+00  2.297e-01  1.095e+00  1.731e+01
ndvi_sq:sin_1_hour  1.824e+00  5.484e-01  1.649e-01  2.016e+01
ndvi_sq:cos_1_hour  4.919e-02  2.033e+01  5.413e-03  4.470e-01
ndvi_sq:sin_2_hour  5.411e+00  1.848e-01  5.400e-01  5.423e+01
ndvi_sq:cos_2_hour  6.015e-01  1.663e+00  6.028e-02  6.002e+00

```

Concordance= 0.549 (se = 0.004)

Likelihood ratio test= 270 on 13 df, p=<2e-16

Wald test = 221.1 on 13 df, p=<2e-16

Score (logrank) test = 221.1 on 13 df, p=<2e-16

```
AIC(ssf_harmonics_quadratic)
```

```
[1] 46265.62
```

Manually create the selection surface

```

ndvi_harmonic_reconstruction <- function(ssf_harmonic_model, ndvi_value, hour) {

  # linear terms
  ndvi_coef <- ssf_harmonic_model$model$coefficients["ndvi"]
  ndvi_sin_1_coef <- ssf_harmonic_model$model$coefficients["ndvi:sin_1_hour"]
  ndvi_cos_1_coef <- ssf_harmonic_model$model$coefficients["ndvi:cos_1_hour"]
  ndvi_sin_2_coef <- ssf_harmonic_model$model$coefficients["ndvi:sin_2_hour"]
  ndvi_cos_2_coef <- ssf_harmonic_model$model$coefficients["ndvi:cos_2_hour"]

  # quadratic terms
  ndvi_sq_coef <- ssf_harmonic_model$model$coefficients["ndvi_sq"]
  ndvi_sq_sin_1_coef <- ssf_harmonic_model$model$coefficients["ndvi_sq:sin_1_hour"]
  ndvi_sq_cos_1_coef <- ssf_harmonic_model$model$coefficients["ndvi_sq:cos_1_hour"]
  ndvi_sq_sin_2_coef <- ssf_harmonic_model$model$coefficients["ndvi_sq:sin_2_hour"]
  ndvi_sq_cos_2_coef <- ssf_harmonic_model$model$coefficients["ndvi_sq:cos_2_hour"]

  sin_1_hour <- sin(2 * pi * hour / 24)
  cos_1_hour <- cos(2 * pi * hour / 24)
  sin_2_hour <- sin(4 * pi * hour / 24)
  cos_2_hour <- cos(4 * pi * hour / 24)

  linear_part <- ndvi_value * ndvi_coef +
    ndvi_value * sin_1_hour * ndvi_sin_1_coef +
    ndvi_value * cos_1_hour * ndvi_cos_1_coef +
    ndvi_value * sin_2_hour * ndvi_sin_2_coef +

```



```

    ndvi_value * cos_2_hour * ndvi_cos_2_coef

ndvi_sq_value <- ndvi_value^2

quadratic_part <- ndvi_sq_value * ndvi_sq_coef +
  ndvi_sq_value * sin_1_hour * ndvi_sq_sin_1_coef +
  ndvi_sq_value * cos_1_hour * ndvi_sq_cos_1_coef +
  ndvi_sq_value * sin_2_hour * ndvi_sq_sin_2_coef +
  ndvi_sq_value * cos_2_hour * ndvi_sq_cos_2_coef

return(as.numeric(linear_part + quadratic_part))
}

# test function
ndvi_harmonic_reconstruction(ssf_harmonics_quadratic, ndvi_values[1], hour = 12)

```

```
[1] 0
```

Create table of NDVI and hour values

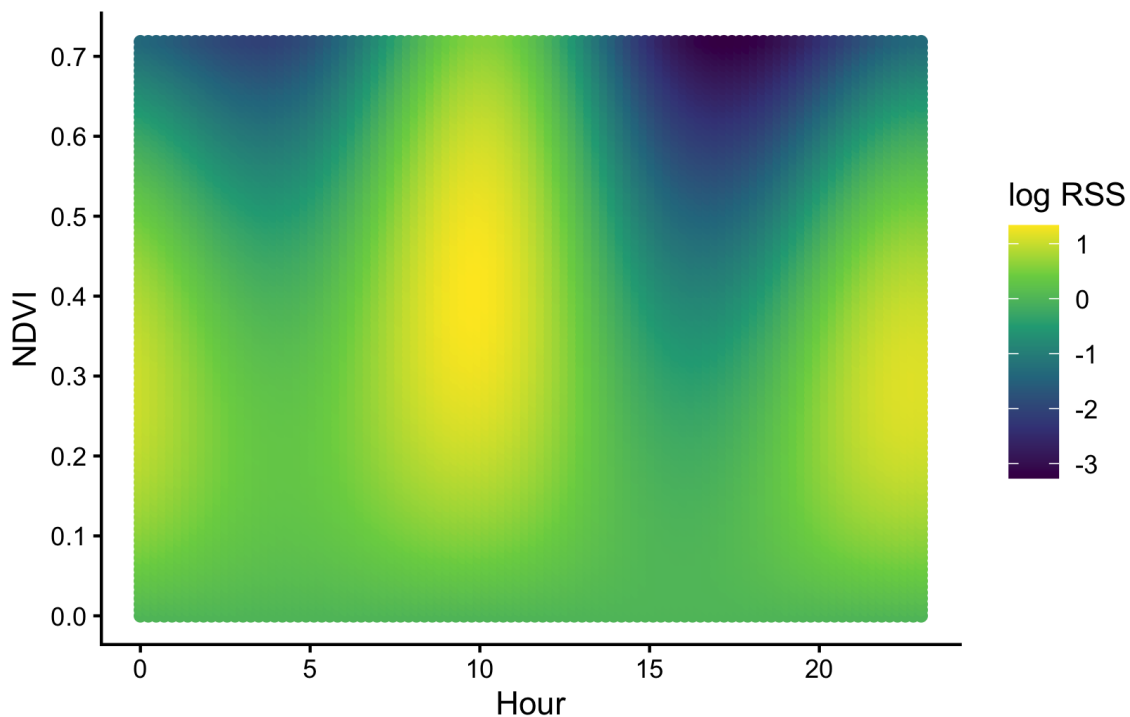
```

ndvi_hour_values <- expand.grid(ndvi = ndvi_values,
                               hour = seq(0, 23, length.out = length(ndvi_values)))

ndvi_hour_values$ssf_quadratic_log_rss <- mapply(ndvi_harmonic_reconstruction,
                                                  ssf_harmonic_model = list(ssf_harmonics_qua
                                                  ndvi_value = ndvi_hour_values$ndvi,
                                                  hour = ndvi_hour_values$hour)

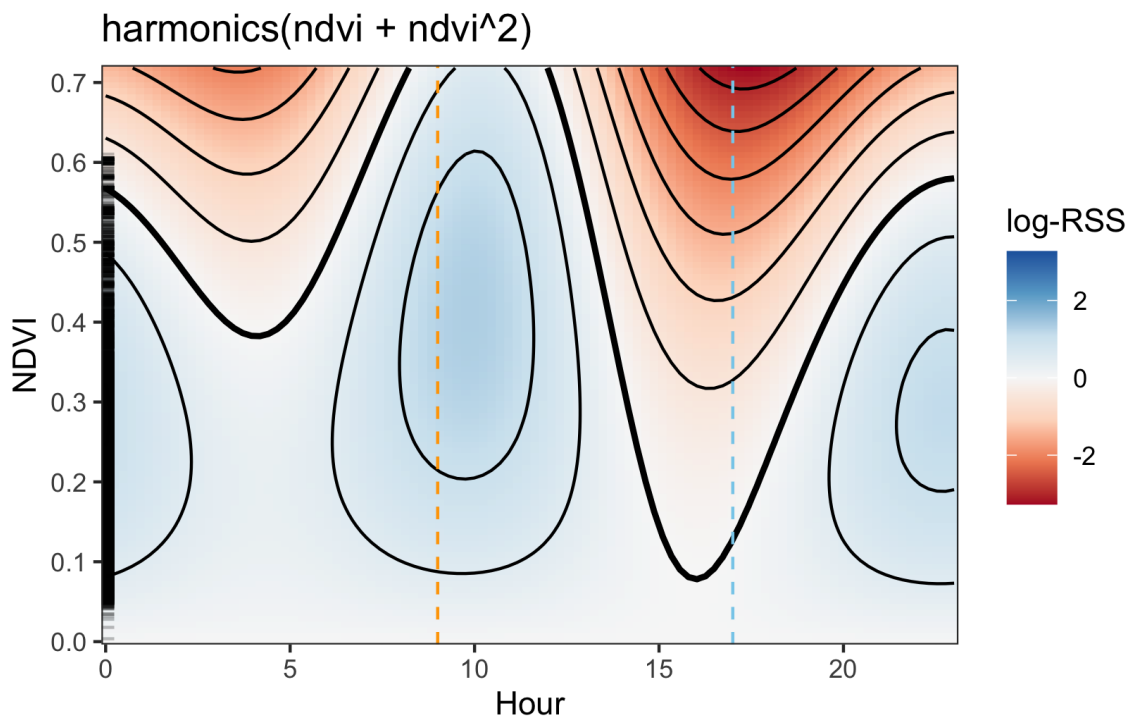
# plot the response curve
ggplot(data = ndvi_hour_values) +
  geom_point(aes(x = hour, y = ndvi, colour = ssf_quadratic_log_rss)) +
  labs(x = "Hour", y = "NDVI", colour = "log RSS") +
  scale_colour_viridis_c() +
  scale_y_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  theme_classic()

```



```
max_abs_colour <- max(abs(range(ndvi_hour_values$ssf_quadratic_log_rss)))

# plot the selection surface
ggplot() +
  geom_raster(data = ndvi_hour_values,
    aes(x = hour, y = ndvi, fill = ssf_quadratic_log_rss)) +
  labs(title = "harmonics(ndvi + ndvi^2)", x = "Hour", y = "NDVI", fill = "log-RSS") +
  geom_contour(data = ndvi_hour_values,
    aes(x = hour, y = ndvi, z = ssf_quadratic_log_rss),
    colour = "black") +
  geom_contour(data = ndvi_hour_values,
    aes(x = hour, y = ndvi, z = ssf_quadratic_log_rss),
    breaks = 0, colour = "black", linetype = "solid", linewidth = 1) +
  geom_vline(xintercept = 9, linetype = "dashed", colour = "orange") +
  geom_vline(xintercept = 17, linetype = "dashed", colour = "skyblue") +
  scale_fill_distiller(palette = "RdBu", type = "div", direction = 1,
    limits = c(-max_abs_colour, max_abs_colour)) +
  coord_cartesian(expand = FALSE) +
  geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
    aes(y = ndvi), alpha = 0.25, length = unit(0.015, "npc"), sides="l") +
  scale_y_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  theme_bw()
```



```
ggsave(paste0(plot_save_path, "/ndvi_harmonic_quadratic_response_id", which_buffalo, ".png"),
       width = 150, height = 80, units = "mm", dpi = 600)
```

Pull out a response curve from one of the hours

```
sample_hours <- c(9, 17)

curve_df <- purrr::map_dfr(sample_hours, function(sample_hour) {

  # Harmonics at sample hour
  s1 <- sin(2 * pi * sample_hour / 24)
  c1 <- cos(2 * pi * sample_hour / 24)
  s2 <- sin(4 * pi * sample_hour / 24)
  c2 <- cos(4 * pi * sample_hour / 24)

  mod <- ssf_harmonics_quadratic$model
  betas <- coef(mod)
  V <- vcov(mod)
  cn <- names(betas)

  # For each NDVI value, build the contrast vector g such that g' = log-RSS
```

```

purrr::map_dfr(ndvi_values, function(x) {

  g <- setNames(numeric(length(betas)), cn)

  # linear term:  $x * (_{ndvi} + \text{harmonics})$ 
  g["ndvi"] <- x
  g["ndvi:sin_1_hour"] <- x * s1
  g["ndvi:cos_1_hour"] <- x * c1
  g["ndvi:sin_2_hour"] <- x * s2
  g["ndvi:cos_2_hour"] <- x * c2

  # quadratic term:  $x^2 * (_{ndvi\_sq} + \text{harmonics})$ 
  g["ndvi_sq"] <- x^2
  g["ndvi_sq:sin_1_hour"] <- x^2 * s1
  g["ndvi_sq:cos_1_hour"] <- x^2 * c1
  g["ndvi_sq:sin_2_hour"] <- x^2 * s2
  g["ndvi_sq:cos_2_hour"] <- x^2 * c2

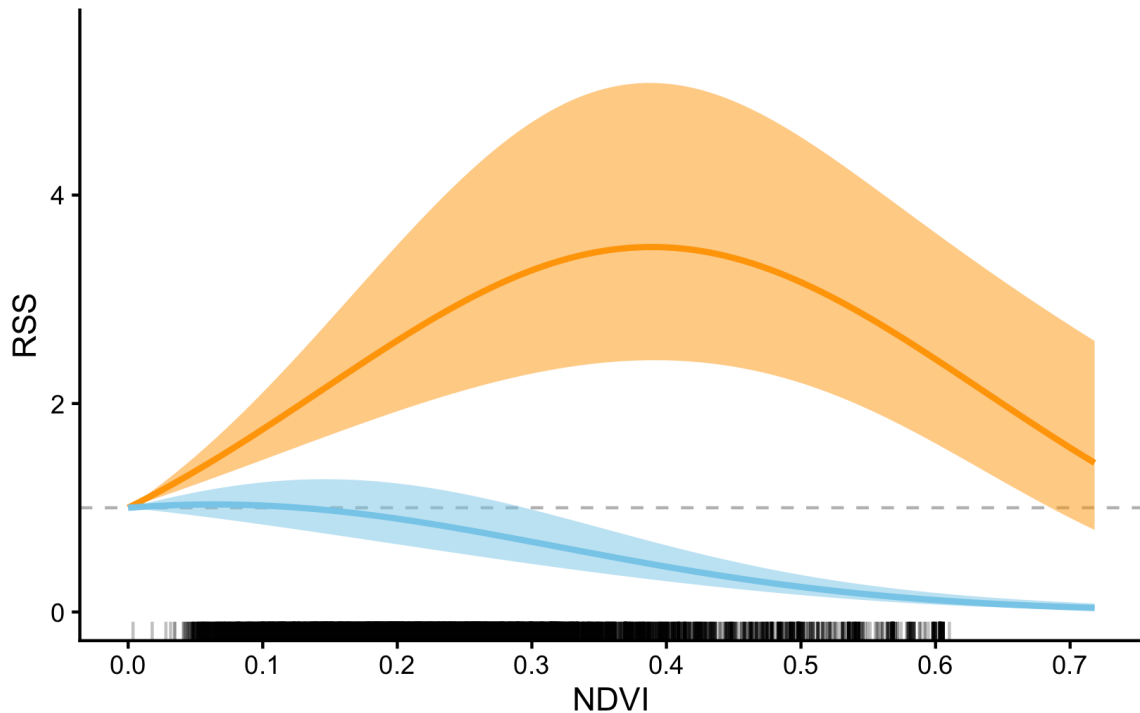
  log_rss <- as.numeric(g %*% betas)
  se_log <- sqrt(as.numeric(t(g) %*% V %*% g))

  data.frame(
    hour = sample_hour,
    ndvi = x,
    log_rss = log_rss,
    rss = exp(log_rss),
    # delta method on the RSS scale: SE_rss = exp(log_rss) * SE_log
    rss_lwr = exp(log_rss - 1.96 * se_log),
    rss_upr = exp(log_rss + 1.96 * se_log)
  )
})

# plot the response curve
ggplot(curve_df, aes(x = ndvi)) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey") +
  geom_ribbon(aes(ymin = rss_lwr, ymax = rss_upr, fill = as.factor(hour)),
    alpha = 0.5) +
  geom_line(aes(y = rss, colour = as.factor(hour)), linewidth = 1) +
  geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
    aes(x = ndvi), alpha = 0.25, sides="b") +
  scale_colour_manual(values = c("orange", "skyblue")) +
  scale_fill_manual(values = c("orange", "skyblue")) +

```

```
scale_x_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
scale_y_continuous(limits = c(0, 5.5)) +
labs(x = "NDVI", y = "RSS") +
theme_classic() +
theme(legend.position = "none")
```



```
ggsave(paste0(plot_save_path, "/ndvi_quadratic_response_sample_hour_id", which_buffalo, ".png"),
        width = 70, height = 80, units = "mm", dpi = 600)
```

Temporal dynamics with smooth terms

```
# fit an ssf gam model
# We set the knots explicitly at 0 and 24. Without this, mgcv places the cyclic
# endpoints at the range of the data (0 and 23), which wraps hour 23 onto hour 0
# and squeezes an hour out of the daily cycle.
ssf_gam_temporal <- mgcv::gam(cbind(times, step_id_) ~
    s(hour, by = ndvi, bs = "cc") + #, k = 5
    sl_ + log_sl + cos_ta,
    knots = list(hour = c(0, 24)),
    data = buffalo_ssf_data,
```

```

        family = cox.ph,
        weight = case_)

summary(ssf_gam_temporal)

```

Family: Cox PH

Link function: identity

Formula:

```

cbind(times, step_id_) ~ s(hour, by = ndvi, bs = "cc") + sl_ +
  log_sl + cos_ta

```

Parametric coefficients:

	Estimate	Std. Error	z value	Pr(> z)
sl_	-1.179e-05	3.196e-05	-0.369	0.712
log_sl	7.320e-06	5.702e-03	0.001	0.999
cos_ta	-5.380e-03	1.504e-02	-0.358	0.721

Approximate significance of smooth terms:

	edf	Ref.df	Chi.sq	p-value
s(hour):ndvi	8.191	8.819	154.5	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Deviance explained = -3.61%

-REML = 23204 Scale est. = 1 n = 106678

```

AIC(ssf_gam_temporal)

```

```

[1] 46362.27

```

```

gam_smooths <- smooth_estimates(ssf_gam_temporal)

```

```

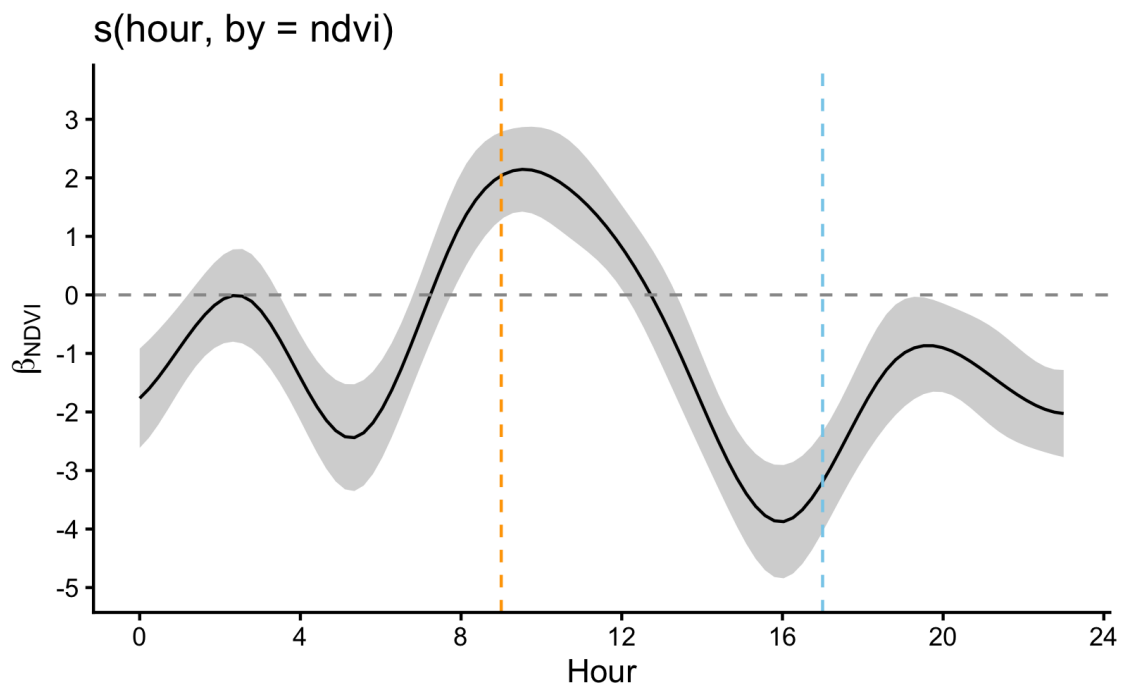
gratia::draw(ssf_gam_temporal, rug = FALSE) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey60") +
  geom_vline(xintercept = 9, linetype = "dashed", colour = "orange") +
  geom_vline(xintercept = 17, linetype = "dashed", colour = "skyblue") +
  labs() +
  scale_x_continuous(breaks = seq(0, 24, by = 4)) +
  scale_y_continuous(limits = c(-5, 3.5), breaks = seq(-5, 4, by = 1)) +
  labs(title = "s(hour, by = ndvi)",

```

```

    subtitle = NULL,
    caption = NULL,
    x = "Hour",
    y = expression(beta[NDVI]), fill = NULL) +
    theme_classic()

```



```

ggsave(paste0(plot_save_path, "/ndvi_gam_coef_response_id", which_buffalo, ".png"),
        width = 150, height = 80, units = "mm", dpi = 600)

```

Pull out a response curve from one of the hours

```

pred_data_hour9 <- pred_data
pred_data_hour9$hour <- 9

pred_data_hour17 <- pred_data
pred_data_hour17$hour <- 17

pred_data_hour9_17 <- bind_rows(pred_data_hour9, pred_data_hour17)

# Get predictions on log-scale with standard errors
predictions <- predict(ssf_gam_temporal,

```

```

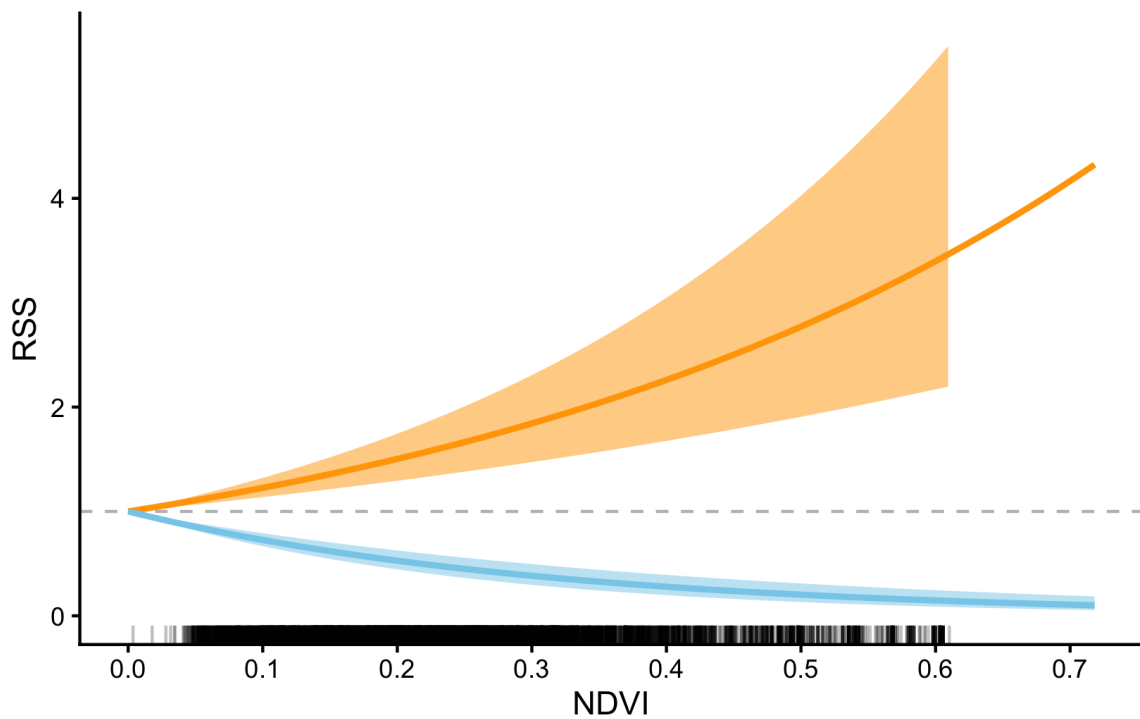
        newdata = pred_data_hour9_17,
        type = "link", # log-scale
        se.fit = TRUE)

pred_data_hour9_17 <- pred_data_hour9_17 %>%
  mutate(
    # Extract log RSS and standard errors
    log_rss = predictions$fit,
    se_log_rss = predictions$se.fit,
    # Calculate confidence intervals on log-scale
    log_rss_lower = log_rss - 1.96 * se_log_rss,
    log_rss_upper = log_rss + 1.96 * se_log_rss,
    # Transform to RSS scale
    rss = exp(log_rss),
    rss_lower = exp(log_rss_lower),
    rss_upper = exp(log_rss_upper)
  )

# plot the response curve
ggplot(data = pred_data_hour9_17) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey") +
  geom_ribbon(aes(x = ndvi,
                 ymin = rss_lower, ymax = rss_upper,
                 fill = as.factor(hour)),
            alpha = 0.5) +
  geom_line(aes(x = ndvi, y = rss, colour = as.factor(hour)),
            linewidth = 1) +
  geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
          aes(x = ndvi), alpha = 0.25, sides="b") +
  scale_colour_manual(values = c("orange", "skyblue")) +
  scale_fill_manual(values = c("orange", "skyblue")) +
  scale_x_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  scale_y_continuous(limits = c(0, 5.5)) +
  labs(x = "NDVI", y = "RSS") +
  theme_classic() +
  theme(legend.position = "none")

```

Warning: Removed 15 rows containing missing values or values outside the scale range (`geom\_ribbon()`).



```
ggsave(paste0(plot_save_path, "/ndvi_gam_response_sample_hour_id", which_buffalo, ".png"),
       width = 70, height = 80, units = "mm", dpi = 600)
```

Warning: Removed 15 rows containing missing values or values outside the scale range (``geom_ribbon()``).

GAM selection surface

```
ndvi_hour_values <- expand.grid(ndvi = ndvi_values,
                               hour = seq(0, 23, length.out = length(ndvi_values)))

ndvi_hour_values <- ndvi_hour_values %>% mutate(
  sl_ = 0,
  log_sl = 0,
  cos_ta = 0
)

ndvi_hour_values
```

```
ndvi      hour sl_ log_sl cos_ta
```

1	0.000000000	0.0000000	0	0	0
2	0.007254508	0.0000000	0	0	0
3	0.014509015	0.0000000	0	0	0
4	0.021763523	0.0000000	0	0	0
5	0.029018030	0.0000000	0	0	0
6	0.036272538	0.0000000	0	0	0
7	0.043527045	0.0000000	0	0	0
8	0.050781553	0.0000000	0	0	0
9	0.058036060	0.0000000	0	0	0
10	0.065290568	0.0000000	0	0	0
11	0.072545076	0.0000000	0	0	0
12	0.079799583	0.0000000	0	0	0
13	0.087054091	0.0000000	0	0	0
14	0.094308598	0.0000000	0	0	0
15	0.101563106	0.0000000	0	0	0
16	0.108817613	0.0000000	0	0	0
17	0.116072121	0.0000000	0	0	0
18	0.123326628	0.0000000	0	0	0
19	0.130581136	0.0000000	0	0	0
20	0.137835643	0.0000000	0	0	0
21	0.145090151	0.0000000	0	0	0
22	0.152344659	0.0000000	0	0	0
23	0.159599166	0.0000000	0	0	0
24	0.166853674	0.0000000	0	0	0
25	0.174108181	0.0000000	0	0	0
26	0.181362689	0.0000000	0	0	0
27	0.188617196	0.0000000	0	0	0
28	0.195871704	0.0000000	0	0	0
29	0.203126211	0.0000000	0	0	0
30	0.210380719	0.0000000	0	0	0
31	0.217635227	0.0000000	0	0	0
32	0.224889734	0.0000000	0	0	0
33	0.232144242	0.0000000	0	0	0
34	0.239398749	0.0000000	0	0	0
35	0.246653257	0.0000000	0	0	0
36	0.253907764	0.0000000	0	0	0
37	0.261162272	0.0000000	0	0	0
38	0.268416779	0.0000000	0	0	0
39	0.275671287	0.0000000	0	0	0
40	0.282925794	0.0000000	0	0	0
41	0.290180302	0.0000000	0	0	0
42	0.297434810	0.0000000	0	0	0
43	0.304689317	0.0000000	0	0	0
44	0.311943825	0.0000000	0	0	0
45	0.319198332	0.0000000	0	0	0

46	0.326452840	0.0000000	0	0	0
47	0.333707347	0.0000000	0	0	0
48	0.340961855	0.0000000	0	0	0
49	0.348216362	0.0000000	0	0	0
50	0.355470870	0.0000000	0	0	0
51	0.362725378	0.0000000	0	0	0
52	0.369979885	0.0000000	0	0	0
53	0.377234393	0.0000000	0	0	0
54	0.384488900	0.0000000	0	0	0
55	0.391743408	0.0000000	0	0	0
56	0.398997915	0.0000000	0	0	0
57	0.406252423	0.0000000	0	0	0
58	0.413506930	0.0000000	0	0	0
59	0.420761438	0.0000000	0	0	0
60	0.428015945	0.0000000	0	0	0
61	0.435270453	0.0000000	0	0	0
62	0.442524961	0.0000000	0	0	0
63	0.449779468	0.0000000	0	0	0
64	0.457033976	0.0000000	0	0	0
65	0.464288483	0.0000000	0	0	0
66	0.471542991	0.0000000	0	0	0
67	0.478797498	0.0000000	0	0	0
68	0.486052006	0.0000000	0	0	0
69	0.493306513	0.0000000	0	0	0
70	0.500561021	0.0000000	0	0	0
71	0.507815529	0.0000000	0	0	0
72	0.515070036	0.0000000	0	0	0
73	0.522324544	0.0000000	0	0	0
74	0.529579051	0.0000000	0	0	0
75	0.536833559	0.0000000	0	0	0
76	0.544088066	0.0000000	0	0	0
77	0.551342574	0.0000000	0	0	0
78	0.558597081	0.0000000	0	0	0
79	0.565851589	0.0000000	0	0	0
80	0.573106097	0.0000000	0	0	0
81	0.580360604	0.0000000	0	0	0
82	0.587615112	0.0000000	0	0	0
83	0.594869619	0.0000000	0	0	0
84	0.602124127	0.0000000	0	0	0
85	0.609378634	0.0000000	0	0	0
86	0.616633142	0.0000000	0	0	0
87	0.623887649	0.0000000	0	0	0
88	0.631142157	0.0000000	0	0	0
89	0.638396664	0.0000000	0	0	0
90	0.645651172	0.0000000	0	0	0

91	0.652905680	0.0000000	0	0	0
92	0.660160187	0.0000000	0	0	0
93	0.667414695	0.0000000	0	0	0
94	0.674669202	0.0000000	0	0	0
95	0.681923710	0.0000000	0	0	0
96	0.689178217	0.0000000	0	0	0
97	0.696432725	0.0000000	0	0	0
98	0.703687232	0.0000000	0	0	0
99	0.710941740	0.0000000	0	0	0
100	0.718196248	0.0000000	0	0	0
101	0.000000000	0.2323232	0	0	0
102	0.007254508	0.2323232	0	0	0
103	0.014509015	0.2323232	0	0	0
104	0.021763523	0.2323232	0	0	0
105	0.029018030	0.2323232	0	0	0
106	0.036272538	0.2323232	0	0	0
107	0.043527045	0.2323232	0	0	0
108	0.050781553	0.2323232	0	0	0
109	0.058036060	0.2323232	0	0	0
110	0.065290568	0.2323232	0	0	0
111	0.072545076	0.2323232	0	0	0
112	0.079799583	0.2323232	0	0	0
113	0.087054091	0.2323232	0	0	0
114	0.094308598	0.2323232	0	0	0
115	0.101563106	0.2323232	0	0	0
116	0.108817613	0.2323232	0	0	0
117	0.116072121	0.2323232	0	0	0
118	0.123326628	0.2323232	0	0	0
119	0.130581136	0.2323232	0	0	0
120	0.137835643	0.2323232	0	0	0
121	0.145090151	0.2323232	0	0	0
122	0.152344659	0.2323232	0	0	0
123	0.159599166	0.2323232	0	0	0
124	0.166853674	0.2323232	0	0	0
125	0.174108181	0.2323232	0	0	0
126	0.181362689	0.2323232	0	0	0
127	0.188617196	0.2323232	0	0	0
128	0.195871704	0.2323232	0	0	0
129	0.203126211	0.2323232	0	0	0
130	0.210380719	0.2323232	0	0	0
131	0.217635227	0.2323232	0	0	0
132	0.224889734	0.2323232	0	0	0
133	0.232144242	0.2323232	0	0	0
134	0.239398749	0.2323232	0	0	0
135	0.246653257	0.2323232	0	0	0

136	0.253907764	0.2323232	0	0	0
137	0.261162272	0.2323232	0	0	0
138	0.268416779	0.2323232	0	0	0
139	0.275671287	0.2323232	0	0	0
140	0.282925794	0.2323232	0	0	0
141	0.290180302	0.2323232	0	0	0
142	0.297434810	0.2323232	0	0	0
143	0.304689317	0.2323232	0	0	0
144	0.311943825	0.2323232	0	0	0
145	0.319198332	0.2323232	0	0	0
146	0.326452840	0.2323232	0	0	0
147	0.333707347	0.2323232	0	0	0
148	0.340961855	0.2323232	0	0	0
149	0.348216362	0.2323232	0	0	0
150	0.355470870	0.2323232	0	0	0
151	0.362725378	0.2323232	0	0	0
152	0.369979885	0.2323232	0	0	0
153	0.377234393	0.2323232	0	0	0
154	0.384488900	0.2323232	0	0	0
155	0.391743408	0.2323232	0	0	0
156	0.398997915	0.2323232	0	0	0
157	0.406252423	0.2323232	0	0	0
158	0.413506930	0.2323232	0	0	0
159	0.420761438	0.2323232	0	0	0
160	0.428015945	0.2323232	0	0	0
161	0.435270453	0.2323232	0	0	0
162	0.442524961	0.2323232	0	0	0
163	0.449779468	0.2323232	0	0	0
164	0.457033976	0.2323232	0	0	0
165	0.464288483	0.2323232	0	0	0
166	0.471542991	0.2323232	0	0	0
167	0.478797498	0.2323232	0	0	0
168	0.486052006	0.2323232	0	0	0
169	0.493306513	0.2323232	0	0	0
170	0.500561021	0.2323232	0	0	0
171	0.507815529	0.2323232	0	0	0
172	0.515070036	0.2323232	0	0	0
173	0.522324544	0.2323232	0	0	0
174	0.529579051	0.2323232	0	0	0
175	0.536833559	0.2323232	0	0	0
176	0.544088066	0.2323232	0	0	0
177	0.551342574	0.2323232	0	0	0
178	0.558597081	0.2323232	0	0	0
179	0.565851589	0.2323232	0	0	0
180	0.573106097	0.2323232	0	0	0

181	0.580360604	0.2323232	0	0	0
182	0.587615112	0.2323232	0	0	0
183	0.594869619	0.2323232	0	0	0
184	0.602124127	0.2323232	0	0	0
185	0.609378634	0.2323232	0	0	0
186	0.616633142	0.2323232	0	0	0
187	0.623887649	0.2323232	0	0	0
188	0.631142157	0.2323232	0	0	0
189	0.638396664	0.2323232	0	0	0
190	0.645651172	0.2323232	0	0	0
191	0.652905680	0.2323232	0	0	0
192	0.660160187	0.2323232	0	0	0
193	0.667414695	0.2323232	0	0	0
194	0.674669202	0.2323232	0	0	0
195	0.681923710	0.2323232	0	0	0
196	0.689178217	0.2323232	0	0	0
197	0.696432725	0.2323232	0	0	0
198	0.703687232	0.2323232	0	0	0
199	0.710941740	0.2323232	0	0	0
200	0.718196248	0.2323232	0	0	0
201	0.000000000	0.4646465	0	0	0
202	0.007254508	0.4646465	0	0	0
203	0.014509015	0.4646465	0	0	0
204	0.021763523	0.4646465	0	0	0
205	0.029018030	0.4646465	0	0	0
206	0.036272538	0.4646465	0	0	0
207	0.043527045	0.4646465	0	0	0
208	0.050781553	0.4646465	0	0	0
209	0.058036060	0.4646465	0	0	0
210	0.065290568	0.4646465	0	0	0
211	0.072545076	0.4646465	0	0	0
212	0.079799583	0.4646465	0	0	0
213	0.087054091	0.4646465	0	0	0
214	0.094308598	0.4646465	0	0	0
215	0.101563106	0.4646465	0	0	0
216	0.108817613	0.4646465	0	0	0
217	0.116072121	0.4646465	0	0	0
218	0.123326628	0.4646465	0	0	0
219	0.130581136	0.4646465	0	0	0
220	0.137835643	0.4646465	0	0	0
221	0.145090151	0.4646465	0	0	0
222	0.152344659	0.4646465	0	0	0
223	0.159599166	0.4646465	0	0	0
224	0.166853674	0.4646465	0	0	0
225	0.174108181	0.4646465	0	0	0

226	0.181362689	0.4646465	0	0	0
227	0.188617196	0.4646465	0	0	0
228	0.195871704	0.4646465	0	0	0
229	0.203126211	0.4646465	0	0	0
230	0.210380719	0.4646465	0	0	0
231	0.217635227	0.4646465	0	0	0
232	0.224889734	0.4646465	0	0	0
233	0.232144242	0.4646465	0	0	0
234	0.239398749	0.4646465	0	0	0
235	0.246653257	0.4646465	0	0	0
236	0.253907764	0.4646465	0	0	0
237	0.261162272	0.4646465	0	0	0
238	0.268416779	0.4646465	0	0	0
239	0.275671287	0.4646465	0	0	0
240	0.282925794	0.4646465	0	0	0
241	0.290180302	0.4646465	0	0	0
242	0.297434810	0.4646465	0	0	0
243	0.304689317	0.4646465	0	0	0
244	0.311943825	0.4646465	0	0	0
245	0.319198332	0.4646465	0	0	0
246	0.326452840	0.4646465	0	0	0
247	0.333707347	0.4646465	0	0	0
248	0.340961855	0.4646465	0	0	0
249	0.348216362	0.4646465	0	0	0
250	0.355470870	0.4646465	0	0	0
251	0.362725378	0.4646465	0	0	0
252	0.369979885	0.4646465	0	0	0
253	0.377234393	0.4646465	0	0	0
254	0.384488900	0.4646465	0	0	0
255	0.391743408	0.4646465	0	0	0
256	0.398997915	0.4646465	0	0	0
257	0.406252423	0.4646465	0	0	0
258	0.413506930	0.4646465	0	0	0
259	0.420761438	0.4646465	0	0	0
260	0.428015945	0.4646465	0	0	0
261	0.435270453	0.4646465	0	0	0
262	0.442524961	0.4646465	0	0	0
263	0.449779468	0.4646465	0	0	0
264	0.457033976	0.4646465	0	0	0
265	0.464288483	0.4646465	0	0	0
266	0.471542991	0.4646465	0	0	0
267	0.478797498	0.4646465	0	0	0
268	0.486052006	0.4646465	0	0	0
269	0.493306513	0.4646465	0	0	0
270	0.500561021	0.4646465	0	0	0

271	0.507815529	0.4646465	0	0	0
272	0.515070036	0.4646465	0	0	0
273	0.522324544	0.4646465	0	0	0
274	0.529579051	0.4646465	0	0	0
275	0.536833559	0.4646465	0	0	0
276	0.544088066	0.4646465	0	0	0
277	0.551342574	0.4646465	0	0	0
278	0.558597081	0.4646465	0	0	0
279	0.565851589	0.4646465	0	0	0
280	0.573106097	0.4646465	0	0	0
281	0.580360604	0.4646465	0	0	0
282	0.587615112	0.4646465	0	0	0
283	0.594869619	0.4646465	0	0	0
284	0.602124127	0.4646465	0	0	0
285	0.609378634	0.4646465	0	0	0
286	0.616633142	0.4646465	0	0	0
287	0.623887649	0.4646465	0	0	0
288	0.631142157	0.4646465	0	0	0
289	0.638396664	0.4646465	0	0	0
290	0.645651172	0.4646465	0	0	0
291	0.652905680	0.4646465	0	0	0
292	0.660160187	0.4646465	0	0	0
293	0.667414695	0.4646465	0	0	0
294	0.674669202	0.4646465	0	0	0
295	0.681923710	0.4646465	0	0	0
296	0.689178217	0.4646465	0	0	0
297	0.696432725	0.4646465	0	0	0
298	0.703687232	0.4646465	0	0	0
299	0.710941740	0.4646465	0	0	0
300	0.718196248	0.4646465	0	0	0
301	0.000000000	0.6969697	0	0	0
302	0.007254508	0.6969697	0	0	0
303	0.014509015	0.6969697	0	0	0
304	0.021763523	0.6969697	0	0	0
305	0.029018030	0.6969697	0	0	0
306	0.036272538	0.6969697	0	0	0
307	0.043527045	0.6969697	0	0	0
308	0.050781553	0.6969697	0	0	0
309	0.058036060	0.6969697	0	0	0
310	0.065290568	0.6969697	0	0	0
311	0.072545076	0.6969697	0	0	0
312	0.079799583	0.6969697	0	0	0
313	0.087054091	0.6969697	0	0	0
314	0.094308598	0.6969697	0	0	0
315	0.101563106	0.6969697	0	0	0

316	0.108817613	0.6969697	0	0	0
317	0.116072121	0.6969697	0	0	0
318	0.123326628	0.6969697	0	0	0
319	0.130581136	0.6969697	0	0	0
320	0.137835643	0.6969697	0	0	0
321	0.145090151	0.6969697	0	0	0
322	0.152344659	0.6969697	0	0	0
323	0.159599166	0.6969697	0	0	0
324	0.166853674	0.6969697	0	0	0
325	0.174108181	0.6969697	0	0	0
326	0.181362689	0.6969697	0	0	0
327	0.188617196	0.6969697	0	0	0
328	0.195871704	0.6969697	0	0	0
329	0.203126211	0.6969697	0	0	0
330	0.210380719	0.6969697	0	0	0
331	0.217635227	0.6969697	0	0	0
332	0.224889734	0.6969697	0	0	0
333	0.232144242	0.6969697	0	0	0
334	0.239398749	0.6969697	0	0	0
335	0.246653257	0.6969697	0	0	0
336	0.253907764	0.6969697	0	0	0
337	0.261162272	0.6969697	0	0	0
338	0.268416779	0.6969697	0	0	0
339	0.275671287	0.6969697	0	0	0
340	0.282925794	0.6969697	0	0	0
341	0.290180302	0.6969697	0	0	0
342	0.297434810	0.6969697	0	0	0
343	0.304689317	0.6969697	0	0	0
344	0.311943825	0.6969697	0	0	0
345	0.319198332	0.6969697	0	0	0
346	0.326452840	0.6969697	0	0	0
347	0.333707347	0.6969697	0	0	0
348	0.340961855	0.6969697	0	0	0
349	0.348216362	0.6969697	0	0	0
350	0.355470870	0.6969697	0	0	0
351	0.362725378	0.6969697	0	0	0
352	0.369979885	0.6969697	0	0	0
353	0.377234393	0.6969697	0	0	0
354	0.384488900	0.6969697	0	0	0
355	0.391743408	0.6969697	0	0	0
356	0.398997915	0.6969697	0	0	0
357	0.406252423	0.6969697	0	0	0
358	0.413506930	0.6969697	0	0	0
359	0.420761438	0.6969697	0	0	0
360	0.428015945	0.6969697	0	0	0

361	0.435270453	0.6969697	0	0	0
362	0.442524961	0.6969697	0	0	0
363	0.449779468	0.6969697	0	0	0
364	0.457033976	0.6969697	0	0	0
365	0.464288483	0.6969697	0	0	0
366	0.471542991	0.6969697	0	0	0
367	0.478797498	0.6969697	0	0	0
368	0.486052006	0.6969697	0	0	0
369	0.493306513	0.6969697	0	0	0
370	0.500561021	0.6969697	0	0	0
371	0.507815529	0.6969697	0	0	0
372	0.515070036	0.6969697	0	0	0
373	0.522324544	0.6969697	0	0	0
374	0.529579051	0.6969697	0	0	0
375	0.536833559	0.6969697	0	0	0
376	0.544088066	0.6969697	0	0	0
377	0.551342574	0.6969697	0	0	0
378	0.558597081	0.6969697	0	0	0
379	0.565851589	0.6969697	0	0	0
380	0.573106097	0.6969697	0	0	0
381	0.580360604	0.6969697	0	0	0
382	0.587615112	0.6969697	0	0	0
383	0.594869619	0.6969697	0	0	0
384	0.602124127	0.6969697	0	0	0
385	0.609378634	0.6969697	0	0	0
386	0.616633142	0.6969697	0	0	0
387	0.623887649	0.6969697	0	0	0
388	0.631142157	0.6969697	0	0	0
389	0.638396664	0.6969697	0	0	0
390	0.645651172	0.6969697	0	0	0
391	0.652905680	0.6969697	0	0	0
392	0.660160187	0.6969697	0	0	0
393	0.667414695	0.6969697	0	0	0
394	0.674669202	0.6969697	0	0	0
395	0.681923710	0.6969697	0	0	0
396	0.689178217	0.6969697	0	0	0
397	0.696432725	0.6969697	0	0	0
398	0.703687232	0.6969697	0	0	0
399	0.710941740	0.6969697	0	0	0
400	0.718196248	0.6969697	0	0	0
401	0.000000000	0.9292929	0	0	0
402	0.007254508	0.9292929	0	0	0
403	0.014509015	0.9292929	0	0	0
404	0.021763523	0.9292929	0	0	0
405	0.029018030	0.9292929	0	0	0

406	0.036272538	0.9292929	0	0	0
407	0.043527045	0.9292929	0	0	0
408	0.050781553	0.9292929	0	0	0
409	0.058036060	0.9292929	0	0	0
410	0.065290568	0.9292929	0	0	0
411	0.072545076	0.9292929	0	0	0
412	0.079799583	0.9292929	0	0	0
413	0.087054091	0.9292929	0	0	0
414	0.094308598	0.9292929	0	0	0
415	0.101563106	0.9292929	0	0	0
416	0.108817613	0.9292929	0	0	0
417	0.116072121	0.9292929	0	0	0
418	0.123326628	0.9292929	0	0	0
419	0.130581136	0.9292929	0	0	0
420	0.137835643	0.9292929	0	0	0
421	0.145090151	0.9292929	0	0	0
422	0.152344659	0.9292929	0	0	0
423	0.159599166	0.9292929	0	0	0
424	0.166853674	0.9292929	0	0	0
425	0.174108181	0.9292929	0	0	0
426	0.181362689	0.9292929	0	0	0
427	0.188617196	0.9292929	0	0	0
428	0.195871704	0.9292929	0	0	0
429	0.203126211	0.9292929	0	0	0
430	0.210380719	0.9292929	0	0	0
431	0.217635227	0.9292929	0	0	0
432	0.224889734	0.9292929	0	0	0
433	0.232144242	0.9292929	0	0	0
434	0.239398749	0.9292929	0	0	0
435	0.246653257	0.9292929	0	0	0
436	0.253907764	0.9292929	0	0	0
437	0.261162272	0.9292929	0	0	0
438	0.268416779	0.9292929	0	0	0
439	0.275671287	0.9292929	0	0	0
440	0.282925794	0.9292929	0	0	0
441	0.290180302	0.9292929	0	0	0
442	0.297434810	0.9292929	0	0	0
443	0.304689317	0.9292929	0	0	0
444	0.311943825	0.9292929	0	0	0
445	0.319198332	0.9292929	0	0	0
446	0.326452840	0.9292929	0	0	0
447	0.333707347	0.9292929	0	0	0
448	0.340961855	0.9292929	0	0	0
449	0.348216362	0.9292929	0	0	0
450	0.355470870	0.9292929	0	0	0

451	0.362725378	0.9292929	0	0	0
452	0.369979885	0.9292929	0	0	0
453	0.377234393	0.9292929	0	0	0
454	0.384488900	0.9292929	0	0	0
455	0.391743408	0.9292929	0	0	0
456	0.398997915	0.9292929	0	0	0
457	0.406252423	0.9292929	0	0	0
458	0.413506930	0.9292929	0	0	0
459	0.420761438	0.9292929	0	0	0
460	0.428015945	0.9292929	0	0	0
461	0.435270453	0.9292929	0	0	0
462	0.442524961	0.9292929	0	0	0
463	0.449779468	0.9292929	0	0	0
464	0.457033976	0.9292929	0	0	0
465	0.464288483	0.9292929	0	0	0
466	0.471542991	0.9292929	0	0	0
467	0.478797498	0.9292929	0	0	0
468	0.486052006	0.9292929	0	0	0
469	0.493306513	0.9292929	0	0	0
470	0.500561021	0.9292929	0	0	0
471	0.507815529	0.9292929	0	0	0
472	0.515070036	0.9292929	0	0	0
473	0.522324544	0.9292929	0	0	0
474	0.529579051	0.9292929	0	0	0
475	0.536833559	0.9292929	0	0	0
476	0.544088066	0.9292929	0	0	0
477	0.551342574	0.9292929	0	0	0
478	0.558597081	0.9292929	0	0	0
479	0.565851589	0.9292929	0	0	0
480	0.573106097	0.9292929	0	0	0
481	0.580360604	0.9292929	0	0	0
482	0.587615112	0.9292929	0	0	0
483	0.594869619	0.9292929	0	0	0
484	0.602124127	0.9292929	0	0	0
485	0.609378634	0.9292929	0	0	0
486	0.616633142	0.9292929	0	0	0
487	0.623887649	0.9292929	0	0	0
488	0.631142157	0.9292929	0	0	0
489	0.638396664	0.9292929	0	0	0
490	0.645651172	0.9292929	0	0	0
491	0.652905680	0.9292929	0	0	0
492	0.660160187	0.9292929	0	0	0
493	0.667414695	0.9292929	0	0	0
494	0.674669202	0.9292929	0	0	0
495	0.681923710	0.9292929	0	0	0

496	0.689178217	0.9292929	0	0	0
497	0.696432725	0.9292929	0	0	0
498	0.703687232	0.9292929	0	0	0
499	0.710941740	0.9292929	0	0	0
500	0.718196248	0.9292929	0	0	0
501	0.000000000	1.1616162	0	0	0
502	0.007254508	1.1616162	0	0	0
503	0.014509015	1.1616162	0	0	0
504	0.021763523	1.1616162	0	0	0
505	0.029018030	1.1616162	0	0	0
506	0.036272538	1.1616162	0	0	0
507	0.043527045	1.1616162	0	0	0
508	0.050781553	1.1616162	0	0	0
509	0.058036060	1.1616162	0	0	0
510	0.065290568	1.1616162	0	0	0
511	0.072545076	1.1616162	0	0	0
512	0.079799583	1.1616162	0	0	0
513	0.087054091	1.1616162	0	0	0
514	0.094308598	1.1616162	0	0	0
515	0.101563106	1.1616162	0	0	0
516	0.108817613	1.1616162	0	0	0
517	0.116072121	1.1616162	0	0	0
518	0.123326628	1.1616162	0	0	0
519	0.130581136	1.1616162	0	0	0
520	0.137835643	1.1616162	0	0	0
521	0.145090151	1.1616162	0	0	0
522	0.152344659	1.1616162	0	0	0
523	0.159599166	1.1616162	0	0	0
524	0.166853674	1.1616162	0	0	0
525	0.174108181	1.1616162	0	0	0
526	0.181362689	1.1616162	0	0	0
527	0.188617196	1.1616162	0	0	0
528	0.195871704	1.1616162	0	0	0
529	0.203126211	1.1616162	0	0	0
530	0.210380719	1.1616162	0	0	0
531	0.217635227	1.1616162	0	0	0
532	0.224889734	1.1616162	0	0	0
533	0.232144242	1.1616162	0	0	0
534	0.239398749	1.1616162	0	0	0
535	0.246653257	1.1616162	0	0	0
536	0.253907764	1.1616162	0	0	0
537	0.261162272	1.1616162	0	0	0
538	0.268416779	1.1616162	0	0	0
539	0.275671287	1.1616162	0	0	0
540	0.282925794	1.1616162	0	0	0

541	0.290180302	1.1616162	0	0	0
542	0.297434810	1.1616162	0	0	0
543	0.304689317	1.1616162	0	0	0
544	0.311943825	1.1616162	0	0	0
545	0.319198332	1.1616162	0	0	0
546	0.326452840	1.1616162	0	0	0
547	0.333707347	1.1616162	0	0	0
548	0.340961855	1.1616162	0	0	0
549	0.348216362	1.1616162	0	0	0
550	0.355470870	1.1616162	0	0	0
551	0.362725378	1.1616162	0	0	0
552	0.369979885	1.1616162	0	0	0
553	0.377234393	1.1616162	0	0	0
554	0.384488900	1.1616162	0	0	0
555	0.391743408	1.1616162	0	0	0
556	0.398997915	1.1616162	0	0	0
557	0.406252423	1.1616162	0	0	0
558	0.413506930	1.1616162	0	0	0
559	0.420761438	1.1616162	0	0	0
560	0.428015945	1.1616162	0	0	0
561	0.435270453	1.1616162	0	0	0
562	0.442524961	1.1616162	0	0	0
563	0.449779468	1.1616162	0	0	0
564	0.457033976	1.1616162	0	0	0
565	0.464288483	1.1616162	0	0	0
566	0.471542991	1.1616162	0	0	0
567	0.478797498	1.1616162	0	0	0
568	0.486052006	1.1616162	0	0	0
569	0.493306513	1.1616162	0	0	0
570	0.500561021	1.1616162	0	0	0
571	0.507815529	1.1616162	0	0	0
572	0.515070036	1.1616162	0	0	0
573	0.522324544	1.1616162	0	0	0
574	0.529579051	1.1616162	0	0	0
575	0.536833559	1.1616162	0	0	0
576	0.544088066	1.1616162	0	0	0
577	0.551342574	1.1616162	0	0	0
578	0.558597081	1.1616162	0	0	0
579	0.565851589	1.1616162	0	0	0
580	0.573106097	1.1616162	0	0	0
581	0.580360604	1.1616162	0	0	0
582	0.587615112	1.1616162	0	0	0
583	0.594869619	1.1616162	0	0	0
584	0.602124127	1.1616162	0	0	0
585	0.609378634	1.1616162	0	0	0

586	0.616633142	1.1616162	0	0	0
587	0.623887649	1.1616162	0	0	0
588	0.631142157	1.1616162	0	0	0
589	0.638396664	1.1616162	0	0	0
590	0.645651172	1.1616162	0	0	0
591	0.652905680	1.1616162	0	0	0
592	0.660160187	1.1616162	0	0	0
593	0.667414695	1.1616162	0	0	0
594	0.674669202	1.1616162	0	0	0
595	0.681923710	1.1616162	0	0	0
596	0.689178217	1.1616162	0	0	0
597	0.696432725	1.1616162	0	0	0
598	0.703687232	1.1616162	0	0	0
599	0.710941740	1.1616162	0	0	0
600	0.718196248	1.1616162	0	0	0
601	0.000000000	1.3939394	0	0	0
602	0.007254508	1.3939394	0	0	0
603	0.014509015	1.3939394	0	0	0
604	0.021763523	1.3939394	0	0	0
605	0.029018030	1.3939394	0	0	0
606	0.036272538	1.3939394	0	0	0
607	0.043527045	1.3939394	0	0	0
608	0.050781553	1.3939394	0	0	0
609	0.058036060	1.3939394	0	0	0
610	0.065290568	1.3939394	0	0	0
611	0.072545076	1.3939394	0	0	0
612	0.079799583	1.3939394	0	0	0
613	0.087054091	1.3939394	0	0	0
614	0.094308598	1.3939394	0	0	0
615	0.101563106	1.3939394	0	0	0
616	0.108817613	1.3939394	0	0	0
617	0.116072121	1.3939394	0	0	0
618	0.123326628	1.3939394	0	0	0
619	0.130581136	1.3939394	0	0	0
620	0.137835643	1.3939394	0	0	0
621	0.145090151	1.3939394	0	0	0
622	0.152344659	1.3939394	0	0	0
623	0.159599166	1.3939394	0	0	0
624	0.166853674	1.3939394	0	0	0
625	0.174108181	1.3939394	0	0	0
626	0.181362689	1.3939394	0	0	0
627	0.188617196	1.3939394	0	0	0
628	0.195871704	1.3939394	0	0	0
629	0.203126211	1.3939394	0	0	0
630	0.210380719	1.3939394	0	0	0

631	0.217635227	1.3939394	0	0	0
632	0.224889734	1.3939394	0	0	0
633	0.232144242	1.3939394	0	0	0
634	0.239398749	1.3939394	0	0	0
635	0.246653257	1.3939394	0	0	0
636	0.253907764	1.3939394	0	0	0
637	0.261162272	1.3939394	0	0	0
638	0.268416779	1.3939394	0	0	0
639	0.275671287	1.3939394	0	0	0
640	0.282925794	1.3939394	0	0	0
641	0.290180302	1.3939394	0	0	0
642	0.297434810	1.3939394	0	0	0
643	0.304689317	1.3939394	0	0	0
644	0.311943825	1.3939394	0	0	0
645	0.319198332	1.3939394	0	0	0
646	0.326452840	1.3939394	0	0	0
647	0.333707347	1.3939394	0	0	0
648	0.340961855	1.3939394	0	0	0
649	0.348216362	1.3939394	0	0	0
650	0.355470870	1.3939394	0	0	0
651	0.362725378	1.3939394	0	0	0
652	0.369979885	1.3939394	0	0	0
653	0.377234393	1.3939394	0	0	0
654	0.384488900	1.3939394	0	0	0
655	0.391743408	1.3939394	0	0	0
656	0.398997915	1.3939394	0	0	0
657	0.406252423	1.3939394	0	0	0
658	0.413506930	1.3939394	0	0	0
659	0.420761438	1.3939394	0	0	0
660	0.428015945	1.3939394	0	0	0
661	0.435270453	1.3939394	0	0	0
662	0.442524961	1.3939394	0	0	0
663	0.449779468	1.3939394	0	0	0
664	0.457033976	1.3939394	0	0	0
665	0.464288483	1.3939394	0	0	0
666	0.471542991	1.3939394	0	0	0
667	0.478797498	1.3939394	0	0	0
668	0.486052006	1.3939394	0	0	0
669	0.493306513	1.3939394	0	0	0
670	0.500561021	1.3939394	0	0	0
671	0.507815529	1.3939394	0	0	0
672	0.515070036	1.3939394	0	0	0
673	0.522324544	1.3939394	0	0	0
674	0.529579051	1.3939394	0	0	0
675	0.536833559	1.3939394	0	0	0

676	0.544088066	1.3939394	0	0	0
677	0.551342574	1.3939394	0	0	0
678	0.558597081	1.3939394	0	0	0
679	0.565851589	1.3939394	0	0	0
680	0.573106097	1.3939394	0	0	0
681	0.580360604	1.3939394	0	0	0
682	0.587615112	1.3939394	0	0	0
683	0.594869619	1.3939394	0	0	0
684	0.602124127	1.3939394	0	0	0
685	0.609378634	1.3939394	0	0	0
686	0.616633142	1.3939394	0	0	0
687	0.623887649	1.3939394	0	0	0
688	0.631142157	1.3939394	0	0	0
689	0.638396664	1.3939394	0	0	0
690	0.645651172	1.3939394	0	0	0
691	0.652905680	1.3939394	0	0	0
692	0.660160187	1.3939394	0	0	0
693	0.667414695	1.3939394	0	0	0
694	0.674669202	1.3939394	0	0	0
695	0.681923710	1.3939394	0	0	0
696	0.689178217	1.3939394	0	0	0
697	0.696432725	1.3939394	0	0	0
698	0.703687232	1.3939394	0	0	0
699	0.710941740	1.3939394	0	0	0
700	0.718196248	1.3939394	0	0	0
701	0.000000000	1.6262626	0	0	0
702	0.007254508	1.6262626	0	0	0
703	0.014509015	1.6262626	0	0	0
704	0.021763523	1.6262626	0	0	0
705	0.029018030	1.6262626	0	0	0
706	0.036272538	1.6262626	0	0	0
707	0.043527045	1.6262626	0	0	0
708	0.050781553	1.6262626	0	0	0
709	0.058036060	1.6262626	0	0	0
710	0.065290568	1.6262626	0	0	0
711	0.072545076	1.6262626	0	0	0
712	0.079799583	1.6262626	0	0	0
713	0.087054091	1.6262626	0	0	0
714	0.094308598	1.6262626	0	0	0
715	0.101563106	1.6262626	0	0	0
716	0.108817613	1.6262626	0	0	0
717	0.116072121	1.6262626	0	0	0
718	0.123326628	1.6262626	0	0	0
719	0.130581136	1.6262626	0	0	0
720	0.137835643	1.6262626	0	0	0

721	0.145090151	1.6262626	0	0	0
722	0.152344659	1.6262626	0	0	0
723	0.159599166	1.6262626	0	0	0
724	0.166853674	1.6262626	0	0	0
725	0.174108181	1.6262626	0	0	0
726	0.181362689	1.6262626	0	0	0
727	0.188617196	1.6262626	0	0	0
728	0.195871704	1.6262626	0	0	0
729	0.203126211	1.6262626	0	0	0
730	0.210380719	1.6262626	0	0	0
731	0.217635227	1.6262626	0	0	0
732	0.224889734	1.6262626	0	0	0
733	0.232144242	1.6262626	0	0	0
734	0.239398749	1.6262626	0	0	0
735	0.246653257	1.6262626	0	0	0
736	0.253907764	1.6262626	0	0	0
737	0.261162272	1.6262626	0	0	0
738	0.268416779	1.6262626	0	0	0
739	0.275671287	1.6262626	0	0	0
740	0.282925794	1.6262626	0	0	0
741	0.290180302	1.6262626	0	0	0
742	0.297434810	1.6262626	0	0	0
743	0.304689317	1.6262626	0	0	0
744	0.311943825	1.6262626	0	0	0
745	0.319198332	1.6262626	0	0	0
746	0.326452840	1.6262626	0	0	0
747	0.333707347	1.6262626	0	0	0
748	0.340961855	1.6262626	0	0	0
749	0.348216362	1.6262626	0	0	0
750	0.355470870	1.6262626	0	0	0
751	0.362725378	1.6262626	0	0	0
752	0.369979885	1.6262626	0	0	0
753	0.377234393	1.6262626	0	0	0
754	0.384488900	1.6262626	0	0	0
755	0.391743408	1.6262626	0	0	0
756	0.398997915	1.6262626	0	0	0
757	0.406252423	1.6262626	0	0	0
758	0.413506930	1.6262626	0	0	0
759	0.420761438	1.6262626	0	0	0
760	0.428015945	1.6262626	0	0	0
761	0.435270453	1.6262626	0	0	0
762	0.442524961	1.6262626	0	0	0
763	0.449779468	1.6262626	0	0	0
764	0.457033976	1.6262626	0	0	0
765	0.464288483	1.6262626	0	0	0

766	0.471542991	1.6262626	0	0	0
767	0.478797498	1.6262626	0	0	0
768	0.486052006	1.6262626	0	0	0
769	0.493306513	1.6262626	0	0	0
770	0.500561021	1.6262626	0	0	0
771	0.507815529	1.6262626	0	0	0
772	0.515070036	1.6262626	0	0	0
773	0.522324544	1.6262626	0	0	0
774	0.529579051	1.6262626	0	0	0
775	0.536833559	1.6262626	0	0	0
776	0.544088066	1.6262626	0	0	0
777	0.551342574	1.6262626	0	0	0
778	0.558597081	1.6262626	0	0	0
779	0.565851589	1.6262626	0	0	0
780	0.573106097	1.6262626	0	0	0
781	0.580360604	1.6262626	0	0	0
782	0.587615112	1.6262626	0	0	0
783	0.594869619	1.6262626	0	0	0
784	0.602124127	1.6262626	0	0	0
785	0.609378634	1.6262626	0	0	0
786	0.616633142	1.6262626	0	0	0
787	0.623887649	1.6262626	0	0	0
788	0.631142157	1.6262626	0	0	0
789	0.638396664	1.6262626	0	0	0
790	0.645651172	1.6262626	0	0	0
791	0.652905680	1.6262626	0	0	0
792	0.660160187	1.6262626	0	0	0
793	0.667414695	1.6262626	0	0	0
794	0.674669202	1.6262626	0	0	0
795	0.681923710	1.6262626	0	0	0
796	0.689178217	1.6262626	0	0	0
797	0.696432725	1.6262626	0	0	0
798	0.703687232	1.6262626	0	0	0
799	0.710941740	1.6262626	0	0	0
800	0.718196248	1.6262626	0	0	0
801	0.000000000	1.8585859	0	0	0
802	0.007254508	1.8585859	0	0	0
803	0.014509015	1.8585859	0	0	0
804	0.021763523	1.8585859	0	0	0
805	0.029018030	1.8585859	0	0	0
806	0.036272538	1.8585859	0	0	0
807	0.043527045	1.8585859	0	0	0
808	0.050781553	1.8585859	0	0	0
809	0.058036060	1.8585859	0	0	0
810	0.065290568	1.8585859	0	0	0

811	0.072545076	1.8585859	0	0	0
812	0.079799583	1.8585859	0	0	0
813	0.087054091	1.8585859	0	0	0
814	0.094308598	1.8585859	0	0	0
815	0.101563106	1.8585859	0	0	0
816	0.108817613	1.8585859	0	0	0
817	0.116072121	1.8585859	0	0	0
818	0.123326628	1.8585859	0	0	0
819	0.130581136	1.8585859	0	0	0
820	0.137835643	1.8585859	0	0	0
821	0.145090151	1.8585859	0	0	0
822	0.152344659	1.8585859	0	0	0
823	0.159599166	1.8585859	0	0	0
824	0.166853674	1.8585859	0	0	0
825	0.174108181	1.8585859	0	0	0
826	0.181362689	1.8585859	0	0	0
827	0.188617196	1.8585859	0	0	0
828	0.195871704	1.8585859	0	0	0
829	0.203126211	1.8585859	0	0	0
830	0.210380719	1.8585859	0	0	0
831	0.217635227	1.8585859	0	0	0
832	0.224889734	1.8585859	0	0	0
833	0.232144242	1.8585859	0	0	0
834	0.239398749	1.8585859	0	0	0
835	0.246653257	1.8585859	0	0	0
836	0.253907764	1.8585859	0	0	0
837	0.261162272	1.8585859	0	0	0
838	0.268416779	1.8585859	0	0	0
839	0.275671287	1.8585859	0	0	0
840	0.282925794	1.8585859	0	0	0
841	0.290180302	1.8585859	0	0	0
842	0.297434810	1.8585859	0	0	0
843	0.304689317	1.8585859	0	0	0
844	0.311943825	1.8585859	0	0	0
845	0.319198332	1.8585859	0	0	0
846	0.326452840	1.8585859	0	0	0
847	0.333707347	1.8585859	0	0	0
848	0.340961855	1.8585859	0	0	0
849	0.348216362	1.8585859	0	0	0
850	0.355470870	1.8585859	0	0	0
851	0.362725378	1.8585859	0	0	0
852	0.369979885	1.8585859	0	0	0
853	0.377234393	1.8585859	0	0	0
854	0.384488900	1.8585859	0	0	0
855	0.391743408	1.8585859	0	0	0

856	0.398997915	1.8585859	0	0	0
857	0.406252423	1.8585859	0	0	0
858	0.413506930	1.8585859	0	0	0
859	0.420761438	1.8585859	0	0	0
860	0.428015945	1.8585859	0	0	0
861	0.435270453	1.8585859	0	0	0
862	0.442524961	1.8585859	0	0	0
863	0.449779468	1.8585859	0	0	0
864	0.457033976	1.8585859	0	0	0
865	0.464288483	1.8585859	0	0	0
866	0.471542991	1.8585859	0	0	0
867	0.478797498	1.8585859	0	0	0
868	0.486052006	1.8585859	0	0	0
869	0.493306513	1.8585859	0	0	0
870	0.500561021	1.8585859	0	0	0
871	0.507815529	1.8585859	0	0	0
872	0.515070036	1.8585859	0	0	0
873	0.522324544	1.8585859	0	0	0
874	0.529579051	1.8585859	0	0	0
875	0.536833559	1.8585859	0	0	0
876	0.544088066	1.8585859	0	0	0
877	0.551342574	1.8585859	0	0	0
878	0.558597081	1.8585859	0	0	0
879	0.565851589	1.8585859	0	0	0
880	0.573106097	1.8585859	0	0	0
881	0.580360604	1.8585859	0	0	0
882	0.587615112	1.8585859	0	0	0
883	0.594869619	1.8585859	0	0	0
884	0.602124127	1.8585859	0	0	0
885	0.609378634	1.8585859	0	0	0
886	0.616633142	1.8585859	0	0	0
887	0.623887649	1.8585859	0	0	0
888	0.631142157	1.8585859	0	0	0
889	0.638396664	1.8585859	0	0	0
890	0.645651172	1.8585859	0	0	0
891	0.652905680	1.8585859	0	0	0
892	0.660160187	1.8585859	0	0	0
893	0.667414695	1.8585859	0	0	0
894	0.674669202	1.8585859	0	0	0
895	0.681923710	1.8585859	0	0	0
896	0.689178217	1.8585859	0	0	0
897	0.696432725	1.8585859	0	0	0
898	0.703687232	1.8585859	0	0	0
899	0.710941740	1.8585859	0	0	0
900	0.718196248	1.8585859	0	0	0

901	0.000000000	2.0909091	0	0	0
902	0.007254508	2.0909091	0	0	0
903	0.014509015	2.0909091	0	0	0
904	0.021763523	2.0909091	0	0	0
905	0.029018030	2.0909091	0	0	0
906	0.036272538	2.0909091	0	0	0
907	0.043527045	2.0909091	0	0	0
908	0.050781553	2.0909091	0	0	0
909	0.058036060	2.0909091	0	0	0
910	0.065290568	2.0909091	0	0	0
911	0.072545076	2.0909091	0	0	0
912	0.079799583	2.0909091	0	0	0
913	0.087054091	2.0909091	0	0	0
914	0.094308598	2.0909091	0	0	0
915	0.101563106	2.0909091	0	0	0
916	0.108817613	2.0909091	0	0	0
917	0.116072121	2.0909091	0	0	0
918	0.123326628	2.0909091	0	0	0
919	0.130581136	2.0909091	0	0	0
920	0.137835643	2.0909091	0	0	0
921	0.145090151	2.0909091	0	0	0
922	0.152344659	2.0909091	0	0	0
923	0.159599166	2.0909091	0	0	0
924	0.166853674	2.0909091	0	0	0
925	0.174108181	2.0909091	0	0	0
926	0.181362689	2.0909091	0	0	0
927	0.188617196	2.0909091	0	0	0
928	0.195871704	2.0909091	0	0	0
929	0.203126211	2.0909091	0	0	0
930	0.210380719	2.0909091	0	0	0
931	0.217635227	2.0909091	0	0	0
932	0.224889734	2.0909091	0	0	0
933	0.232144242	2.0909091	0	0	0
934	0.239398749	2.0909091	0	0	0
935	0.246653257	2.0909091	0	0	0
936	0.253907764	2.0909091	0	0	0
937	0.261162272	2.0909091	0	0	0
938	0.268416779	2.0909091	0	0	0
939	0.275671287	2.0909091	0	0	0
940	0.282925794	2.0909091	0	0	0
941	0.290180302	2.0909091	0	0	0
942	0.297434810	2.0909091	0	0	0
943	0.304689317	2.0909091	0	0	0
944	0.311943825	2.0909091	0	0	0
945	0.319198332	2.0909091	0	0	0

946	0.326452840	2.0909091	0	0	0
947	0.333707347	2.0909091	0	0	0
948	0.340961855	2.0909091	0	0	0
949	0.348216362	2.0909091	0	0	0
950	0.355470870	2.0909091	0	0	0
951	0.362725378	2.0909091	0	0	0
952	0.369979885	2.0909091	0	0	0
953	0.377234393	2.0909091	0	0	0
954	0.384488900	2.0909091	0	0	0
955	0.391743408	2.0909091	0	0	0
956	0.398997915	2.0909091	0	0	0
957	0.406252423	2.0909091	0	0	0
958	0.413506930	2.0909091	0	0	0
959	0.420761438	2.0909091	0	0	0
960	0.428015945	2.0909091	0	0	0
961	0.435270453	2.0909091	0	0	0
962	0.442524961	2.0909091	0	0	0
963	0.449779468	2.0909091	0	0	0
964	0.457033976	2.0909091	0	0	0
965	0.464288483	2.0909091	0	0	0
966	0.471542991	2.0909091	0	0	0
967	0.478797498	2.0909091	0	0	0
968	0.486052006	2.0909091	0	0	0
969	0.493306513	2.0909091	0	0	0
970	0.500561021	2.0909091	0	0	0
971	0.507815529	2.0909091	0	0	0
972	0.515070036	2.0909091	0	0	0
973	0.522324544	2.0909091	0	0	0
974	0.529579051	2.0909091	0	0	0
975	0.536833559	2.0909091	0	0	0
976	0.544088066	2.0909091	0	0	0
977	0.551342574	2.0909091	0	0	0
978	0.558597081	2.0909091	0	0	0
979	0.565851589	2.0909091	0	0	0
980	0.573106097	2.0909091	0	0	0
981	0.580360604	2.0909091	0	0	0
982	0.587615112	2.0909091	0	0	0
983	0.594869619	2.0909091	0	0	0
984	0.602124127	2.0909091	0	0	0
985	0.609378634	2.0909091	0	0	0
986	0.616633142	2.0909091	0	0	0
987	0.623887649	2.0909091	0	0	0
988	0.631142157	2.0909091	0	0	0
989	0.638396664	2.0909091	0	0	0
990	0.645651172	2.0909091	0	0	0

991	0.652905680	2.0909091	0	0	0
992	0.660160187	2.0909091	0	0	0
993	0.667414695	2.0909091	0	0	0
994	0.674669202	2.0909091	0	0	0
995	0.681923710	2.0909091	0	0	0
996	0.689178217	2.0909091	0	0	0
997	0.696432725	2.0909091	0	0	0
998	0.703687232	2.0909091	0	0	0
999	0.710941740	2.0909091	0	0	0
1000	0.718196248	2.0909091	0	0	0
1001	0.000000000	2.3232323	0	0	0
1002	0.007254508	2.3232323	0	0	0
1003	0.014509015	2.3232323	0	0	0
1004	0.021763523	2.3232323	0	0	0
1005	0.029018030	2.3232323	0	0	0
1006	0.036272538	2.3232323	0	0	0
1007	0.043527045	2.3232323	0	0	0
1008	0.050781553	2.3232323	0	0	0
1009	0.058036060	2.3232323	0	0	0
1010	0.065290568	2.3232323	0	0	0
1011	0.072545076	2.3232323	0	0	0
1012	0.079799583	2.3232323	0	0	0
1013	0.087054091	2.3232323	0	0	0
1014	0.094308598	2.3232323	0	0	0
1015	0.101563106	2.3232323	0	0	0
1016	0.108817613	2.3232323	0	0	0
1017	0.116072121	2.3232323	0	0	0
1018	0.123326628	2.3232323	0	0	0
1019	0.130581136	2.3232323	0	0	0
1020	0.137835643	2.3232323	0	0	0
1021	0.145090151	2.3232323	0	0	0
1022	0.152344659	2.3232323	0	0	0
1023	0.159599166	2.3232323	0	0	0
1024	0.166853674	2.3232323	0	0	0
1025	0.174108181	2.3232323	0	0	0
1026	0.181362689	2.3232323	0	0	0
1027	0.188617196	2.3232323	0	0	0
1028	0.195871704	2.3232323	0	0	0
1029	0.203126211	2.3232323	0	0	0
1030	0.210380719	2.3232323	0	0	0
1031	0.217635227	2.3232323	0	0	0
1032	0.224889734	2.3232323	0	0	0
1033	0.232144242	2.3232323	0	0	0
1034	0.239398749	2.3232323	0	0	0
1035	0.246653257	2.3232323	0	0	0

1036	0.253907764	2.3232323	0	0	0
1037	0.261162272	2.3232323	0	0	0
1038	0.268416779	2.3232323	0	0	0
1039	0.275671287	2.3232323	0	0	0
1040	0.282925794	2.3232323	0	0	0
1041	0.290180302	2.3232323	0	0	0
1042	0.297434810	2.3232323	0	0	0
1043	0.304689317	2.3232323	0	0	0
1044	0.311943825	2.3232323	0	0	0
1045	0.319198332	2.3232323	0	0	0
1046	0.326452840	2.3232323	0	0	0
1047	0.333707347	2.3232323	0	0	0
1048	0.340961855	2.3232323	0	0	0
1049	0.348216362	2.3232323	0	0	0
1050	0.355470870	2.3232323	0	0	0
1051	0.362725378	2.3232323	0	0	0
1052	0.369979885	2.3232323	0	0	0
1053	0.377234393	2.3232323	0	0	0
1054	0.384488900	2.3232323	0	0	0
1055	0.391743408	2.3232323	0	0	0
1056	0.398997915	2.3232323	0	0	0
1057	0.406252423	2.3232323	0	0	0
1058	0.413506930	2.3232323	0	0	0
1059	0.420761438	2.3232323	0	0	0
1060	0.428015945	2.3232323	0	0	0
1061	0.435270453	2.3232323	0	0	0
1062	0.442524961	2.3232323	0	0	0
1063	0.449779468	2.3232323	0	0	0
1064	0.457033976	2.3232323	0	0	0
1065	0.464288483	2.3232323	0	0	0
1066	0.471542991	2.3232323	0	0	0
1067	0.478797498	2.3232323	0	0	0
1068	0.486052006	2.3232323	0	0	0
1069	0.493306513	2.3232323	0	0	0
1070	0.500561021	2.3232323	0	0	0
1071	0.507815529	2.3232323	0	0	0
1072	0.515070036	2.3232323	0	0	0
1073	0.522324544	2.3232323	0	0	0
1074	0.529579051	2.3232323	0	0	0
1075	0.536833559	2.3232323	0	0	0
1076	0.544088066	2.3232323	0	0	0
1077	0.551342574	2.3232323	0	0	0
1078	0.558597081	2.3232323	0	0	0
1079	0.565851589	2.3232323	0	0	0
1080	0.573106097	2.3232323	0	0	0

1081	0.580360604	2.3232323	0	0	0
1082	0.587615112	2.3232323	0	0	0
1083	0.594869619	2.3232323	0	0	0
1084	0.602124127	2.3232323	0	0	0
1085	0.609378634	2.3232323	0	0	0
1086	0.616633142	2.3232323	0	0	0
1087	0.623887649	2.3232323	0	0	0
1088	0.631142157	2.3232323	0	0	0
1089	0.638396664	2.3232323	0	0	0
1090	0.645651172	2.3232323	0	0	0
1091	0.652905680	2.3232323	0	0	0
1092	0.660160187	2.3232323	0	0	0
1093	0.667414695	2.3232323	0	0	0
1094	0.674669202	2.3232323	0	0	0
1095	0.681923710	2.3232323	0	0	0
1096	0.689178217	2.3232323	0	0	0
1097	0.696432725	2.3232323	0	0	0
1098	0.703687232	2.3232323	0	0	0
1099	0.710941740	2.3232323	0	0	0
1100	0.718196248	2.3232323	0	0	0
1101	0.000000000	2.5555556	0	0	0
1102	0.007254508	2.5555556	0	0	0
1103	0.014509015	2.5555556	0	0	0
1104	0.021763523	2.5555556	0	0	0
1105	0.029018030	2.5555556	0	0	0
1106	0.036272538	2.5555556	0	0	0
1107	0.043527045	2.5555556	0	0	0
1108	0.050781553	2.5555556	0	0	0
1109	0.058036060	2.5555556	0	0	0
1110	0.065290568	2.5555556	0	0	0
1111	0.072545076	2.5555556	0	0	0
1112	0.079799583	2.5555556	0	0	0
1113	0.087054091	2.5555556	0	0	0
1114	0.094308598	2.5555556	0	0	0
1115	0.101563106	2.5555556	0	0	0
1116	0.108817613	2.5555556	0	0	0
1117	0.116072121	2.5555556	0	0	0
1118	0.123326628	2.5555556	0	0	0
1119	0.130581136	2.5555556	0	0	0
1120	0.137835643	2.5555556	0	0	0
1121	0.145090151	2.5555556	0	0	0
1122	0.152344659	2.5555556	0	0	0
1123	0.159599166	2.5555556	0	0	0
1124	0.166853674	2.5555556	0	0	0
1125	0.174108181	2.5555556	0	0	0

1126	0.181362689	2.5555556	0	0	0
1127	0.188617196	2.5555556	0	0	0
1128	0.195871704	2.5555556	0	0	0
1129	0.203126211	2.5555556	0	0	0
1130	0.210380719	2.5555556	0	0	0
1131	0.217635227	2.5555556	0	0	0
1132	0.224889734	2.5555556	0	0	0
1133	0.232144242	2.5555556	0	0	0
1134	0.239398749	2.5555556	0	0	0
1135	0.246653257	2.5555556	0	0	0
1136	0.253907764	2.5555556	0	0	0
1137	0.261162272	2.5555556	0	0	0
1138	0.268416779	2.5555556	0	0	0
1139	0.275671287	2.5555556	0	0	0
1140	0.282925794	2.5555556	0	0	0
1141	0.290180302	2.5555556	0	0	0
1142	0.297434810	2.5555556	0	0	0
1143	0.304689317	2.5555556	0	0	0
1144	0.311943825	2.5555556	0	0	0
1145	0.319198332	2.5555556	0	0	0
1146	0.326452840	2.5555556	0	0	0
1147	0.333707347	2.5555556	0	0	0
1148	0.340961855	2.5555556	0	0	0
1149	0.348216362	2.5555556	0	0	0
1150	0.355470870	2.5555556	0	0	0
1151	0.362725378	2.5555556	0	0	0
1152	0.369979885	2.5555556	0	0	0
1153	0.377234393	2.5555556	0	0	0
1154	0.384488900	2.5555556	0	0	0
1155	0.391743408	2.5555556	0	0	0
1156	0.398997915	2.5555556	0	0	0
1157	0.406252423	2.5555556	0	0	0
1158	0.413506930	2.5555556	0	0	0
1159	0.420761438	2.5555556	0	0	0
1160	0.428015945	2.5555556	0	0	0
1161	0.435270453	2.5555556	0	0	0
1162	0.442524961	2.5555556	0	0	0
1163	0.449779468	2.5555556	0	0	0
1164	0.457033976	2.5555556	0	0	0
1165	0.464288483	2.5555556	0	0	0
1166	0.471542991	2.5555556	0	0	0
1167	0.478797498	2.5555556	0	0	0
1168	0.486052006	2.5555556	0	0	0
1169	0.493306513	2.5555556	0	0	0
1170	0.500561021	2.5555556	0	0	0

1171	0.507815529	2.5555556	0	0	0
1172	0.515070036	2.5555556	0	0	0
1173	0.522324544	2.5555556	0	0	0
1174	0.529579051	2.5555556	0	0	0
1175	0.536833559	2.5555556	0	0	0
1176	0.544088066	2.5555556	0	0	0
1177	0.551342574	2.5555556	0	0	0
1178	0.558597081	2.5555556	0	0	0
1179	0.565851589	2.5555556	0	0	0
1180	0.573106097	2.5555556	0	0	0
1181	0.580360604	2.5555556	0	0	0
1182	0.587615112	2.5555556	0	0	0
1183	0.594869619	2.5555556	0	0	0
1184	0.602124127	2.5555556	0	0	0
1185	0.609378634	2.5555556	0	0	0
1186	0.616633142	2.5555556	0	0	0
1187	0.623887649	2.5555556	0	0	0
1188	0.631142157	2.5555556	0	0	0
1189	0.638396664	2.5555556	0	0	0
1190	0.645651172	2.5555556	0	0	0
1191	0.652905680	2.5555556	0	0	0
1192	0.660160187	2.5555556	0	0	0
1193	0.667414695	2.5555556	0	0	0
1194	0.674669202	2.5555556	0	0	0
1195	0.681923710	2.5555556	0	0	0
1196	0.689178217	2.5555556	0	0	0
1197	0.696432725	2.5555556	0	0	0
1198	0.703687232	2.5555556	0	0	0
1199	0.710941740	2.5555556	0	0	0
1200	0.718196248	2.5555556	0	0	0
1201	0.000000000	2.7878788	0	0	0
1202	0.007254508	2.7878788	0	0	0
1203	0.014509015	2.7878788	0	0	0
1204	0.021763523	2.7878788	0	0	0
1205	0.029018030	2.7878788	0	0	0
1206	0.036272538	2.7878788	0	0	0
1207	0.043527045	2.7878788	0	0	0
1208	0.050781553	2.7878788	0	0	0
1209	0.058036060	2.7878788	0	0	0
1210	0.065290568	2.7878788	0	0	0
1211	0.072545076	2.7878788	0	0	0
1212	0.079799583	2.7878788	0	0	0
1213	0.087054091	2.7878788	0	0	0
1214	0.094308598	2.7878788	0	0	0
1215	0.101563106	2.7878788	0	0	0

1216	0.108817613	2.7878788	0	0	0
1217	0.116072121	2.7878788	0	0	0
1218	0.123326628	2.7878788	0	0	0
1219	0.130581136	2.7878788	0	0	0
1220	0.137835643	2.7878788	0	0	0
1221	0.145090151	2.7878788	0	0	0
1222	0.152344659	2.7878788	0	0	0
1223	0.159599166	2.7878788	0	0	0
1224	0.166853674	2.7878788	0	0	0
1225	0.174108181	2.7878788	0	0	0
1226	0.181362689	2.7878788	0	0	0
1227	0.188617196	2.7878788	0	0	0
1228	0.195871704	2.7878788	0	0	0
1229	0.203126211	2.7878788	0	0	0
1230	0.210380719	2.7878788	0	0	0
1231	0.217635227	2.7878788	0	0	0
1232	0.224889734	2.7878788	0	0	0
1233	0.232144242	2.7878788	0	0	0
1234	0.239398749	2.7878788	0	0	0
1235	0.246653257	2.7878788	0	0	0
1236	0.253907764	2.7878788	0	0	0
1237	0.261162272	2.7878788	0	0	0
1238	0.268416779	2.7878788	0	0	0
1239	0.275671287	2.7878788	0	0	0
1240	0.282925794	2.7878788	0	0	0
1241	0.290180302	2.7878788	0	0	0
1242	0.297434810	2.7878788	0	0	0
1243	0.304689317	2.7878788	0	0	0
1244	0.311943825	2.7878788	0	0	0
1245	0.319198332	2.7878788	0	0	0
1246	0.326452840	2.7878788	0	0	0
1247	0.333707347	2.7878788	0	0	0
1248	0.340961855	2.7878788	0	0	0
1249	0.348216362	2.7878788	0	0	0
1250	0.355470870	2.7878788	0	0	0
1251	0.362725378	2.7878788	0	0	0
1252	0.369979885	2.7878788	0	0	0
1253	0.377234393	2.7878788	0	0	0
1254	0.384488900	2.7878788	0	0	0
1255	0.391743408	2.7878788	0	0	0
1256	0.398997915	2.7878788	0	0	0
1257	0.406252423	2.7878788	0	0	0
1258	0.413506930	2.7878788	0	0	0
1259	0.420761438	2.7878788	0	0	0
1260	0.428015945	2.7878788	0	0	0

1261	0.435270453	2.7878788	0	0	0
1262	0.442524961	2.7878788	0	0	0
1263	0.449779468	2.7878788	0	0	0
1264	0.457033976	2.7878788	0	0	0
1265	0.464288483	2.7878788	0	0	0
1266	0.471542991	2.7878788	0	0	0
1267	0.478797498	2.7878788	0	0	0
1268	0.486052006	2.7878788	0	0	0
1269	0.493306513	2.7878788	0	0	0
1270	0.500561021	2.7878788	0	0	0
1271	0.507815529	2.7878788	0	0	0
1272	0.515070036	2.7878788	0	0	0
1273	0.522324544	2.7878788	0	0	0
1274	0.529579051	2.7878788	0	0	0
1275	0.536833559	2.7878788	0	0	0
1276	0.544088066	2.7878788	0	0	0
1277	0.551342574	2.7878788	0	0	0
1278	0.558597081	2.7878788	0	0	0
1279	0.565851589	2.7878788	0	0	0
1280	0.573106097	2.7878788	0	0	0
1281	0.580360604	2.7878788	0	0	0
1282	0.587615112	2.7878788	0	0	0
1283	0.594869619	2.7878788	0	0	0
1284	0.602124127	2.7878788	0	0	0
1285	0.609378634	2.7878788	0	0	0
1286	0.616633142	2.7878788	0	0	0
1287	0.623887649	2.7878788	0	0	0
1288	0.631142157	2.7878788	0	0	0
1289	0.638396664	2.7878788	0	0	0
1290	0.645651172	2.7878788	0	0	0
1291	0.652905680	2.7878788	0	0	0
1292	0.660160187	2.7878788	0	0	0
1293	0.667414695	2.7878788	0	0	0
1294	0.674669202	2.7878788	0	0	0
1295	0.681923710	2.7878788	0	0	0
1296	0.689178217	2.7878788	0	0	0
1297	0.696432725	2.7878788	0	0	0
1298	0.703687232	2.7878788	0	0	0
1299	0.710941740	2.7878788	0	0	0
1300	0.718196248	2.7878788	0	0	0
1301	0.000000000	3.0202020	0	0	0
1302	0.007254508	3.0202020	0	0	0
1303	0.014509015	3.0202020	0	0	0
1304	0.021763523	3.0202020	0	0	0
1305	0.029018030	3.0202020	0	0	0

1306	0.036272538	3.0202020	0	0	0
1307	0.043527045	3.0202020	0	0	0
1308	0.050781553	3.0202020	0	0	0
1309	0.058036060	3.0202020	0	0	0
1310	0.065290568	3.0202020	0	0	0
1311	0.072545076	3.0202020	0	0	0
1312	0.079799583	3.0202020	0	0	0
1313	0.087054091	3.0202020	0	0	0
1314	0.094308598	3.0202020	0	0	0
1315	0.101563106	3.0202020	0	0	0
1316	0.108817613	3.0202020	0	0	0
1317	0.116072121	3.0202020	0	0	0
1318	0.123326628	3.0202020	0	0	0
1319	0.130581136	3.0202020	0	0	0
1320	0.137835643	3.0202020	0	0	0
1321	0.145090151	3.0202020	0	0	0
1322	0.152344659	3.0202020	0	0	0
1323	0.159599166	3.0202020	0	0	0
1324	0.166853674	3.0202020	0	0	0
1325	0.174108181	3.0202020	0	0	0
1326	0.181362689	3.0202020	0	0	0
1327	0.188617196	3.0202020	0	0	0
1328	0.195871704	3.0202020	0	0	0
1329	0.203126211	3.0202020	0	0	0
1330	0.210380719	3.0202020	0	0	0
1331	0.217635227	3.0202020	0	0	0
1332	0.224889734	3.0202020	0	0	0
1333	0.232144242	3.0202020	0	0	0
1334	0.239398749	3.0202020	0	0	0
1335	0.246653257	3.0202020	0	0	0
1336	0.253907764	3.0202020	0	0	0
1337	0.261162272	3.0202020	0	0	0
1338	0.268416779	3.0202020	0	0	0
1339	0.275671287	3.0202020	0	0	0
1340	0.282925794	3.0202020	0	0	0
1341	0.290180302	3.0202020	0	0	0
1342	0.297434810	3.0202020	0	0	0
1343	0.304689317	3.0202020	0	0	0
1344	0.311943825	3.0202020	0	0	0
1345	0.319198332	3.0202020	0	0	0
1346	0.326452840	3.0202020	0	0	0
1347	0.333707347	3.0202020	0	0	0
1348	0.340961855	3.0202020	0	0	0
1349	0.348216362	3.0202020	0	0	0
1350	0.355470870	3.0202020	0	0	0

1351	0.362725378	3.0202020	0	0	0
1352	0.369979885	3.0202020	0	0	0
1353	0.377234393	3.0202020	0	0	0
1354	0.384488900	3.0202020	0	0	0
1355	0.391743408	3.0202020	0	0	0
1356	0.398997915	3.0202020	0	0	0
1357	0.406252423	3.0202020	0	0	0
1358	0.413506930	3.0202020	0	0	0
1359	0.420761438	3.0202020	0	0	0
1360	0.428015945	3.0202020	0	0	0
1361	0.435270453	3.0202020	0	0	0
1362	0.442524961	3.0202020	0	0	0
1363	0.449779468	3.0202020	0	0	0
1364	0.457033976	3.0202020	0	0	0
1365	0.464288483	3.0202020	0	0	0
1366	0.471542991	3.0202020	0	0	0
1367	0.478797498	3.0202020	0	0	0
1368	0.486052006	3.0202020	0	0	0
1369	0.493306513	3.0202020	0	0	0
1370	0.500561021	3.0202020	0	0	0
1371	0.507815529	3.0202020	0	0	0
1372	0.515070036	3.0202020	0	0	0
1373	0.522324544	3.0202020	0	0	0
1374	0.529579051	3.0202020	0	0	0
1375	0.536833559	3.0202020	0	0	0
1376	0.544088066	3.0202020	0	0	0
1377	0.551342574	3.0202020	0	0	0
1378	0.558597081	3.0202020	0	0	0
1379	0.565851589	3.0202020	0	0	0
1380	0.573106097	3.0202020	0	0	0
1381	0.580360604	3.0202020	0	0	0
1382	0.587615112	3.0202020	0	0	0
1383	0.594869619	3.0202020	0	0	0
1384	0.602124127	3.0202020	0	0	0
1385	0.609378634	3.0202020	0	0	0
1386	0.616633142	3.0202020	0	0	0
1387	0.623887649	3.0202020	0	0	0
1388	0.631142157	3.0202020	0	0	0
1389	0.638396664	3.0202020	0	0	0
1390	0.645651172	3.0202020	0	0	0
1391	0.652905680	3.0202020	0	0	0
1392	0.660160187	3.0202020	0	0	0
1393	0.667414695	3.0202020	0	0	0
1394	0.674669202	3.0202020	0	0	0
1395	0.681923710	3.0202020	0	0	0

1396	0.689178217	3.0202020	0	0	0
1397	0.696432725	3.0202020	0	0	0
1398	0.703687232	3.0202020	0	0	0
1399	0.710941740	3.0202020	0	0	0
1400	0.718196248	3.0202020	0	0	0
1401	0.000000000	3.2525253	0	0	0
1402	0.007254508	3.2525253	0	0	0
1403	0.014509015	3.2525253	0	0	0
1404	0.021763523	3.2525253	0	0	0
1405	0.029018030	3.2525253	0	0	0
1406	0.036272538	3.2525253	0	0	0
1407	0.043527045	3.2525253	0	0	0
1408	0.050781553	3.2525253	0	0	0
1409	0.058036060	3.2525253	0	0	0
1410	0.065290568	3.2525253	0	0	0
1411	0.072545076	3.2525253	0	0	0
1412	0.079799583	3.2525253	0	0	0
1413	0.087054091	3.2525253	0	0	0
1414	0.094308598	3.2525253	0	0	0
1415	0.101563106	3.2525253	0	0	0
1416	0.108817613	3.2525253	0	0	0
1417	0.116072121	3.2525253	0	0	0
1418	0.123326628	3.2525253	0	0	0
1419	0.130581136	3.2525253	0	0	0
1420	0.137835643	3.2525253	0	0	0
1421	0.145090151	3.2525253	0	0	0
1422	0.152344659	3.2525253	0	0	0
1423	0.159599166	3.2525253	0	0	0
1424	0.166853674	3.2525253	0	0	0
1425	0.174108181	3.2525253	0	0	0
1426	0.181362689	3.2525253	0	0	0
1427	0.188617196	3.2525253	0	0	0
1428	0.195871704	3.2525253	0	0	0
1429	0.203126211	3.2525253	0	0	0
1430	0.210380719	3.2525253	0	0	0
1431	0.217635227	3.2525253	0	0	0
1432	0.224889734	3.2525253	0	0	0
1433	0.232144242	3.2525253	0	0	0
1434	0.239398749	3.2525253	0	0	0
1435	0.246653257	3.2525253	0	0	0
1436	0.253907764	3.2525253	0	0	0
1437	0.261162272	3.2525253	0	0	0
1438	0.268416779	3.2525253	0	0	0
1439	0.275671287	3.2525253	0	0	0
1440	0.282925794	3.2525253	0	0	0

1441	0.290180302	3.2525253	0	0	0
1442	0.297434810	3.2525253	0	0	0
1443	0.304689317	3.2525253	0	0	0
1444	0.311943825	3.2525253	0	0	0
1445	0.319198332	3.2525253	0	0	0
1446	0.326452840	3.2525253	0	0	0
1447	0.333707347	3.2525253	0	0	0
1448	0.340961855	3.2525253	0	0	0
1449	0.348216362	3.2525253	0	0	0
1450	0.355470870	3.2525253	0	0	0
1451	0.362725378	3.2525253	0	0	0
1452	0.369979885	3.2525253	0	0	0
1453	0.377234393	3.2525253	0	0	0
1454	0.384488900	3.2525253	0	0	0
1455	0.391743408	3.2525253	0	0	0
1456	0.398997915	3.2525253	0	0	0
1457	0.406252423	3.2525253	0	0	0
1458	0.413506930	3.2525253	0	0	0
1459	0.420761438	3.2525253	0	0	0
1460	0.428015945	3.2525253	0	0	0
1461	0.435270453	3.2525253	0	0	0
1462	0.442524961	3.2525253	0	0	0
1463	0.449779468	3.2525253	0	0	0
1464	0.457033976	3.2525253	0	0	0
1465	0.464288483	3.2525253	0	0	0
1466	0.471542991	3.2525253	0	0	0
1467	0.478797498	3.2525253	0	0	0
1468	0.486052006	3.2525253	0	0	0
1469	0.493306513	3.2525253	0	0	0
1470	0.500561021	3.2525253	0	0	0
1471	0.507815529	3.2525253	0	0	0
1472	0.515070036	3.2525253	0	0	0
1473	0.522324544	3.2525253	0	0	0
1474	0.529579051	3.2525253	0	0	0
1475	0.536833559	3.2525253	0	0	0
1476	0.544088066	3.2525253	0	0	0
1477	0.551342574	3.2525253	0	0	0
1478	0.558597081	3.2525253	0	0	0
1479	0.565851589	3.2525253	0	0	0
1480	0.573106097	3.2525253	0	0	0
1481	0.580360604	3.2525253	0	0	0
1482	0.587615112	3.2525253	0	0	0
1483	0.594869619	3.2525253	0	0	0
1484	0.602124127	3.2525253	0	0	0
1485	0.609378634	3.2525253	0	0	0

1486	0.616633142	3.2525253	0	0	0
1487	0.623887649	3.2525253	0	0	0
1488	0.631142157	3.2525253	0	0	0
1489	0.638396664	3.2525253	0	0	0
1490	0.645651172	3.2525253	0	0	0
1491	0.652905680	3.2525253	0	0	0
1492	0.660160187	3.2525253	0	0	0
1493	0.667414695	3.2525253	0	0	0
1494	0.674669202	3.2525253	0	0	0
1495	0.681923710	3.2525253	0	0	0
1496	0.689178217	3.2525253	0	0	0
1497	0.696432725	3.2525253	0	0	0
1498	0.703687232	3.2525253	0	0	0
1499	0.710941740	3.2525253	0	0	0
1500	0.718196248	3.2525253	0	0	0
1501	0.000000000	3.4848485	0	0	0
1502	0.007254508	3.4848485	0	0	0
1503	0.014509015	3.4848485	0	0	0
1504	0.021763523	3.4848485	0	0	0
1505	0.029018030	3.4848485	0	0	0
1506	0.036272538	3.4848485	0	0	0
1507	0.043527045	3.4848485	0	0	0
1508	0.050781553	3.4848485	0	0	0
1509	0.058036060	3.4848485	0	0	0
1510	0.065290568	3.4848485	0	0	0
1511	0.072545076	3.4848485	0	0	0
1512	0.079799583	3.4848485	0	0	0
1513	0.087054091	3.4848485	0	0	0
1514	0.094308598	3.4848485	0	0	0
1515	0.101563106	3.4848485	0	0	0
1516	0.108817613	3.4848485	0	0	0
1517	0.116072121	3.4848485	0	0	0
1518	0.123326628	3.4848485	0	0	0
1519	0.130581136	3.4848485	0	0	0
1520	0.137835643	3.4848485	0	0	0
1521	0.145090151	3.4848485	0	0	0
1522	0.152344659	3.4848485	0	0	0
1523	0.159599166	3.4848485	0	0	0
1524	0.166853674	3.4848485	0	0	0
1525	0.174108181	3.4848485	0	0	0
1526	0.181362689	3.4848485	0	0	0
1527	0.188617196	3.4848485	0	0	0
1528	0.195871704	3.4848485	0	0	0
1529	0.203126211	3.4848485	0	0	0
1530	0.210380719	3.4848485	0	0	0

1531	0.217635227	3.4848485	0	0	0
1532	0.224889734	3.4848485	0	0	0
1533	0.232144242	3.4848485	0	0	0
1534	0.239398749	3.4848485	0	0	0
1535	0.246653257	3.4848485	0	0	0
1536	0.253907764	3.4848485	0	0	0
1537	0.261162272	3.4848485	0	0	0
1538	0.268416779	3.4848485	0	0	0
1539	0.275671287	3.4848485	0	0	0
1540	0.282925794	3.4848485	0	0	0
1541	0.290180302	3.4848485	0	0	0
1542	0.297434810	3.4848485	0	0	0
1543	0.304689317	3.4848485	0	0	0
1544	0.311943825	3.4848485	0	0	0
1545	0.319198332	3.4848485	0	0	0
1546	0.326452840	3.4848485	0	0	0
1547	0.333707347	3.4848485	0	0	0
1548	0.340961855	3.4848485	0	0	0
1549	0.348216362	3.4848485	0	0	0
1550	0.355470870	3.4848485	0	0	0
1551	0.362725378	3.4848485	0	0	0
1552	0.369979885	3.4848485	0	0	0
1553	0.377234393	3.4848485	0	0	0
1554	0.384488900	3.4848485	0	0	0
1555	0.391743408	3.4848485	0	0	0
1556	0.398997915	3.4848485	0	0	0
1557	0.406252423	3.4848485	0	0	0
1558	0.413506930	3.4848485	0	0	0
1559	0.420761438	3.4848485	0	0	0
1560	0.428015945	3.4848485	0	0	0
1561	0.435270453	3.4848485	0	0	0
1562	0.442524961	3.4848485	0	0	0
1563	0.449779468	3.4848485	0	0	0
1564	0.457033976	3.4848485	0	0	0
1565	0.464288483	3.4848485	0	0	0
1566	0.471542991	3.4848485	0	0	0
1567	0.478797498	3.4848485	0	0	0
1568	0.486052006	3.4848485	0	0	0
1569	0.493306513	3.4848485	0	0	0
1570	0.500561021	3.4848485	0	0	0
1571	0.507815529	3.4848485	0	0	0
1572	0.515070036	3.4848485	0	0	0
1573	0.522324544	3.4848485	0	0	0
1574	0.529579051	3.4848485	0	0	0
1575	0.536833559	3.4848485	0	0	0

1576	0.544088066	3.4848485	0	0	0
1577	0.551342574	3.4848485	0	0	0
1578	0.558597081	3.4848485	0	0	0
1579	0.565851589	3.4848485	0	0	0
1580	0.573106097	3.4848485	0	0	0
1581	0.580360604	3.4848485	0	0	0
1582	0.587615112	3.4848485	0	0	0
1583	0.594869619	3.4848485	0	0	0
1584	0.602124127	3.4848485	0	0	0
1585	0.609378634	3.4848485	0	0	0
1586	0.616633142	3.4848485	0	0	0
1587	0.623887649	3.4848485	0	0	0
1588	0.631142157	3.4848485	0	0	0
1589	0.638396664	3.4848485	0	0	0
1590	0.645651172	3.4848485	0	0	0
1591	0.652905680	3.4848485	0	0	0
1592	0.660160187	3.4848485	0	0	0
1593	0.667414695	3.4848485	0	0	0
1594	0.674669202	3.4848485	0	0	0
1595	0.681923710	3.4848485	0	0	0
1596	0.689178217	3.4848485	0	0	0
1597	0.696432725	3.4848485	0	0	0
1598	0.703687232	3.4848485	0	0	0
1599	0.710941740	3.4848485	0	0	0
1600	0.718196248	3.4848485	0	0	0
1601	0.000000000	3.7171717	0	0	0
1602	0.007254508	3.7171717	0	0	0
1603	0.014509015	3.7171717	0	0	0
1604	0.021763523	3.7171717	0	0	0
1605	0.029018030	3.7171717	0	0	0
1606	0.036272538	3.7171717	0	0	0
1607	0.043527045	3.7171717	0	0	0
1608	0.050781553	3.7171717	0	0	0
1609	0.058036060	3.7171717	0	0	0
1610	0.065290568	3.7171717	0	0	0
1611	0.072545076	3.7171717	0	0	0
1612	0.079799583	3.7171717	0	0	0
1613	0.087054091	3.7171717	0	0	0
1614	0.094308598	3.7171717	0	0	0
1615	0.101563106	3.7171717	0	0	0
1616	0.108817613	3.7171717	0	0	0
1617	0.116072121	3.7171717	0	0	0
1618	0.123326628	3.7171717	0	0	0
1619	0.130581136	3.7171717	0	0	0
1620	0.137835643	3.7171717	0	0	0

1621	0.145090151	3.7171717	0	0	0
1622	0.152344659	3.7171717	0	0	0
1623	0.159599166	3.7171717	0	0	0
1624	0.166853674	3.7171717	0	0	0
1625	0.174108181	3.7171717	0	0	0
1626	0.181362689	3.7171717	0	0	0
1627	0.188617196	3.7171717	0	0	0
1628	0.195871704	3.7171717	0	0	0
1629	0.203126211	3.7171717	0	0	0
1630	0.210380719	3.7171717	0	0	0
1631	0.217635227	3.7171717	0	0	0
1632	0.224889734	3.7171717	0	0	0
1633	0.232144242	3.7171717	0	0	0
1634	0.239398749	3.7171717	0	0	0
1635	0.246653257	3.7171717	0	0	0
1636	0.253907764	3.7171717	0	0	0
1637	0.261162272	3.7171717	0	0	0
1638	0.268416779	3.7171717	0	0	0
1639	0.275671287	3.7171717	0	0	0
1640	0.282925794	3.7171717	0	0	0
1641	0.290180302	3.7171717	0	0	0
1642	0.297434810	3.7171717	0	0	0
1643	0.304689317	3.7171717	0	0	0
1644	0.311943825	3.7171717	0	0	0
1645	0.319198332	3.7171717	0	0	0
1646	0.326452840	3.7171717	0	0	0
1647	0.333707347	3.7171717	0	0	0
1648	0.340961855	3.7171717	0	0	0
1649	0.348216362	3.7171717	0	0	0
1650	0.355470870	3.7171717	0	0	0
1651	0.362725378	3.7171717	0	0	0
1652	0.369979885	3.7171717	0	0	0
1653	0.377234393	3.7171717	0	0	0
1654	0.384488900	3.7171717	0	0	0
1655	0.391743408	3.7171717	0	0	0
1656	0.398997915	3.7171717	0	0	0
1657	0.406252423	3.7171717	0	0	0
1658	0.413506930	3.7171717	0	0	0
1659	0.420761438	3.7171717	0	0	0
1660	0.428015945	3.7171717	0	0	0
1661	0.435270453	3.7171717	0	0	0
1662	0.442524961	3.7171717	0	0	0
1663	0.449779468	3.7171717	0	0	0
1664	0.457033976	3.7171717	0	0	0
1665	0.464288483	3.7171717	0	0	0

1666	0.471542991	3.7171717	0	0	0
1667	0.478797498	3.7171717	0	0	0
1668	0.486052006	3.7171717	0	0	0
1669	0.493306513	3.7171717	0	0	0
1670	0.500561021	3.7171717	0	0	0
1671	0.507815529	3.7171717	0	0	0
1672	0.515070036	3.7171717	0	0	0
1673	0.522324544	3.7171717	0	0	0
1674	0.529579051	3.7171717	0	0	0
1675	0.536833559	3.7171717	0	0	0
1676	0.544088066	3.7171717	0	0	0
1677	0.551342574	3.7171717	0	0	0
1678	0.558597081	3.7171717	0	0	0
1679	0.565851589	3.7171717	0	0	0
1680	0.573106097	3.7171717	0	0	0
1681	0.580360604	3.7171717	0	0	0
1682	0.587615112	3.7171717	0	0	0
1683	0.594869619	3.7171717	0	0	0
1684	0.602124127	3.7171717	0	0	0
1685	0.609378634	3.7171717	0	0	0
1686	0.616633142	3.7171717	0	0	0
1687	0.623887649	3.7171717	0	0	0
1688	0.631142157	3.7171717	0	0	0
1689	0.638396664	3.7171717	0	0	0
1690	0.645651172	3.7171717	0	0	0
1691	0.652905680	3.7171717	0	0	0
1692	0.660160187	3.7171717	0	0	0
1693	0.667414695	3.7171717	0	0	0
1694	0.674669202	3.7171717	0	0	0
1695	0.681923710	3.7171717	0	0	0
1696	0.689178217	3.7171717	0	0	0
1697	0.696432725	3.7171717	0	0	0
1698	0.703687232	3.7171717	0	0	0
1699	0.710941740	3.7171717	0	0	0
1700	0.718196248	3.7171717	0	0	0
1701	0.000000000	3.9494949	0	0	0
1702	0.007254508	3.9494949	0	0	0
1703	0.014509015	3.9494949	0	0	0
1704	0.021763523	3.9494949	0	0	0
1705	0.029018030	3.9494949	0	0	0
1706	0.036272538	3.9494949	0	0	0
1707	0.043527045	3.9494949	0	0	0
1708	0.050781553	3.9494949	0	0	0
1709	0.058036060	3.9494949	0	0	0
1710	0.065290568	3.9494949	0	0	0

1711	0.072545076	3.9494949	0	0	0
1712	0.079799583	3.9494949	0	0	0
1713	0.087054091	3.9494949	0	0	0
1714	0.094308598	3.9494949	0	0	0
1715	0.101563106	3.9494949	0	0	0
1716	0.108817613	3.9494949	0	0	0
1717	0.116072121	3.9494949	0	0	0
1718	0.123326628	3.9494949	0	0	0
1719	0.130581136	3.9494949	0	0	0
1720	0.137835643	3.9494949	0	0	0
1721	0.145090151	3.9494949	0	0	0
1722	0.152344659	3.9494949	0	0	0
1723	0.159599166	3.9494949	0	0	0
1724	0.166853674	3.9494949	0	0	0
1725	0.174108181	3.9494949	0	0	0
1726	0.181362689	3.9494949	0	0	0
1727	0.188617196	3.9494949	0	0	0
1728	0.195871704	3.9494949	0	0	0
1729	0.203126211	3.9494949	0	0	0
1730	0.210380719	3.9494949	0	0	0
1731	0.217635227	3.9494949	0	0	0
1732	0.224889734	3.9494949	0	0	0
1733	0.232144242	3.9494949	0	0	0
1734	0.239398749	3.9494949	0	0	0
1735	0.246653257	3.9494949	0	0	0
1736	0.253907764	3.9494949	0	0	0
1737	0.261162272	3.9494949	0	0	0
1738	0.268416779	3.9494949	0	0	0
1739	0.275671287	3.9494949	0	0	0
1740	0.282925794	3.9494949	0	0	0
1741	0.290180302	3.9494949	0	0	0
1742	0.297434810	3.9494949	0	0	0
1743	0.304689317	3.9494949	0	0	0
1744	0.311943825	3.9494949	0	0	0
1745	0.319198332	3.9494949	0	0	0
1746	0.326452840	3.9494949	0	0	0
1747	0.333707347	3.9494949	0	0	0
1748	0.340961855	3.9494949	0	0	0
1749	0.348216362	3.9494949	0	0	0
1750	0.355470870	3.9494949	0	0	0
1751	0.362725378	3.9494949	0	0	0
1752	0.369979885	3.9494949	0	0	0
1753	0.377234393	3.9494949	0	0	0
1754	0.384488900	3.9494949	0	0	0
1755	0.391743408	3.9494949	0	0	0

1756	0.398997915	3.9494949	0	0	0
1757	0.406252423	3.9494949	0	0	0
1758	0.413506930	3.9494949	0	0	0
1759	0.420761438	3.9494949	0	0	0
1760	0.428015945	3.9494949	0	0	0
1761	0.435270453	3.9494949	0	0	0
1762	0.442524961	3.9494949	0	0	0
1763	0.449779468	3.9494949	0	0	0
1764	0.457033976	3.9494949	0	0	0
1765	0.464288483	3.9494949	0	0	0
1766	0.471542991	3.9494949	0	0	0
1767	0.478797498	3.9494949	0	0	0
1768	0.486052006	3.9494949	0	0	0
1769	0.493306513	3.9494949	0	0	0
1770	0.500561021	3.9494949	0	0	0
1771	0.507815529	3.9494949	0	0	0
1772	0.515070036	3.9494949	0	0	0
1773	0.522324544	3.9494949	0	0	0
1774	0.529579051	3.9494949	0	0	0
1775	0.536833559	3.9494949	0	0	0
1776	0.544088066	3.9494949	0	0	0
1777	0.551342574	3.9494949	0	0	0
1778	0.558597081	3.9494949	0	0	0
1779	0.565851589	3.9494949	0	0	0
1780	0.573106097	3.9494949	0	0	0
1781	0.580360604	3.9494949	0	0	0
1782	0.587615112	3.9494949	0	0	0
1783	0.594869619	3.9494949	0	0	0
1784	0.602124127	3.9494949	0	0	0
1785	0.609378634	3.9494949	0	0	0
1786	0.616633142	3.9494949	0	0	0
1787	0.623887649	3.9494949	0	0	0
1788	0.631142157	3.9494949	0	0	0
1789	0.638396664	3.9494949	0	0	0
1790	0.645651172	3.9494949	0	0	0
1791	0.652905680	3.9494949	0	0	0
1792	0.660160187	3.9494949	0	0	0
1793	0.667414695	3.9494949	0	0	0
1794	0.674669202	3.9494949	0	0	0
1795	0.681923710	3.9494949	0	0	0
1796	0.689178217	3.9494949	0	0	0
1797	0.696432725	3.9494949	0	0	0
1798	0.703687232	3.9494949	0	0	0
1799	0.710941740	3.9494949	0	0	0
1800	0.718196248	3.9494949	0	0	0

1801	0.000000000	4.1818182	0	0	0
1802	0.007254508	4.1818182	0	0	0
1803	0.014509015	4.1818182	0	0	0
1804	0.021763523	4.1818182	0	0	0
1805	0.029018030	4.1818182	0	0	0
1806	0.036272538	4.1818182	0	0	0
1807	0.043527045	4.1818182	0	0	0
1808	0.050781553	4.1818182	0	0	0
1809	0.058036060	4.1818182	0	0	0
1810	0.065290568	4.1818182	0	0	0
1811	0.072545076	4.1818182	0	0	0
1812	0.079799583	4.1818182	0	0	0
1813	0.087054091	4.1818182	0	0	0
1814	0.094308598	4.1818182	0	0	0
1815	0.101563106	4.1818182	0	0	0
1816	0.108817613	4.1818182	0	0	0
1817	0.116072121	4.1818182	0	0	0
1818	0.123326628	4.1818182	0	0	0
1819	0.130581136	4.1818182	0	0	0
1820	0.137835643	4.1818182	0	0	0
1821	0.145090151	4.1818182	0	0	0
1822	0.152344659	4.1818182	0	0	0
1823	0.159599166	4.1818182	0	0	0
1824	0.166853674	4.1818182	0	0	0
1825	0.174108181	4.1818182	0	0	0
1826	0.181362689	4.1818182	0	0	0
1827	0.188617196	4.1818182	0	0	0
1828	0.195871704	4.1818182	0	0	0
1829	0.203126211	4.1818182	0	0	0
1830	0.210380719	4.1818182	0	0	0
1831	0.217635227	4.1818182	0	0	0
1832	0.224889734	4.1818182	0	0	0
1833	0.232144242	4.1818182	0	0	0
1834	0.239398749	4.1818182	0	0	0
1835	0.246653257	4.1818182	0	0	0
1836	0.253907764	4.1818182	0	0	0
1837	0.261162272	4.1818182	0	0	0
1838	0.268416779	4.1818182	0	0	0
1839	0.275671287	4.1818182	0	0	0
1840	0.282925794	4.1818182	0	0	0
1841	0.290180302	4.1818182	0	0	0
1842	0.297434810	4.1818182	0	0	0
1843	0.304689317	4.1818182	0	0	0
1844	0.311943825	4.1818182	0	0	0
1845	0.319198332	4.1818182	0	0	0

1846	0.326452840	4.1818182	0	0	0
1847	0.333707347	4.1818182	0	0	0
1848	0.340961855	4.1818182	0	0	0
1849	0.348216362	4.1818182	0	0	0
1850	0.355470870	4.1818182	0	0	0
1851	0.362725378	4.1818182	0	0	0
1852	0.369979885	4.1818182	0	0	0
1853	0.377234393	4.1818182	0	0	0
1854	0.384488900	4.1818182	0	0	0
1855	0.391743408	4.1818182	0	0	0
1856	0.398997915	4.1818182	0	0	0
1857	0.406252423	4.1818182	0	0	0
1858	0.413506930	4.1818182	0	0	0
1859	0.420761438	4.1818182	0	0	0
1860	0.428015945	4.1818182	0	0	0
1861	0.435270453	4.1818182	0	0	0
1862	0.442524961	4.1818182	0	0	0
1863	0.449779468	4.1818182	0	0	0
1864	0.457033976	4.1818182	0	0	0
1865	0.464288483	4.1818182	0	0	0
1866	0.471542991	4.1818182	0	0	0
1867	0.478797498	4.1818182	0	0	0
1868	0.486052006	4.1818182	0	0	0
1869	0.493306513	4.1818182	0	0	0
1870	0.500561021	4.1818182	0	0	0
1871	0.507815529	4.1818182	0	0	0
1872	0.515070036	4.1818182	0	0	0
1873	0.522324544	4.1818182	0	0	0
1874	0.529579051	4.1818182	0	0	0
1875	0.536833559	4.1818182	0	0	0
1876	0.544088066	4.1818182	0	0	0
1877	0.551342574	4.1818182	0	0	0
1878	0.558597081	4.1818182	0	0	0
1879	0.565851589	4.1818182	0	0	0
1880	0.573106097	4.1818182	0	0	0
1881	0.580360604	4.1818182	0	0	0
1882	0.587615112	4.1818182	0	0	0
1883	0.594869619	4.1818182	0	0	0
1884	0.602124127	4.1818182	0	0	0
1885	0.609378634	4.1818182	0	0	0
1886	0.616633142	4.1818182	0	0	0
1887	0.623887649	4.1818182	0	0	0
1888	0.631142157	4.1818182	0	0	0
1889	0.638396664	4.1818182	0	0	0
1890	0.645651172	4.1818182	0	0	0

1891	0.652905680	4.1818182	0	0	0
1892	0.660160187	4.1818182	0	0	0
1893	0.667414695	4.1818182	0	0	0
1894	0.674669202	4.1818182	0	0	0
1895	0.681923710	4.1818182	0	0	0
1896	0.689178217	4.1818182	0	0	0
1897	0.696432725	4.1818182	0	0	0
1898	0.703687232	4.1818182	0	0	0
1899	0.710941740	4.1818182	0	0	0
1900	0.718196248	4.1818182	0	0	0
1901	0.000000000	4.4141414	0	0	0
1902	0.007254508	4.4141414	0	0	0
1903	0.014509015	4.4141414	0	0	0
1904	0.021763523	4.4141414	0	0	0
1905	0.029018030	4.4141414	0	0	0
1906	0.036272538	4.4141414	0	0	0
1907	0.043527045	4.4141414	0	0	0
1908	0.050781553	4.4141414	0	0	0
1909	0.058036060	4.4141414	0	0	0
1910	0.065290568	4.4141414	0	0	0
1911	0.072545076	4.4141414	0	0	0
1912	0.079799583	4.4141414	0	0	0
1913	0.087054091	4.4141414	0	0	0
1914	0.094308598	4.4141414	0	0	0
1915	0.101563106	4.4141414	0	0	0
1916	0.108817613	4.4141414	0	0	0
1917	0.116072121	4.4141414	0	0	0
1918	0.123326628	4.4141414	0	0	0
1919	0.130581136	4.4141414	0	0	0
1920	0.137835643	4.4141414	0	0	0
1921	0.145090151	4.4141414	0	0	0
1922	0.152344659	4.4141414	0	0	0
1923	0.159599166	4.4141414	0	0	0
1924	0.166853674	4.4141414	0	0	0
1925	0.174108181	4.4141414	0	0	0
1926	0.181362689	4.4141414	0	0	0
1927	0.188617196	4.4141414	0	0	0
1928	0.195871704	4.4141414	0	0	0
1929	0.203126211	4.4141414	0	0	0
1930	0.210380719	4.4141414	0	0	0
1931	0.217635227	4.4141414	0	0	0
1932	0.224889734	4.4141414	0	0	0
1933	0.232144242	4.4141414	0	0	0
1934	0.239398749	4.4141414	0	0	0
1935	0.246653257	4.4141414	0	0	0

1936	0.253907764	4.4141414	0	0	0
1937	0.261162272	4.4141414	0	0	0
1938	0.268416779	4.4141414	0	0	0
1939	0.275671287	4.4141414	0	0	0
1940	0.282925794	4.4141414	0	0	0
1941	0.290180302	4.4141414	0	0	0
1942	0.297434810	4.4141414	0	0	0
1943	0.304689317	4.4141414	0	0	0
1944	0.311943825	4.4141414	0	0	0
1945	0.319198332	4.4141414	0	0	0
1946	0.326452840	4.4141414	0	0	0
1947	0.333707347	4.4141414	0	0	0
1948	0.340961855	4.4141414	0	0	0
1949	0.348216362	4.4141414	0	0	0
1950	0.355470870	4.4141414	0	0	0
1951	0.362725378	4.4141414	0	0	0
1952	0.369979885	4.4141414	0	0	0
1953	0.377234393	4.4141414	0	0	0
1954	0.384488900	4.4141414	0	0	0
1955	0.391743408	4.4141414	0	0	0
1956	0.398997915	4.4141414	0	0	0
1957	0.406252423	4.4141414	0	0	0
1958	0.413506930	4.4141414	0	0	0
1959	0.420761438	4.4141414	0	0	0
1960	0.428015945	4.4141414	0	0	0
1961	0.435270453	4.4141414	0	0	0
1962	0.442524961	4.4141414	0	0	0
1963	0.449779468	4.4141414	0	0	0
1964	0.457033976	4.4141414	0	0	0
1965	0.464288483	4.4141414	0	0	0
1966	0.471542991	4.4141414	0	0	0
1967	0.478797498	4.4141414	0	0	0
1968	0.486052006	4.4141414	0	0	0
1969	0.493306513	4.4141414	0	0	0
1970	0.500561021	4.4141414	0	0	0
1971	0.507815529	4.4141414	0	0	0
1972	0.515070036	4.4141414	0	0	0
1973	0.522324544	4.4141414	0	0	0
1974	0.529579051	4.4141414	0	0	0
1975	0.536833559	4.4141414	0	0	0
1976	0.544088066	4.4141414	0	0	0
1977	0.551342574	4.4141414	0	0	0
1978	0.558597081	4.4141414	0	0	0
1979	0.565851589	4.4141414	0	0	0
1980	0.573106097	4.4141414	0	0	0

1981	0.580360604	4.4141414	0	0	0
1982	0.587615112	4.4141414	0	0	0
1983	0.594869619	4.4141414	0	0	0
1984	0.602124127	4.4141414	0	0	0
1985	0.609378634	4.4141414	0	0	0
1986	0.616633142	4.4141414	0	0	0
1987	0.623887649	4.4141414	0	0	0
1988	0.631142157	4.4141414	0	0	0
1989	0.638396664	4.4141414	0	0	0
1990	0.645651172	4.4141414	0	0	0
1991	0.652905680	4.4141414	0	0	0
1992	0.660160187	4.4141414	0	0	0
1993	0.667414695	4.4141414	0	0	0
1994	0.674669202	4.4141414	0	0	0
1995	0.681923710	4.4141414	0	0	0
1996	0.689178217	4.4141414	0	0	0
1997	0.696432725	4.4141414	0	0	0
1998	0.703687232	4.4141414	0	0	0
1999	0.710941740	4.4141414	0	0	0
2000	0.718196248	4.4141414	0	0	0
2001	0.000000000	4.6464646	0	0	0
2002	0.007254508	4.6464646	0	0	0
2003	0.014509015	4.6464646	0	0	0
2004	0.021763523	4.6464646	0	0	0
2005	0.029018030	4.6464646	0	0	0
2006	0.036272538	4.6464646	0	0	0
2007	0.043527045	4.6464646	0	0	0
2008	0.050781553	4.6464646	0	0	0
2009	0.058036060	4.6464646	0	0	0
2010	0.065290568	4.6464646	0	0	0
2011	0.072545076	4.6464646	0	0	0
2012	0.079799583	4.6464646	0	0	0
2013	0.087054091	4.6464646	0	0	0
2014	0.094308598	4.6464646	0	0	0
2015	0.101563106	4.6464646	0	0	0
2016	0.108817613	4.6464646	0	0	0
2017	0.116072121	4.6464646	0	0	0
2018	0.123326628	4.6464646	0	0	0
2019	0.130581136	4.6464646	0	0	0
2020	0.137835643	4.6464646	0	0	0
2021	0.145090151	4.6464646	0	0	0
2022	0.152344659	4.6464646	0	0	0
2023	0.159599166	4.6464646	0	0	0
2024	0.166853674	4.6464646	0	0	0
2025	0.174108181	4.6464646	0	0	0

2026	0.181362689	4.6464646	0	0	0
2027	0.188617196	4.6464646	0	0	0
2028	0.195871704	4.6464646	0	0	0
2029	0.203126211	4.6464646	0	0	0
2030	0.210380719	4.6464646	0	0	0
2031	0.217635227	4.6464646	0	0	0
2032	0.224889734	4.6464646	0	0	0
2033	0.232144242	4.6464646	0	0	0
2034	0.239398749	4.6464646	0	0	0
2035	0.246653257	4.6464646	0	0	0
2036	0.253907764	4.6464646	0	0	0
2037	0.261162272	4.6464646	0	0	0
2038	0.268416779	4.6464646	0	0	0
2039	0.275671287	4.6464646	0	0	0
2040	0.282925794	4.6464646	0	0	0
2041	0.290180302	4.6464646	0	0	0
2042	0.297434810	4.6464646	0	0	0
2043	0.304689317	4.6464646	0	0	0
2044	0.311943825	4.6464646	0	0	0
2045	0.319198332	4.6464646	0	0	0
2046	0.326452840	4.6464646	0	0	0
2047	0.333707347	4.6464646	0	0	0
2048	0.340961855	4.6464646	0	0	0
2049	0.348216362	4.6464646	0	0	0
2050	0.355470870	4.6464646	0	0	0
2051	0.362725378	4.6464646	0	0	0
2052	0.369979885	4.6464646	0	0	0
2053	0.377234393	4.6464646	0	0	0
2054	0.384488900	4.6464646	0	0	0
2055	0.391743408	4.6464646	0	0	0
2056	0.398997915	4.6464646	0	0	0
2057	0.406252423	4.6464646	0	0	0
2058	0.413506930	4.6464646	0	0	0
2059	0.420761438	4.6464646	0	0	0
2060	0.428015945	4.6464646	0	0	0
2061	0.435270453	4.6464646	0	0	0
2062	0.442524961	4.6464646	0	0	0
2063	0.449779468	4.6464646	0	0	0
2064	0.457033976	4.6464646	0	0	0
2065	0.464288483	4.6464646	0	0	0
2066	0.471542991	4.6464646	0	0	0
2067	0.478797498	4.6464646	0	0	0
2068	0.486052006	4.6464646	0	0	0
2069	0.493306513	4.6464646	0	0	0
2070	0.500561021	4.6464646	0	0	0

2071	0.507815529	4.6464646	0	0	0
2072	0.515070036	4.6464646	0	0	0
2073	0.522324544	4.6464646	0	0	0
2074	0.529579051	4.6464646	0	0	0
2075	0.536833559	4.6464646	0	0	0
2076	0.544088066	4.6464646	0	0	0
2077	0.551342574	4.6464646	0	0	0
2078	0.558597081	4.6464646	0	0	0
2079	0.565851589	4.6464646	0	0	0
2080	0.573106097	4.6464646	0	0	0
2081	0.580360604	4.6464646	0	0	0
2082	0.587615112	4.6464646	0	0	0
2083	0.594869619	4.6464646	0	0	0
2084	0.602124127	4.6464646	0	0	0
2085	0.609378634	4.6464646	0	0	0
2086	0.616633142	4.6464646	0	0	0
2087	0.623887649	4.6464646	0	0	0
2088	0.631142157	4.6464646	0	0	0
2089	0.638396664	4.6464646	0	0	0
2090	0.645651172	4.6464646	0	0	0
2091	0.652905680	4.6464646	0	0	0
2092	0.660160187	4.6464646	0	0	0
2093	0.667414695	4.6464646	0	0	0
2094	0.674669202	4.6464646	0	0	0
2095	0.681923710	4.6464646	0	0	0
2096	0.689178217	4.6464646	0	0	0
2097	0.696432725	4.6464646	0	0	0
2098	0.703687232	4.6464646	0	0	0
2099	0.710941740	4.6464646	0	0	0
2100	0.718196248	4.6464646	0	0	0
2101	0.000000000	4.8787879	0	0	0
2102	0.007254508	4.8787879	0	0	0
2103	0.014509015	4.8787879	0	0	0
2104	0.021763523	4.8787879	0	0	0
2105	0.029018030	4.8787879	0	0	0
2106	0.036272538	4.8787879	0	0	0
2107	0.043527045	4.8787879	0	0	0
2108	0.050781553	4.8787879	0	0	0
2109	0.058036060	4.8787879	0	0	0
2110	0.065290568	4.8787879	0	0	0
2111	0.072545076	4.8787879	0	0	0
2112	0.079799583	4.8787879	0	0	0
2113	0.087054091	4.8787879	0	0	0
2114	0.094308598	4.8787879	0	0	0
2115	0.101563106	4.8787879	0	0	0

2116	0.108817613	4.8787879	0	0	0
2117	0.116072121	4.8787879	0	0	0
2118	0.123326628	4.8787879	0	0	0
2119	0.130581136	4.8787879	0	0	0
2120	0.137835643	4.8787879	0	0	0
2121	0.145090151	4.8787879	0	0	0
2122	0.152344659	4.8787879	0	0	0
2123	0.159599166	4.8787879	0	0	0
2124	0.166853674	4.8787879	0	0	0
2125	0.174108181	4.8787879	0	0	0
2126	0.181362689	4.8787879	0	0	0
2127	0.188617196	4.8787879	0	0	0
2128	0.195871704	4.8787879	0	0	0
2129	0.203126211	4.8787879	0	0	0
2130	0.210380719	4.8787879	0	0	0
2131	0.217635227	4.8787879	0	0	0
2132	0.224889734	4.8787879	0	0	0
2133	0.232144242	4.8787879	0	0	0
2134	0.239398749	4.8787879	0	0	0
2135	0.246653257	4.8787879	0	0	0
2136	0.253907764	4.8787879	0	0	0
2137	0.261162272	4.8787879	0	0	0
2138	0.268416779	4.8787879	0	0	0
2139	0.275671287	4.8787879	0	0	0
2140	0.282925794	4.8787879	0	0	0
2141	0.290180302	4.8787879	0	0	0
2142	0.297434810	4.8787879	0	0	0
2143	0.304689317	4.8787879	0	0	0
2144	0.311943825	4.8787879	0	0	0
2145	0.319198332	4.8787879	0	0	0
2146	0.326452840	4.8787879	0	0	0
2147	0.333707347	4.8787879	0	0	0
2148	0.340961855	4.8787879	0	0	0
2149	0.348216362	4.8787879	0	0	0
2150	0.355470870	4.8787879	0	0	0
2151	0.362725378	4.8787879	0	0	0
2152	0.369979885	4.8787879	0	0	0
2153	0.377234393	4.8787879	0	0	0
2154	0.384488900	4.8787879	0	0	0
2155	0.391743408	4.8787879	0	0	0
2156	0.398997915	4.8787879	0	0	0
2157	0.406252423	4.8787879	0	0	0
2158	0.413506930	4.8787879	0	0	0
2159	0.420761438	4.8787879	0	0	0
2160	0.428015945	4.8787879	0	0	0

2161	0.435270453	4.8787879	0	0	0
2162	0.442524961	4.8787879	0	0	0
2163	0.449779468	4.8787879	0	0	0
2164	0.457033976	4.8787879	0	0	0
2165	0.464288483	4.8787879	0	0	0
2166	0.471542991	4.8787879	0	0	0
2167	0.478797498	4.8787879	0	0	0
2168	0.486052006	4.8787879	0	0	0
2169	0.493306513	4.8787879	0	0	0
2170	0.500561021	4.8787879	0	0	0
2171	0.507815529	4.8787879	0	0	0
2172	0.515070036	4.8787879	0	0	0
2173	0.522324544	4.8787879	0	0	0
2174	0.529579051	4.8787879	0	0	0
2175	0.536833559	4.8787879	0	0	0
2176	0.544088066	4.8787879	0	0	0
2177	0.551342574	4.8787879	0	0	0
2178	0.558597081	4.8787879	0	0	0
2179	0.565851589	4.8787879	0	0	0
2180	0.573106097	4.8787879	0	0	0
2181	0.580360604	4.8787879	0	0	0
2182	0.587615112	4.8787879	0	0	0
2183	0.594869619	4.8787879	0	0	0
2184	0.602124127	4.8787879	0	0	0
2185	0.609378634	4.8787879	0	0	0
2186	0.616633142	4.8787879	0	0	0
2187	0.623887649	4.8787879	0	0	0
2188	0.631142157	4.8787879	0	0	0
2189	0.638396664	4.8787879	0	0	0
2190	0.645651172	4.8787879	0	0	0
2191	0.652905680	4.8787879	0	0	0
2192	0.660160187	4.8787879	0	0	0
2193	0.667414695	4.8787879	0	0	0
2194	0.674669202	4.8787879	0	0	0
2195	0.681923710	4.8787879	0	0	0
2196	0.689178217	4.8787879	0	0	0
2197	0.696432725	4.8787879	0	0	0
2198	0.703687232	4.8787879	0	0	0
2199	0.710941740	4.8787879	0	0	0
2200	0.718196248	4.8787879	0	0	0
2201	0.000000000	5.1111111	0	0	0
2202	0.007254508	5.1111111	0	0	0
2203	0.014509015	5.1111111	0	0	0
2204	0.021763523	5.1111111	0	0	0
2205	0.029018030	5.1111111	0	0	0

2206	0.036272538	5.1111111	0	0	0
2207	0.043527045	5.1111111	0	0	0
2208	0.050781553	5.1111111	0	0	0
2209	0.058036060	5.1111111	0	0	0
2210	0.065290568	5.1111111	0	0	0
2211	0.072545076	5.1111111	0	0	0
2212	0.079799583	5.1111111	0	0	0
2213	0.087054091	5.1111111	0	0	0
2214	0.094308598	5.1111111	0	0	0
2215	0.101563106	5.1111111	0	0	0
2216	0.108817613	5.1111111	0	0	0
2217	0.116072121	5.1111111	0	0	0
2218	0.123326628	5.1111111	0	0	0
2219	0.130581136	5.1111111	0	0	0
2220	0.137835643	5.1111111	0	0	0
2221	0.145090151	5.1111111	0	0	0
2222	0.152344659	5.1111111	0	0	0
2223	0.159599166	5.1111111	0	0	0
2224	0.166853674	5.1111111	0	0	0
2225	0.174108181	5.1111111	0	0	0
2226	0.181362689	5.1111111	0	0	0
2227	0.188617196	5.1111111	0	0	0
2228	0.195871704	5.1111111	0	0	0
2229	0.203126211	5.1111111	0	0	0
2230	0.210380719	5.1111111	0	0	0
2231	0.217635227	5.1111111	0	0	0
2232	0.224889734	5.1111111	0	0	0
2233	0.232144242	5.1111111	0	0	0
2234	0.239398749	5.1111111	0	0	0
2235	0.246653257	5.1111111	0	0	0
2236	0.253907764	5.1111111	0	0	0
2237	0.261162272	5.1111111	0	0	0
2238	0.268416779	5.1111111	0	0	0
2239	0.275671287	5.1111111	0	0	0
2240	0.282925794	5.1111111	0	0	0
2241	0.290180302	5.1111111	0	0	0
2242	0.297434810	5.1111111	0	0	0
2243	0.304689317	5.1111111	0	0	0
2244	0.311943825	5.1111111	0	0	0
2245	0.319198332	5.1111111	0	0	0
2246	0.326452840	5.1111111	0	0	0
2247	0.333707347	5.1111111	0	0	0
2248	0.340961855	5.1111111	0	0	0
2249	0.348216362	5.1111111	0	0	0
2250	0.355470870	5.1111111	0	0	0

2251	0.362725378	5.1111111	0	0	0
2252	0.369979885	5.1111111	0	0	0
2253	0.377234393	5.1111111	0	0	0
2254	0.384488900	5.1111111	0	0	0
2255	0.391743408	5.1111111	0	0	0
2256	0.398997915	5.1111111	0	0	0
2257	0.406252423	5.1111111	0	0	0
2258	0.413506930	5.1111111	0	0	0
2259	0.420761438	5.1111111	0	0	0
2260	0.428015945	5.1111111	0	0	0
2261	0.435270453	5.1111111	0	0	0
2262	0.442524961	5.1111111	0	0	0
2263	0.449779468	5.1111111	0	0	0
2264	0.457033976	5.1111111	0	0	0
2265	0.464288483	5.1111111	0	0	0
2266	0.471542991	5.1111111	0	0	0
2267	0.478797498	5.1111111	0	0	0
2268	0.486052006	5.1111111	0	0	0
2269	0.493306513	5.1111111	0	0	0
2270	0.500561021	5.1111111	0	0	0
2271	0.507815529	5.1111111	0	0	0
2272	0.515070036	5.1111111	0	0	0
2273	0.522324544	5.1111111	0	0	0
2274	0.529579051	5.1111111	0	0	0
2275	0.536833559	5.1111111	0	0	0
2276	0.544088066	5.1111111	0	0	0
2277	0.551342574	5.1111111	0	0	0
2278	0.558597081	5.1111111	0	0	0
2279	0.565851589	5.1111111	0	0	0
2280	0.573106097	5.1111111	0	0	0
2281	0.580360604	5.1111111	0	0	0
2282	0.587615112	5.1111111	0	0	0
2283	0.594869619	5.1111111	0	0	0
2284	0.602124127	5.1111111	0	0	0
2285	0.609378634	5.1111111	0	0	0
2286	0.616633142	5.1111111	0	0	0
2287	0.623887649	5.1111111	0	0	0
2288	0.631142157	5.1111111	0	0	0
2289	0.638396664	5.1111111	0	0	0
2290	0.645651172	5.1111111	0	0	0
2291	0.652905680	5.1111111	0	0	0
2292	0.660160187	5.1111111	0	0	0
2293	0.667414695	5.1111111	0	0	0
2294	0.674669202	5.1111111	0	0	0
2295	0.681923710	5.1111111	0	0	0

2296	0.689178217	5.1111111	0	0	0
2297	0.696432725	5.1111111	0	0	0
2298	0.703687232	5.1111111	0	0	0
2299	0.710941740	5.1111111	0	0	0
2300	0.718196248	5.1111111	0	0	0
2301	0.000000000	5.3434343	0	0	0
2302	0.007254508	5.3434343	0	0	0
2303	0.014509015	5.3434343	0	0	0
2304	0.021763523	5.3434343	0	0	0
2305	0.029018030	5.3434343	0	0	0
2306	0.036272538	5.3434343	0	0	0
2307	0.043527045	5.3434343	0	0	0
2308	0.050781553	5.3434343	0	0	0
2309	0.058036060	5.3434343	0	0	0
2310	0.065290568	5.3434343	0	0	0
2311	0.072545076	5.3434343	0	0	0
2312	0.079799583	5.3434343	0	0	0
2313	0.087054091	5.3434343	0	0	0
2314	0.094308598	5.3434343	0	0	0
2315	0.101563106	5.3434343	0	0	0
2316	0.108817613	5.3434343	0	0	0
2317	0.116072121	5.3434343	0	0	0
2318	0.123326628	5.3434343	0	0	0
2319	0.130581136	5.3434343	0	0	0
2320	0.137835643	5.3434343	0	0	0
2321	0.145090151	5.3434343	0	0	0
2322	0.152344659	5.3434343	0	0	0
2323	0.159599166	5.3434343	0	0	0
2324	0.166853674	5.3434343	0	0	0
2325	0.174108181	5.3434343	0	0	0
2326	0.181362689	5.3434343	0	0	0
2327	0.188617196	5.3434343	0	0	0
2328	0.195871704	5.3434343	0	0	0
2329	0.203126211	5.3434343	0	0	0
2330	0.210380719	5.3434343	0	0	0
2331	0.217635227	5.3434343	0	0	0
2332	0.224889734	5.3434343	0	0	0
2333	0.232144242	5.3434343	0	0	0
2334	0.239398749	5.3434343	0	0	0
2335	0.246653257	5.3434343	0	0	0
2336	0.253907764	5.3434343	0	0	0
2337	0.261162272	5.3434343	0	0	0
2338	0.268416779	5.3434343	0	0	0
2339	0.275671287	5.3434343	0	0	0
2340	0.282925794	5.3434343	0	0	0

2341	0.290180302	5.3434343	0	0	0
2342	0.297434810	5.3434343	0	0	0
2343	0.304689317	5.3434343	0	0	0
2344	0.311943825	5.3434343	0	0	0
2345	0.319198332	5.3434343	0	0	0
2346	0.326452840	5.3434343	0	0	0
2347	0.333707347	5.3434343	0	0	0
2348	0.340961855	5.3434343	0	0	0
2349	0.348216362	5.3434343	0	0	0
2350	0.355470870	5.3434343	0	0	0
2351	0.362725378	5.3434343	0	0	0
2352	0.369979885	5.3434343	0	0	0
2353	0.377234393	5.3434343	0	0	0
2354	0.384488900	5.3434343	0	0	0
2355	0.391743408	5.3434343	0	0	0
2356	0.398997915	5.3434343	0	0	0
2357	0.406252423	5.3434343	0	0	0
2358	0.413506930	5.3434343	0	0	0
2359	0.420761438	5.3434343	0	0	0
2360	0.428015945	5.3434343	0	0	0
2361	0.435270453	5.3434343	0	0	0
2362	0.442524961	5.3434343	0	0	0
2363	0.449779468	5.3434343	0	0	0
2364	0.457033976	5.3434343	0	0	0
2365	0.464288483	5.3434343	0	0	0
2366	0.471542991	5.3434343	0	0	0
2367	0.478797498	5.3434343	0	0	0
2368	0.486052006	5.3434343	0	0	0
2369	0.493306513	5.3434343	0	0	0
2370	0.500561021	5.3434343	0	0	0
2371	0.507815529	5.3434343	0	0	0
2372	0.515070036	5.3434343	0	0	0
2373	0.522324544	5.3434343	0	0	0
2374	0.529579051	5.3434343	0	0	0
2375	0.536833559	5.3434343	0	0	0
2376	0.544088066	5.3434343	0	0	0
2377	0.551342574	5.3434343	0	0	0
2378	0.558597081	5.3434343	0	0	0
2379	0.565851589	5.3434343	0	0	0
2380	0.573106097	5.3434343	0	0	0
2381	0.580360604	5.3434343	0	0	0
2382	0.587615112	5.3434343	0	0	0
2383	0.594869619	5.3434343	0	0	0
2384	0.602124127	5.3434343	0	0	0
2385	0.609378634	5.3434343	0	0	0

2386	0.616633142	5.3434343	0	0	0
2387	0.623887649	5.3434343	0	0	0
2388	0.631142157	5.3434343	0	0	0
2389	0.638396664	5.3434343	0	0	0
2390	0.645651172	5.3434343	0	0	0
2391	0.652905680	5.3434343	0	0	0
2392	0.660160187	5.3434343	0	0	0
2393	0.667414695	5.3434343	0	0	0
2394	0.674669202	5.3434343	0	0	0
2395	0.681923710	5.3434343	0	0	0
2396	0.689178217	5.3434343	0	0	0
2397	0.696432725	5.3434343	0	0	0
2398	0.703687232	5.3434343	0	0	0
2399	0.710941740	5.3434343	0	0	0
2400	0.718196248	5.3434343	0	0	0
2401	0.000000000	5.5757576	0	0	0
2402	0.007254508	5.5757576	0	0	0
2403	0.014509015	5.5757576	0	0	0
2404	0.021763523	5.5757576	0	0	0
2405	0.029018030	5.5757576	0	0	0
2406	0.036272538	5.5757576	0	0	0
2407	0.043527045	5.5757576	0	0	0
2408	0.050781553	5.5757576	0	0	0
2409	0.058036060	5.5757576	0	0	0
2410	0.065290568	5.5757576	0	0	0
2411	0.072545076	5.5757576	0	0	0
2412	0.079799583	5.5757576	0	0	0
2413	0.087054091	5.5757576	0	0	0
2414	0.094308598	5.5757576	0	0	0
2415	0.101563106	5.5757576	0	0	0
2416	0.108817613	5.5757576	0	0	0
2417	0.116072121	5.5757576	0	0	0
2418	0.123326628	5.5757576	0	0	0
2419	0.130581136	5.5757576	0	0	0
2420	0.137835643	5.5757576	0	0	0
2421	0.145090151	5.5757576	0	0	0
2422	0.152344659	5.5757576	0	0	0
2423	0.159599166	5.5757576	0	0	0
2424	0.166853674	5.5757576	0	0	0
2425	0.174108181	5.5757576	0	0	0
2426	0.181362689	5.5757576	0	0	0
2427	0.188617196	5.5757576	0	0	0
2428	0.195871704	5.5757576	0	0	0
2429	0.203126211	5.5757576	0	0	0
2430	0.210380719	5.5757576	0	0	0

2431	0.217635227	5.5757576	0	0	0
2432	0.224889734	5.5757576	0	0	0
2433	0.232144242	5.5757576	0	0	0
2434	0.239398749	5.5757576	0	0	0
2435	0.246653257	5.5757576	0	0	0
2436	0.253907764	5.5757576	0	0	0
2437	0.261162272	5.5757576	0	0	0
2438	0.268416779	5.5757576	0	0	0
2439	0.275671287	5.5757576	0	0	0
2440	0.282925794	5.5757576	0	0	0
2441	0.290180302	5.5757576	0	0	0
2442	0.297434810	5.5757576	0	0	0
2443	0.304689317	5.5757576	0	0	0
2444	0.311943825	5.5757576	0	0	0
2445	0.319198332	5.5757576	0	0	0
2446	0.326452840	5.5757576	0	0	0
2447	0.333707347	5.5757576	0	0	0
2448	0.340961855	5.5757576	0	0	0
2449	0.348216362	5.5757576	0	0	0
2450	0.355470870	5.5757576	0	0	0
2451	0.362725378	5.5757576	0	0	0
2452	0.369979885	5.5757576	0	0	0
2453	0.377234393	5.5757576	0	0	0
2454	0.384488900	5.5757576	0	0	0
2455	0.391743408	5.5757576	0	0	0
2456	0.398997915	5.5757576	0	0	0
2457	0.406252423	5.5757576	0	0	0
2458	0.413506930	5.5757576	0	0	0
2459	0.420761438	5.5757576	0	0	0
2460	0.428015945	5.5757576	0	0	0
2461	0.435270453	5.5757576	0	0	0
2462	0.442524961	5.5757576	0	0	0
2463	0.449779468	5.5757576	0	0	0
2464	0.457033976	5.5757576	0	0	0
2465	0.464288483	5.5757576	0	0	0
2466	0.471542991	5.5757576	0	0	0
2467	0.478797498	5.5757576	0	0	0
2468	0.486052006	5.5757576	0	0	0
2469	0.493306513	5.5757576	0	0	0
2470	0.500561021	5.5757576	0	0	0
2471	0.507815529	5.5757576	0	0	0
2472	0.515070036	5.5757576	0	0	0
2473	0.522324544	5.5757576	0	0	0
2474	0.529579051	5.5757576	0	0	0
2475	0.536833559	5.5757576	0	0	0

2476	0.544088066	5.5757576	0	0	0
2477	0.551342574	5.5757576	0	0	0
2478	0.558597081	5.5757576	0	0	0
2479	0.565851589	5.5757576	0	0	0
2480	0.573106097	5.5757576	0	0	0
2481	0.580360604	5.5757576	0	0	0
2482	0.587615112	5.5757576	0	0	0
2483	0.594869619	5.5757576	0	0	0
2484	0.602124127	5.5757576	0	0	0
2485	0.609378634	5.5757576	0	0	0
2486	0.616633142	5.5757576	0	0	0
2487	0.623887649	5.5757576	0	0	0
2488	0.631142157	5.5757576	0	0	0
2489	0.638396664	5.5757576	0	0	0
2490	0.645651172	5.5757576	0	0	0
2491	0.652905680	5.5757576	0	0	0
2492	0.660160187	5.5757576	0	0	0
2493	0.667414695	5.5757576	0	0	0
2494	0.674669202	5.5757576	0	0	0
2495	0.681923710	5.5757576	0	0	0
2496	0.689178217	5.5757576	0	0	0
2497	0.696432725	5.5757576	0	0	0
2498	0.703687232	5.5757576	0	0	0
2499	0.710941740	5.5757576	0	0	0
2500	0.718196248	5.5757576	0	0	0
2501	0.000000000	5.8080808	0	0	0
2502	0.007254508	5.8080808	0	0	0
2503	0.014509015	5.8080808	0	0	0
2504	0.021763523	5.8080808	0	0	0
2505	0.029018030	5.8080808	0	0	0
2506	0.036272538	5.8080808	0	0	0
2507	0.043527045	5.8080808	0	0	0
2508	0.050781553	5.8080808	0	0	0
2509	0.058036060	5.8080808	0	0	0
2510	0.065290568	5.8080808	0	0	0
2511	0.072545076	5.8080808	0	0	0
2512	0.079799583	5.8080808	0	0	0
2513	0.087054091	5.8080808	0	0	0
2514	0.094308598	5.8080808	0	0	0
2515	0.101563106	5.8080808	0	0	0
2516	0.108817613	5.8080808	0	0	0
2517	0.116072121	5.8080808	0	0	0
2518	0.123326628	5.8080808	0	0	0
2519	0.130581136	5.8080808	0	0	0
2520	0.137835643	5.8080808	0	0	0

2521	0.145090151	5.8080808	0	0	0
2522	0.152344659	5.8080808	0	0	0
2523	0.159599166	5.8080808	0	0	0
2524	0.166853674	5.8080808	0	0	0
2525	0.174108181	5.8080808	0	0	0
2526	0.181362689	5.8080808	0	0	0
2527	0.188617196	5.8080808	0	0	0
2528	0.195871704	5.8080808	0	0	0
2529	0.203126211	5.8080808	0	0	0
2530	0.210380719	5.8080808	0	0	0
2531	0.217635227	5.8080808	0	0	0
2532	0.224889734	5.8080808	0	0	0
2533	0.232144242	5.8080808	0	0	0
2534	0.239398749	5.8080808	0	0	0
2535	0.246653257	5.8080808	0	0	0
2536	0.253907764	5.8080808	0	0	0
2537	0.261162272	5.8080808	0	0	0
2538	0.268416779	5.8080808	0	0	0
2539	0.275671287	5.8080808	0	0	0
2540	0.282925794	5.8080808	0	0	0
2541	0.290180302	5.8080808	0	0	0
2542	0.297434810	5.8080808	0	0	0
2543	0.304689317	5.8080808	0	0	0
2544	0.311943825	5.8080808	0	0	0
2545	0.319198332	5.8080808	0	0	0
2546	0.326452840	5.8080808	0	0	0
2547	0.333707347	5.8080808	0	0	0
2548	0.340961855	5.8080808	0	0	0
2549	0.348216362	5.8080808	0	0	0
2550	0.355470870	5.8080808	0	0	0
2551	0.362725378	5.8080808	0	0	0
2552	0.369979885	5.8080808	0	0	0
2553	0.377234393	5.8080808	0	0	0
2554	0.384488900	5.8080808	0	0	0
2555	0.391743408	5.8080808	0	0	0
2556	0.398997915	5.8080808	0	0	0
2557	0.406252423	5.8080808	0	0	0
2558	0.413506930	5.8080808	0	0	0
2559	0.420761438	5.8080808	0	0	0
2560	0.428015945	5.8080808	0	0	0
2561	0.435270453	5.8080808	0	0	0
2562	0.442524961	5.8080808	0	0	0
2563	0.449779468	5.8080808	0	0	0
2564	0.457033976	5.8080808	0	0	0
2565	0.464288483	5.8080808	0	0	0

2566	0.471542991	5.8080808	0	0	0
2567	0.478797498	5.8080808	0	0	0
2568	0.486052006	5.8080808	0	0	0
2569	0.493306513	5.8080808	0	0	0
2570	0.500561021	5.8080808	0	0	0
2571	0.507815529	5.8080808	0	0	0
2572	0.515070036	5.8080808	0	0	0
2573	0.522324544	5.8080808	0	0	0
2574	0.529579051	5.8080808	0	0	0
2575	0.536833559	5.8080808	0	0	0
2576	0.544088066	5.8080808	0	0	0
2577	0.551342574	5.8080808	0	0	0
2578	0.558597081	5.8080808	0	0	0
2579	0.565851589	5.8080808	0	0	0
2580	0.573106097	5.8080808	0	0	0
2581	0.580360604	5.8080808	0	0	0
2582	0.587615112	5.8080808	0	0	0
2583	0.594869619	5.8080808	0	0	0
2584	0.602124127	5.8080808	0	0	0
2585	0.609378634	5.8080808	0	0	0
2586	0.616633142	5.8080808	0	0	0
2587	0.623887649	5.8080808	0	0	0
2588	0.631142157	5.8080808	0	0	0
2589	0.638396664	5.8080808	0	0	0
2590	0.645651172	5.8080808	0	0	0
2591	0.652905680	5.8080808	0	0	0
2592	0.660160187	5.8080808	0	0	0
2593	0.667414695	5.8080808	0	0	0
2594	0.674669202	5.8080808	0	0	0
2595	0.681923710	5.8080808	0	0	0
2596	0.689178217	5.8080808	0	0	0
2597	0.696432725	5.8080808	0	0	0
2598	0.703687232	5.8080808	0	0	0
2599	0.710941740	5.8080808	0	0	0
2600	0.718196248	5.8080808	0	0	0
2601	0.000000000	6.0404040	0	0	0
2602	0.007254508	6.0404040	0	0	0
2603	0.014509015	6.0404040	0	0	0
2604	0.021763523	6.0404040	0	0	0
2605	0.029018030	6.0404040	0	0	0
2606	0.036272538	6.0404040	0	0	0
2607	0.043527045	6.0404040	0	0	0
2608	0.050781553	6.0404040	0	0	0
2609	0.058036060	6.0404040	0	0	0
2610	0.065290568	6.0404040	0	0	0

2611	0.072545076	6.0404040	0	0	0
2612	0.079799583	6.0404040	0	0	0
2613	0.087054091	6.0404040	0	0	0
2614	0.094308598	6.0404040	0	0	0
2615	0.101563106	6.0404040	0	0	0
2616	0.108817613	6.0404040	0	0	0
2617	0.116072121	6.0404040	0	0	0
2618	0.123326628	6.0404040	0	0	0
2619	0.130581136	6.0404040	0	0	0
2620	0.137835643	6.0404040	0	0	0
2621	0.145090151	6.0404040	0	0	0
2622	0.152344659	6.0404040	0	0	0
2623	0.159599166	6.0404040	0	0	0
2624	0.166853674	6.0404040	0	0	0
2625	0.174108181	6.0404040	0	0	0
2626	0.181362689	6.0404040	0	0	0
2627	0.188617196	6.0404040	0	0	0
2628	0.195871704	6.0404040	0	0	0
2629	0.203126211	6.0404040	0	0	0
2630	0.210380719	6.0404040	0	0	0
2631	0.217635227	6.0404040	0	0	0
2632	0.224889734	6.0404040	0	0	0
2633	0.232144242	6.0404040	0	0	0
2634	0.239398749	6.0404040	0	0	0
2635	0.246653257	6.0404040	0	0	0
2636	0.253907764	6.0404040	0	0	0
2637	0.261162272	6.0404040	0	0	0
2638	0.268416779	6.0404040	0	0	0
2639	0.275671287	6.0404040	0	0	0
2640	0.282925794	6.0404040	0	0	0
2641	0.290180302	6.0404040	0	0	0
2642	0.297434810	6.0404040	0	0	0
2643	0.304689317	6.0404040	0	0	0
2644	0.311943825	6.0404040	0	0	0
2645	0.319198332	6.0404040	0	0	0
2646	0.326452840	6.0404040	0	0	0
2647	0.333707347	6.0404040	0	0	0
2648	0.340961855	6.0404040	0	0	0
2649	0.348216362	6.0404040	0	0	0
2650	0.355470870	6.0404040	0	0	0
2651	0.362725378	6.0404040	0	0	0
2652	0.369979885	6.0404040	0	0	0
2653	0.377234393	6.0404040	0	0	0
2654	0.384488900	6.0404040	0	0	0
2655	0.391743408	6.0404040	0	0	0

2656	0.398997915	6.0404040	0	0	0
2657	0.406252423	6.0404040	0	0	0
2658	0.413506930	6.0404040	0	0	0
2659	0.420761438	6.0404040	0	0	0
2660	0.428015945	6.0404040	0	0	0
2661	0.435270453	6.0404040	0	0	0
2662	0.442524961	6.0404040	0	0	0
2663	0.449779468	6.0404040	0	0	0
2664	0.457033976	6.0404040	0	0	0
2665	0.464288483	6.0404040	0	0	0
2666	0.471542991	6.0404040	0	0	0
2667	0.478797498	6.0404040	0	0	0
2668	0.486052006	6.0404040	0	0	0
2669	0.493306513	6.0404040	0	0	0
2670	0.500561021	6.0404040	0	0	0
2671	0.507815529	6.0404040	0	0	0
2672	0.515070036	6.0404040	0	0	0
2673	0.522324544	6.0404040	0	0	0
2674	0.529579051	6.0404040	0	0	0
2675	0.536833559	6.0404040	0	0	0
2676	0.544088066	6.0404040	0	0	0
2677	0.551342574	6.0404040	0	0	0
2678	0.558597081	6.0404040	0	0	0
2679	0.565851589	6.0404040	0	0	0
2680	0.573106097	6.0404040	0	0	0
2681	0.580360604	6.0404040	0	0	0
2682	0.587615112	6.0404040	0	0	0
2683	0.594869619	6.0404040	0	0	0
2684	0.602124127	6.0404040	0	0	0
2685	0.609378634	6.0404040	0	0	0
2686	0.616633142	6.0404040	0	0	0
2687	0.623887649	6.0404040	0	0	0
2688	0.631142157	6.0404040	0	0	0
2689	0.638396664	6.0404040	0	0	0
2690	0.645651172	6.0404040	0	0	0
2691	0.652905680	6.0404040	0	0	0
2692	0.660160187	6.0404040	0	0	0
2693	0.667414695	6.0404040	0	0	0
2694	0.674669202	6.0404040	0	0	0
2695	0.681923710	6.0404040	0	0	0
2696	0.689178217	6.0404040	0	0	0
2697	0.696432725	6.0404040	0	0	0
2698	0.703687232	6.0404040	0	0	0
2699	0.710941740	6.0404040	0	0	0
2700	0.718196248	6.0404040	0	0	0

2701	0.000000000	6.2727273	0	0	0
2702	0.007254508	6.2727273	0	0	0
2703	0.014509015	6.2727273	0	0	0
2704	0.021763523	6.2727273	0	0	0
2705	0.029018030	6.2727273	0	0	0
2706	0.036272538	6.2727273	0	0	0
2707	0.043527045	6.2727273	0	0	0
2708	0.050781553	6.2727273	0	0	0
2709	0.058036060	6.2727273	0	0	0
2710	0.065290568	6.2727273	0	0	0
2711	0.072545076	6.2727273	0	0	0
2712	0.079799583	6.2727273	0	0	0
2713	0.087054091	6.2727273	0	0	0
2714	0.094308598	6.2727273	0	0	0
2715	0.101563106	6.2727273	0	0	0
2716	0.108817613	6.2727273	0	0	0
2717	0.116072121	6.2727273	0	0	0
2718	0.123326628	6.2727273	0	0	0
2719	0.130581136	6.2727273	0	0	0
2720	0.137835643	6.2727273	0	0	0
2721	0.145090151	6.2727273	0	0	0
2722	0.152344659	6.2727273	0	0	0
2723	0.159599166	6.2727273	0	0	0
2724	0.166853674	6.2727273	0	0	0
2725	0.174108181	6.2727273	0	0	0
2726	0.181362689	6.2727273	0	0	0
2727	0.188617196	6.2727273	0	0	0
2728	0.195871704	6.2727273	0	0	0
2729	0.203126211	6.2727273	0	0	0
2730	0.210380719	6.2727273	0	0	0
2731	0.217635227	6.2727273	0	0	0
2732	0.224889734	6.2727273	0	0	0
2733	0.232144242	6.2727273	0	0	0
2734	0.239398749	6.2727273	0	0	0
2735	0.246653257	6.2727273	0	0	0
2736	0.253907764	6.2727273	0	0	0
2737	0.261162272	6.2727273	0	0	0
2738	0.268416779	6.2727273	0	0	0
2739	0.275671287	6.2727273	0	0	0
2740	0.282925794	6.2727273	0	0	0
2741	0.290180302	6.2727273	0	0	0
2742	0.297434810	6.2727273	0	0	0
2743	0.304689317	6.2727273	0	0	0
2744	0.311943825	6.2727273	0	0	0
2745	0.319198332	6.2727273	0	0	0

2746	0.326452840	6.2727273	0	0	0
2747	0.333707347	6.2727273	0	0	0
2748	0.340961855	6.2727273	0	0	0
2749	0.348216362	6.2727273	0	0	0
2750	0.355470870	6.2727273	0	0	0
2751	0.362725378	6.2727273	0	0	0
2752	0.369979885	6.2727273	0	0	0
2753	0.377234393	6.2727273	0	0	0
2754	0.384488900	6.2727273	0	0	0
2755	0.391743408	6.2727273	0	0	0
2756	0.398997915	6.2727273	0	0	0
2757	0.406252423	6.2727273	0	0	0
2758	0.413506930	6.2727273	0	0	0
2759	0.420761438	6.2727273	0	0	0
2760	0.428015945	6.2727273	0	0	0
2761	0.435270453	6.2727273	0	0	0
2762	0.442524961	6.2727273	0	0	0
2763	0.449779468	6.2727273	0	0	0
2764	0.457033976	6.2727273	0	0	0
2765	0.464288483	6.2727273	0	0	0
2766	0.471542991	6.2727273	0	0	0
2767	0.478797498	6.2727273	0	0	0
2768	0.486052006	6.2727273	0	0	0
2769	0.493306513	6.2727273	0	0	0
2770	0.500561021	6.2727273	0	0	0
2771	0.507815529	6.2727273	0	0	0
2772	0.515070036	6.2727273	0	0	0
2773	0.522324544	6.2727273	0	0	0
2774	0.529579051	6.2727273	0	0	0
2775	0.536833559	6.2727273	0	0	0
2776	0.544088066	6.2727273	0	0	0
2777	0.551342574	6.2727273	0	0	0
2778	0.558597081	6.2727273	0	0	0
2779	0.565851589	6.2727273	0	0	0
2780	0.573106097	6.2727273	0	0	0
2781	0.580360604	6.2727273	0	0	0
2782	0.587615112	6.2727273	0	0	0
2783	0.594869619	6.2727273	0	0	0
2784	0.602124127	6.2727273	0	0	0
2785	0.609378634	6.2727273	0	0	0
2786	0.616633142	6.2727273	0	0	0
2787	0.623887649	6.2727273	0	0	0
2788	0.631142157	6.2727273	0	0	0
2789	0.638396664	6.2727273	0	0	0
2790	0.645651172	6.2727273	0	0	0

2791	0.652905680	6.2727273	0	0	0
2792	0.660160187	6.2727273	0	0	0
2793	0.667414695	6.2727273	0	0	0
2794	0.674669202	6.2727273	0	0	0
2795	0.681923710	6.2727273	0	0	0
2796	0.689178217	6.2727273	0	0	0
2797	0.696432725	6.2727273	0	0	0
2798	0.703687232	6.2727273	0	0	0
2799	0.710941740	6.2727273	0	0	0
2800	0.718196248	6.2727273	0	0	0
2801	0.000000000	6.5050505	0	0	0
2802	0.007254508	6.5050505	0	0	0
2803	0.014509015	6.5050505	0	0	0
2804	0.021763523	6.5050505	0	0	0
2805	0.029018030	6.5050505	0	0	0
2806	0.036272538	6.5050505	0	0	0
2807	0.043527045	6.5050505	0	0	0
2808	0.050781553	6.5050505	0	0	0
2809	0.058036060	6.5050505	0	0	0
2810	0.065290568	6.5050505	0	0	0
2811	0.072545076	6.5050505	0	0	0
2812	0.079799583	6.5050505	0	0	0
2813	0.087054091	6.5050505	0	0	0
2814	0.094308598	6.5050505	0	0	0
2815	0.101563106	6.5050505	0	0	0
2816	0.108817613	6.5050505	0	0	0
2817	0.116072121	6.5050505	0	0	0
2818	0.123326628	6.5050505	0	0	0
2819	0.130581136	6.5050505	0	0	0
2820	0.137835643	6.5050505	0	0	0
2821	0.145090151	6.5050505	0	0	0
2822	0.152344659	6.5050505	0	0	0
2823	0.159599166	6.5050505	0	0	0
2824	0.166853674	6.5050505	0	0	0
2825	0.174108181	6.5050505	0	0	0
2826	0.181362689	6.5050505	0	0	0
2827	0.188617196	6.5050505	0	0	0
2828	0.195871704	6.5050505	0	0	0
2829	0.203126211	6.5050505	0	0	0
2830	0.210380719	6.5050505	0	0	0
2831	0.217635227	6.5050505	0	0	0
2832	0.224889734	6.5050505	0	0	0
2833	0.232144242	6.5050505	0	0	0
2834	0.239398749	6.5050505	0	0	0
2835	0.246653257	6.5050505	0	0	0

2836	0.253907764	6.5050505	0	0	0
2837	0.261162272	6.5050505	0	0	0
2838	0.268416779	6.5050505	0	0	0
2839	0.275671287	6.5050505	0	0	0
2840	0.282925794	6.5050505	0	0	0
2841	0.290180302	6.5050505	0	0	0
2842	0.297434810	6.5050505	0	0	0
2843	0.304689317	6.5050505	0	0	0
2844	0.311943825	6.5050505	0	0	0
2845	0.319198332	6.5050505	0	0	0
2846	0.326452840	6.5050505	0	0	0
2847	0.333707347	6.5050505	0	0	0
2848	0.340961855	6.5050505	0	0	0
2849	0.348216362	6.5050505	0	0	0
2850	0.355470870	6.5050505	0	0	0
2851	0.362725378	6.5050505	0	0	0
2852	0.369979885	6.5050505	0	0	0
2853	0.377234393	6.5050505	0	0	0
2854	0.384488900	6.5050505	0	0	0
2855	0.391743408	6.5050505	0	0	0
2856	0.398997915	6.5050505	0	0	0
2857	0.406252423	6.5050505	0	0	0
2858	0.413506930	6.5050505	0	0	0
2859	0.420761438	6.5050505	0	0	0
2860	0.428015945	6.5050505	0	0	0
2861	0.435270453	6.5050505	0	0	0
2862	0.442524961	6.5050505	0	0	0
2863	0.449779468	6.5050505	0	0	0
2864	0.457033976	6.5050505	0	0	0
2865	0.464288483	6.5050505	0	0	0
2866	0.471542991	6.5050505	0	0	0
2867	0.478797498	6.5050505	0	0	0
2868	0.486052006	6.5050505	0	0	0
2869	0.493306513	6.5050505	0	0	0
2870	0.500561021	6.5050505	0	0	0
2871	0.507815529	6.5050505	0	0	0
2872	0.515070036	6.5050505	0	0	0
2873	0.522324544	6.5050505	0	0	0
2874	0.529579051	6.5050505	0	0	0
2875	0.536833559	6.5050505	0	0	0
2876	0.544088066	6.5050505	0	0	0
2877	0.551342574	6.5050505	0	0	0
2878	0.558597081	6.5050505	0	0	0
2879	0.565851589	6.5050505	0	0	0
2880	0.573106097	6.5050505	0	0	0

2881	0.580360604	6.5050505	0	0	0
2882	0.587615112	6.5050505	0	0	0
2883	0.594869619	6.5050505	0	0	0
2884	0.602124127	6.5050505	0	0	0
2885	0.609378634	6.5050505	0	0	0
2886	0.616633142	6.5050505	0	0	0
2887	0.623887649	6.5050505	0	0	0
2888	0.631142157	6.5050505	0	0	0
2889	0.638396664	6.5050505	0	0	0
2890	0.645651172	6.5050505	0	0	0
2891	0.652905680	6.5050505	0	0	0
2892	0.660160187	6.5050505	0	0	0
2893	0.667414695	6.5050505	0	0	0
2894	0.674669202	6.5050505	0	0	0
2895	0.681923710	6.5050505	0	0	0
2896	0.689178217	6.5050505	0	0	0
2897	0.696432725	6.5050505	0	0	0
2898	0.703687232	6.5050505	0	0	0
2899	0.710941740	6.5050505	0	0	0
2900	0.718196248	6.5050505	0	0	0
2901	0.000000000	6.7373737	0	0	0
2902	0.007254508	6.7373737	0	0	0
2903	0.014509015	6.7373737	0	0	0
2904	0.021763523	6.7373737	0	0	0
2905	0.029018030	6.7373737	0	0	0
2906	0.036272538	6.7373737	0	0	0
2907	0.043527045	6.7373737	0	0	0
2908	0.050781553	6.7373737	0	0	0
2909	0.058036060	6.7373737	0	0	0
2910	0.065290568	6.7373737	0	0	0
2911	0.072545076	6.7373737	0	0	0
2912	0.079799583	6.7373737	0	0	0
2913	0.087054091	6.7373737	0	0	0
2914	0.094308598	6.7373737	0	0	0
2915	0.101563106	6.7373737	0	0	0
2916	0.108817613	6.7373737	0	0	0
2917	0.116072121	6.7373737	0	0	0
2918	0.123326628	6.7373737	0	0	0
2919	0.130581136	6.7373737	0	0	0
2920	0.137835643	6.7373737	0	0	0
2921	0.145090151	6.7373737	0	0	0
2922	0.152344659	6.7373737	0	0	0
2923	0.159599166	6.7373737	0	0	0
2924	0.166853674	6.7373737	0	0	0
2925	0.174108181	6.7373737	0	0	0

2926	0.181362689	6.7373737	0	0	0
2927	0.188617196	6.7373737	0	0	0
2928	0.195871704	6.7373737	0	0	0
2929	0.203126211	6.7373737	0	0	0
2930	0.210380719	6.7373737	0	0	0
2931	0.217635227	6.7373737	0	0	0
2932	0.224889734	6.7373737	0	0	0
2933	0.232144242	6.7373737	0	0	0
2934	0.239398749	6.7373737	0	0	0
2935	0.246653257	6.7373737	0	0	0
2936	0.253907764	6.7373737	0	0	0
2937	0.261162272	6.7373737	0	0	0
2938	0.268416779	6.7373737	0	0	0
2939	0.275671287	6.7373737	0	0	0
2940	0.282925794	6.7373737	0	0	0
2941	0.290180302	6.7373737	0	0	0
2942	0.297434810	6.7373737	0	0	0
2943	0.304689317	6.7373737	0	0	0
2944	0.311943825	6.7373737	0	0	0
2945	0.319198332	6.7373737	0	0	0
2946	0.326452840	6.7373737	0	0	0
2947	0.333707347	6.7373737	0	0	0
2948	0.340961855	6.7373737	0	0	0
2949	0.348216362	6.7373737	0	0	0
2950	0.355470870	6.7373737	0	0	0
2951	0.362725378	6.7373737	0	0	0
2952	0.369979885	6.7373737	0	0	0
2953	0.377234393	6.7373737	0	0	0
2954	0.384488900	6.7373737	0	0	0
2955	0.391743408	6.7373737	0	0	0
2956	0.398997915	6.7373737	0	0	0
2957	0.406252423	6.7373737	0	0	0
2958	0.413506930	6.7373737	0	0	0
2959	0.420761438	6.7373737	0	0	0
2960	0.428015945	6.7373737	0	0	0
2961	0.435270453	6.7373737	0	0	0
2962	0.442524961	6.7373737	0	0	0
2963	0.449779468	6.7373737	0	0	0
2964	0.457033976	6.7373737	0	0	0
2965	0.464288483	6.7373737	0	0	0
2966	0.471542991	6.7373737	0	0	0
2967	0.478797498	6.7373737	0	0	0
2968	0.486052006	6.7373737	0	0	0
2969	0.493306513	6.7373737	0	0	0
2970	0.500561021	6.7373737	0	0	0

2971	0.507815529	6.7373737	0	0	0
2972	0.515070036	6.7373737	0	0	0
2973	0.522324544	6.7373737	0	0	0
2974	0.529579051	6.7373737	0	0	0
2975	0.536833559	6.7373737	0	0	0
2976	0.544088066	6.7373737	0	0	0
2977	0.551342574	6.7373737	0	0	0
2978	0.558597081	6.7373737	0	0	0
2979	0.565851589	6.7373737	0	0	0
2980	0.573106097	6.7373737	0	0	0
2981	0.580360604	6.7373737	0	0	0
2982	0.587615112	6.7373737	0	0	0
2983	0.594869619	6.7373737	0	0	0
2984	0.602124127	6.7373737	0	0	0
2985	0.609378634	6.7373737	0	0	0
2986	0.616633142	6.7373737	0	0	0
2987	0.623887649	6.7373737	0	0	0
2988	0.631142157	6.7373737	0	0	0
2989	0.638396664	6.7373737	0	0	0
2990	0.645651172	6.7373737	0	0	0
2991	0.652905680	6.7373737	0	0	0
2992	0.660160187	6.7373737	0	0	0
2993	0.667414695	6.7373737	0	0	0
2994	0.674669202	6.7373737	0	0	0
2995	0.681923710	6.7373737	0	0	0
2996	0.689178217	6.7373737	0	0	0
2997	0.696432725	6.7373737	0	0	0
2998	0.703687232	6.7373737	0	0	0
2999	0.710941740	6.7373737	0	0	0
3000	0.718196248	6.7373737	0	0	0
3001	0.000000000	6.9696970	0	0	0
3002	0.007254508	6.9696970	0	0	0
3003	0.014509015	6.9696970	0	0	0
3004	0.021763523	6.9696970	0	0	0
3005	0.029018030	6.9696970	0	0	0
3006	0.036272538	6.9696970	0	0	0
3007	0.043527045	6.9696970	0	0	0
3008	0.050781553	6.9696970	0	0	0
3009	0.058036060	6.9696970	0	0	0
3010	0.065290568	6.9696970	0	0	0
3011	0.072545076	6.9696970	0	0	0
3012	0.079799583	6.9696970	0	0	0
3013	0.087054091	6.9696970	0	0	0
3014	0.094308598	6.9696970	0	0	0
3015	0.101563106	6.9696970	0	0	0

3016	0.108817613	6.9696970	0	0	0
3017	0.116072121	6.9696970	0	0	0
3018	0.123326628	6.9696970	0	0	0
3019	0.130581136	6.9696970	0	0	0
3020	0.137835643	6.9696970	0	0	0
3021	0.145090151	6.9696970	0	0	0
3022	0.152344659	6.9696970	0	0	0
3023	0.159599166	6.9696970	0	0	0
3024	0.166853674	6.9696970	0	0	0
3025	0.174108181	6.9696970	0	0	0
3026	0.181362689	6.9696970	0	0	0
3027	0.188617196	6.9696970	0	0	0
3028	0.195871704	6.9696970	0	0	0
3029	0.203126211	6.9696970	0	0	0
3030	0.210380719	6.9696970	0	0	0
3031	0.217635227	6.9696970	0	0	0
3032	0.224889734	6.9696970	0	0	0
3033	0.232144242	6.9696970	0	0	0
3034	0.239398749	6.9696970	0	0	0
3035	0.246653257	6.9696970	0	0	0
3036	0.253907764	6.9696970	0	0	0
3037	0.261162272	6.9696970	0	0	0
3038	0.268416779	6.9696970	0	0	0
3039	0.275671287	6.9696970	0	0	0
3040	0.282925794	6.9696970	0	0	0
3041	0.290180302	6.9696970	0	0	0
3042	0.297434810	6.9696970	0	0	0
3043	0.304689317	6.9696970	0	0	0
3044	0.311943825	6.9696970	0	0	0
3045	0.319198332	6.9696970	0	0	0
3046	0.326452840	6.9696970	0	0	0
3047	0.333707347	6.9696970	0	0	0
3048	0.340961855	6.9696970	0	0	0
3049	0.348216362	6.9696970	0	0	0
3050	0.355470870	6.9696970	0	0	0
3051	0.362725378	6.9696970	0	0	0
3052	0.369979885	6.9696970	0	0	0
3053	0.377234393	6.9696970	0	0	0
3054	0.384488900	6.9696970	0	0	0
3055	0.391743408	6.9696970	0	0	0
3056	0.398997915	6.9696970	0	0	0
3057	0.406252423	6.9696970	0	0	0
3058	0.413506930	6.9696970	0	0	0
3059	0.420761438	6.9696970	0	0	0
3060	0.428015945	6.9696970	0	0	0

3061	0.435270453	6.9696970	0	0	0
3062	0.442524961	6.9696970	0	0	0
3063	0.449779468	6.9696970	0	0	0
3064	0.457033976	6.9696970	0	0	0
3065	0.464288483	6.9696970	0	0	0
3066	0.471542991	6.9696970	0	0	0
3067	0.478797498	6.9696970	0	0	0
3068	0.486052006	6.9696970	0	0	0
3069	0.493306513	6.9696970	0	0	0
3070	0.500561021	6.9696970	0	0	0
3071	0.507815529	6.9696970	0	0	0
3072	0.515070036	6.9696970	0	0	0
3073	0.522324544	6.9696970	0	0	0
3074	0.529579051	6.9696970	0	0	0
3075	0.536833559	6.9696970	0	0	0
3076	0.544088066	6.9696970	0	0	0
3077	0.551342574	6.9696970	0	0	0
3078	0.558597081	6.9696970	0	0	0
3079	0.565851589	6.9696970	0	0	0
3080	0.573106097	6.9696970	0	0	0
3081	0.580360604	6.9696970	0	0	0
3082	0.587615112	6.9696970	0	0	0
3083	0.594869619	6.9696970	0	0	0
3084	0.602124127	6.9696970	0	0	0
3085	0.609378634	6.9696970	0	0	0
3086	0.616633142	6.9696970	0	0	0
3087	0.623887649	6.9696970	0	0	0
3088	0.631142157	6.9696970	0	0	0
3089	0.638396664	6.9696970	0	0	0
3090	0.645651172	6.9696970	0	0	0
3091	0.652905680	6.9696970	0	0	0
3092	0.660160187	6.9696970	0	0	0
3093	0.667414695	6.9696970	0	0	0
3094	0.674669202	6.9696970	0	0	0
3095	0.681923710	6.9696970	0	0	0
3096	0.689178217	6.9696970	0	0	0
3097	0.696432725	6.9696970	0	0	0
3098	0.703687232	6.9696970	0	0	0
3099	0.710941740	6.9696970	0	0	0
3100	0.718196248	6.9696970	0	0	0
3101	0.000000000	7.2020202	0	0	0
3102	0.007254508	7.2020202	0	0	0
3103	0.014509015	7.2020202	0	0	0
3104	0.021763523	7.2020202	0	0	0
3105	0.029018030	7.2020202	0	0	0

3106	0.036272538	7.2020202	0	0	0
3107	0.043527045	7.2020202	0	0	0
3108	0.050781553	7.2020202	0	0	0
3109	0.058036060	7.2020202	0	0	0
3110	0.065290568	7.2020202	0	0	0
3111	0.072545076	7.2020202	0	0	0
3112	0.079799583	7.2020202	0	0	0
3113	0.087054091	7.2020202	0	0	0
3114	0.094308598	7.2020202	0	0	0
3115	0.101563106	7.2020202	0	0	0
3116	0.108817613	7.2020202	0	0	0
3117	0.116072121	7.2020202	0	0	0
3118	0.123326628	7.2020202	0	0	0
3119	0.130581136	7.2020202	0	0	0
3120	0.137835643	7.2020202	0	0	0
3121	0.145090151	7.2020202	0	0	0
3122	0.152344659	7.2020202	0	0	0
3123	0.159599166	7.2020202	0	0	0
3124	0.166853674	7.2020202	0	0	0
3125	0.174108181	7.2020202	0	0	0
3126	0.181362689	7.2020202	0	0	0
3127	0.188617196	7.2020202	0	0	0
3128	0.195871704	7.2020202	0	0	0
3129	0.203126211	7.2020202	0	0	0
3130	0.210380719	7.2020202	0	0	0
3131	0.217635227	7.2020202	0	0	0
3132	0.224889734	7.2020202	0	0	0
3133	0.232144242	7.2020202	0	0	0
3134	0.239398749	7.2020202	0	0	0
3135	0.246653257	7.2020202	0	0	0
3136	0.253907764	7.2020202	0	0	0
3137	0.261162272	7.2020202	0	0	0
3138	0.268416779	7.2020202	0	0	0
3139	0.275671287	7.2020202	0	0	0
3140	0.282925794	7.2020202	0	0	0
3141	0.290180302	7.2020202	0	0	0
3142	0.297434810	7.2020202	0	0	0
3143	0.304689317	7.2020202	0	0	0
3144	0.311943825	7.2020202	0	0	0
3145	0.319198332	7.2020202	0	0	0
3146	0.326452840	7.2020202	0	0	0
3147	0.333707347	7.2020202	0	0	0
3148	0.340961855	7.2020202	0	0	0
3149	0.348216362	7.2020202	0	0	0
3150	0.355470870	7.2020202	0	0	0

3151	0.362725378	7.2020202	0	0	0
3152	0.369979885	7.2020202	0	0	0
3153	0.377234393	7.2020202	0	0	0
3154	0.384488900	7.2020202	0	0	0
3155	0.391743408	7.2020202	0	0	0
3156	0.398997915	7.2020202	0	0	0
3157	0.406252423	7.2020202	0	0	0
3158	0.413506930	7.2020202	0	0	0
3159	0.420761438	7.2020202	0	0	0
3160	0.428015945	7.2020202	0	0	0
3161	0.435270453	7.2020202	0	0	0
3162	0.442524961	7.2020202	0	0	0
3163	0.449779468	7.2020202	0	0	0
3164	0.457033976	7.2020202	0	0	0
3165	0.464288483	7.2020202	0	0	0
3166	0.471542991	7.2020202	0	0	0
3167	0.478797498	7.2020202	0	0	0
3168	0.486052006	7.2020202	0	0	0
3169	0.493306513	7.2020202	0	0	0
3170	0.500561021	7.2020202	0	0	0
3171	0.507815529	7.2020202	0	0	0
3172	0.515070036	7.2020202	0	0	0
3173	0.522324544	7.2020202	0	0	0
3174	0.529579051	7.2020202	0	0	0
3175	0.536833559	7.2020202	0	0	0
3176	0.544088066	7.2020202	0	0	0
3177	0.551342574	7.2020202	0	0	0
3178	0.558597081	7.2020202	0	0	0
3179	0.565851589	7.2020202	0	0	0
3180	0.573106097	7.2020202	0	0	0
3181	0.580360604	7.2020202	0	0	0
3182	0.587615112	7.2020202	0	0	0
3183	0.594869619	7.2020202	0	0	0
3184	0.602124127	7.2020202	0	0	0
3185	0.609378634	7.2020202	0	0	0
3186	0.616633142	7.2020202	0	0	0
3187	0.623887649	7.2020202	0	0	0
3188	0.631142157	7.2020202	0	0	0
3189	0.638396664	7.2020202	0	0	0
3190	0.645651172	7.2020202	0	0	0
3191	0.652905680	7.2020202	0	0	0
3192	0.660160187	7.2020202	0	0	0
3193	0.667414695	7.2020202	0	0	0
3194	0.674669202	7.2020202	0	0	0
3195	0.681923710	7.2020202	0	0	0

3196	0.689178217	7.2020202	0	0	0
3197	0.696432725	7.2020202	0	0	0
3198	0.703687232	7.2020202	0	0	0
3199	0.710941740	7.2020202	0	0	0
3200	0.718196248	7.2020202	0	0	0
3201	0.000000000	7.4343434	0	0	0
3202	0.007254508	7.4343434	0	0	0
3203	0.014509015	7.4343434	0	0	0
3204	0.021763523	7.4343434	0	0	0
3205	0.029018030	7.4343434	0	0	0
3206	0.036272538	7.4343434	0	0	0
3207	0.043527045	7.4343434	0	0	0
3208	0.050781553	7.4343434	0	0	0
3209	0.058036060	7.4343434	0	0	0
3210	0.065290568	7.4343434	0	0	0
3211	0.072545076	7.4343434	0	0	0
3212	0.079799583	7.4343434	0	0	0
3213	0.087054091	7.4343434	0	0	0
3214	0.094308598	7.4343434	0	0	0
3215	0.101563106	7.4343434	0	0	0
3216	0.108817613	7.4343434	0	0	0
3217	0.116072121	7.4343434	0	0	0
3218	0.123326628	7.4343434	0	0	0
3219	0.130581136	7.4343434	0	0	0
3220	0.137835643	7.4343434	0	0	0
3221	0.145090151	7.4343434	0	0	0
3222	0.152344659	7.4343434	0	0	0
3223	0.159599166	7.4343434	0	0	0
3224	0.166853674	7.4343434	0	0	0
3225	0.174108181	7.4343434	0	0	0
3226	0.181362689	7.4343434	0	0	0
3227	0.188617196	7.4343434	0	0	0
3228	0.195871704	7.4343434	0	0	0
3229	0.203126211	7.4343434	0	0	0
3230	0.210380719	7.4343434	0	0	0
3231	0.217635227	7.4343434	0	0	0
3232	0.224889734	7.4343434	0	0	0
3233	0.232144242	7.4343434	0	0	0
3234	0.239398749	7.4343434	0	0	0
3235	0.246653257	7.4343434	0	0	0
3236	0.253907764	7.4343434	0	0	0
3237	0.261162272	7.4343434	0	0	0
3238	0.268416779	7.4343434	0	0	0
3239	0.275671287	7.4343434	0	0	0
3240	0.282925794	7.4343434	0	0	0

3241	0.290180302	7.4343434	0	0	0
3242	0.297434810	7.4343434	0	0	0
3243	0.304689317	7.4343434	0	0	0
3244	0.311943825	7.4343434	0	0	0
3245	0.319198332	7.4343434	0	0	0
3246	0.326452840	7.4343434	0	0	0
3247	0.333707347	7.4343434	0	0	0
3248	0.340961855	7.4343434	0	0	0
3249	0.348216362	7.4343434	0	0	0
3250	0.355470870	7.4343434	0	0	0
3251	0.362725378	7.4343434	0	0	0
3252	0.369979885	7.4343434	0	0	0
3253	0.377234393	7.4343434	0	0	0
3254	0.384488900	7.4343434	0	0	0
3255	0.391743408	7.4343434	0	0	0
3256	0.398997915	7.4343434	0	0	0
3257	0.406252423	7.4343434	0	0	0
3258	0.413506930	7.4343434	0	0	0
3259	0.420761438	7.4343434	0	0	0
3260	0.428015945	7.4343434	0	0	0
3261	0.435270453	7.4343434	0	0	0
3262	0.442524961	7.4343434	0	0	0
3263	0.449779468	7.4343434	0	0	0
3264	0.457033976	7.4343434	0	0	0
3265	0.464288483	7.4343434	0	0	0
3266	0.471542991	7.4343434	0	0	0
3267	0.478797498	7.4343434	0	0	0
3268	0.486052006	7.4343434	0	0	0
3269	0.493306513	7.4343434	0	0	0
3270	0.500561021	7.4343434	0	0	0
3271	0.507815529	7.4343434	0	0	0
3272	0.515070036	7.4343434	0	0	0
3273	0.522324544	7.4343434	0	0	0
3274	0.529579051	7.4343434	0	0	0
3275	0.536833559	7.4343434	0	0	0
3276	0.544088066	7.4343434	0	0	0
3277	0.551342574	7.4343434	0	0	0
3278	0.558597081	7.4343434	0	0	0
3279	0.565851589	7.4343434	0	0	0
3280	0.573106097	7.4343434	0	0	0
3281	0.580360604	7.4343434	0	0	0
3282	0.587615112	7.4343434	0	0	0
3283	0.594869619	7.4343434	0	0	0
3284	0.602124127	7.4343434	0	0	0
3285	0.609378634	7.4343434	0	0	0

3286	0.616633142	7.4343434	0	0	0
3287	0.623887649	7.4343434	0	0	0
3288	0.631142157	7.4343434	0	0	0
3289	0.638396664	7.4343434	0	0	0
3290	0.645651172	7.4343434	0	0	0
3291	0.652905680	7.4343434	0	0	0
3292	0.660160187	7.4343434	0	0	0
3293	0.667414695	7.4343434	0	0	0
3294	0.674669202	7.4343434	0	0	0
3295	0.681923710	7.4343434	0	0	0
3296	0.689178217	7.4343434	0	0	0
3297	0.696432725	7.4343434	0	0	0
3298	0.703687232	7.4343434	0	0	0
3299	0.710941740	7.4343434	0	0	0
3300	0.718196248	7.4343434	0	0	0
3301	0.000000000	7.6666667	0	0	0
3302	0.007254508	7.6666667	0	0	0
3303	0.014509015	7.6666667	0	0	0
3304	0.021763523	7.6666667	0	0	0
3305	0.029018030	7.6666667	0	0	0
3306	0.036272538	7.6666667	0	0	0
3307	0.043527045	7.6666667	0	0	0
3308	0.050781553	7.6666667	0	0	0
3309	0.058036060	7.6666667	0	0	0
3310	0.065290568	7.6666667	0	0	0
3311	0.072545076	7.6666667	0	0	0
3312	0.079799583	7.6666667	0	0	0
3313	0.087054091	7.6666667	0	0	0
3314	0.094308598	7.6666667	0	0	0
3315	0.101563106	7.6666667	0	0	0
3316	0.108817613	7.6666667	0	0	0
3317	0.116072121	7.6666667	0	0	0
3318	0.123326628	7.6666667	0	0	0
3319	0.130581136	7.6666667	0	0	0
3320	0.137835643	7.6666667	0	0	0
3321	0.145090151	7.6666667	0	0	0
3322	0.152344659	7.6666667	0	0	0
3323	0.159599166	7.6666667	0	0	0
3324	0.166853674	7.6666667	0	0	0
3325	0.174108181	7.6666667	0	0	0
3326	0.181362689	7.6666667	0	0	0
3327	0.188617196	7.6666667	0	0	0
3328	0.195871704	7.6666667	0	0	0
3329	0.203126211	7.6666667	0	0	0
3330	0.210380719	7.6666667	0	0	0

3331	0.217635227	7.6666667	0	0	0
3332	0.224889734	7.6666667	0	0	0
3333	0.232144242	7.6666667	0	0	0
3334	0.239398749	7.6666667	0	0	0
3335	0.246653257	7.6666667	0	0	0
3336	0.253907764	7.6666667	0	0	0
3337	0.261162272	7.6666667	0	0	0
3338	0.268416779	7.6666667	0	0	0
3339	0.275671287	7.6666667	0	0	0
3340	0.282925794	7.6666667	0	0	0
3341	0.290180302	7.6666667	0	0	0
3342	0.297434810	7.6666667	0	0	0
3343	0.304689317	7.6666667	0	0	0
3344	0.311943825	7.6666667	0	0	0
3345	0.319198332	7.6666667	0	0	0
3346	0.326452840	7.6666667	0	0	0
3347	0.333707347	7.6666667	0	0	0
3348	0.340961855	7.6666667	0	0	0
3349	0.348216362	7.6666667	0	0	0
3350	0.355470870	7.6666667	0	0	0
3351	0.362725378	7.6666667	0	0	0
3352	0.369979885	7.6666667	0	0	0
3353	0.377234393	7.6666667	0	0	0
3354	0.384488900	7.6666667	0	0	0
3355	0.391743408	7.6666667	0	0	0
3356	0.398997915	7.6666667	0	0	0
3357	0.406252423	7.6666667	0	0	0
3358	0.413506930	7.6666667	0	0	0
3359	0.420761438	7.6666667	0	0	0
3360	0.428015945	7.6666667	0	0	0
3361	0.435270453	7.6666667	0	0	0
3362	0.442524961	7.6666667	0	0	0
3363	0.449779468	7.6666667	0	0	0
3364	0.457033976	7.6666667	0	0	0
3365	0.464288483	7.6666667	0	0	0
3366	0.471542991	7.6666667	0	0	0
3367	0.478797498	7.6666667	0	0	0
3368	0.486052006	7.6666667	0	0	0
3369	0.493306513	7.6666667	0	0	0
3370	0.500561021	7.6666667	0	0	0
3371	0.507815529	7.6666667	0	0	0
3372	0.515070036	7.6666667	0	0	0
3373	0.522324544	7.6666667	0	0	0
3374	0.529579051	7.6666667	0	0	0
3375	0.536833559	7.6666667	0	0	0

3376	0.544088066	7.6666667	0	0	0
3377	0.551342574	7.6666667	0	0	0
3378	0.558597081	7.6666667	0	0	0
3379	0.565851589	7.6666667	0	0	0
3380	0.573106097	7.6666667	0	0	0
3381	0.580360604	7.6666667	0	0	0
3382	0.587615112	7.6666667	0	0	0
3383	0.594869619	7.6666667	0	0	0
3384	0.602124127	7.6666667	0	0	0
3385	0.609378634	7.6666667	0	0	0
3386	0.616633142	7.6666667	0	0	0
3387	0.623887649	7.6666667	0	0	0
3388	0.631142157	7.6666667	0	0	0
3389	0.638396664	7.6666667	0	0	0
3390	0.645651172	7.6666667	0	0	0
3391	0.652905680	7.6666667	0	0	0
3392	0.660160187	7.6666667	0	0	0
3393	0.667414695	7.6666667	0	0	0
3394	0.674669202	7.6666667	0	0	0
3395	0.681923710	7.6666667	0	0	0
3396	0.689178217	7.6666667	0	0	0
3397	0.696432725	7.6666667	0	0	0
3398	0.703687232	7.6666667	0	0	0
3399	0.710941740	7.6666667	0	0	0
3400	0.718196248	7.6666667	0	0	0
3401	0.000000000	7.8989899	0	0	0
3402	0.007254508	7.8989899	0	0	0
3403	0.014509015	7.8989899	0	0	0
3404	0.021763523	7.8989899	0	0	0
3405	0.029018030	7.8989899	0	0	0
3406	0.036272538	7.8989899	0	0	0
3407	0.043527045	7.8989899	0	0	0
3408	0.050781553	7.8989899	0	0	0
3409	0.058036060	7.8989899	0	0	0
3410	0.065290568	7.8989899	0	0	0
3411	0.072545076	7.8989899	0	0	0
3412	0.079799583	7.8989899	0	0	0
3413	0.087054091	7.8989899	0	0	0
3414	0.094308598	7.8989899	0	0	0
3415	0.101563106	7.8989899	0	0	0
3416	0.108817613	7.8989899	0	0	0
3417	0.116072121	7.8989899	0	0	0
3418	0.123326628	7.8989899	0	0	0
3419	0.130581136	7.8989899	0	0	0
3420	0.137835643	7.8989899	0	0	0

3421	0.145090151	7.8989899	0	0	0
3422	0.152344659	7.8989899	0	0	0
3423	0.159599166	7.8989899	0	0	0
3424	0.166853674	7.8989899	0	0	0
3425	0.174108181	7.8989899	0	0	0
3426	0.181362689	7.8989899	0	0	0
3427	0.188617196	7.8989899	0	0	0
3428	0.195871704	7.8989899	0	0	0
3429	0.203126211	7.8989899	0	0	0
3430	0.210380719	7.8989899	0	0	0
3431	0.217635227	7.8989899	0	0	0
3432	0.224889734	7.8989899	0	0	0
3433	0.232144242	7.8989899	0	0	0
3434	0.239398749	7.8989899	0	0	0
3435	0.246653257	7.8989899	0	0	0
3436	0.253907764	7.8989899	0	0	0
3437	0.261162272	7.8989899	0	0	0
3438	0.268416779	7.8989899	0	0	0
3439	0.275671287	7.8989899	0	0	0
3440	0.282925794	7.8989899	0	0	0
3441	0.290180302	7.8989899	0	0	0
3442	0.297434810	7.8989899	0	0	0
3443	0.304689317	7.8989899	0	0	0
3444	0.311943825	7.8989899	0	0	0
3445	0.319198332	7.8989899	0	0	0
3446	0.326452840	7.8989899	0	0	0
3447	0.333707347	7.8989899	0	0	0
3448	0.340961855	7.8989899	0	0	0
3449	0.348216362	7.8989899	0	0	0
3450	0.355470870	7.8989899	0	0	0
3451	0.362725378	7.8989899	0	0	0
3452	0.369979885	7.8989899	0	0	0
3453	0.377234393	7.8989899	0	0	0
3454	0.384488900	7.8989899	0	0	0
3455	0.391743408	7.8989899	0	0	0
3456	0.398997915	7.8989899	0	0	0
3457	0.406252423	7.8989899	0	0	0
3458	0.413506930	7.8989899	0	0	0
3459	0.420761438	7.8989899	0	0	0
3460	0.428015945	7.8989899	0	0	0
3461	0.435270453	7.8989899	0	0	0
3462	0.442524961	7.8989899	0	0	0
3463	0.449779468	7.8989899	0	0	0
3464	0.457033976	7.8989899	0	0	0
3465	0.464288483	7.8989899	0	0	0

3466	0.471542991	7.8989899	0	0	0
3467	0.478797498	7.8989899	0	0	0
3468	0.486052006	7.8989899	0	0	0
3469	0.493306513	7.8989899	0	0	0
3470	0.500561021	7.8989899	0	0	0
3471	0.507815529	7.8989899	0	0	0
3472	0.515070036	7.8989899	0	0	0
3473	0.522324544	7.8989899	0	0	0
3474	0.529579051	7.8989899	0	0	0
3475	0.536833559	7.8989899	0	0	0
3476	0.544088066	7.8989899	0	0	0
3477	0.551342574	7.8989899	0	0	0
3478	0.558597081	7.8989899	0	0	0
3479	0.565851589	7.8989899	0	0	0
3480	0.573106097	7.8989899	0	0	0
3481	0.580360604	7.8989899	0	0	0
3482	0.587615112	7.8989899	0	0	0
3483	0.594869619	7.8989899	0	0	0
3484	0.602124127	7.8989899	0	0	0
3485	0.609378634	7.8989899	0	0	0
3486	0.616633142	7.8989899	0	0	0
3487	0.623887649	7.8989899	0	0	0
3488	0.631142157	7.8989899	0	0	0
3489	0.638396664	7.8989899	0	0	0
3490	0.645651172	7.8989899	0	0	0
3491	0.652905680	7.8989899	0	0	0
3492	0.660160187	7.8989899	0	0	0
3493	0.667414695	7.8989899	0	0	0
3494	0.674669202	7.8989899	0	0	0
3495	0.681923710	7.8989899	0	0	0
3496	0.689178217	7.8989899	0	0	0
3497	0.696432725	7.8989899	0	0	0
3498	0.703687232	7.8989899	0	0	0
3499	0.710941740	7.8989899	0	0	0
3500	0.718196248	7.8989899	0	0	0
3501	0.000000000	8.1313131	0	0	0
3502	0.007254508	8.1313131	0	0	0
3503	0.014509015	8.1313131	0	0	0
3504	0.021763523	8.1313131	0	0	0
3505	0.029018030	8.1313131	0	0	0
3506	0.036272538	8.1313131	0	0	0
3507	0.043527045	8.1313131	0	0	0
3508	0.050781553	8.1313131	0	0	0
3509	0.058036060	8.1313131	0	0	0
3510	0.065290568	8.1313131	0	0	0

3511	0.072545076	8.1313131	0	0	0
3512	0.079799583	8.1313131	0	0	0
3513	0.087054091	8.1313131	0	0	0
3514	0.094308598	8.1313131	0	0	0
3515	0.101563106	8.1313131	0	0	0
3516	0.108817613	8.1313131	0	0	0
3517	0.116072121	8.1313131	0	0	0
3518	0.123326628	8.1313131	0	0	0
3519	0.130581136	8.1313131	0	0	0
3520	0.137835643	8.1313131	0	0	0
3521	0.145090151	8.1313131	0	0	0
3522	0.152344659	8.1313131	0	0	0
3523	0.159599166	8.1313131	0	0	0
3524	0.166853674	8.1313131	0	0	0
3525	0.174108181	8.1313131	0	0	0
3526	0.181362689	8.1313131	0	0	0
3527	0.188617196	8.1313131	0	0	0
3528	0.195871704	8.1313131	0	0	0
3529	0.203126211	8.1313131	0	0	0
3530	0.210380719	8.1313131	0	0	0
3531	0.217635227	8.1313131	0	0	0
3532	0.224889734	8.1313131	0	0	0
3533	0.232144242	8.1313131	0	0	0
3534	0.239398749	8.1313131	0	0	0
3535	0.246653257	8.1313131	0	0	0
3536	0.253907764	8.1313131	0	0	0
3537	0.261162272	8.1313131	0	0	0
3538	0.268416779	8.1313131	0	0	0
3539	0.275671287	8.1313131	0	0	0
3540	0.282925794	8.1313131	0	0	0
3541	0.290180302	8.1313131	0	0	0
3542	0.297434810	8.1313131	0	0	0
3543	0.304689317	8.1313131	0	0	0
3544	0.311943825	8.1313131	0	0	0
3545	0.319198332	8.1313131	0	0	0
3546	0.326452840	8.1313131	0	0	0
3547	0.333707347	8.1313131	0	0	0
3548	0.340961855	8.1313131	0	0	0
3549	0.348216362	8.1313131	0	0	0
3550	0.355470870	8.1313131	0	0	0
3551	0.362725378	8.1313131	0	0	0
3552	0.369979885	8.1313131	0	0	0
3553	0.377234393	8.1313131	0	0	0
3554	0.384488900	8.1313131	0	0	0
3555	0.391743408	8.1313131	0	0	0

3556	0.398997915	8.1313131	0	0	0
3557	0.406252423	8.1313131	0	0	0
3558	0.413506930	8.1313131	0	0	0
3559	0.420761438	8.1313131	0	0	0
3560	0.428015945	8.1313131	0	0	0
3561	0.435270453	8.1313131	0	0	0
3562	0.442524961	8.1313131	0	0	0
3563	0.449779468	8.1313131	0	0	0
3564	0.457033976	8.1313131	0	0	0
3565	0.464288483	8.1313131	0	0	0
3566	0.471542991	8.1313131	0	0	0
3567	0.478797498	8.1313131	0	0	0
3568	0.486052006	8.1313131	0	0	0
3569	0.493306513	8.1313131	0	0	0
3570	0.500561021	8.1313131	0	0	0
3571	0.507815529	8.1313131	0	0	0
3572	0.515070036	8.1313131	0	0	0
3573	0.522324544	8.1313131	0	0	0
3574	0.529579051	8.1313131	0	0	0
3575	0.536833559	8.1313131	0	0	0
3576	0.544088066	8.1313131	0	0	0
3577	0.551342574	8.1313131	0	0	0
3578	0.558597081	8.1313131	0	0	0
3579	0.565851589	8.1313131	0	0	0
3580	0.573106097	8.1313131	0	0	0
3581	0.580360604	8.1313131	0	0	0
3582	0.587615112	8.1313131	0	0	0
3583	0.594869619	8.1313131	0	0	0
3584	0.602124127	8.1313131	0	0	0
3585	0.609378634	8.1313131	0	0	0
3586	0.616633142	8.1313131	0	0	0
3587	0.623887649	8.1313131	0	0	0
3588	0.631142157	8.1313131	0	0	0
3589	0.638396664	8.1313131	0	0	0
3590	0.645651172	8.1313131	0	0	0
3591	0.652905680	8.1313131	0	0	0
3592	0.660160187	8.1313131	0	0	0
3593	0.667414695	8.1313131	0	0	0
3594	0.674669202	8.1313131	0	0	0
3595	0.681923710	8.1313131	0	0	0
3596	0.689178217	8.1313131	0	0	0
3597	0.696432725	8.1313131	0	0	0
3598	0.703687232	8.1313131	0	0	0
3599	0.710941740	8.1313131	0	0	0
3600	0.718196248	8.1313131	0	0	0

3601	0.000000000	8.3636364	0	0	0
3602	0.007254508	8.3636364	0	0	0
3603	0.014509015	8.3636364	0	0	0
3604	0.021763523	8.3636364	0	0	0
3605	0.029018030	8.3636364	0	0	0
3606	0.036272538	8.3636364	0	0	0
3607	0.043527045	8.3636364	0	0	0
3608	0.050781553	8.3636364	0	0	0
3609	0.058036060	8.3636364	0	0	0
3610	0.065290568	8.3636364	0	0	0
3611	0.072545076	8.3636364	0	0	0
3612	0.079799583	8.3636364	0	0	0
3613	0.087054091	8.3636364	0	0	0
3614	0.094308598	8.3636364	0	0	0
3615	0.101563106	8.3636364	0	0	0
3616	0.108817613	8.3636364	0	0	0
3617	0.116072121	8.3636364	0	0	0
3618	0.123326628	8.3636364	0	0	0
3619	0.130581136	8.3636364	0	0	0
3620	0.137835643	8.3636364	0	0	0
3621	0.145090151	8.3636364	0	0	0
3622	0.152344659	8.3636364	0	0	0
3623	0.159599166	8.3636364	0	0	0
3624	0.166853674	8.3636364	0	0	0
3625	0.174108181	8.3636364	0	0	0
3626	0.181362689	8.3636364	0	0	0
3627	0.188617196	8.3636364	0	0	0
3628	0.195871704	8.3636364	0	0	0
3629	0.203126211	8.3636364	0	0	0
3630	0.210380719	8.3636364	0	0	0
3631	0.217635227	8.3636364	0	0	0
3632	0.224889734	8.3636364	0	0	0
3633	0.232144242	8.3636364	0	0	0
3634	0.239398749	8.3636364	0	0	0
3635	0.246653257	8.3636364	0	0	0
3636	0.253907764	8.3636364	0	0	0
3637	0.261162272	8.3636364	0	0	0
3638	0.268416779	8.3636364	0	0	0
3639	0.275671287	8.3636364	0	0	0
3640	0.282925794	8.3636364	0	0	0
3641	0.290180302	8.3636364	0	0	0
3642	0.297434810	8.3636364	0	0	0
3643	0.304689317	8.3636364	0	0	0
3644	0.311943825	8.3636364	0	0	0
3645	0.319198332	8.3636364	0	0	0

3646	0.326452840	8.3636364	0	0	0
3647	0.333707347	8.3636364	0	0	0
3648	0.340961855	8.3636364	0	0	0
3649	0.348216362	8.3636364	0	0	0
3650	0.355470870	8.3636364	0	0	0
3651	0.362725378	8.3636364	0	0	0
3652	0.369979885	8.3636364	0	0	0
3653	0.377234393	8.3636364	0	0	0
3654	0.384488900	8.3636364	0	0	0
3655	0.391743408	8.3636364	0	0	0
3656	0.398997915	8.3636364	0	0	0
3657	0.406252423	8.3636364	0	0	0
3658	0.413506930	8.3636364	0	0	0
3659	0.420761438	8.3636364	0	0	0
3660	0.428015945	8.3636364	0	0	0
3661	0.435270453	8.3636364	0	0	0
3662	0.442524961	8.3636364	0	0	0
3663	0.449779468	8.3636364	0	0	0
3664	0.457033976	8.3636364	0	0	0
3665	0.464288483	8.3636364	0	0	0
3666	0.471542991	8.3636364	0	0	0
3667	0.478797498	8.3636364	0	0	0
3668	0.486052006	8.3636364	0	0	0
3669	0.493306513	8.3636364	0	0	0
3670	0.500561021	8.3636364	0	0	0
3671	0.507815529	8.3636364	0	0	0
3672	0.515070036	8.3636364	0	0	0
3673	0.522324544	8.3636364	0	0	0
3674	0.529579051	8.3636364	0	0	0
3675	0.536833559	8.3636364	0	0	0
3676	0.544088066	8.3636364	0	0	0
3677	0.551342574	8.3636364	0	0	0
3678	0.558597081	8.3636364	0	0	0
3679	0.565851589	8.3636364	0	0	0
3680	0.573106097	8.3636364	0	0	0
3681	0.580360604	8.3636364	0	0	0
3682	0.587615112	8.3636364	0	0	0
3683	0.594869619	8.3636364	0	0	0
3684	0.602124127	8.3636364	0	0	0
3685	0.609378634	8.3636364	0	0	0
3686	0.616633142	8.3636364	0	0	0
3687	0.623887649	8.3636364	0	0	0
3688	0.631142157	8.3636364	0	0	0
3689	0.638396664	8.3636364	0	0	0
3690	0.645651172	8.3636364	0	0	0

3691	0.652905680	8.3636364	0	0	0
3692	0.660160187	8.3636364	0	0	0
3693	0.667414695	8.3636364	0	0	0
3694	0.674669202	8.3636364	0	0	0
3695	0.681923710	8.3636364	0	0	0
3696	0.689178217	8.3636364	0	0	0
3697	0.696432725	8.3636364	0	0	0
3698	0.703687232	8.3636364	0	0	0
3699	0.710941740	8.3636364	0	0	0
3700	0.718196248	8.3636364	0	0	0
3701	0.000000000	8.5959596	0	0	0
3702	0.007254508	8.5959596	0	0	0
3703	0.014509015	8.5959596	0	0	0
3704	0.021763523	8.5959596	0	0	0
3705	0.029018030	8.5959596	0	0	0
3706	0.036272538	8.5959596	0	0	0
3707	0.043527045	8.5959596	0	0	0
3708	0.050781553	8.5959596	0	0	0
3709	0.058036060	8.5959596	0	0	0
3710	0.065290568	8.5959596	0	0	0
3711	0.072545076	8.5959596	0	0	0
3712	0.079799583	8.5959596	0	0	0
3713	0.087054091	8.5959596	0	0	0
3714	0.094308598	8.5959596	0	0	0
3715	0.101563106	8.5959596	0	0	0
3716	0.108817613	8.5959596	0	0	0
3717	0.116072121	8.5959596	0	0	0
3718	0.123326628	8.5959596	0	0	0
3719	0.130581136	8.5959596	0	0	0
3720	0.137835643	8.5959596	0	0	0
3721	0.145090151	8.5959596	0	0	0
3722	0.152344659	8.5959596	0	0	0
3723	0.159599166	8.5959596	0	0	0
3724	0.166853674	8.5959596	0	0	0
3725	0.174108181	8.5959596	0	0	0
3726	0.181362689	8.5959596	0	0	0
3727	0.188617196	8.5959596	0	0	0
3728	0.195871704	8.5959596	0	0	0
3729	0.203126211	8.5959596	0	0	0
3730	0.210380719	8.5959596	0	0	0
3731	0.217635227	8.5959596	0	0	0
3732	0.224889734	8.5959596	0	0	0
3733	0.232144242	8.5959596	0	0	0
3734	0.239398749	8.5959596	0	0	0
3735	0.246653257	8.5959596	0	0	0

3736	0.253907764	8.5959596	0	0	0
3737	0.261162272	8.5959596	0	0	0
3738	0.268416779	8.5959596	0	0	0
3739	0.275671287	8.5959596	0	0	0
3740	0.282925794	8.5959596	0	0	0
3741	0.290180302	8.5959596	0	0	0
3742	0.297434810	8.5959596	0	0	0
3743	0.304689317	8.5959596	0	0	0
3744	0.311943825	8.5959596	0	0	0
3745	0.319198332	8.5959596	0	0	0
3746	0.326452840	8.5959596	0	0	0
3747	0.333707347	8.5959596	0	0	0
3748	0.340961855	8.5959596	0	0	0
3749	0.348216362	8.5959596	0	0	0
3750	0.355470870	8.5959596	0	0	0
3751	0.362725378	8.5959596	0	0	0
3752	0.369979885	8.5959596	0	0	0
3753	0.377234393	8.5959596	0	0	0
3754	0.384488900	8.5959596	0	0	0
3755	0.391743408	8.5959596	0	0	0
3756	0.398997915	8.5959596	0	0	0
3757	0.406252423	8.5959596	0	0	0
3758	0.413506930	8.5959596	0	0	0
3759	0.420761438	8.5959596	0	0	0
3760	0.428015945	8.5959596	0	0	0
3761	0.435270453	8.5959596	0	0	0
3762	0.442524961	8.5959596	0	0	0
3763	0.449779468	8.5959596	0	0	0
3764	0.457033976	8.5959596	0	0	0
3765	0.464288483	8.5959596	0	0	0
3766	0.471542991	8.5959596	0	0	0
3767	0.478797498	8.5959596	0	0	0
3768	0.486052006	8.5959596	0	0	0
3769	0.493306513	8.5959596	0	0	0
3770	0.500561021	8.5959596	0	0	0
3771	0.507815529	8.5959596	0	0	0
3772	0.515070036	8.5959596	0	0	0
3773	0.522324544	8.5959596	0	0	0
3774	0.529579051	8.5959596	0	0	0
3775	0.536833559	8.5959596	0	0	0
3776	0.544088066	8.5959596	0	0	0
3777	0.551342574	8.5959596	0	0	0
3778	0.558597081	8.5959596	0	0	0
3779	0.565851589	8.5959596	0	0	0
3780	0.573106097	8.5959596	0	0	0

3781	0.580360604	8.5959596	0	0	0
3782	0.587615112	8.5959596	0	0	0
3783	0.594869619	8.5959596	0	0	0
3784	0.602124127	8.5959596	0	0	0
3785	0.609378634	8.5959596	0	0	0
3786	0.616633142	8.5959596	0	0	0
3787	0.623887649	8.5959596	0	0	0
3788	0.631142157	8.5959596	0	0	0
3789	0.638396664	8.5959596	0	0	0
3790	0.645651172	8.5959596	0	0	0
3791	0.652905680	8.5959596	0	0	0
3792	0.660160187	8.5959596	0	0	0
3793	0.667414695	8.5959596	0	0	0
3794	0.674669202	8.5959596	0	0	0
3795	0.681923710	8.5959596	0	0	0
3796	0.689178217	8.5959596	0	0	0
3797	0.696432725	8.5959596	0	0	0
3798	0.703687232	8.5959596	0	0	0
3799	0.710941740	8.5959596	0	0	0
3800	0.718196248	8.5959596	0	0	0
3801	0.000000000	8.8282828	0	0	0
3802	0.007254508	8.8282828	0	0	0
3803	0.014509015	8.8282828	0	0	0
3804	0.021763523	8.8282828	0	0	0
3805	0.029018030	8.8282828	0	0	0
3806	0.036272538	8.8282828	0	0	0
3807	0.043527045	8.8282828	0	0	0
3808	0.050781553	8.8282828	0	0	0
3809	0.058036060	8.8282828	0	0	0
3810	0.065290568	8.8282828	0	0	0
3811	0.072545076	8.8282828	0	0	0
3812	0.079799583	8.8282828	0	0	0
3813	0.087054091	8.8282828	0	0	0
3814	0.094308598	8.8282828	0	0	0
3815	0.101563106	8.8282828	0	0	0
3816	0.108817613	8.8282828	0	0	0
3817	0.116072121	8.8282828	0	0	0
3818	0.123326628	8.8282828	0	0	0
3819	0.130581136	8.8282828	0	0	0
3820	0.137835643	8.8282828	0	0	0
3821	0.145090151	8.8282828	0	0	0
3822	0.152344659	8.8282828	0	0	0
3823	0.159599166	8.8282828	0	0	0
3824	0.166853674	8.8282828	0	0	0
3825	0.174108181	8.8282828	0	0	0

3826	0.181362689	8.8282828	0	0	0
3827	0.188617196	8.8282828	0	0	0
3828	0.195871704	8.8282828	0	0	0
3829	0.203126211	8.8282828	0	0	0
3830	0.210380719	8.8282828	0	0	0
3831	0.217635227	8.8282828	0	0	0
3832	0.224889734	8.8282828	0	0	0
3833	0.232144242	8.8282828	0	0	0
3834	0.239398749	8.8282828	0	0	0
3835	0.246653257	8.8282828	0	0	0
3836	0.253907764	8.8282828	0	0	0
3837	0.261162272	8.8282828	0	0	0
3838	0.268416779	8.8282828	0	0	0
3839	0.275671287	8.8282828	0	0	0
3840	0.282925794	8.8282828	0	0	0
3841	0.290180302	8.8282828	0	0	0
3842	0.297434810	8.8282828	0	0	0
3843	0.304689317	8.8282828	0	0	0
3844	0.311943825	8.8282828	0	0	0
3845	0.319198332	8.8282828	0	0	0
3846	0.326452840	8.8282828	0	0	0
3847	0.333707347	8.8282828	0	0	0
3848	0.340961855	8.8282828	0	0	0
3849	0.348216362	8.8282828	0	0	0
3850	0.355470870	8.8282828	0	0	0
3851	0.362725378	8.8282828	0	0	0
3852	0.369979885	8.8282828	0	0	0
3853	0.377234393	8.8282828	0	0	0
3854	0.384488900	8.8282828	0	0	0
3855	0.391743408	8.8282828	0	0	0
3856	0.398997915	8.8282828	0	0	0
3857	0.406252423	8.8282828	0	0	0
3858	0.413506930	8.8282828	0	0	0
3859	0.420761438	8.8282828	0	0	0
3860	0.428015945	8.8282828	0	0	0
3861	0.435270453	8.8282828	0	0	0
3862	0.442524961	8.8282828	0	0	0
3863	0.449779468	8.8282828	0	0	0
3864	0.457033976	8.8282828	0	0	0
3865	0.464288483	8.8282828	0	0	0
3866	0.471542991	8.8282828	0	0	0
3867	0.478797498	8.8282828	0	0	0
3868	0.486052006	8.8282828	0	0	0
3869	0.493306513	8.8282828	0	0	0
3870	0.500561021	8.8282828	0	0	0

3871	0.507815529	8.8282828	0	0	0
3872	0.515070036	8.8282828	0	0	0
3873	0.522324544	8.8282828	0	0	0
3874	0.529579051	8.8282828	0	0	0
3875	0.536833559	8.8282828	0	0	0
3876	0.544088066	8.8282828	0	0	0
3877	0.551342574	8.8282828	0	0	0
3878	0.558597081	8.8282828	0	0	0
3879	0.565851589	8.8282828	0	0	0
3880	0.573106097	8.8282828	0	0	0
3881	0.580360604	8.8282828	0	0	0
3882	0.587615112	8.8282828	0	0	0
3883	0.594869619	8.8282828	0	0	0
3884	0.602124127	8.8282828	0	0	0
3885	0.609378634	8.8282828	0	0	0
3886	0.616633142	8.8282828	0	0	0
3887	0.623887649	8.8282828	0	0	0
3888	0.631142157	8.8282828	0	0	0
3889	0.638396664	8.8282828	0	0	0
3890	0.645651172	8.8282828	0	0	0
3891	0.652905680	8.8282828	0	0	0
3892	0.660160187	8.8282828	0	0	0
3893	0.667414695	8.8282828	0	0	0
3894	0.674669202	8.8282828	0	0	0
3895	0.681923710	8.8282828	0	0	0
3896	0.689178217	8.8282828	0	0	0
3897	0.696432725	8.8282828	0	0	0
3898	0.703687232	8.8282828	0	0	0
3899	0.710941740	8.8282828	0	0	0
3900	0.718196248	8.8282828	0	0	0
3901	0.000000000	9.0606061	0	0	0
3902	0.007254508	9.0606061	0	0	0
3903	0.014509015	9.0606061	0	0	0
3904	0.021763523	9.0606061	0	0	0
3905	0.029018030	9.0606061	0	0	0
3906	0.036272538	9.0606061	0	0	0
3907	0.043527045	9.0606061	0	0	0
3908	0.050781553	9.0606061	0	0	0
3909	0.058036060	9.0606061	0	0	0
3910	0.065290568	9.0606061	0	0	0
3911	0.072545076	9.0606061	0	0	0
3912	0.079799583	9.0606061	0	0	0
3913	0.087054091	9.0606061	0	0	0
3914	0.094308598	9.0606061	0	0	0
3915	0.101563106	9.0606061	0	0	0

3916	0.108817613	9.0606061	0	0	0
3917	0.116072121	9.0606061	0	0	0
3918	0.123326628	9.0606061	0	0	0
3919	0.130581136	9.0606061	0	0	0
3920	0.137835643	9.0606061	0	0	0
3921	0.145090151	9.0606061	0	0	0
3922	0.152344659	9.0606061	0	0	0
3923	0.159599166	9.0606061	0	0	0
3924	0.166853674	9.0606061	0	0	0
3925	0.174108181	9.0606061	0	0	0
3926	0.181362689	9.0606061	0	0	0
3927	0.188617196	9.0606061	0	0	0
3928	0.195871704	9.0606061	0	0	0
3929	0.203126211	9.0606061	0	0	0
3930	0.210380719	9.0606061	0	0	0
3931	0.217635227	9.0606061	0	0	0
3932	0.224889734	9.0606061	0	0	0
3933	0.232144242	9.0606061	0	0	0
3934	0.239398749	9.0606061	0	0	0
3935	0.246653257	9.0606061	0	0	0
3936	0.253907764	9.0606061	0	0	0
3937	0.261162272	9.0606061	0	0	0
3938	0.268416779	9.0606061	0	0	0
3939	0.275671287	9.0606061	0	0	0
3940	0.282925794	9.0606061	0	0	0
3941	0.290180302	9.0606061	0	0	0
3942	0.297434810	9.0606061	0	0	0
3943	0.304689317	9.0606061	0	0	0
3944	0.311943825	9.0606061	0	0	0
3945	0.319198332	9.0606061	0	0	0
3946	0.326452840	9.0606061	0	0	0
3947	0.333707347	9.0606061	0	0	0
3948	0.340961855	9.0606061	0	0	0
3949	0.348216362	9.0606061	0	0	0
3950	0.355470870	9.0606061	0	0	0
3951	0.362725378	9.0606061	0	0	0
3952	0.369979885	9.0606061	0	0	0
3953	0.377234393	9.0606061	0	0	0
3954	0.384488900	9.0606061	0	0	0
3955	0.391743408	9.0606061	0	0	0
3956	0.398997915	9.0606061	0	0	0
3957	0.406252423	9.0606061	0	0	0
3958	0.413506930	9.0606061	0	0	0
3959	0.420761438	9.0606061	0	0	0
3960	0.428015945	9.0606061	0	0	0

3961	0.435270453	9.0606061	0	0	0
3962	0.442524961	9.0606061	0	0	0
3963	0.449779468	9.0606061	0	0	0
3964	0.457033976	9.0606061	0	0	0
3965	0.464288483	9.0606061	0	0	0
3966	0.471542991	9.0606061	0	0	0
3967	0.478797498	9.0606061	0	0	0
3968	0.486052006	9.0606061	0	0	0
3969	0.493306513	9.0606061	0	0	0
3970	0.500561021	9.0606061	0	0	0
3971	0.507815529	9.0606061	0	0	0
3972	0.515070036	9.0606061	0	0	0
3973	0.522324544	9.0606061	0	0	0
3974	0.529579051	9.0606061	0	0	0
3975	0.536833559	9.0606061	0	0	0
3976	0.544088066	9.0606061	0	0	0
3977	0.551342574	9.0606061	0	0	0
3978	0.558597081	9.0606061	0	0	0
3979	0.565851589	9.0606061	0	0	0
3980	0.573106097	9.0606061	0	0	0
3981	0.580360604	9.0606061	0	0	0
3982	0.587615112	9.0606061	0	0	0
3983	0.594869619	9.0606061	0	0	0
3984	0.602124127	9.0606061	0	0	0
3985	0.609378634	9.0606061	0	0	0
3986	0.616633142	9.0606061	0	0	0
3987	0.623887649	9.0606061	0	0	0
3988	0.631142157	9.0606061	0	0	0
3989	0.638396664	9.0606061	0	0	0
3990	0.645651172	9.0606061	0	0	0
3991	0.652905680	9.0606061	0	0	0
3992	0.660160187	9.0606061	0	0	0
3993	0.667414695	9.0606061	0	0	0
3994	0.674669202	9.0606061	0	0	0
3995	0.681923710	9.0606061	0	0	0
3996	0.689178217	9.0606061	0	0	0
3997	0.696432725	9.0606061	0	0	0
3998	0.703687232	9.0606061	0	0	0
3999	0.710941740	9.0606061	0	0	0
4000	0.718196248	9.0606061	0	0	0
4001	0.000000000	9.2929293	0	0	0
4002	0.007254508	9.2929293	0	0	0
4003	0.014509015	9.2929293	0	0	0
4004	0.021763523	9.2929293	0	0	0
4005	0.029018030	9.2929293	0	0	0

4006	0.036272538	9.2929293	0	0	0
4007	0.043527045	9.2929293	0	0	0
4008	0.050781553	9.2929293	0	0	0
4009	0.058036060	9.2929293	0	0	0
4010	0.065290568	9.2929293	0	0	0
4011	0.072545076	9.2929293	0	0	0
4012	0.079799583	9.2929293	0	0	0
4013	0.087054091	9.2929293	0	0	0
4014	0.094308598	9.2929293	0	0	0
4015	0.101563106	9.2929293	0	0	0
4016	0.108817613	9.2929293	0	0	0
4017	0.116072121	9.2929293	0	0	0
4018	0.123326628	9.2929293	0	0	0
4019	0.130581136	9.2929293	0	0	0
4020	0.137835643	9.2929293	0	0	0
4021	0.145090151	9.2929293	0	0	0
4022	0.152344659	9.2929293	0	0	0
4023	0.159599166	9.2929293	0	0	0
4024	0.166853674	9.2929293	0	0	0
4025	0.174108181	9.2929293	0	0	0
4026	0.181362689	9.2929293	0	0	0
4027	0.188617196	9.2929293	0	0	0
4028	0.195871704	9.2929293	0	0	0
4029	0.203126211	9.2929293	0	0	0
4030	0.210380719	9.2929293	0	0	0
4031	0.217635227	9.2929293	0	0	0
4032	0.224889734	9.2929293	0	0	0
4033	0.232144242	9.2929293	0	0	0
4034	0.239398749	9.2929293	0	0	0
4035	0.246653257	9.2929293	0	0	0
4036	0.253907764	9.2929293	0	0	0
4037	0.261162272	9.2929293	0	0	0
4038	0.268416779	9.2929293	0	0	0
4039	0.275671287	9.2929293	0	0	0
4040	0.282925794	9.2929293	0	0	0
4041	0.290180302	9.2929293	0	0	0
4042	0.297434810	9.2929293	0	0	0
4043	0.304689317	9.2929293	0	0	0
4044	0.311943825	9.2929293	0	0	0
4045	0.319198332	9.2929293	0	0	0
4046	0.326452840	9.2929293	0	0	0
4047	0.333707347	9.2929293	0	0	0
4048	0.340961855	9.2929293	0	0	0
4049	0.348216362	9.2929293	0	0	0
4050	0.355470870	9.2929293	0	0	0

4051	0.362725378	9.2929293	0	0	0
4052	0.369979885	9.2929293	0	0	0
4053	0.377234393	9.2929293	0	0	0
4054	0.384488900	9.2929293	0	0	0
4055	0.391743408	9.2929293	0	0	0
4056	0.398997915	9.2929293	0	0	0
4057	0.406252423	9.2929293	0	0	0
4058	0.413506930	9.2929293	0	0	0
4059	0.420761438	9.2929293	0	0	0
4060	0.428015945	9.2929293	0	0	0
4061	0.435270453	9.2929293	0	0	0
4062	0.442524961	9.2929293	0	0	0
4063	0.449779468	9.2929293	0	0	0
4064	0.457033976	9.2929293	0	0	0
4065	0.464288483	9.2929293	0	0	0
4066	0.471542991	9.2929293	0	0	0
4067	0.478797498	9.2929293	0	0	0
4068	0.486052006	9.2929293	0	0	0
4069	0.493306513	9.2929293	0	0	0
4070	0.500561021	9.2929293	0	0	0
4071	0.507815529	9.2929293	0	0	0
4072	0.515070036	9.2929293	0	0	0
4073	0.522324544	9.2929293	0	0	0
4074	0.529579051	9.2929293	0	0	0
4075	0.536833559	9.2929293	0	0	0
4076	0.544088066	9.2929293	0	0	0
4077	0.551342574	9.2929293	0	0	0
4078	0.558597081	9.2929293	0	0	0
4079	0.565851589	9.2929293	0	0	0
4080	0.573106097	9.2929293	0	0	0
4081	0.580360604	9.2929293	0	0	0
4082	0.587615112	9.2929293	0	0	0
4083	0.594869619	9.2929293	0	0	0
4084	0.602124127	9.2929293	0	0	0
4085	0.609378634	9.2929293	0	0	0
4086	0.616633142	9.2929293	0	0	0
4087	0.623887649	9.2929293	0	0	0
4088	0.631142157	9.2929293	0	0	0
4089	0.638396664	9.2929293	0	0	0
4090	0.645651172	9.2929293	0	0	0
4091	0.652905680	9.2929293	0	0	0
4092	0.660160187	9.2929293	0	0	0
4093	0.667414695	9.2929293	0	0	0
4094	0.674669202	9.2929293	0	0	0
4095	0.681923710	9.2929293	0	0	0

4096	0.689178217	9.2929293	0	0	0
4097	0.696432725	9.2929293	0	0	0
4098	0.703687232	9.2929293	0	0	0
4099	0.710941740	9.2929293	0	0	0
4100	0.718196248	9.2929293	0	0	0
4101	0.000000000	9.5252525	0	0	0
4102	0.007254508	9.5252525	0	0	0
4103	0.014509015	9.5252525	0	0	0
4104	0.021763523	9.5252525	0	0	0
4105	0.029018030	9.5252525	0	0	0
4106	0.036272538	9.5252525	0	0	0
4107	0.043527045	9.5252525	0	0	0
4108	0.050781553	9.5252525	0	0	0
4109	0.058036060	9.5252525	0	0	0
4110	0.065290568	9.5252525	0	0	0
4111	0.072545076	9.5252525	0	0	0
4112	0.079799583	9.5252525	0	0	0
4113	0.087054091	9.5252525	0	0	0
4114	0.094308598	9.5252525	0	0	0
4115	0.101563106	9.5252525	0	0	0
4116	0.108817613	9.5252525	0	0	0
4117	0.116072121	9.5252525	0	0	0
4118	0.123326628	9.5252525	0	0	0
4119	0.130581136	9.5252525	0	0	0
4120	0.137835643	9.5252525	0	0	0
4121	0.145090151	9.5252525	0	0	0
4122	0.152344659	9.5252525	0	0	0
4123	0.159599166	9.5252525	0	0	0
4124	0.166853674	9.5252525	0	0	0
4125	0.174108181	9.5252525	0	0	0
4126	0.181362689	9.5252525	0	0	0
4127	0.188617196	9.5252525	0	0	0
4128	0.195871704	9.5252525	0	0	0
4129	0.203126211	9.5252525	0	0	0
4130	0.210380719	9.5252525	0	0	0
4131	0.217635227	9.5252525	0	0	0
4132	0.224889734	9.5252525	0	0	0
4133	0.232144242	9.5252525	0	0	0
4134	0.239398749	9.5252525	0	0	0
4135	0.246653257	9.5252525	0	0	0
4136	0.253907764	9.5252525	0	0	0
4137	0.261162272	9.5252525	0	0	0
4138	0.268416779	9.5252525	0	0	0
4139	0.275671287	9.5252525	0	0	0
4140	0.282925794	9.5252525	0	0	0

4141	0.290180302	9.5252525	0	0	0
4142	0.297434810	9.5252525	0	0	0
4143	0.304689317	9.5252525	0	0	0
4144	0.311943825	9.5252525	0	0	0
4145	0.319198332	9.5252525	0	0	0
4146	0.326452840	9.5252525	0	0	0
4147	0.333707347	9.5252525	0	0	0
4148	0.340961855	9.5252525	0	0	0
4149	0.348216362	9.5252525	0	0	0
4150	0.355470870	9.5252525	0	0	0
4151	0.362725378	9.5252525	0	0	0
4152	0.369979885	9.5252525	0	0	0
4153	0.377234393	9.5252525	0	0	0
4154	0.384488900	9.5252525	0	0	0
4155	0.391743408	9.5252525	0	0	0
4156	0.398997915	9.5252525	0	0	0
4157	0.406252423	9.5252525	0	0	0
4158	0.413506930	9.5252525	0	0	0
4159	0.420761438	9.5252525	0	0	0
4160	0.428015945	9.5252525	0	0	0
4161	0.435270453	9.5252525	0	0	0
4162	0.442524961	9.5252525	0	0	0
4163	0.449779468	9.5252525	0	0	0
4164	0.457033976	9.5252525	0	0	0
4165	0.464288483	9.5252525	0	0	0
4166	0.471542991	9.5252525	0	0	0
4167	0.478797498	9.5252525	0	0	0
4168	0.486052006	9.5252525	0	0	0
4169	0.493306513	9.5252525	0	0	0
4170	0.500561021	9.5252525	0	0	0
4171	0.507815529	9.5252525	0	0	0
4172	0.515070036	9.5252525	0	0	0
4173	0.522324544	9.5252525	0	0	0
4174	0.529579051	9.5252525	0	0	0
4175	0.536833559	9.5252525	0	0	0
4176	0.544088066	9.5252525	0	0	0
4177	0.551342574	9.5252525	0	0	0
4178	0.558597081	9.5252525	0	0	0
4179	0.565851589	9.5252525	0	0	0
4180	0.573106097	9.5252525	0	0	0
4181	0.580360604	9.5252525	0	0	0
4182	0.587615112	9.5252525	0	0	0
4183	0.594869619	9.5252525	0	0	0
4184	0.602124127	9.5252525	0	0	0
4185	0.609378634	9.5252525	0	0	0

4186	0.616633142	9.5252525	0	0	0
4187	0.623887649	9.5252525	0	0	0
4188	0.631142157	9.5252525	0	0	0
4189	0.638396664	9.5252525	0	0	0
4190	0.645651172	9.5252525	0	0	0
4191	0.652905680	9.5252525	0	0	0
4192	0.660160187	9.5252525	0	0	0
4193	0.667414695	9.5252525	0	0	0
4194	0.674669202	9.5252525	0	0	0
4195	0.681923710	9.5252525	0	0	0
4196	0.689178217	9.5252525	0	0	0
4197	0.696432725	9.5252525	0	0	0
4198	0.703687232	9.5252525	0	0	0
4199	0.710941740	9.5252525	0	0	0
4200	0.718196248	9.5252525	0	0	0
4201	0.000000000	9.7575758	0	0	0
4202	0.007254508	9.7575758	0	0	0
4203	0.014509015	9.7575758	0	0	0
4204	0.021763523	9.7575758	0	0	0
4205	0.029018030	9.7575758	0	0	0
4206	0.036272538	9.7575758	0	0	0
4207	0.043527045	9.7575758	0	0	0
4208	0.050781553	9.7575758	0	0	0
4209	0.058036060	9.7575758	0	0	0
4210	0.065290568	9.7575758	0	0	0
4211	0.072545076	9.7575758	0	0	0
4212	0.079799583	9.7575758	0	0	0
4213	0.087054091	9.7575758	0	0	0
4214	0.094308598	9.7575758	0	0	0
4215	0.101563106	9.7575758	0	0	0
4216	0.108817613	9.7575758	0	0	0
4217	0.116072121	9.7575758	0	0	0
4218	0.123326628	9.7575758	0	0	0
4219	0.130581136	9.7575758	0	0	0
4220	0.137835643	9.7575758	0	0	0
4221	0.145090151	9.7575758	0	0	0
4222	0.152344659	9.7575758	0	0	0
4223	0.159599166	9.7575758	0	0	0
4224	0.166853674	9.7575758	0	0	0
4225	0.174108181	9.7575758	0	0	0
4226	0.181362689	9.7575758	0	0	0
4227	0.188617196	9.7575758	0	0	0
4228	0.195871704	9.7575758	0	0	0
4229	0.203126211	9.7575758	0	0	0
4230	0.210380719	9.7575758	0	0	0

4231	0.217635227	9.7575758	0	0	0
4232	0.224889734	9.7575758	0	0	0
4233	0.232144242	9.7575758	0	0	0
4234	0.239398749	9.7575758	0	0	0
4235	0.246653257	9.7575758	0	0	0
4236	0.253907764	9.7575758	0	0	0
4237	0.261162272	9.7575758	0	0	0
4238	0.268416779	9.7575758	0	0	0
4239	0.275671287	9.7575758	0	0	0
4240	0.282925794	9.7575758	0	0	0
4241	0.290180302	9.7575758	0	0	0
4242	0.297434810	9.7575758	0	0	0
4243	0.304689317	9.7575758	0	0	0
4244	0.311943825	9.7575758	0	0	0
4245	0.319198332	9.7575758	0	0	0
4246	0.326452840	9.7575758	0	0	0
4247	0.333707347	9.7575758	0	0	0
4248	0.340961855	9.7575758	0	0	0
4249	0.348216362	9.7575758	0	0	0
4250	0.355470870	9.7575758	0	0	0
4251	0.362725378	9.7575758	0	0	0
4252	0.369979885	9.7575758	0	0	0
4253	0.377234393	9.7575758	0	0	0
4254	0.384488900	9.7575758	0	0	0
4255	0.391743408	9.7575758	0	0	0
4256	0.398997915	9.7575758	0	0	0
4257	0.406252423	9.7575758	0	0	0
4258	0.413506930	9.7575758	0	0	0
4259	0.420761438	9.7575758	0	0	0
4260	0.428015945	9.7575758	0	0	0
4261	0.435270453	9.7575758	0	0	0
4262	0.442524961	9.7575758	0	0	0
4263	0.449779468	9.7575758	0	0	0
4264	0.457033976	9.7575758	0	0	0
4265	0.464288483	9.7575758	0	0	0
4266	0.471542991	9.7575758	0	0	0
4267	0.478797498	9.7575758	0	0	0
4268	0.486052006	9.7575758	0	0	0
4269	0.493306513	9.7575758	0	0	0
4270	0.500561021	9.7575758	0	0	0
4271	0.507815529	9.7575758	0	0	0
4272	0.515070036	9.7575758	0	0	0
4273	0.522324544	9.7575758	0	0	0
4274	0.529579051	9.7575758	0	0	0
4275	0.536833559	9.7575758	0	0	0

4276	0.544088066	9.7575758	0	0	0
4277	0.551342574	9.7575758	0	0	0
4278	0.558597081	9.7575758	0	0	0
4279	0.565851589	9.7575758	0	0	0
4280	0.573106097	9.7575758	0	0	0
4281	0.580360604	9.7575758	0	0	0
4282	0.587615112	9.7575758	0	0	0
4283	0.594869619	9.7575758	0	0	0
4284	0.602124127	9.7575758	0	0	0
4285	0.609378634	9.7575758	0	0	0
4286	0.616633142	9.7575758	0	0	0
4287	0.623887649	9.7575758	0	0	0
4288	0.631142157	9.7575758	0	0	0
4289	0.638396664	9.7575758	0	0	0
4290	0.645651172	9.7575758	0	0	0
4291	0.652905680	9.7575758	0	0	0
4292	0.660160187	9.7575758	0	0	0
4293	0.667414695	9.7575758	0	0	0
4294	0.674669202	9.7575758	0	0	0
4295	0.681923710	9.7575758	0	0	0
4296	0.689178217	9.7575758	0	0	0
4297	0.696432725	9.7575758	0	0	0
4298	0.703687232	9.7575758	0	0	0
4299	0.710941740	9.7575758	0	0	0
4300	0.718196248	9.7575758	0	0	0
4301	0.000000000	9.9898990	0	0	0
4302	0.007254508	9.9898990	0	0	0
4303	0.014509015	9.9898990	0	0	0
4304	0.021763523	9.9898990	0	0	0
4305	0.029018030	9.9898990	0	0	0
4306	0.036272538	9.9898990	0	0	0
4307	0.043527045	9.9898990	0	0	0
4308	0.050781553	9.9898990	0	0	0
4309	0.058036060	9.9898990	0	0	0
4310	0.065290568	9.9898990	0	0	0
4311	0.072545076	9.9898990	0	0	0
4312	0.079799583	9.9898990	0	0	0
4313	0.087054091	9.9898990	0	0	0
4314	0.094308598	9.9898990	0	0	0
4315	0.101563106	9.9898990	0	0	0
4316	0.108817613	9.9898990	0	0	0
4317	0.116072121	9.9898990	0	0	0
4318	0.123326628	9.9898990	0	0	0
4319	0.130581136	9.9898990	0	0	0
4320	0.137835643	9.9898990	0	0	0

4321	0.145090151	9.9898990	0	0	0
4322	0.152344659	9.9898990	0	0	0
4323	0.159599166	9.9898990	0	0	0
4324	0.166853674	9.9898990	0	0	0
4325	0.174108181	9.9898990	0	0	0
4326	0.181362689	9.9898990	0	0	0
4327	0.188617196	9.9898990	0	0	0
4328	0.195871704	9.9898990	0	0	0
4329	0.203126211	9.9898990	0	0	0
4330	0.210380719	9.9898990	0	0	0
4331	0.217635227	9.9898990	0	0	0
4332	0.224889734	9.9898990	0	0	0
4333	0.232144242	9.9898990	0	0	0
4334	0.239398749	9.9898990	0	0	0
4335	0.246653257	9.9898990	0	0	0
4336	0.253907764	9.9898990	0	0	0
4337	0.261162272	9.9898990	0	0	0
4338	0.268416779	9.9898990	0	0	0
4339	0.275671287	9.9898990	0	0	0
4340	0.282925794	9.9898990	0	0	0
4341	0.290180302	9.9898990	0	0	0
4342	0.297434810	9.9898990	0	0	0
4343	0.304689317	9.9898990	0	0	0
4344	0.311943825	9.9898990	0	0	0
4345	0.319198332	9.9898990	0	0	0
4346	0.326452840	9.9898990	0	0	0
4347	0.333707347	9.9898990	0	0	0
4348	0.340961855	9.9898990	0	0	0
4349	0.348216362	9.9898990	0	0	0
4350	0.355470870	9.9898990	0	0	0
4351	0.362725378	9.9898990	0	0	0
4352	0.369979885	9.9898990	0	0	0
4353	0.377234393	9.9898990	0	0	0
4354	0.384488900	9.9898990	0	0	0
4355	0.391743408	9.9898990	0	0	0
4356	0.398997915	9.9898990	0	0	0
4357	0.406252423	9.9898990	0	0	0
4358	0.413506930	9.9898990	0	0	0
4359	0.420761438	9.9898990	0	0	0
4360	0.428015945	9.9898990	0	0	0
4361	0.435270453	9.9898990	0	0	0
4362	0.442524961	9.9898990	0	0	0
4363	0.449779468	9.9898990	0	0	0
4364	0.457033976	9.9898990	0	0	0
4365	0.464288483	9.9898990	0	0	0

4366	0.471542991	9.9898990	0	0	0
4367	0.478797498	9.9898990	0	0	0
4368	0.486052006	9.9898990	0	0	0
4369	0.493306513	9.9898990	0	0	0
4370	0.500561021	9.9898990	0	0	0
4371	0.507815529	9.9898990	0	0	0
4372	0.515070036	9.9898990	0	0	0
4373	0.522324544	9.9898990	0	0	0
4374	0.529579051	9.9898990	0	0	0
4375	0.536833559	9.9898990	0	0	0
4376	0.544088066	9.9898990	0	0	0
4377	0.551342574	9.9898990	0	0	0
4378	0.558597081	9.9898990	0	0	0
4379	0.565851589	9.9898990	0	0	0
4380	0.573106097	9.9898990	0	0	0
4381	0.580360604	9.9898990	0	0	0
4382	0.587615112	9.9898990	0	0	0
4383	0.594869619	9.9898990	0	0	0
4384	0.602124127	9.9898990	0	0	0
4385	0.609378634	9.9898990	0	0	0
4386	0.616633142	9.9898990	0	0	0
4387	0.623887649	9.9898990	0	0	0
4388	0.631142157	9.9898990	0	0	0
4389	0.638396664	9.9898990	0	0	0
4390	0.645651172	9.9898990	0	0	0
4391	0.652905680	9.9898990	0	0	0
4392	0.660160187	9.9898990	0	0	0
4393	0.667414695	9.9898990	0	0	0
4394	0.674669202	9.9898990	0	0	0
4395	0.681923710	9.9898990	0	0	0
4396	0.689178217	9.9898990	0	0	0
4397	0.696432725	9.9898990	0	0	0
4398	0.703687232	9.9898990	0	0	0
4399	0.710941740	9.9898990	0	0	0
4400	0.718196248	9.9898990	0	0	0
4401	0.000000000	10.2222222	0	0	0
4402	0.007254508	10.2222222	0	0	0
4403	0.014509015	10.2222222	0	0	0
4404	0.021763523	10.2222222	0	0	0
4405	0.029018030	10.2222222	0	0	0
4406	0.036272538	10.2222222	0	0	0
4407	0.043527045	10.2222222	0	0	0
4408	0.050781553	10.2222222	0	0	0
4409	0.058036060	10.2222222	0	0	0
4410	0.065290568	10.2222222	0	0	0

4411	0.072545076	10.2222222	0	0	0
4412	0.079799583	10.2222222	0	0	0
4413	0.087054091	10.2222222	0	0	0
4414	0.094308598	10.2222222	0	0	0
4415	0.101563106	10.2222222	0	0	0
4416	0.108817613	10.2222222	0	0	0
4417	0.116072121	10.2222222	0	0	0
4418	0.123326628	10.2222222	0	0	0
4419	0.130581136	10.2222222	0	0	0
4420	0.137835643	10.2222222	0	0	0
4421	0.145090151	10.2222222	0	0	0
4422	0.152344659	10.2222222	0	0	0
4423	0.159599166	10.2222222	0	0	0
4424	0.166853674	10.2222222	0	0	0
4425	0.174108181	10.2222222	0	0	0
4426	0.181362689	10.2222222	0	0	0
4427	0.188617196	10.2222222	0	0	0
4428	0.195871704	10.2222222	0	0	0
4429	0.203126211	10.2222222	0	0	0
4430	0.210380719	10.2222222	0	0	0
4431	0.217635227	10.2222222	0	0	0
4432	0.224889734	10.2222222	0	0	0
4433	0.232144242	10.2222222	0	0	0
4434	0.239398749	10.2222222	0	0	0
4435	0.246653257	10.2222222	0	0	0
4436	0.253907764	10.2222222	0	0	0
4437	0.261162272	10.2222222	0	0	0
4438	0.268416779	10.2222222	0	0	0
4439	0.275671287	10.2222222	0	0	0
4440	0.282925794	10.2222222	0	0	0
4441	0.290180302	10.2222222	0	0	0
4442	0.297434810	10.2222222	0	0	0
4443	0.304689317	10.2222222	0	0	0
4444	0.311943825	10.2222222	0	0	0
4445	0.319198332	10.2222222	0	0	0
4446	0.326452840	10.2222222	0	0	0
4447	0.333707347	10.2222222	0	0	0
4448	0.340961855	10.2222222	0	0	0
4449	0.348216362	10.2222222	0	0	0
4450	0.355470870	10.2222222	0	0	0
4451	0.362725378	10.2222222	0	0	0
4452	0.369979885	10.2222222	0	0	0
4453	0.377234393	10.2222222	0	0	0
4454	0.384488900	10.2222222	0	0	0
4455	0.391743408	10.2222222	0	0	0

4456	0.398997915	10.2222222	0	0	0
4457	0.406252423	10.2222222	0	0	0
4458	0.413506930	10.2222222	0	0	0
4459	0.420761438	10.2222222	0	0	0
4460	0.428015945	10.2222222	0	0	0
4461	0.435270453	10.2222222	0	0	0
4462	0.442524961	10.2222222	0	0	0
4463	0.449779468	10.2222222	0	0	0
4464	0.457033976	10.2222222	0	0	0
4465	0.464288483	10.2222222	0	0	0
4466	0.471542991	10.2222222	0	0	0
4467	0.478797498	10.2222222	0	0	0
4468	0.486052006	10.2222222	0	0	0
4469	0.493306513	10.2222222	0	0	0
4470	0.500561021	10.2222222	0	0	0
4471	0.507815529	10.2222222	0	0	0
4472	0.515070036	10.2222222	0	0	0
4473	0.522324544	10.2222222	0	0	0
4474	0.529579051	10.2222222	0	0	0
4475	0.536833559	10.2222222	0	0	0
4476	0.544088066	10.2222222	0	0	0
4477	0.551342574	10.2222222	0	0	0
4478	0.558597081	10.2222222	0	0	0
4479	0.565851589	10.2222222	0	0	0
4480	0.573106097	10.2222222	0	0	0
4481	0.580360604	10.2222222	0	0	0
4482	0.587615112	10.2222222	0	0	0
4483	0.594869619	10.2222222	0	0	0
4484	0.602124127	10.2222222	0	0	0
4485	0.609378634	10.2222222	0	0	0
4486	0.616633142	10.2222222	0	0	0
4487	0.623887649	10.2222222	0	0	0
4488	0.631142157	10.2222222	0	0	0
4489	0.638396664	10.2222222	0	0	0
4490	0.645651172	10.2222222	0	0	0
4491	0.652905680	10.2222222	0	0	0
4492	0.660160187	10.2222222	0	0	0
4493	0.667414695	10.2222222	0	0	0
4494	0.674669202	10.2222222	0	0	0
4495	0.681923710	10.2222222	0	0	0
4496	0.689178217	10.2222222	0	0	0
4497	0.696432725	10.2222222	0	0	0
4498	0.703687232	10.2222222	0	0	0
4499	0.710941740	10.2222222	0	0	0
4500	0.718196248	10.2222222	0	0	0

4501	0.000000000	10.4545455	0	0	0
4502	0.007254508	10.4545455	0	0	0
4503	0.014509015	10.4545455	0	0	0
4504	0.021763523	10.4545455	0	0	0
4505	0.029018030	10.4545455	0	0	0
4506	0.036272538	10.4545455	0	0	0
4507	0.043527045	10.4545455	0	0	0
4508	0.050781553	10.4545455	0	0	0
4509	0.058036060	10.4545455	0	0	0
4510	0.065290568	10.4545455	0	0	0
4511	0.072545076	10.4545455	0	0	0
4512	0.079799583	10.4545455	0	0	0
4513	0.087054091	10.4545455	0	0	0
4514	0.094308598	10.4545455	0	0	0
4515	0.101563106	10.4545455	0	0	0
4516	0.108817613	10.4545455	0	0	0
4517	0.116072121	10.4545455	0	0	0
4518	0.123326628	10.4545455	0	0	0
4519	0.130581136	10.4545455	0	0	0
4520	0.137835643	10.4545455	0	0	0
4521	0.145090151	10.4545455	0	0	0
4522	0.152344659	10.4545455	0	0	0
4523	0.159599166	10.4545455	0	0	0
4524	0.166853674	10.4545455	0	0	0
4525	0.174108181	10.4545455	0	0	0
4526	0.181362689	10.4545455	0	0	0
4527	0.188617196	10.4545455	0	0	0
4528	0.195871704	10.4545455	0	0	0
4529	0.203126211	10.4545455	0	0	0
4530	0.210380719	10.4545455	0	0	0
4531	0.217635227	10.4545455	0	0	0
4532	0.224889734	10.4545455	0	0	0
4533	0.232144242	10.4545455	0	0	0
4534	0.239398749	10.4545455	0	0	0
4535	0.246653257	10.4545455	0	0	0
4536	0.253907764	10.4545455	0	0	0
4537	0.261162272	10.4545455	0	0	0
4538	0.268416779	10.4545455	0	0	0
4539	0.275671287	10.4545455	0	0	0
4540	0.282925794	10.4545455	0	0	0
4541	0.290180302	10.4545455	0	0	0
4542	0.297434810	10.4545455	0	0	0
4543	0.304689317	10.4545455	0	0	0
4544	0.311943825	10.4545455	0	0	0
4545	0.319198332	10.4545455	0	0	0

4546	0.326452840	10.4545455	0	0	0
4547	0.333707347	10.4545455	0	0	0
4548	0.340961855	10.4545455	0	0	0
4549	0.348216362	10.4545455	0	0	0
4550	0.355470870	10.4545455	0	0	0
4551	0.362725378	10.4545455	0	0	0
4552	0.369979885	10.4545455	0	0	0
4553	0.377234393	10.4545455	0	0	0
4554	0.384488900	10.4545455	0	0	0
4555	0.391743408	10.4545455	0	0	0
4556	0.398997915	10.4545455	0	0	0
4557	0.406252423	10.4545455	0	0	0
4558	0.413506930	10.4545455	0	0	0
4559	0.420761438	10.4545455	0	0	0
4560	0.428015945	10.4545455	0	0	0
4561	0.435270453	10.4545455	0	0	0
4562	0.442524961	10.4545455	0	0	0
4563	0.449779468	10.4545455	0	0	0
4564	0.457033976	10.4545455	0	0	0
4565	0.464288483	10.4545455	0	0	0
4566	0.471542991	10.4545455	0	0	0
4567	0.478797498	10.4545455	0	0	0
4568	0.486052006	10.4545455	0	0	0
4569	0.493306513	10.4545455	0	0	0
4570	0.500561021	10.4545455	0	0	0
4571	0.507815529	10.4545455	0	0	0
4572	0.515070036	10.4545455	0	0	0
4573	0.522324544	10.4545455	0	0	0
4574	0.529579051	10.4545455	0	0	0
4575	0.536833559	10.4545455	0	0	0
4576	0.544088066	10.4545455	0	0	0
4577	0.551342574	10.4545455	0	0	0
4578	0.558597081	10.4545455	0	0	0
4579	0.565851589	10.4545455	0	0	0
4580	0.573106097	10.4545455	0	0	0
4581	0.580360604	10.4545455	0	0	0
4582	0.587615112	10.4545455	0	0	0
4583	0.594869619	10.4545455	0	0	0
4584	0.602124127	10.4545455	0	0	0
4585	0.609378634	10.4545455	0	0	0
4586	0.616633142	10.4545455	0	0	0
4587	0.623887649	10.4545455	0	0	0
4588	0.631142157	10.4545455	0	0	0
4589	0.638396664	10.4545455	0	0	0
4590	0.645651172	10.4545455	0	0	0

4591	0.652905680	10.4545455	0	0	0
4592	0.660160187	10.4545455	0	0	0
4593	0.667414695	10.4545455	0	0	0
4594	0.674669202	10.4545455	0	0	0
4595	0.681923710	10.4545455	0	0	0
4596	0.689178217	10.4545455	0	0	0
4597	0.696432725	10.4545455	0	0	0
4598	0.703687232	10.4545455	0	0	0
4599	0.710941740	10.4545455	0	0	0
4600	0.718196248	10.4545455	0	0	0
4601	0.000000000	10.6868687	0	0	0
4602	0.007254508	10.6868687	0	0	0
4603	0.014509015	10.6868687	0	0	0
4604	0.021763523	10.6868687	0	0	0
4605	0.029018030	10.6868687	0	0	0
4606	0.036272538	10.6868687	0	0	0
4607	0.043527045	10.6868687	0	0	0
4608	0.050781553	10.6868687	0	0	0
4609	0.058036060	10.6868687	0	0	0
4610	0.065290568	10.6868687	0	0	0
4611	0.072545076	10.6868687	0	0	0
4612	0.079799583	10.6868687	0	0	0
4613	0.087054091	10.6868687	0	0	0
4614	0.094308598	10.6868687	0	0	0
4615	0.101563106	10.6868687	0	0	0
4616	0.108817613	10.6868687	0	0	0
4617	0.116072121	10.6868687	0	0	0
4618	0.123326628	10.6868687	0	0	0
4619	0.130581136	10.6868687	0	0	0
4620	0.137835643	10.6868687	0	0	0
4621	0.145090151	10.6868687	0	0	0
4622	0.152344659	10.6868687	0	0	0
4623	0.159599166	10.6868687	0	0	0
4624	0.166853674	10.6868687	0	0	0
4625	0.174108181	10.6868687	0	0	0
4626	0.181362689	10.6868687	0	0	0
4627	0.188617196	10.6868687	0	0	0
4628	0.195871704	10.6868687	0	0	0
4629	0.203126211	10.6868687	0	0	0
4630	0.210380719	10.6868687	0	0	0
4631	0.217635227	10.6868687	0	0	0
4632	0.224889734	10.6868687	0	0	0
4633	0.232144242	10.6868687	0	0	0
4634	0.239398749	10.6868687	0	0	0
4635	0.246653257	10.6868687	0	0	0

4636	0.253907764	10.6868687	0	0	0
4637	0.261162272	10.6868687	0	0	0
4638	0.268416779	10.6868687	0	0	0
4639	0.275671287	10.6868687	0	0	0
4640	0.282925794	10.6868687	0	0	0
4641	0.290180302	10.6868687	0	0	0
4642	0.297434810	10.6868687	0	0	0
4643	0.304689317	10.6868687	0	0	0
4644	0.311943825	10.6868687	0	0	0
4645	0.319198332	10.6868687	0	0	0
4646	0.326452840	10.6868687	0	0	0
4647	0.333707347	10.6868687	0	0	0
4648	0.340961855	10.6868687	0	0	0
4649	0.348216362	10.6868687	0	0	0
4650	0.355470870	10.6868687	0	0	0
4651	0.362725378	10.6868687	0	0	0
4652	0.369979885	10.6868687	0	0	0
4653	0.377234393	10.6868687	0	0	0
4654	0.384488900	10.6868687	0	0	0
4655	0.391743408	10.6868687	0	0	0
4656	0.398997915	10.6868687	0	0	0
4657	0.406252423	10.6868687	0	0	0
4658	0.413506930	10.6868687	0	0	0
4659	0.420761438	10.6868687	0	0	0
4660	0.428015945	10.6868687	0	0	0
4661	0.435270453	10.6868687	0	0	0
4662	0.442524961	10.6868687	0	0	0
4663	0.449779468	10.6868687	0	0	0
4664	0.457033976	10.6868687	0	0	0
4665	0.464288483	10.6868687	0	0	0
4666	0.471542991	10.6868687	0	0	0
4667	0.478797498	10.6868687	0	0	0
4668	0.486052006	10.6868687	0	0	0
4669	0.493306513	10.6868687	0	0	0
4670	0.500561021	10.6868687	0	0	0
4671	0.507815529	10.6868687	0	0	0
4672	0.515070036	10.6868687	0	0	0
4673	0.522324544	10.6868687	0	0	0
4674	0.529579051	10.6868687	0	0	0
4675	0.536833559	10.6868687	0	0	0
4676	0.544088066	10.6868687	0	0	0
4677	0.551342574	10.6868687	0	0	0
4678	0.558597081	10.6868687	0	0	0
4679	0.565851589	10.6868687	0	0	0
4680	0.573106097	10.6868687	0	0	0

4681	0.580360604	10.6868687	0	0	0
4682	0.587615112	10.6868687	0	0	0
4683	0.594869619	10.6868687	0	0	0
4684	0.602124127	10.6868687	0	0	0
4685	0.609378634	10.6868687	0	0	0
4686	0.616633142	10.6868687	0	0	0
4687	0.623887649	10.6868687	0	0	0
4688	0.631142157	10.6868687	0	0	0
4689	0.638396664	10.6868687	0	0	0
4690	0.645651172	10.6868687	0	0	0
4691	0.652905680	10.6868687	0	0	0
4692	0.660160187	10.6868687	0	0	0
4693	0.667414695	10.6868687	0	0	0
4694	0.674669202	10.6868687	0	0	0
4695	0.681923710	10.6868687	0	0	0
4696	0.689178217	10.6868687	0	0	0
4697	0.696432725	10.6868687	0	0	0
4698	0.703687232	10.6868687	0	0	0
4699	0.710941740	10.6868687	0	0	0
4700	0.718196248	10.6868687	0	0	0
4701	0.000000000	10.9191919	0	0	0
4702	0.007254508	10.9191919	0	0	0
4703	0.014509015	10.9191919	0	0	0
4704	0.021763523	10.9191919	0	0	0
4705	0.029018030	10.9191919	0	0	0
4706	0.036272538	10.9191919	0	0	0
4707	0.043527045	10.9191919	0	0	0
4708	0.050781553	10.9191919	0	0	0
4709	0.058036060	10.9191919	0	0	0
4710	0.065290568	10.9191919	0	0	0
4711	0.072545076	10.9191919	0	0	0
4712	0.079799583	10.9191919	0	0	0
4713	0.087054091	10.9191919	0	0	0
4714	0.094308598	10.9191919	0	0	0
4715	0.101563106	10.9191919	0	0	0
4716	0.108817613	10.9191919	0	0	0
4717	0.116072121	10.9191919	0	0	0
4718	0.123326628	10.9191919	0	0	0
4719	0.130581136	10.9191919	0	0	0
4720	0.137835643	10.9191919	0	0	0
4721	0.145090151	10.9191919	0	0	0
4722	0.152344659	10.9191919	0	0	0
4723	0.159599166	10.9191919	0	0	0
4724	0.166853674	10.9191919	0	0	0
4725	0.174108181	10.9191919	0	0	0

4726	0.181362689	10.9191919	0	0	0
4727	0.188617196	10.9191919	0	0	0
4728	0.195871704	10.9191919	0	0	0
4729	0.203126211	10.9191919	0	0	0
4730	0.210380719	10.9191919	0	0	0
4731	0.217635227	10.9191919	0	0	0
4732	0.224889734	10.9191919	0	0	0
4733	0.232144242	10.9191919	0	0	0
4734	0.239398749	10.9191919	0	0	0
4735	0.246653257	10.9191919	0	0	0
4736	0.253907764	10.9191919	0	0	0
4737	0.261162272	10.9191919	0	0	0
4738	0.268416779	10.9191919	0	0	0
4739	0.275671287	10.9191919	0	0	0
4740	0.282925794	10.9191919	0	0	0
4741	0.290180302	10.9191919	0	0	0
4742	0.297434810	10.9191919	0	0	0
4743	0.304689317	10.9191919	0	0	0
4744	0.311943825	10.9191919	0	0	0
4745	0.319198332	10.9191919	0	0	0
4746	0.326452840	10.9191919	0	0	0
4747	0.333707347	10.9191919	0	0	0
4748	0.340961855	10.9191919	0	0	0
4749	0.348216362	10.9191919	0	0	0
4750	0.355470870	10.9191919	0	0	0
4751	0.362725378	10.9191919	0	0	0
4752	0.369979885	10.9191919	0	0	0
4753	0.377234393	10.9191919	0	0	0
4754	0.384488900	10.9191919	0	0	0
4755	0.391743408	10.9191919	0	0	0
4756	0.398997915	10.9191919	0	0	0
4757	0.406252423	10.9191919	0	0	0
4758	0.413506930	10.9191919	0	0	0
4759	0.420761438	10.9191919	0	0	0
4760	0.428015945	10.9191919	0	0	0
4761	0.435270453	10.9191919	0	0	0
4762	0.442524961	10.9191919	0	0	0
4763	0.449779468	10.9191919	0	0	0
4764	0.457033976	10.9191919	0	0	0
4765	0.464288483	10.9191919	0	0	0
4766	0.471542991	10.9191919	0	0	0
4767	0.478797498	10.9191919	0	0	0
4768	0.486052006	10.9191919	0	0	0
4769	0.493306513	10.9191919	0	0	0
4770	0.500561021	10.9191919	0	0	0

4771	0.507815529	10.9191919	0	0	0
4772	0.515070036	10.9191919	0	0	0
4773	0.522324544	10.9191919	0	0	0
4774	0.529579051	10.9191919	0	0	0
4775	0.536833559	10.9191919	0	0	0
4776	0.544088066	10.9191919	0	0	0
4777	0.551342574	10.9191919	0	0	0
4778	0.558597081	10.9191919	0	0	0
4779	0.565851589	10.9191919	0	0	0
4780	0.573106097	10.9191919	0	0	0
4781	0.580360604	10.9191919	0	0	0
4782	0.587615112	10.9191919	0	0	0
4783	0.594869619	10.9191919	0	0	0
4784	0.602124127	10.9191919	0	0	0
4785	0.609378634	10.9191919	0	0	0
4786	0.616633142	10.9191919	0	0	0
4787	0.623887649	10.9191919	0	0	0
4788	0.631142157	10.9191919	0	0	0
4789	0.638396664	10.9191919	0	0	0
4790	0.645651172	10.9191919	0	0	0
4791	0.652905680	10.9191919	0	0	0
4792	0.660160187	10.9191919	0	0	0
4793	0.667414695	10.9191919	0	0	0
4794	0.674669202	10.9191919	0	0	0
4795	0.681923710	10.9191919	0	0	0
4796	0.689178217	10.9191919	0	0	0
4797	0.696432725	10.9191919	0	0	0
4798	0.703687232	10.9191919	0	0	0
4799	0.710941740	10.9191919	0	0	0
4800	0.718196248	10.9191919	0	0	0
4801	0.000000000	11.1515152	0	0	0
4802	0.007254508	11.1515152	0	0	0
4803	0.014509015	11.1515152	0	0	0
4804	0.021763523	11.1515152	0	0	0
4805	0.029018030	11.1515152	0	0	0
4806	0.036272538	11.1515152	0	0	0
4807	0.043527045	11.1515152	0	0	0
4808	0.050781553	11.1515152	0	0	0
4809	0.058036060	11.1515152	0	0	0
4810	0.065290568	11.1515152	0	0	0
4811	0.072545076	11.1515152	0	0	0
4812	0.079799583	11.1515152	0	0	0
4813	0.087054091	11.1515152	0	0	0
4814	0.094308598	11.1515152	0	0	0
4815	0.101563106	11.1515152	0	0	0

4816	0.108817613	11.1515152	0	0	0
4817	0.116072121	11.1515152	0	0	0
4818	0.123326628	11.1515152	0	0	0
4819	0.130581136	11.1515152	0	0	0
4820	0.137835643	11.1515152	0	0	0
4821	0.145090151	11.1515152	0	0	0
4822	0.152344659	11.1515152	0	0	0
4823	0.159599166	11.1515152	0	0	0
4824	0.166853674	11.1515152	0	0	0
4825	0.174108181	11.1515152	0	0	0
4826	0.181362689	11.1515152	0	0	0
4827	0.188617196	11.1515152	0	0	0
4828	0.195871704	11.1515152	0	0	0
4829	0.203126211	11.1515152	0	0	0
4830	0.210380719	11.1515152	0	0	0
4831	0.217635227	11.1515152	0	0	0
4832	0.224889734	11.1515152	0	0	0
4833	0.232144242	11.1515152	0	0	0
4834	0.239398749	11.1515152	0	0	0
4835	0.246653257	11.1515152	0	0	0
4836	0.253907764	11.1515152	0	0	0
4837	0.261162272	11.1515152	0	0	0
4838	0.268416779	11.1515152	0	0	0
4839	0.275671287	11.1515152	0	0	0
4840	0.282925794	11.1515152	0	0	0
4841	0.290180302	11.1515152	0	0	0
4842	0.297434810	11.1515152	0	0	0
4843	0.304689317	11.1515152	0	0	0
4844	0.311943825	11.1515152	0	0	0
4845	0.319198332	11.1515152	0	0	0
4846	0.326452840	11.1515152	0	0	0
4847	0.333707347	11.1515152	0	0	0
4848	0.340961855	11.1515152	0	0	0
4849	0.348216362	11.1515152	0	0	0
4850	0.355470870	11.1515152	0	0	0
4851	0.362725378	11.1515152	0	0	0
4852	0.369979885	11.1515152	0	0	0
4853	0.377234393	11.1515152	0	0	0
4854	0.384488900	11.1515152	0	0	0
4855	0.391743408	11.1515152	0	0	0
4856	0.398997915	11.1515152	0	0	0
4857	0.406252423	11.1515152	0	0	0
4858	0.413506930	11.1515152	0	0	0
4859	0.420761438	11.1515152	0	0	0
4860	0.428015945	11.1515152	0	0	0

4861	0.435270453	11.1515152	0	0	0
4862	0.442524961	11.1515152	0	0	0
4863	0.449779468	11.1515152	0	0	0
4864	0.457033976	11.1515152	0	0	0
4865	0.464288483	11.1515152	0	0	0
4866	0.471542991	11.1515152	0	0	0
4867	0.478797498	11.1515152	0	0	0
4868	0.486052006	11.1515152	0	0	0
4869	0.493306513	11.1515152	0	0	0
4870	0.500561021	11.1515152	0	0	0
4871	0.507815529	11.1515152	0	0	0
4872	0.515070036	11.1515152	0	0	0
4873	0.522324544	11.1515152	0	0	0
4874	0.529579051	11.1515152	0	0	0
4875	0.536833559	11.1515152	0	0	0
4876	0.544088066	11.1515152	0	0	0
4877	0.551342574	11.1515152	0	0	0
4878	0.558597081	11.1515152	0	0	0
4879	0.565851589	11.1515152	0	0	0
4880	0.573106097	11.1515152	0	0	0
4881	0.580360604	11.1515152	0	0	0
4882	0.587615112	11.1515152	0	0	0
4883	0.594869619	11.1515152	0	0	0
4884	0.602124127	11.1515152	0	0	0
4885	0.609378634	11.1515152	0	0	0
4886	0.616633142	11.1515152	0	0	0
4887	0.623887649	11.1515152	0	0	0
4888	0.631142157	11.1515152	0	0	0
4889	0.638396664	11.1515152	0	0	0
4890	0.645651172	11.1515152	0	0	0
4891	0.652905680	11.1515152	0	0	0
4892	0.660160187	11.1515152	0	0	0
4893	0.667414695	11.1515152	0	0	0
4894	0.674669202	11.1515152	0	0	0
4895	0.681923710	11.1515152	0	0	0
4896	0.689178217	11.1515152	0	0	0
4897	0.696432725	11.1515152	0	0	0
4898	0.703687232	11.1515152	0	0	0
4899	0.710941740	11.1515152	0	0	0
4900	0.718196248	11.1515152	0	0	0
4901	0.000000000	11.3838384	0	0	0
4902	0.007254508	11.3838384	0	0	0
4903	0.014509015	11.3838384	0	0	0
4904	0.021763523	11.3838384	0	0	0
4905	0.029018030	11.3838384	0	0	0

4906	0.036272538	11.3838384	0	0	0
4907	0.043527045	11.3838384	0	0	0
4908	0.050781553	11.3838384	0	0	0
4909	0.058036060	11.3838384	0	0	0
4910	0.065290568	11.3838384	0	0	0
4911	0.072545076	11.3838384	0	0	0
4912	0.079799583	11.3838384	0	0	0
4913	0.087054091	11.3838384	0	0	0
4914	0.094308598	11.3838384	0	0	0
4915	0.101563106	11.3838384	0	0	0
4916	0.108817613	11.3838384	0	0	0
4917	0.116072121	11.3838384	0	0	0
4918	0.123326628	11.3838384	0	0	0
4919	0.130581136	11.3838384	0	0	0
4920	0.137835643	11.3838384	0	0	0
4921	0.145090151	11.3838384	0	0	0
4922	0.152344659	11.3838384	0	0	0
4923	0.159599166	11.3838384	0	0	0
4924	0.166853674	11.3838384	0	0	0
4925	0.174108181	11.3838384	0	0	0
4926	0.181362689	11.3838384	0	0	0
4927	0.188617196	11.3838384	0	0	0
4928	0.195871704	11.3838384	0	0	0
4929	0.203126211	11.3838384	0	0	0
4930	0.210380719	11.3838384	0	0	0
4931	0.217635227	11.3838384	0	0	0
4932	0.224889734	11.3838384	0	0	0
4933	0.232144242	11.3838384	0	0	0
4934	0.239398749	11.3838384	0	0	0
4935	0.246653257	11.3838384	0	0	0
4936	0.253907764	11.3838384	0	0	0
4937	0.261162272	11.3838384	0	0	0
4938	0.268416779	11.3838384	0	0	0
4939	0.275671287	11.3838384	0	0	0
4940	0.282925794	11.3838384	0	0	0
4941	0.290180302	11.3838384	0	0	0
4942	0.297434810	11.3838384	0	0	0
4943	0.304689317	11.3838384	0	0	0
4944	0.311943825	11.3838384	0	0	0
4945	0.319198332	11.3838384	0	0	0
4946	0.326452840	11.3838384	0	0	0
4947	0.333707347	11.3838384	0	0	0
4948	0.340961855	11.3838384	0	0	0
4949	0.348216362	11.3838384	0	0	0
4950	0.355470870	11.3838384	0	0	0

4951	0.362725378	11.3838384	0	0	0
4952	0.369979885	11.3838384	0	0	0
4953	0.377234393	11.3838384	0	0	0
4954	0.384488900	11.3838384	0	0	0
4955	0.391743408	11.3838384	0	0	0
4956	0.398997915	11.3838384	0	0	0
4957	0.406252423	11.3838384	0	0	0
4958	0.413506930	11.3838384	0	0	0
4959	0.420761438	11.3838384	0	0	0
4960	0.428015945	11.3838384	0	0	0
4961	0.435270453	11.3838384	0	0	0
4962	0.442524961	11.3838384	0	0	0
4963	0.449779468	11.3838384	0	0	0
4964	0.457033976	11.3838384	0	0	0
4965	0.464288483	11.3838384	0	0	0
4966	0.471542991	11.3838384	0	0	0
4967	0.478797498	11.3838384	0	0	0
4968	0.486052006	11.3838384	0	0	0
4969	0.493306513	11.3838384	0	0	0
4970	0.500561021	11.3838384	0	0	0
4971	0.507815529	11.3838384	0	0	0
4972	0.515070036	11.3838384	0	0	0
4973	0.522324544	11.3838384	0	0	0
4974	0.529579051	11.3838384	0	0	0
4975	0.536833559	11.3838384	0	0	0
4976	0.544088066	11.3838384	0	0	0
4977	0.551342574	11.3838384	0	0	0
4978	0.558597081	11.3838384	0	0	0
4979	0.565851589	11.3838384	0	0	0
4980	0.573106097	11.3838384	0	0	0
4981	0.580360604	11.3838384	0	0	0
4982	0.587615112	11.3838384	0	0	0
4983	0.594869619	11.3838384	0	0	0
4984	0.602124127	11.3838384	0	0	0
4985	0.609378634	11.3838384	0	0	0
4986	0.616633142	11.3838384	0	0	0
4987	0.623887649	11.3838384	0	0	0
4988	0.631142157	11.3838384	0	0	0
4989	0.638396664	11.3838384	0	0	0
4990	0.645651172	11.3838384	0	0	0
4991	0.652905680	11.3838384	0	0	0
4992	0.660160187	11.3838384	0	0	0
4993	0.667414695	11.3838384	0	0	0
4994	0.674669202	11.3838384	0	0	0
4995	0.681923710	11.3838384	0	0	0

4996	0.689178217	11.3838384	0	0	0
4997	0.696432725	11.3838384	0	0	0
4998	0.703687232	11.3838384	0	0	0
4999	0.710941740	11.3838384	0	0	0
5000	0.718196248	11.3838384	0	0	0
5001	0.000000000	11.6161616	0	0	0
5002	0.007254508	11.6161616	0	0	0
5003	0.014509015	11.6161616	0	0	0
5004	0.021763523	11.6161616	0	0	0
5005	0.029018030	11.6161616	0	0	0
5006	0.036272538	11.6161616	0	0	0
5007	0.043527045	11.6161616	0	0	0
5008	0.050781553	11.6161616	0	0	0
5009	0.058036060	11.6161616	0	0	0
5010	0.065290568	11.6161616	0	0	0
5011	0.072545076	11.6161616	0	0	0
5012	0.079799583	11.6161616	0	0	0
5013	0.087054091	11.6161616	0	0	0
5014	0.094308598	11.6161616	0	0	0
5015	0.101563106	11.6161616	0	0	0
5016	0.108817613	11.6161616	0	0	0
5017	0.116072121	11.6161616	0	0	0
5018	0.123326628	11.6161616	0	0	0
5019	0.130581136	11.6161616	0	0	0
5020	0.137835643	11.6161616	0	0	0
5021	0.145090151	11.6161616	0	0	0
5022	0.152344659	11.6161616	0	0	0
5023	0.159599166	11.6161616	0	0	0
5024	0.166853674	11.6161616	0	0	0
5025	0.174108181	11.6161616	0	0	0
5026	0.181362689	11.6161616	0	0	0
5027	0.188617196	11.6161616	0	0	0
5028	0.195871704	11.6161616	0	0	0
5029	0.203126211	11.6161616	0	0	0
5030	0.210380719	11.6161616	0	0	0
5031	0.217635227	11.6161616	0	0	0
5032	0.224889734	11.6161616	0	0	0
5033	0.232144242	11.6161616	0	0	0
5034	0.239398749	11.6161616	0	0	0
5035	0.246653257	11.6161616	0	0	0
5036	0.253907764	11.6161616	0	0	0
5037	0.261162272	11.6161616	0	0	0
5038	0.268416779	11.6161616	0	0	0
5039	0.275671287	11.6161616	0	0	0
5040	0.282925794	11.6161616	0	0	0

5041	0.290180302	11.6161616	0	0	0
5042	0.297434810	11.6161616	0	0	0
5043	0.304689317	11.6161616	0	0	0
5044	0.311943825	11.6161616	0	0	0
5045	0.319198332	11.6161616	0	0	0
5046	0.326452840	11.6161616	0	0	0
5047	0.333707347	11.6161616	0	0	0
5048	0.340961855	11.6161616	0	0	0
5049	0.348216362	11.6161616	0	0	0
5050	0.355470870	11.6161616	0	0	0
5051	0.362725378	11.6161616	0	0	0
5052	0.369979885	11.6161616	0	0	0
5053	0.377234393	11.6161616	0	0	0
5054	0.384488900	11.6161616	0	0	0
5055	0.391743408	11.6161616	0	0	0
5056	0.398997915	11.6161616	0	0	0
5057	0.406252423	11.6161616	0	0	0
5058	0.413506930	11.6161616	0	0	0
5059	0.420761438	11.6161616	0	0	0
5060	0.428015945	11.6161616	0	0	0
5061	0.435270453	11.6161616	0	0	0
5062	0.442524961	11.6161616	0	0	0
5063	0.449779468	11.6161616	0	0	0
5064	0.457033976	11.6161616	0	0	0
5065	0.464288483	11.6161616	0	0	0
5066	0.471542991	11.6161616	0	0	0
5067	0.478797498	11.6161616	0	0	0
5068	0.486052006	11.6161616	0	0	0
5069	0.493306513	11.6161616	0	0	0
5070	0.500561021	11.6161616	0	0	0
5071	0.507815529	11.6161616	0	0	0
5072	0.515070036	11.6161616	0	0	0
5073	0.522324544	11.6161616	0	0	0
5074	0.529579051	11.6161616	0	0	0
5075	0.536833559	11.6161616	0	0	0
5076	0.544088066	11.6161616	0	0	0
5077	0.551342574	11.6161616	0	0	0
5078	0.558597081	11.6161616	0	0	0
5079	0.565851589	11.6161616	0	0	0
5080	0.573106097	11.6161616	0	0	0
5081	0.580360604	11.6161616	0	0	0
5082	0.587615112	11.6161616	0	0	0
5083	0.594869619	11.6161616	0	0	0
5084	0.602124127	11.6161616	0	0	0
5085	0.609378634	11.6161616	0	0	0

5086	0.616633142	11.6161616	0	0	0
5087	0.623887649	11.6161616	0	0	0
5088	0.631142157	11.6161616	0	0	0
5089	0.638396664	11.6161616	0	0	0
5090	0.645651172	11.6161616	0	0	0
5091	0.652905680	11.6161616	0	0	0
5092	0.660160187	11.6161616	0	0	0
5093	0.667414695	11.6161616	0	0	0
5094	0.674669202	11.6161616	0	0	0
5095	0.681923710	11.6161616	0	0	0
5096	0.689178217	11.6161616	0	0	0
5097	0.696432725	11.6161616	0	0	0
5098	0.703687232	11.6161616	0	0	0
5099	0.710941740	11.6161616	0	0	0
5100	0.718196248	11.6161616	0	0	0
5101	0.000000000	11.8484848	0	0	0
5102	0.007254508	11.8484848	0	0	0
5103	0.014509015	11.8484848	0	0	0
5104	0.021763523	11.8484848	0	0	0
5105	0.029018030	11.8484848	0	0	0
5106	0.036272538	11.8484848	0	0	0
5107	0.043527045	11.8484848	0	0	0
5108	0.050781553	11.8484848	0	0	0
5109	0.058036060	11.8484848	0	0	0
5110	0.065290568	11.8484848	0	0	0
5111	0.072545076	11.8484848	0	0	0
5112	0.079799583	11.8484848	0	0	0
5113	0.087054091	11.8484848	0	0	0
5114	0.094308598	11.8484848	0	0	0
5115	0.101563106	11.8484848	0	0	0
5116	0.108817613	11.8484848	0	0	0
5117	0.116072121	11.8484848	0	0	0
5118	0.123326628	11.8484848	0	0	0
5119	0.130581136	11.8484848	0	0	0
5120	0.137835643	11.8484848	0	0	0
5121	0.145090151	11.8484848	0	0	0
5122	0.152344659	11.8484848	0	0	0
5123	0.159599166	11.8484848	0	0	0
5124	0.166853674	11.8484848	0	0	0
5125	0.174108181	11.8484848	0	0	0
5126	0.181362689	11.8484848	0	0	0
5127	0.188617196	11.8484848	0	0	0
5128	0.195871704	11.8484848	0	0	0
5129	0.203126211	11.8484848	0	0	0
5130	0.210380719	11.8484848	0	0	0

5131	0.217635227	11.8484848	0	0	0
5132	0.224889734	11.8484848	0	0	0
5133	0.232144242	11.8484848	0	0	0
5134	0.239398749	11.8484848	0	0	0
5135	0.246653257	11.8484848	0	0	0
5136	0.253907764	11.8484848	0	0	0
5137	0.261162272	11.8484848	0	0	0
5138	0.268416779	11.8484848	0	0	0
5139	0.275671287	11.8484848	0	0	0
5140	0.282925794	11.8484848	0	0	0
5141	0.290180302	11.8484848	0	0	0
5142	0.297434810	11.8484848	0	0	0
5143	0.304689317	11.8484848	0	0	0
5144	0.311943825	11.8484848	0	0	0
5145	0.319198332	11.8484848	0	0	0
5146	0.326452840	11.8484848	0	0	0
5147	0.333707347	11.8484848	0	0	0
5148	0.340961855	11.8484848	0	0	0
5149	0.348216362	11.8484848	0	0	0
5150	0.355470870	11.8484848	0	0	0
5151	0.362725378	11.8484848	0	0	0
5152	0.369979885	11.8484848	0	0	0
5153	0.377234393	11.8484848	0	0	0
5154	0.384488900	11.8484848	0	0	0
5155	0.391743408	11.8484848	0	0	0
5156	0.398997915	11.8484848	0	0	0
5157	0.406252423	11.8484848	0	0	0
5158	0.413506930	11.8484848	0	0	0
5159	0.420761438	11.8484848	0	0	0
5160	0.428015945	11.8484848	0	0	0
5161	0.435270453	11.8484848	0	0	0
5162	0.442524961	11.8484848	0	0	0
5163	0.449779468	11.8484848	0	0	0
5164	0.457033976	11.8484848	0	0	0
5165	0.464288483	11.8484848	0	0	0
5166	0.471542991	11.8484848	0	0	0
5167	0.478797498	11.8484848	0	0	0
5168	0.486052006	11.8484848	0	0	0
5169	0.493306513	11.8484848	0	0	0
5170	0.500561021	11.8484848	0	0	0
5171	0.507815529	11.8484848	0	0	0
5172	0.515070036	11.8484848	0	0	0
5173	0.522324544	11.8484848	0	0	0
5174	0.529579051	11.8484848	0	0	0
5175	0.536833559	11.8484848	0	0	0

5176	0.544088066	11.8484848	0	0	0
5177	0.551342574	11.8484848	0	0	0
5178	0.558597081	11.8484848	0	0	0
5179	0.565851589	11.8484848	0	0	0
5180	0.573106097	11.8484848	0	0	0
5181	0.580360604	11.8484848	0	0	0
5182	0.587615112	11.8484848	0	0	0
5183	0.594869619	11.8484848	0	0	0
5184	0.602124127	11.8484848	0	0	0
5185	0.609378634	11.8484848	0	0	0
5186	0.616633142	11.8484848	0	0	0
5187	0.623887649	11.8484848	0	0	0
5188	0.631142157	11.8484848	0	0	0
5189	0.638396664	11.8484848	0	0	0
5190	0.645651172	11.8484848	0	0	0
5191	0.652905680	11.8484848	0	0	0
5192	0.660160187	11.8484848	0	0	0
5193	0.667414695	11.8484848	0	0	0
5194	0.674669202	11.8484848	0	0	0
5195	0.681923710	11.8484848	0	0	0
5196	0.689178217	11.8484848	0	0	0
5197	0.696432725	11.8484848	0	0	0
5198	0.703687232	11.8484848	0	0	0
5199	0.710941740	11.8484848	0	0	0
5200	0.718196248	11.8484848	0	0	0
5201	0.000000000	12.0808081	0	0	0
5202	0.007254508	12.0808081	0	0	0
5203	0.014509015	12.0808081	0	0	0
5204	0.021763523	12.0808081	0	0	0
5205	0.029018030	12.0808081	0	0	0
5206	0.036272538	12.0808081	0	0	0
5207	0.043527045	12.0808081	0	0	0
5208	0.050781553	12.0808081	0	0	0
5209	0.058036060	12.0808081	0	0	0
5210	0.065290568	12.0808081	0	0	0
5211	0.072545076	12.0808081	0	0	0
5212	0.079799583	12.0808081	0	0	0
5213	0.087054091	12.0808081	0	0	0
5214	0.094308598	12.0808081	0	0	0
5215	0.101563106	12.0808081	0	0	0
5216	0.108817613	12.0808081	0	0	0
5217	0.116072121	12.0808081	0	0	0
5218	0.123326628	12.0808081	0	0	0
5219	0.130581136	12.0808081	0	0	0
5220	0.137835643	12.0808081	0	0	0

5221	0.145090151	12.0808081	0	0	0
5222	0.152344659	12.0808081	0	0	0
5223	0.159599166	12.0808081	0	0	0
5224	0.166853674	12.0808081	0	0	0
5225	0.174108181	12.0808081	0	0	0
5226	0.181362689	12.0808081	0	0	0
5227	0.188617196	12.0808081	0	0	0
5228	0.195871704	12.0808081	0	0	0
5229	0.203126211	12.0808081	0	0	0
5230	0.210380719	12.0808081	0	0	0
5231	0.217635227	12.0808081	0	0	0
5232	0.224889734	12.0808081	0	0	0
5233	0.232144242	12.0808081	0	0	0
5234	0.239398749	12.0808081	0	0	0
5235	0.246653257	12.0808081	0	0	0
5236	0.253907764	12.0808081	0	0	0
5237	0.261162272	12.0808081	0	0	0
5238	0.268416779	12.0808081	0	0	0
5239	0.275671287	12.0808081	0	0	0
5240	0.282925794	12.0808081	0	0	0
5241	0.290180302	12.0808081	0	0	0
5242	0.297434810	12.0808081	0	0	0
5243	0.304689317	12.0808081	0	0	0
5244	0.311943825	12.0808081	0	0	0
5245	0.319198332	12.0808081	0	0	0
5246	0.326452840	12.0808081	0	0	0
5247	0.333707347	12.0808081	0	0	0
5248	0.340961855	12.0808081	0	0	0
5249	0.348216362	12.0808081	0	0	0
5250	0.355470870	12.0808081	0	0	0
5251	0.362725378	12.0808081	0	0	0
5252	0.369979885	12.0808081	0	0	0
5253	0.377234393	12.0808081	0	0	0
5254	0.384488900	12.0808081	0	0	0
5255	0.391743408	12.0808081	0	0	0
5256	0.398997915	12.0808081	0	0	0
5257	0.406252423	12.0808081	0	0	0
5258	0.413506930	12.0808081	0	0	0
5259	0.420761438	12.0808081	0	0	0
5260	0.428015945	12.0808081	0	0	0
5261	0.435270453	12.0808081	0	0	0
5262	0.442524961	12.0808081	0	0	0
5263	0.449779468	12.0808081	0	0	0
5264	0.457033976	12.0808081	0	0	0
5265	0.464288483	12.0808081	0	0	0

5266	0.471542991	12.0808081	0	0	0
5267	0.478797498	12.0808081	0	0	0
5268	0.486052006	12.0808081	0	0	0
5269	0.493306513	12.0808081	0	0	0
5270	0.500561021	12.0808081	0	0	0
5271	0.507815529	12.0808081	0	0	0
5272	0.515070036	12.0808081	0	0	0
5273	0.522324544	12.0808081	0	0	0
5274	0.529579051	12.0808081	0	0	0
5275	0.536833559	12.0808081	0	0	0
5276	0.544088066	12.0808081	0	0	0
5277	0.551342574	12.0808081	0	0	0
5278	0.558597081	12.0808081	0	0	0
5279	0.565851589	12.0808081	0	0	0
5280	0.573106097	12.0808081	0	0	0
5281	0.580360604	12.0808081	0	0	0
5282	0.587615112	12.0808081	0	0	0
5283	0.594869619	12.0808081	0	0	0
5284	0.602124127	12.0808081	0	0	0
5285	0.609378634	12.0808081	0	0	0
5286	0.616633142	12.0808081	0	0	0
5287	0.623887649	12.0808081	0	0	0
5288	0.631142157	12.0808081	0	0	0
5289	0.638396664	12.0808081	0	0	0
5290	0.645651172	12.0808081	0	0	0
5291	0.652905680	12.0808081	0	0	0
5292	0.660160187	12.0808081	0	0	0
5293	0.667414695	12.0808081	0	0	0
5294	0.674669202	12.0808081	0	0	0
5295	0.681923710	12.0808081	0	0	0
5296	0.689178217	12.0808081	0	0	0
5297	0.696432725	12.0808081	0	0	0
5298	0.703687232	12.0808081	0	0	0
5299	0.710941740	12.0808081	0	0	0
5300	0.718196248	12.0808081	0	0	0
5301	0.000000000	12.3131313	0	0	0
5302	0.007254508	12.3131313	0	0	0
5303	0.014509015	12.3131313	0	0	0
5304	0.021763523	12.3131313	0	0	0
5305	0.029018030	12.3131313	0	0	0
5306	0.036272538	12.3131313	0	0	0
5307	0.043527045	12.3131313	0	0	0
5308	0.050781553	12.3131313	0	0	0
5309	0.058036060	12.3131313	0	0	0
5310	0.065290568	12.3131313	0	0	0

5311	0.072545076	12.3131313	0	0	0
5312	0.079799583	12.3131313	0	0	0
5313	0.087054091	12.3131313	0	0	0
5314	0.094308598	12.3131313	0	0	0
5315	0.101563106	12.3131313	0	0	0
5316	0.108817613	12.3131313	0	0	0
5317	0.116072121	12.3131313	0	0	0
5318	0.123326628	12.3131313	0	0	0
5319	0.130581136	12.3131313	0	0	0
5320	0.137835643	12.3131313	0	0	0
5321	0.145090151	12.3131313	0	0	0
5322	0.152344659	12.3131313	0	0	0
5323	0.159599166	12.3131313	0	0	0
5324	0.166853674	12.3131313	0	0	0
5325	0.174108181	12.3131313	0	0	0
5326	0.181362689	12.3131313	0	0	0
5327	0.188617196	12.3131313	0	0	0
5328	0.195871704	12.3131313	0	0	0
5329	0.203126211	12.3131313	0	0	0
5330	0.210380719	12.3131313	0	0	0
5331	0.217635227	12.3131313	0	0	0
5332	0.224889734	12.3131313	0	0	0
5333	0.232144242	12.3131313	0	0	0
5334	0.239398749	12.3131313	0	0	0
5335	0.246653257	12.3131313	0	0	0
5336	0.253907764	12.3131313	0	0	0
5337	0.261162272	12.3131313	0	0	0
5338	0.268416779	12.3131313	0	0	0
5339	0.275671287	12.3131313	0	0	0
5340	0.282925794	12.3131313	0	0	0
5341	0.290180302	12.3131313	0	0	0
5342	0.297434810	12.3131313	0	0	0
5343	0.304689317	12.3131313	0	0	0
5344	0.311943825	12.3131313	0	0	0
5345	0.319198332	12.3131313	0	0	0
5346	0.326452840	12.3131313	0	0	0
5347	0.333707347	12.3131313	0	0	0
5348	0.340961855	12.3131313	0	0	0
5349	0.348216362	12.3131313	0	0	0
5350	0.355470870	12.3131313	0	0	0
5351	0.362725378	12.3131313	0	0	0
5352	0.369979885	12.3131313	0	0	0
5353	0.377234393	12.3131313	0	0	0
5354	0.384488900	12.3131313	0	0	0
5355	0.391743408	12.3131313	0	0	0

5356	0.398997915	12.3131313	0	0	0
5357	0.406252423	12.3131313	0	0	0
5358	0.413506930	12.3131313	0	0	0
5359	0.420761438	12.3131313	0	0	0
5360	0.428015945	12.3131313	0	0	0
5361	0.435270453	12.3131313	0	0	0
5362	0.442524961	12.3131313	0	0	0
5363	0.449779468	12.3131313	0	0	0
5364	0.457033976	12.3131313	0	0	0
5365	0.464288483	12.3131313	0	0	0
5366	0.471542991	12.3131313	0	0	0
5367	0.478797498	12.3131313	0	0	0
5368	0.486052006	12.3131313	0	0	0
5369	0.493306513	12.3131313	0	0	0
5370	0.500561021	12.3131313	0	0	0
5371	0.507815529	12.3131313	0	0	0
5372	0.515070036	12.3131313	0	0	0
5373	0.522324544	12.3131313	0	0	0
5374	0.529579051	12.3131313	0	0	0
5375	0.536833559	12.3131313	0	0	0
5376	0.544088066	12.3131313	0	0	0
5377	0.551342574	12.3131313	0	0	0
5378	0.558597081	12.3131313	0	0	0
5379	0.565851589	12.3131313	0	0	0
5380	0.573106097	12.3131313	0	0	0
5381	0.580360604	12.3131313	0	0	0
5382	0.587615112	12.3131313	0	0	0
5383	0.594869619	12.3131313	0	0	0
5384	0.602124127	12.3131313	0	0	0
5385	0.609378634	12.3131313	0	0	0
5386	0.616633142	12.3131313	0	0	0
5387	0.623887649	12.3131313	0	0	0
5388	0.631142157	12.3131313	0	0	0
5389	0.638396664	12.3131313	0	0	0
5390	0.645651172	12.3131313	0	0	0
5391	0.652905680	12.3131313	0	0	0
5392	0.660160187	12.3131313	0	0	0
5393	0.667414695	12.3131313	0	0	0
5394	0.674669202	12.3131313	0	0	0
5395	0.681923710	12.3131313	0	0	0
5396	0.689178217	12.3131313	0	0	0
5397	0.696432725	12.3131313	0	0	0
5398	0.703687232	12.3131313	0	0	0
5399	0.710941740	12.3131313	0	0	0
5400	0.718196248	12.3131313	0	0	0

5401	0.000000000	12.5454545	0	0	0
5402	0.007254508	12.5454545	0	0	0
5403	0.014509015	12.5454545	0	0	0
5404	0.021763523	12.5454545	0	0	0
5405	0.029018030	12.5454545	0	0	0
5406	0.036272538	12.5454545	0	0	0
5407	0.043527045	12.5454545	0	0	0
5408	0.050781553	12.5454545	0	0	0
5409	0.058036060	12.5454545	0	0	0
5410	0.065290568	12.5454545	0	0	0
5411	0.072545076	12.5454545	0	0	0
5412	0.079799583	12.5454545	0	0	0
5413	0.087054091	12.5454545	0	0	0
5414	0.094308598	12.5454545	0	0	0
5415	0.101563106	12.5454545	0	0	0
5416	0.108817613	12.5454545	0	0	0
5417	0.116072121	12.5454545	0	0	0
5418	0.123326628	12.5454545	0	0	0
5419	0.130581136	12.5454545	0	0	0
5420	0.137835643	12.5454545	0	0	0
5421	0.145090151	12.5454545	0	0	0
5422	0.152344659	12.5454545	0	0	0
5423	0.159599166	12.5454545	0	0	0
5424	0.166853674	12.5454545	0	0	0
5425	0.174108181	12.5454545	0	0	0
5426	0.181362689	12.5454545	0	0	0
5427	0.188617196	12.5454545	0	0	0
5428	0.195871704	12.5454545	0	0	0
5429	0.203126211	12.5454545	0	0	0
5430	0.210380719	12.5454545	0	0	0
5431	0.217635227	12.5454545	0	0	0
5432	0.224889734	12.5454545	0	0	0
5433	0.232144242	12.5454545	0	0	0
5434	0.239398749	12.5454545	0	0	0
5435	0.246653257	12.5454545	0	0	0
5436	0.253907764	12.5454545	0	0	0
5437	0.261162272	12.5454545	0	0	0
5438	0.268416779	12.5454545	0	0	0
5439	0.275671287	12.5454545	0	0	0
5440	0.282925794	12.5454545	0	0	0
5441	0.290180302	12.5454545	0	0	0
5442	0.297434810	12.5454545	0	0	0
5443	0.304689317	12.5454545	0	0	0
5444	0.311943825	12.5454545	0	0	0
5445	0.319198332	12.5454545	0	0	0

5446	0.326452840	12.5454545	0	0	0
5447	0.333707347	12.5454545	0	0	0
5448	0.340961855	12.5454545	0	0	0
5449	0.348216362	12.5454545	0	0	0
5450	0.355470870	12.5454545	0	0	0
5451	0.362725378	12.5454545	0	0	0
5452	0.369979885	12.5454545	0	0	0
5453	0.377234393	12.5454545	0	0	0
5454	0.384488900	12.5454545	0	0	0
5455	0.391743408	12.5454545	0	0	0
5456	0.398997915	12.5454545	0	0	0
5457	0.406252423	12.5454545	0	0	0
5458	0.413506930	12.5454545	0	0	0
5459	0.420761438	12.5454545	0	0	0
5460	0.428015945	12.5454545	0	0	0
5461	0.435270453	12.5454545	0	0	0
5462	0.442524961	12.5454545	0	0	0
5463	0.449779468	12.5454545	0	0	0
5464	0.457033976	12.5454545	0	0	0
5465	0.464288483	12.5454545	0	0	0
5466	0.471542991	12.5454545	0	0	0
5467	0.478797498	12.5454545	0	0	0
5468	0.486052006	12.5454545	0	0	0
5469	0.493306513	12.5454545	0	0	0
5470	0.500561021	12.5454545	0	0	0
5471	0.507815529	12.5454545	0	0	0
5472	0.515070036	12.5454545	0	0	0
5473	0.522324544	12.5454545	0	0	0
5474	0.529579051	12.5454545	0	0	0
5475	0.536833559	12.5454545	0	0	0
5476	0.544088066	12.5454545	0	0	0
5477	0.551342574	12.5454545	0	0	0
5478	0.558597081	12.5454545	0	0	0
5479	0.565851589	12.5454545	0	0	0
5480	0.573106097	12.5454545	0	0	0
5481	0.580360604	12.5454545	0	0	0
5482	0.587615112	12.5454545	0	0	0
5483	0.594869619	12.5454545	0	0	0
5484	0.602124127	12.5454545	0	0	0
5485	0.609378634	12.5454545	0	0	0
5486	0.616633142	12.5454545	0	0	0
5487	0.623887649	12.5454545	0	0	0
5488	0.631142157	12.5454545	0	0	0
5489	0.638396664	12.5454545	0	0	0
5490	0.645651172	12.5454545	0	0	0

5491	0.652905680	12.5454545	0	0	0
5492	0.660160187	12.5454545	0	0	0
5493	0.667414695	12.5454545	0	0	0
5494	0.674669202	12.5454545	0	0	0
5495	0.681923710	12.5454545	0	0	0
5496	0.689178217	12.5454545	0	0	0
5497	0.696432725	12.5454545	0	0	0
5498	0.703687232	12.5454545	0	0	0
5499	0.710941740	12.5454545	0	0	0
5500	0.718196248	12.5454545	0	0	0
5501	0.000000000	12.7777778	0	0	0
5502	0.007254508	12.7777778	0	0	0
5503	0.014509015	12.7777778	0	0	0
5504	0.021763523	12.7777778	0	0	0
5505	0.029018030	12.7777778	0	0	0
5506	0.036272538	12.7777778	0	0	0
5507	0.043527045	12.7777778	0	0	0
5508	0.050781553	12.7777778	0	0	0
5509	0.058036060	12.7777778	0	0	0
5510	0.065290568	12.7777778	0	0	0
5511	0.072545076	12.7777778	0	0	0
5512	0.079799583	12.7777778	0	0	0
5513	0.087054091	12.7777778	0	0	0
5514	0.094308598	12.7777778	0	0	0
5515	0.101563106	12.7777778	0	0	0
5516	0.108817613	12.7777778	0	0	0
5517	0.116072121	12.7777778	0	0	0
5518	0.123326628	12.7777778	0	0	0
5519	0.130581136	12.7777778	0	0	0
5520	0.137835643	12.7777778	0	0	0
5521	0.145090151	12.7777778	0	0	0
5522	0.152344659	12.7777778	0	0	0
5523	0.159599166	12.7777778	0	0	0
5524	0.166853674	12.7777778	0	0	0
5525	0.174108181	12.7777778	0	0	0
5526	0.181362689	12.7777778	0	0	0
5527	0.188617196	12.7777778	0	0	0
5528	0.195871704	12.7777778	0	0	0
5529	0.203126211	12.7777778	0	0	0
5530	0.210380719	12.7777778	0	0	0
5531	0.217635227	12.7777778	0	0	0
5532	0.224889734	12.7777778	0	0	0
5533	0.232144242	12.7777778	0	0	0
5534	0.239398749	12.7777778	0	0	0
5535	0.246653257	12.7777778	0	0	0

5536	0.253907764	12.7777778	0	0	0
5537	0.261162272	12.7777778	0	0	0
5538	0.268416779	12.7777778	0	0	0
5539	0.275671287	12.7777778	0	0	0
5540	0.282925794	12.7777778	0	0	0
5541	0.290180302	12.7777778	0	0	0
5542	0.297434810	12.7777778	0	0	0
5543	0.304689317	12.7777778	0	0	0
5544	0.311943825	12.7777778	0	0	0
5545	0.319198332	12.7777778	0	0	0
5546	0.326452840	12.7777778	0	0	0
5547	0.333707347	12.7777778	0	0	0
5548	0.340961855	12.7777778	0	0	0
5549	0.348216362	12.7777778	0	0	0
5550	0.355470870	12.7777778	0	0	0
5551	0.362725378	12.7777778	0	0	0
5552	0.369979885	12.7777778	0	0	0
5553	0.377234393	12.7777778	0	0	0
5554	0.384488900	12.7777778	0	0	0
5555	0.391743408	12.7777778	0	0	0
5556	0.398997915	12.7777778	0	0	0
5557	0.406252423	12.7777778	0	0	0
5558	0.413506930	12.7777778	0	0	0
5559	0.420761438	12.7777778	0	0	0
5560	0.428015945	12.7777778	0	0	0
5561	0.435270453	12.7777778	0	0	0
5562	0.442524961	12.7777778	0	0	0
5563	0.449779468	12.7777778	0	0	0
5564	0.457033976	12.7777778	0	0	0
5565	0.464288483	12.7777778	0	0	0
5566	0.471542991	12.7777778	0	0	0
5567	0.478797498	12.7777778	0	0	0
5568	0.486052006	12.7777778	0	0	0
5569	0.493306513	12.7777778	0	0	0
5570	0.500561021	12.7777778	0	0	0
5571	0.507815529	12.7777778	0	0	0
5572	0.515070036	12.7777778	0	0	0
5573	0.522324544	12.7777778	0	0	0
5574	0.529579051	12.7777778	0	0	0
5575	0.536833559	12.7777778	0	0	0
5576	0.544088066	12.7777778	0	0	0
5577	0.551342574	12.7777778	0	0	0
5578	0.558597081	12.7777778	0	0	0
5579	0.565851589	12.7777778	0	0	0
5580	0.573106097	12.7777778	0	0	0

5581	0.580360604	12.7777778	0	0	0
5582	0.587615112	12.7777778	0	0	0
5583	0.594869619	12.7777778	0	0	0
5584	0.602124127	12.7777778	0	0	0
5585	0.609378634	12.7777778	0	0	0
5586	0.616633142	12.7777778	0	0	0
5587	0.623887649	12.7777778	0	0	0
5588	0.631142157	12.7777778	0	0	0
5589	0.638396664	12.7777778	0	0	0
5590	0.645651172	12.7777778	0	0	0
5591	0.652905680	12.7777778	0	0	0
5592	0.660160187	12.7777778	0	0	0
5593	0.667414695	12.7777778	0	0	0
5594	0.674669202	12.7777778	0	0	0
5595	0.681923710	12.7777778	0	0	0
5596	0.689178217	12.7777778	0	0	0
5597	0.696432725	12.7777778	0	0	0
5598	0.703687232	12.7777778	0	0	0
5599	0.710941740	12.7777778	0	0	0
5600	0.718196248	12.7777778	0	0	0
5601	0.000000000	13.0101010	0	0	0
5602	0.007254508	13.0101010	0	0	0
5603	0.014509015	13.0101010	0	0	0
5604	0.021763523	13.0101010	0	0	0
5605	0.029018030	13.0101010	0	0	0
5606	0.036272538	13.0101010	0	0	0
5607	0.043527045	13.0101010	0	0	0
5608	0.050781553	13.0101010	0	0	0
5609	0.058036060	13.0101010	0	0	0
5610	0.065290568	13.0101010	0	0	0
5611	0.072545076	13.0101010	0	0	0
5612	0.079799583	13.0101010	0	0	0
5613	0.087054091	13.0101010	0	0	0
5614	0.094308598	13.0101010	0	0	0
5615	0.101563106	13.0101010	0	0	0
5616	0.108817613	13.0101010	0	0	0
5617	0.116072121	13.0101010	0	0	0
5618	0.123326628	13.0101010	0	0	0
5619	0.130581136	13.0101010	0	0	0
5620	0.137835643	13.0101010	0	0	0
5621	0.145090151	13.0101010	0	0	0
5622	0.152344659	13.0101010	0	0	0
5623	0.159599166	13.0101010	0	0	0
5624	0.166853674	13.0101010	0	0	0
5625	0.174108181	13.0101010	0	0	0

5626	0.181362689	13.0101010	0	0	0
5627	0.188617196	13.0101010	0	0	0
5628	0.195871704	13.0101010	0	0	0
5629	0.203126211	13.0101010	0	0	0
5630	0.210380719	13.0101010	0	0	0
5631	0.217635227	13.0101010	0	0	0
5632	0.224889734	13.0101010	0	0	0
5633	0.232144242	13.0101010	0	0	0
5634	0.239398749	13.0101010	0	0	0
5635	0.246653257	13.0101010	0	0	0
5636	0.253907764	13.0101010	0	0	0
5637	0.261162272	13.0101010	0	0	0
5638	0.268416779	13.0101010	0	0	0
5639	0.275671287	13.0101010	0	0	0
5640	0.282925794	13.0101010	0	0	0
5641	0.290180302	13.0101010	0	0	0
5642	0.297434810	13.0101010	0	0	0
5643	0.304689317	13.0101010	0	0	0
5644	0.311943825	13.0101010	0	0	0
5645	0.319198332	13.0101010	0	0	0
5646	0.326452840	13.0101010	0	0	0
5647	0.333707347	13.0101010	0	0	0
5648	0.340961855	13.0101010	0	0	0
5649	0.348216362	13.0101010	0	0	0
5650	0.355470870	13.0101010	0	0	0
5651	0.362725378	13.0101010	0	0	0
5652	0.369979885	13.0101010	0	0	0
5653	0.377234393	13.0101010	0	0	0
5654	0.384488900	13.0101010	0	0	0
5655	0.391743408	13.0101010	0	0	0
5656	0.398997915	13.0101010	0	0	0
5657	0.406252423	13.0101010	0	0	0
5658	0.413506930	13.0101010	0	0	0
5659	0.420761438	13.0101010	0	0	0
5660	0.428015945	13.0101010	0	0	0
5661	0.435270453	13.0101010	0	0	0
5662	0.442524961	13.0101010	0	0	0
5663	0.449779468	13.0101010	0	0	0
5664	0.457033976	13.0101010	0	0	0
5665	0.464288483	13.0101010	0	0	0
5666	0.471542991	13.0101010	0	0	0
5667	0.478797498	13.0101010	0	0	0
5668	0.486052006	13.0101010	0	0	0
5669	0.493306513	13.0101010	0	0	0
5670	0.500561021	13.0101010	0	0	0

5671	0.507815529	13.0101010	0	0	0
5672	0.515070036	13.0101010	0	0	0
5673	0.522324544	13.0101010	0	0	0
5674	0.529579051	13.0101010	0	0	0
5675	0.536833559	13.0101010	0	0	0
5676	0.544088066	13.0101010	0	0	0
5677	0.551342574	13.0101010	0	0	0
5678	0.558597081	13.0101010	0	0	0
5679	0.565851589	13.0101010	0	0	0
5680	0.573106097	13.0101010	0	0	0
5681	0.580360604	13.0101010	0	0	0
5682	0.587615112	13.0101010	0	0	0
5683	0.594869619	13.0101010	0	0	0
5684	0.602124127	13.0101010	0	0	0
5685	0.609378634	13.0101010	0	0	0
5686	0.616633142	13.0101010	0	0	0
5687	0.623887649	13.0101010	0	0	0
5688	0.631142157	13.0101010	0	0	0
5689	0.638396664	13.0101010	0	0	0
5690	0.645651172	13.0101010	0	0	0
5691	0.652905680	13.0101010	0	0	0
5692	0.660160187	13.0101010	0	0	0
5693	0.667414695	13.0101010	0	0	0
5694	0.674669202	13.0101010	0	0	0
5695	0.681923710	13.0101010	0	0	0
5696	0.689178217	13.0101010	0	0	0
5697	0.696432725	13.0101010	0	0	0
5698	0.703687232	13.0101010	0	0	0
5699	0.710941740	13.0101010	0	0	0
5700	0.718196248	13.0101010	0	0	0
5701	0.000000000	13.2424242	0	0	0
5702	0.007254508	13.2424242	0	0	0
5703	0.014509015	13.2424242	0	0	0
5704	0.021763523	13.2424242	0	0	0
5705	0.029018030	13.2424242	0	0	0
5706	0.036272538	13.2424242	0	0	0
5707	0.043527045	13.2424242	0	0	0
5708	0.050781553	13.2424242	0	0	0
5709	0.058036060	13.2424242	0	0	0
5710	0.065290568	13.2424242	0	0	0
5711	0.072545076	13.2424242	0	0	0
5712	0.079799583	13.2424242	0	0	0
5713	0.087054091	13.2424242	0	0	0
5714	0.094308598	13.2424242	0	0	0
5715	0.101563106	13.2424242	0	0	0

5716	0.108817613	13.2424242	0	0	0
5717	0.116072121	13.2424242	0	0	0
5718	0.123326628	13.2424242	0	0	0
5719	0.130581136	13.2424242	0	0	0
5720	0.137835643	13.2424242	0	0	0
5721	0.145090151	13.2424242	0	0	0
5722	0.152344659	13.2424242	0	0	0
5723	0.159599166	13.2424242	0	0	0
5724	0.166853674	13.2424242	0	0	0
5725	0.174108181	13.2424242	0	0	0
5726	0.181362689	13.2424242	0	0	0
5727	0.188617196	13.2424242	0	0	0
5728	0.195871704	13.2424242	0	0	0
5729	0.203126211	13.2424242	0	0	0
5730	0.210380719	13.2424242	0	0	0
5731	0.217635227	13.2424242	0	0	0
5732	0.224889734	13.2424242	0	0	0
5733	0.232144242	13.2424242	0	0	0
5734	0.239398749	13.2424242	0	0	0
5735	0.246653257	13.2424242	0	0	0
5736	0.253907764	13.2424242	0	0	0
5737	0.261162272	13.2424242	0	0	0
5738	0.268416779	13.2424242	0	0	0
5739	0.275671287	13.2424242	0	0	0
5740	0.282925794	13.2424242	0	0	0
5741	0.290180302	13.2424242	0	0	0
5742	0.297434810	13.2424242	0	0	0
5743	0.304689317	13.2424242	0	0	0
5744	0.311943825	13.2424242	0	0	0
5745	0.319198332	13.2424242	0	0	0
5746	0.326452840	13.2424242	0	0	0
5747	0.333707347	13.2424242	0	0	0
5748	0.340961855	13.2424242	0	0	0
5749	0.348216362	13.2424242	0	0	0
5750	0.355470870	13.2424242	0	0	0
5751	0.362725378	13.2424242	0	0	0
5752	0.369979885	13.2424242	0	0	0
5753	0.377234393	13.2424242	0	0	0
5754	0.384488900	13.2424242	0	0	0
5755	0.391743408	13.2424242	0	0	0
5756	0.398997915	13.2424242	0	0	0
5757	0.406252423	13.2424242	0	0	0
5758	0.413506930	13.2424242	0	0	0
5759	0.420761438	13.2424242	0	0	0
5760	0.428015945	13.2424242	0	0	0

5761	0.435270453	13.2424242	0	0	0
5762	0.442524961	13.2424242	0	0	0
5763	0.449779468	13.2424242	0	0	0
5764	0.457033976	13.2424242	0	0	0
5765	0.464288483	13.2424242	0	0	0
5766	0.471542991	13.2424242	0	0	0
5767	0.478797498	13.2424242	0	0	0
5768	0.486052006	13.2424242	0	0	0
5769	0.493306513	13.2424242	0	0	0
5770	0.500561021	13.2424242	0	0	0
5771	0.507815529	13.2424242	0	0	0
5772	0.515070036	13.2424242	0	0	0
5773	0.522324544	13.2424242	0	0	0
5774	0.529579051	13.2424242	0	0	0
5775	0.536833559	13.2424242	0	0	0
5776	0.544088066	13.2424242	0	0	0
5777	0.551342574	13.2424242	0	0	0
5778	0.558597081	13.2424242	0	0	0
5779	0.565851589	13.2424242	0	0	0
5780	0.573106097	13.2424242	0	0	0
5781	0.580360604	13.2424242	0	0	0
5782	0.587615112	13.2424242	0	0	0
5783	0.594869619	13.2424242	0	0	0
5784	0.602124127	13.2424242	0	0	0
5785	0.609378634	13.2424242	0	0	0
5786	0.616633142	13.2424242	0	0	0
5787	0.623887649	13.2424242	0	0	0
5788	0.631142157	13.2424242	0	0	0
5789	0.638396664	13.2424242	0	0	0
5790	0.645651172	13.2424242	0	0	0
5791	0.652905680	13.2424242	0	0	0
5792	0.660160187	13.2424242	0	0	0
5793	0.667414695	13.2424242	0	0	0
5794	0.674669202	13.2424242	0	0	0
5795	0.681923710	13.2424242	0	0	0
5796	0.689178217	13.2424242	0	0	0
5797	0.696432725	13.2424242	0	0	0
5798	0.703687232	13.2424242	0	0	0
5799	0.710941740	13.2424242	0	0	0
5800	0.718196248	13.2424242	0	0	0
5801	0.000000000	13.4747475	0	0	0
5802	0.007254508	13.4747475	0	0	0
5803	0.014509015	13.4747475	0	0	0
5804	0.021763523	13.4747475	0	0	0
5805	0.029018030	13.4747475	0	0	0

5806	0.036272538	13.4747475	0	0	0
5807	0.043527045	13.4747475	0	0	0
5808	0.050781553	13.4747475	0	0	0
5809	0.058036060	13.4747475	0	0	0
5810	0.065290568	13.4747475	0	0	0
5811	0.072545076	13.4747475	0	0	0
5812	0.079799583	13.4747475	0	0	0
5813	0.087054091	13.4747475	0	0	0
5814	0.094308598	13.4747475	0	0	0
5815	0.101563106	13.4747475	0	0	0
5816	0.108817613	13.4747475	0	0	0
5817	0.116072121	13.4747475	0	0	0
5818	0.123326628	13.4747475	0	0	0
5819	0.130581136	13.4747475	0	0	0
5820	0.137835643	13.4747475	0	0	0
5821	0.145090151	13.4747475	0	0	0
5822	0.152344659	13.4747475	0	0	0
5823	0.159599166	13.4747475	0	0	0
5824	0.166853674	13.4747475	0	0	0
5825	0.174108181	13.4747475	0	0	0
5826	0.181362689	13.4747475	0	0	0
5827	0.188617196	13.4747475	0	0	0
5828	0.195871704	13.4747475	0	0	0
5829	0.203126211	13.4747475	0	0	0
5830	0.210380719	13.4747475	0	0	0
5831	0.217635227	13.4747475	0	0	0
5832	0.224889734	13.4747475	0	0	0
5833	0.232144242	13.4747475	0	0	0
5834	0.239398749	13.4747475	0	0	0
5835	0.246653257	13.4747475	0	0	0
5836	0.253907764	13.4747475	0	0	0
5837	0.261162272	13.4747475	0	0	0
5838	0.268416779	13.4747475	0	0	0
5839	0.275671287	13.4747475	0	0	0
5840	0.282925794	13.4747475	0	0	0
5841	0.290180302	13.4747475	0	0	0
5842	0.297434810	13.4747475	0	0	0
5843	0.304689317	13.4747475	0	0	0
5844	0.311943825	13.4747475	0	0	0
5845	0.319198332	13.4747475	0	0	0
5846	0.326452840	13.4747475	0	0	0
5847	0.333707347	13.4747475	0	0	0
5848	0.340961855	13.4747475	0	0	0
5849	0.348216362	13.4747475	0	0	0
5850	0.355470870	13.4747475	0	0	0

5851	0.362725378	13.4747475	0	0	0
5852	0.369979885	13.4747475	0	0	0
5853	0.377234393	13.4747475	0	0	0
5854	0.384488900	13.4747475	0	0	0
5855	0.391743408	13.4747475	0	0	0
5856	0.398997915	13.4747475	0	0	0
5857	0.406252423	13.4747475	0	0	0
5858	0.413506930	13.4747475	0	0	0
5859	0.420761438	13.4747475	0	0	0
5860	0.428015945	13.4747475	0	0	0
5861	0.435270453	13.4747475	0	0	0
5862	0.442524961	13.4747475	0	0	0
5863	0.449779468	13.4747475	0	0	0
5864	0.457033976	13.4747475	0	0	0
5865	0.464288483	13.4747475	0	0	0
5866	0.471542991	13.4747475	0	0	0
5867	0.478797498	13.4747475	0	0	0
5868	0.486052006	13.4747475	0	0	0
5869	0.493306513	13.4747475	0	0	0
5870	0.500561021	13.4747475	0	0	0
5871	0.507815529	13.4747475	0	0	0
5872	0.515070036	13.4747475	0	0	0
5873	0.522324544	13.4747475	0	0	0
5874	0.529579051	13.4747475	0	0	0
5875	0.536833559	13.4747475	0	0	0
5876	0.544088066	13.4747475	0	0	0
5877	0.551342574	13.4747475	0	0	0
5878	0.558597081	13.4747475	0	0	0
5879	0.565851589	13.4747475	0	0	0
5880	0.573106097	13.4747475	0	0	0
5881	0.580360604	13.4747475	0	0	0
5882	0.587615112	13.4747475	0	0	0
5883	0.594869619	13.4747475	0	0	0
5884	0.602124127	13.4747475	0	0	0
5885	0.609378634	13.4747475	0	0	0
5886	0.616633142	13.4747475	0	0	0
5887	0.623887649	13.4747475	0	0	0
5888	0.631142157	13.4747475	0	0	0
5889	0.638396664	13.4747475	0	0	0
5890	0.645651172	13.4747475	0	0	0
5891	0.652905680	13.4747475	0	0	0
5892	0.660160187	13.4747475	0	0	0
5893	0.667414695	13.4747475	0	0	0
5894	0.674669202	13.4747475	0	0	0
5895	0.681923710	13.4747475	0	0	0

5896	0.689178217	13.4747475	0	0	0
5897	0.696432725	13.4747475	0	0	0
5898	0.703687232	13.4747475	0	0	0
5899	0.710941740	13.4747475	0	0	0
5900	0.718196248	13.4747475	0	0	0
5901	0.000000000	13.7070707	0	0	0
5902	0.007254508	13.7070707	0	0	0
5903	0.014509015	13.7070707	0	0	0
5904	0.021763523	13.7070707	0	0	0
5905	0.029018030	13.7070707	0	0	0
5906	0.036272538	13.7070707	0	0	0
5907	0.043527045	13.7070707	0	0	0
5908	0.050781553	13.7070707	0	0	0
5909	0.058036060	13.7070707	0	0	0
5910	0.065290568	13.7070707	0	0	0
5911	0.072545076	13.7070707	0	0	0
5912	0.079799583	13.7070707	0	0	0
5913	0.087054091	13.7070707	0	0	0
5914	0.094308598	13.7070707	0	0	0
5915	0.101563106	13.7070707	0	0	0
5916	0.108817613	13.7070707	0	0	0
5917	0.116072121	13.7070707	0	0	0
5918	0.123326628	13.7070707	0	0	0
5919	0.130581136	13.7070707	0	0	0
5920	0.137835643	13.7070707	0	0	0
5921	0.145090151	13.7070707	0	0	0
5922	0.152344659	13.7070707	0	0	0
5923	0.159599166	13.7070707	0	0	0
5924	0.166853674	13.7070707	0	0	0
5925	0.174108181	13.7070707	0	0	0
5926	0.181362689	13.7070707	0	0	0
5927	0.188617196	13.7070707	0	0	0
5928	0.195871704	13.7070707	0	0	0
5929	0.203126211	13.7070707	0	0	0
5930	0.210380719	13.7070707	0	0	0
5931	0.217635227	13.7070707	0	0	0
5932	0.224889734	13.7070707	0	0	0
5933	0.232144242	13.7070707	0	0	0
5934	0.239398749	13.7070707	0	0	0
5935	0.246653257	13.7070707	0	0	0
5936	0.253907764	13.7070707	0	0	0
5937	0.261162272	13.7070707	0	0	0
5938	0.268416779	13.7070707	0	0	0
5939	0.275671287	13.7070707	0	0	0
5940	0.282925794	13.7070707	0	0	0

5941	0.290180302	13.7070707	0	0	0
5942	0.297434810	13.7070707	0	0	0
5943	0.304689317	13.7070707	0	0	0
5944	0.311943825	13.7070707	0	0	0
5945	0.319198332	13.7070707	0	0	0
5946	0.326452840	13.7070707	0	0	0
5947	0.333707347	13.7070707	0	0	0
5948	0.340961855	13.7070707	0	0	0
5949	0.348216362	13.7070707	0	0	0
5950	0.355470870	13.7070707	0	0	0
5951	0.362725378	13.7070707	0	0	0
5952	0.369979885	13.7070707	0	0	0
5953	0.377234393	13.7070707	0	0	0
5954	0.384488900	13.7070707	0	0	0
5955	0.391743408	13.7070707	0	0	0
5956	0.398997915	13.7070707	0	0	0
5957	0.406252423	13.7070707	0	0	0
5958	0.413506930	13.7070707	0	0	0
5959	0.420761438	13.7070707	0	0	0
5960	0.428015945	13.7070707	0	0	0
5961	0.435270453	13.7070707	0	0	0
5962	0.442524961	13.7070707	0	0	0
5963	0.449779468	13.7070707	0	0	0
5964	0.457033976	13.7070707	0	0	0
5965	0.464288483	13.7070707	0	0	0
5966	0.471542991	13.7070707	0	0	0
5967	0.478797498	13.7070707	0	0	0
5968	0.486052006	13.7070707	0	0	0
5969	0.493306513	13.7070707	0	0	0
5970	0.500561021	13.7070707	0	0	0
5971	0.507815529	13.7070707	0	0	0
5972	0.515070036	13.7070707	0	0	0
5973	0.522324544	13.7070707	0	0	0
5974	0.529579051	13.7070707	0	0	0
5975	0.536833559	13.7070707	0	0	0
5976	0.544088066	13.7070707	0	0	0
5977	0.551342574	13.7070707	0	0	0
5978	0.558597081	13.7070707	0	0	0
5979	0.565851589	13.7070707	0	0	0
5980	0.573106097	13.7070707	0	0	0
5981	0.580360604	13.7070707	0	0	0
5982	0.587615112	13.7070707	0	0	0
5983	0.594869619	13.7070707	0	0	0
5984	0.602124127	13.7070707	0	0	0
5985	0.609378634	13.7070707	0	0	0

5986	0.616633142	13.7070707	0	0	0
5987	0.623887649	13.7070707	0	0	0
5988	0.631142157	13.7070707	0	0	0
5989	0.638396664	13.7070707	0	0	0
5990	0.645651172	13.7070707	0	0	0
5991	0.652905680	13.7070707	0	0	0
5992	0.660160187	13.7070707	0	0	0
5993	0.667414695	13.7070707	0	0	0
5994	0.674669202	13.7070707	0	0	0
5995	0.681923710	13.7070707	0	0	0
5996	0.689178217	13.7070707	0	0	0
5997	0.696432725	13.7070707	0	0	0
5998	0.703687232	13.7070707	0	0	0
5999	0.710941740	13.7070707	0	0	0
6000	0.718196248	13.7070707	0	0	0
6001	0.000000000	13.9393939	0	0	0
6002	0.007254508	13.9393939	0	0	0
6003	0.014509015	13.9393939	0	0	0
6004	0.021763523	13.9393939	0	0	0
6005	0.029018030	13.9393939	0	0	0
6006	0.036272538	13.9393939	0	0	0
6007	0.043527045	13.9393939	0	0	0
6008	0.050781553	13.9393939	0	0	0
6009	0.058036060	13.9393939	0	0	0
6010	0.065290568	13.9393939	0	0	0
6011	0.072545076	13.9393939	0	0	0
6012	0.079799583	13.9393939	0	0	0
6013	0.087054091	13.9393939	0	0	0
6014	0.094308598	13.9393939	0	0	0
6015	0.101563106	13.9393939	0	0	0
6016	0.108817613	13.9393939	0	0	0
6017	0.116072121	13.9393939	0	0	0
6018	0.123326628	13.9393939	0	0	0
6019	0.130581136	13.9393939	0	0	0
6020	0.137835643	13.9393939	0	0	0
6021	0.145090151	13.9393939	0	0	0
6022	0.152344659	13.9393939	0	0	0
6023	0.159599166	13.9393939	0	0	0
6024	0.166853674	13.9393939	0	0	0
6025	0.174108181	13.9393939	0	0	0
6026	0.181362689	13.9393939	0	0	0
6027	0.188617196	13.9393939	0	0	0
6028	0.195871704	13.9393939	0	0	0
6029	0.203126211	13.9393939	0	0	0
6030	0.210380719	13.9393939	0	0	0

6031	0.217635227	13.9393939	0	0	0
6032	0.224889734	13.9393939	0	0	0
6033	0.232144242	13.9393939	0	0	0
6034	0.239398749	13.9393939	0	0	0
6035	0.246653257	13.9393939	0	0	0
6036	0.253907764	13.9393939	0	0	0
6037	0.261162272	13.9393939	0	0	0
6038	0.268416779	13.9393939	0	0	0
6039	0.275671287	13.9393939	0	0	0
6040	0.282925794	13.9393939	0	0	0
6041	0.290180302	13.9393939	0	0	0
6042	0.297434810	13.9393939	0	0	0
6043	0.304689317	13.9393939	0	0	0
6044	0.311943825	13.9393939	0	0	0
6045	0.319198332	13.9393939	0	0	0
6046	0.326452840	13.9393939	0	0	0
6047	0.333707347	13.9393939	0	0	0
6048	0.340961855	13.9393939	0	0	0
6049	0.348216362	13.9393939	0	0	0
6050	0.355470870	13.9393939	0	0	0
6051	0.362725378	13.9393939	0	0	0
6052	0.369979885	13.9393939	0	0	0
6053	0.377234393	13.9393939	0	0	0
6054	0.384488900	13.9393939	0	0	0
6055	0.391743408	13.9393939	0	0	0
6056	0.398997915	13.9393939	0	0	0
6057	0.406252423	13.9393939	0	0	0
6058	0.413506930	13.9393939	0	0	0
6059	0.420761438	13.9393939	0	0	0
6060	0.428015945	13.9393939	0	0	0
6061	0.435270453	13.9393939	0	0	0
6062	0.442524961	13.9393939	0	0	0
6063	0.449779468	13.9393939	0	0	0
6064	0.457033976	13.9393939	0	0	0
6065	0.464288483	13.9393939	0	0	0
6066	0.471542991	13.9393939	0	0	0
6067	0.478797498	13.9393939	0	0	0
6068	0.486052006	13.9393939	0	0	0
6069	0.493306513	13.9393939	0	0	0
6070	0.500561021	13.9393939	0	0	0
6071	0.507815529	13.9393939	0	0	0
6072	0.515070036	13.9393939	0	0	0
6073	0.522324544	13.9393939	0	0	0
6074	0.529579051	13.9393939	0	0	0
6075	0.536833559	13.9393939	0	0	0

6076	0.544088066	13.9393939	0	0	0
6077	0.551342574	13.9393939	0	0	0
6078	0.558597081	13.9393939	0	0	0
6079	0.565851589	13.9393939	0	0	0
6080	0.573106097	13.9393939	0	0	0
6081	0.580360604	13.9393939	0	0	0
6082	0.587615112	13.9393939	0	0	0
6083	0.594869619	13.9393939	0	0	0
6084	0.602124127	13.9393939	0	0	0
6085	0.609378634	13.9393939	0	0	0
6086	0.616633142	13.9393939	0	0	0
6087	0.623887649	13.9393939	0	0	0
6088	0.631142157	13.9393939	0	0	0
6089	0.638396664	13.9393939	0	0	0
6090	0.645651172	13.9393939	0	0	0
6091	0.652905680	13.9393939	0	0	0
6092	0.660160187	13.9393939	0	0	0
6093	0.667414695	13.9393939	0	0	0
6094	0.674669202	13.9393939	0	0	0
6095	0.681923710	13.9393939	0	0	0
6096	0.689178217	13.9393939	0	0	0
6097	0.696432725	13.9393939	0	0	0
6098	0.703687232	13.9393939	0	0	0
6099	0.710941740	13.9393939	0	0	0
6100	0.718196248	13.9393939	0	0	0
6101	0.000000000	14.1717172	0	0	0
6102	0.007254508	14.1717172	0	0	0
6103	0.014509015	14.1717172	0	0	0
6104	0.021763523	14.1717172	0	0	0
6105	0.029018030	14.1717172	0	0	0
6106	0.036272538	14.1717172	0	0	0
6107	0.043527045	14.1717172	0	0	0
6108	0.050781553	14.1717172	0	0	0
6109	0.058036060	14.1717172	0	0	0
6110	0.065290568	14.1717172	0	0	0
6111	0.072545076	14.1717172	0	0	0
6112	0.079799583	14.1717172	0	0	0
6113	0.087054091	14.1717172	0	0	0
6114	0.094308598	14.1717172	0	0	0
6115	0.101563106	14.1717172	0	0	0
6116	0.108817613	14.1717172	0	0	0
6117	0.116072121	14.1717172	0	0	0
6118	0.123326628	14.1717172	0	0	0
6119	0.130581136	14.1717172	0	0	0
6120	0.137835643	14.1717172	0	0	0

6121	0.145090151	14.1717172	0	0	0
6122	0.152344659	14.1717172	0	0	0
6123	0.159599166	14.1717172	0	0	0
6124	0.166853674	14.1717172	0	0	0
6125	0.174108181	14.1717172	0	0	0
6126	0.181362689	14.1717172	0	0	0
6127	0.188617196	14.1717172	0	0	0
6128	0.195871704	14.1717172	0	0	0
6129	0.203126211	14.1717172	0	0	0
6130	0.210380719	14.1717172	0	0	0
6131	0.217635227	14.1717172	0	0	0
6132	0.224889734	14.1717172	0	0	0
6133	0.232144242	14.1717172	0	0	0
6134	0.239398749	14.1717172	0	0	0
6135	0.246653257	14.1717172	0	0	0
6136	0.253907764	14.1717172	0	0	0
6137	0.261162272	14.1717172	0	0	0
6138	0.268416779	14.1717172	0	0	0
6139	0.275671287	14.1717172	0	0	0
6140	0.282925794	14.1717172	0	0	0
6141	0.290180302	14.1717172	0	0	0
6142	0.297434810	14.1717172	0	0	0
6143	0.304689317	14.1717172	0	0	0
6144	0.311943825	14.1717172	0	0	0
6145	0.319198332	14.1717172	0	0	0
6146	0.326452840	14.1717172	0	0	0
6147	0.333707347	14.1717172	0	0	0
6148	0.340961855	14.1717172	0	0	0
6149	0.348216362	14.1717172	0	0	0
6150	0.355470870	14.1717172	0	0	0
6151	0.362725378	14.1717172	0	0	0
6152	0.369979885	14.1717172	0	0	0
6153	0.377234393	14.1717172	0	0	0
6154	0.384488900	14.1717172	0	0	0
6155	0.391743408	14.1717172	0	0	0
6156	0.398997915	14.1717172	0	0	0
6157	0.406252423	14.1717172	0	0	0
6158	0.413506930	14.1717172	0	0	0
6159	0.420761438	14.1717172	0	0	0
6160	0.428015945	14.1717172	0	0	0
6161	0.435270453	14.1717172	0	0	0
6162	0.442524961	14.1717172	0	0	0
6163	0.449779468	14.1717172	0	0	0
6164	0.457033976	14.1717172	0	0	0
6165	0.464288483	14.1717172	0	0	0

6166	0.471542991	14.1717172	0	0	0
6167	0.478797498	14.1717172	0	0	0
6168	0.486052006	14.1717172	0	0	0
6169	0.493306513	14.1717172	0	0	0
6170	0.500561021	14.1717172	0	0	0
6171	0.507815529	14.1717172	0	0	0
6172	0.515070036	14.1717172	0	0	0
6173	0.522324544	14.1717172	0	0	0
6174	0.529579051	14.1717172	0	0	0
6175	0.536833559	14.1717172	0	0	0
6176	0.544088066	14.1717172	0	0	0
6177	0.551342574	14.1717172	0	0	0
6178	0.558597081	14.1717172	0	0	0
6179	0.565851589	14.1717172	0	0	0
6180	0.573106097	14.1717172	0	0	0
6181	0.580360604	14.1717172	0	0	0
6182	0.587615112	14.1717172	0	0	0
6183	0.594869619	14.1717172	0	0	0
6184	0.602124127	14.1717172	0	0	0
6185	0.609378634	14.1717172	0	0	0
6186	0.616633142	14.1717172	0	0	0
6187	0.623887649	14.1717172	0	0	0
6188	0.631142157	14.1717172	0	0	0
6189	0.638396664	14.1717172	0	0	0
6190	0.645651172	14.1717172	0	0	0
6191	0.652905680	14.1717172	0	0	0
6192	0.660160187	14.1717172	0	0	0
6193	0.667414695	14.1717172	0	0	0
6194	0.674669202	14.1717172	0	0	0
6195	0.681923710	14.1717172	0	0	0
6196	0.689178217	14.1717172	0	0	0
6197	0.696432725	14.1717172	0	0	0
6198	0.703687232	14.1717172	0	0	0
6199	0.710941740	14.1717172	0	0	0
6200	0.718196248	14.1717172	0	0	0
6201	0.000000000	14.4040404	0	0	0
6202	0.007254508	14.4040404	0	0	0
6203	0.014509015	14.4040404	0	0	0
6204	0.021763523	14.4040404	0	0	0
6205	0.029018030	14.4040404	0	0	0
6206	0.036272538	14.4040404	0	0	0
6207	0.043527045	14.4040404	0	0	0
6208	0.050781553	14.4040404	0	0	0
6209	0.058036060	14.4040404	0	0	0
6210	0.065290568	14.4040404	0	0	0

6211	0.072545076	14.4040404	0	0	0
6212	0.079799583	14.4040404	0	0	0
6213	0.087054091	14.4040404	0	0	0
6214	0.094308598	14.4040404	0	0	0
6215	0.101563106	14.4040404	0	0	0
6216	0.108817613	14.4040404	0	0	0
6217	0.116072121	14.4040404	0	0	0
6218	0.123326628	14.4040404	0	0	0
6219	0.130581136	14.4040404	0	0	0
6220	0.137835643	14.4040404	0	0	0
6221	0.145090151	14.4040404	0	0	0
6222	0.152344659	14.4040404	0	0	0
6223	0.159599166	14.4040404	0	0	0
6224	0.166853674	14.4040404	0	0	0
6225	0.174108181	14.4040404	0	0	0
6226	0.181362689	14.4040404	0	0	0
6227	0.188617196	14.4040404	0	0	0
6228	0.195871704	14.4040404	0	0	0
6229	0.203126211	14.4040404	0	0	0
6230	0.210380719	14.4040404	0	0	0
6231	0.217635227	14.4040404	0	0	0
6232	0.224889734	14.4040404	0	0	0
6233	0.232144242	14.4040404	0	0	0
6234	0.239398749	14.4040404	0	0	0
6235	0.246653257	14.4040404	0	0	0
6236	0.253907764	14.4040404	0	0	0
6237	0.261162272	14.4040404	0	0	0
6238	0.268416779	14.4040404	0	0	0
6239	0.275671287	14.4040404	0	0	0
6240	0.282925794	14.4040404	0	0	0
6241	0.290180302	14.4040404	0	0	0
6242	0.297434810	14.4040404	0	0	0
6243	0.304689317	14.4040404	0	0	0
6244	0.311943825	14.4040404	0	0	0
6245	0.319198332	14.4040404	0	0	0
6246	0.326452840	14.4040404	0	0	0
6247	0.333707347	14.4040404	0	0	0
6248	0.340961855	14.4040404	0	0	0
6249	0.348216362	14.4040404	0	0	0
6250	0.355470870	14.4040404	0	0	0
6251	0.362725378	14.4040404	0	0	0
6252	0.369979885	14.4040404	0	0	0
6253	0.377234393	14.4040404	0	0	0
6254	0.384488900	14.4040404	0	0	0
6255	0.391743408	14.4040404	0	0	0

6256	0.398997915	14.4040404	0	0	0
6257	0.406252423	14.4040404	0	0	0
6258	0.413506930	14.4040404	0	0	0
6259	0.420761438	14.4040404	0	0	0
6260	0.428015945	14.4040404	0	0	0
6261	0.435270453	14.4040404	0	0	0
6262	0.442524961	14.4040404	0	0	0
6263	0.449779468	14.4040404	0	0	0
6264	0.457033976	14.4040404	0	0	0
6265	0.464288483	14.4040404	0	0	0
6266	0.471542991	14.4040404	0	0	0
6267	0.478797498	14.4040404	0	0	0
6268	0.486052006	14.4040404	0	0	0
6269	0.493306513	14.4040404	0	0	0
6270	0.500561021	14.4040404	0	0	0
6271	0.507815529	14.4040404	0	0	0
6272	0.515070036	14.4040404	0	0	0
6273	0.522324544	14.4040404	0	0	0
6274	0.529579051	14.4040404	0	0	0
6275	0.536833559	14.4040404	0	0	0
6276	0.544088066	14.4040404	0	0	0
6277	0.551342574	14.4040404	0	0	0
6278	0.558597081	14.4040404	0	0	0
6279	0.565851589	14.4040404	0	0	0
6280	0.573106097	14.4040404	0	0	0
6281	0.580360604	14.4040404	0	0	0
6282	0.587615112	14.4040404	0	0	0
6283	0.594869619	14.4040404	0	0	0
6284	0.602124127	14.4040404	0	0	0
6285	0.609378634	14.4040404	0	0	0
6286	0.616633142	14.4040404	0	0	0
6287	0.623887649	14.4040404	0	0	0
6288	0.631142157	14.4040404	0	0	0
6289	0.638396664	14.4040404	0	0	0
6290	0.645651172	14.4040404	0	0	0
6291	0.652905680	14.4040404	0	0	0
6292	0.660160187	14.4040404	0	0	0
6293	0.667414695	14.4040404	0	0	0
6294	0.674669202	14.4040404	0	0	0
6295	0.681923710	14.4040404	0	0	0
6296	0.689178217	14.4040404	0	0	0
6297	0.696432725	14.4040404	0	0	0
6298	0.703687232	14.4040404	0	0	0
6299	0.710941740	14.4040404	0	0	0
6300	0.718196248	14.4040404	0	0	0

6301	0.000000000	14.6363636	0	0	0
6302	0.007254508	14.6363636	0	0	0
6303	0.014509015	14.6363636	0	0	0
6304	0.021763523	14.6363636	0	0	0
6305	0.029018030	14.6363636	0	0	0
6306	0.036272538	14.6363636	0	0	0
6307	0.043527045	14.6363636	0	0	0
6308	0.050781553	14.6363636	0	0	0
6309	0.058036060	14.6363636	0	0	0
6310	0.065290568	14.6363636	0	0	0
6311	0.072545076	14.6363636	0	0	0
6312	0.079799583	14.6363636	0	0	0
6313	0.087054091	14.6363636	0	0	0
6314	0.094308598	14.6363636	0	0	0
6315	0.101563106	14.6363636	0	0	0
6316	0.108817613	14.6363636	0	0	0
6317	0.116072121	14.6363636	0	0	0
6318	0.123326628	14.6363636	0	0	0
6319	0.130581136	14.6363636	0	0	0
6320	0.137835643	14.6363636	0	0	0
6321	0.145090151	14.6363636	0	0	0
6322	0.152344659	14.6363636	0	0	0
6323	0.159599166	14.6363636	0	0	0
6324	0.166853674	14.6363636	0	0	0
6325	0.174108181	14.6363636	0	0	0
6326	0.181362689	14.6363636	0	0	0
6327	0.188617196	14.6363636	0	0	0
6328	0.195871704	14.6363636	0	0	0
6329	0.203126211	14.6363636	0	0	0
6330	0.210380719	14.6363636	0	0	0
6331	0.217635227	14.6363636	0	0	0
6332	0.224889734	14.6363636	0	0	0
6333	0.232144242	14.6363636	0	0	0
6334	0.239398749	14.6363636	0	0	0
6335	0.246653257	14.6363636	0	0	0
6336	0.253907764	14.6363636	0	0	0
6337	0.261162272	14.6363636	0	0	0
6338	0.268416779	14.6363636	0	0	0
6339	0.275671287	14.6363636	0	0	0
6340	0.282925794	14.6363636	0	0	0
6341	0.290180302	14.6363636	0	0	0
6342	0.297434810	14.6363636	0	0	0
6343	0.304689317	14.6363636	0	0	0
6344	0.311943825	14.6363636	0	0	0
6345	0.319198332	14.6363636	0	0	0

6346	0.326452840	14.6363636	0	0	0
6347	0.333707347	14.6363636	0	0	0
6348	0.340961855	14.6363636	0	0	0
6349	0.348216362	14.6363636	0	0	0
6350	0.355470870	14.6363636	0	0	0
6351	0.362725378	14.6363636	0	0	0
6352	0.369979885	14.6363636	0	0	0
6353	0.377234393	14.6363636	0	0	0
6354	0.384488900	14.6363636	0	0	0
6355	0.391743408	14.6363636	0	0	0
6356	0.398997915	14.6363636	0	0	0
6357	0.406252423	14.6363636	0	0	0
6358	0.413506930	14.6363636	0	0	0
6359	0.420761438	14.6363636	0	0	0
6360	0.428015945	14.6363636	0	0	0
6361	0.435270453	14.6363636	0	0	0
6362	0.442524961	14.6363636	0	0	0
6363	0.449779468	14.6363636	0	0	0
6364	0.457033976	14.6363636	0	0	0
6365	0.464288483	14.6363636	0	0	0
6366	0.471542991	14.6363636	0	0	0
6367	0.478797498	14.6363636	0	0	0
6368	0.486052006	14.6363636	0	0	0
6369	0.493306513	14.6363636	0	0	0
6370	0.500561021	14.6363636	0	0	0
6371	0.507815529	14.6363636	0	0	0
6372	0.515070036	14.6363636	0	0	0
6373	0.522324544	14.6363636	0	0	0
6374	0.529579051	14.6363636	0	0	0
6375	0.536833559	14.6363636	0	0	0
6376	0.544088066	14.6363636	0	0	0
6377	0.551342574	14.6363636	0	0	0
6378	0.558597081	14.6363636	0	0	0
6379	0.565851589	14.6363636	0	0	0
6380	0.573106097	14.6363636	0	0	0
6381	0.580360604	14.6363636	0	0	0
6382	0.587615112	14.6363636	0	0	0
6383	0.594869619	14.6363636	0	0	0
6384	0.602124127	14.6363636	0	0	0
6385	0.609378634	14.6363636	0	0	0
6386	0.616633142	14.6363636	0	0	0
6387	0.623887649	14.6363636	0	0	0
6388	0.631142157	14.6363636	0	0	0
6389	0.638396664	14.6363636	0	0	0
6390	0.645651172	14.6363636	0	0	0

6391	0.652905680	14.6363636	0	0	0
6392	0.660160187	14.6363636	0	0	0
6393	0.667414695	14.6363636	0	0	0
6394	0.674669202	14.6363636	0	0	0
6395	0.681923710	14.6363636	0	0	0
6396	0.689178217	14.6363636	0	0	0
6397	0.696432725	14.6363636	0	0	0
6398	0.703687232	14.6363636	0	0	0
6399	0.710941740	14.6363636	0	0	0
6400	0.718196248	14.6363636	0	0	0
6401	0.000000000	14.8686869	0	0	0
6402	0.007254508	14.8686869	0	0	0
6403	0.014509015	14.8686869	0	0	0
6404	0.021763523	14.8686869	0	0	0
6405	0.029018030	14.8686869	0	0	0
6406	0.036272538	14.8686869	0	0	0
6407	0.043527045	14.8686869	0	0	0
6408	0.050781553	14.8686869	0	0	0
6409	0.058036060	14.8686869	0	0	0
6410	0.065290568	14.8686869	0	0	0
6411	0.072545076	14.8686869	0	0	0
6412	0.079799583	14.8686869	0	0	0
6413	0.087054091	14.8686869	0	0	0
6414	0.094308598	14.8686869	0	0	0
6415	0.101563106	14.8686869	0	0	0
6416	0.108817613	14.8686869	0	0	0
6417	0.116072121	14.8686869	0	0	0
6418	0.123326628	14.8686869	0	0	0
6419	0.130581136	14.8686869	0	0	0
6420	0.137835643	14.8686869	0	0	0
6421	0.145090151	14.8686869	0	0	0
6422	0.152344659	14.8686869	0	0	0
6423	0.159599166	14.8686869	0	0	0
6424	0.166853674	14.8686869	0	0	0
6425	0.174108181	14.8686869	0	0	0
6426	0.181362689	14.8686869	0	0	0
6427	0.188617196	14.8686869	0	0	0
6428	0.195871704	14.8686869	0	0	0
6429	0.203126211	14.8686869	0	0	0
6430	0.210380719	14.8686869	0	0	0
6431	0.217635227	14.8686869	0	0	0
6432	0.224889734	14.8686869	0	0	0
6433	0.232144242	14.8686869	0	0	0
6434	0.239398749	14.8686869	0	0	0
6435	0.246653257	14.8686869	0	0	0

6436	0.253907764	14.8686869	0	0	0
6437	0.261162272	14.8686869	0	0	0
6438	0.268416779	14.8686869	0	0	0
6439	0.275671287	14.8686869	0	0	0
6440	0.282925794	14.8686869	0	0	0
6441	0.290180302	14.8686869	0	0	0
6442	0.297434810	14.8686869	0	0	0
6443	0.304689317	14.8686869	0	0	0
6444	0.311943825	14.8686869	0	0	0
6445	0.319198332	14.8686869	0	0	0
6446	0.326452840	14.8686869	0	0	0
6447	0.333707347	14.8686869	0	0	0
6448	0.340961855	14.8686869	0	0	0
6449	0.348216362	14.8686869	0	0	0
6450	0.355470870	14.8686869	0	0	0
6451	0.362725378	14.8686869	0	0	0
6452	0.369979885	14.8686869	0	0	0
6453	0.377234393	14.8686869	0	0	0
6454	0.384488900	14.8686869	0	0	0
6455	0.391743408	14.8686869	0	0	0
6456	0.398997915	14.8686869	0	0	0
6457	0.406252423	14.8686869	0	0	0
6458	0.413506930	14.8686869	0	0	0
6459	0.420761438	14.8686869	0	0	0
6460	0.428015945	14.8686869	0	0	0
6461	0.435270453	14.8686869	0	0	0
6462	0.442524961	14.8686869	0	0	0
6463	0.449779468	14.8686869	0	0	0
6464	0.457033976	14.8686869	0	0	0
6465	0.464288483	14.8686869	0	0	0
6466	0.471542991	14.8686869	0	0	0
6467	0.478797498	14.8686869	0	0	0
6468	0.486052006	14.8686869	0	0	0
6469	0.493306513	14.8686869	0	0	0
6470	0.500561021	14.8686869	0	0	0
6471	0.507815529	14.8686869	0	0	0
6472	0.515070036	14.8686869	0	0	0
6473	0.522324544	14.8686869	0	0	0
6474	0.529579051	14.8686869	0	0	0
6475	0.536833559	14.8686869	0	0	0
6476	0.544088066	14.8686869	0	0	0
6477	0.551342574	14.8686869	0	0	0
6478	0.558597081	14.8686869	0	0	0
6479	0.565851589	14.8686869	0	0	0
6480	0.573106097	14.8686869	0	0	0

6481	0.580360604	14.8686869	0	0	0
6482	0.587615112	14.8686869	0	0	0
6483	0.594869619	14.8686869	0	0	0
6484	0.602124127	14.8686869	0	0	0
6485	0.609378634	14.8686869	0	0	0
6486	0.616633142	14.8686869	0	0	0
6487	0.623887649	14.8686869	0	0	0
6488	0.631142157	14.8686869	0	0	0
6489	0.638396664	14.8686869	0	0	0
6490	0.645651172	14.8686869	0	0	0
6491	0.652905680	14.8686869	0	0	0
6492	0.660160187	14.8686869	0	0	0
6493	0.667414695	14.8686869	0	0	0
6494	0.674669202	14.8686869	0	0	0
6495	0.681923710	14.8686869	0	0	0
6496	0.689178217	14.8686869	0	0	0
6497	0.696432725	14.8686869	0	0	0
6498	0.703687232	14.8686869	0	0	0
6499	0.710941740	14.8686869	0	0	0
6500	0.718196248	14.8686869	0	0	0
6501	0.000000000	15.1010101	0	0	0
6502	0.007254508	15.1010101	0	0	0
6503	0.014509015	15.1010101	0	0	0
6504	0.021763523	15.1010101	0	0	0
6505	0.029018030	15.1010101	0	0	0
6506	0.036272538	15.1010101	0	0	0
6507	0.043527045	15.1010101	0	0	0
6508	0.050781553	15.1010101	0	0	0
6509	0.058036060	15.1010101	0	0	0
6510	0.065290568	15.1010101	0	0	0
6511	0.072545076	15.1010101	0	0	0
6512	0.079799583	15.1010101	0	0	0
6513	0.087054091	15.1010101	0	0	0
6514	0.094308598	15.1010101	0	0	0
6515	0.101563106	15.1010101	0	0	0
6516	0.108817613	15.1010101	0	0	0
6517	0.116072121	15.1010101	0	0	0
6518	0.123326628	15.1010101	0	0	0
6519	0.130581136	15.1010101	0	0	0
6520	0.137835643	15.1010101	0	0	0
6521	0.145090151	15.1010101	0	0	0
6522	0.152344659	15.1010101	0	0	0
6523	0.159599166	15.1010101	0	0	0
6524	0.166853674	15.1010101	0	0	0
6525	0.174108181	15.1010101	0	0	0

6526	0.181362689	15.1010101	0	0	0
6527	0.188617196	15.1010101	0	0	0
6528	0.195871704	15.1010101	0	0	0
6529	0.203126211	15.1010101	0	0	0
6530	0.210380719	15.1010101	0	0	0
6531	0.217635227	15.1010101	0	0	0
6532	0.224889734	15.1010101	0	0	0
6533	0.232144242	15.1010101	0	0	0
6534	0.239398749	15.1010101	0	0	0
6535	0.246653257	15.1010101	0	0	0
6536	0.253907764	15.1010101	0	0	0
6537	0.261162272	15.1010101	0	0	0
6538	0.268416779	15.1010101	0	0	0
6539	0.275671287	15.1010101	0	0	0
6540	0.282925794	15.1010101	0	0	0
6541	0.290180302	15.1010101	0	0	0
6542	0.297434810	15.1010101	0	0	0
6543	0.304689317	15.1010101	0	0	0
6544	0.311943825	15.1010101	0	0	0
6545	0.319198332	15.1010101	0	0	0
6546	0.326452840	15.1010101	0	0	0
6547	0.333707347	15.1010101	0	0	0
6548	0.340961855	15.1010101	0	0	0
6549	0.348216362	15.1010101	0	0	0
6550	0.355470870	15.1010101	0	0	0
6551	0.362725378	15.1010101	0	0	0
6552	0.369979885	15.1010101	0	0	0
6553	0.377234393	15.1010101	0	0	0
6554	0.384488900	15.1010101	0	0	0
6555	0.391743408	15.1010101	0	0	0
6556	0.398997915	15.1010101	0	0	0
6557	0.406252423	15.1010101	0	0	0
6558	0.413506930	15.1010101	0	0	0
6559	0.420761438	15.1010101	0	0	0
6560	0.428015945	15.1010101	0	0	0
6561	0.435270453	15.1010101	0	0	0
6562	0.442524961	15.1010101	0	0	0
6563	0.449779468	15.1010101	0	0	0
6564	0.457033976	15.1010101	0	0	0
6565	0.464288483	15.1010101	0	0	0
6566	0.471542991	15.1010101	0	0	0
6567	0.478797498	15.1010101	0	0	0
6568	0.486052006	15.1010101	0	0	0
6569	0.493306513	15.1010101	0	0	0
6570	0.500561021	15.1010101	0	0	0

6571	0.507815529	15.1010101	0	0	0
6572	0.515070036	15.1010101	0	0	0
6573	0.522324544	15.1010101	0	0	0
6574	0.529579051	15.1010101	0	0	0
6575	0.536833559	15.1010101	0	0	0
6576	0.544088066	15.1010101	0	0	0
6577	0.551342574	15.1010101	0	0	0
6578	0.558597081	15.1010101	0	0	0
6579	0.565851589	15.1010101	0	0	0
6580	0.573106097	15.1010101	0	0	0
6581	0.580360604	15.1010101	0	0	0
6582	0.587615112	15.1010101	0	0	0
6583	0.594869619	15.1010101	0	0	0
6584	0.602124127	15.1010101	0	0	0
6585	0.609378634	15.1010101	0	0	0
6586	0.616633142	15.1010101	0	0	0
6587	0.623887649	15.1010101	0	0	0
6588	0.631142157	15.1010101	0	0	0
6589	0.638396664	15.1010101	0	0	0
6590	0.645651172	15.1010101	0	0	0
6591	0.652905680	15.1010101	0	0	0
6592	0.660160187	15.1010101	0	0	0
6593	0.667414695	15.1010101	0	0	0
6594	0.674669202	15.1010101	0	0	0
6595	0.681923710	15.1010101	0	0	0
6596	0.689178217	15.1010101	0	0	0
6597	0.696432725	15.1010101	0	0	0
6598	0.703687232	15.1010101	0	0	0
6599	0.710941740	15.1010101	0	0	0
6600	0.718196248	15.1010101	0	0	0
6601	0.000000000	15.3333333	0	0	0
6602	0.007254508	15.3333333	0	0	0
6603	0.014509015	15.3333333	0	0	0
6604	0.021763523	15.3333333	0	0	0
6605	0.029018030	15.3333333	0	0	0
6606	0.036272538	15.3333333	0	0	0
6607	0.043527045	15.3333333	0	0	0
6608	0.050781553	15.3333333	0	0	0
6609	0.058036060	15.3333333	0	0	0
6610	0.065290568	15.3333333	0	0	0
6611	0.072545076	15.3333333	0	0	0
6612	0.079799583	15.3333333	0	0	0
6613	0.087054091	15.3333333	0	0	0
6614	0.094308598	15.3333333	0	0	0
6615	0.101563106	15.3333333	0	0	0

6616	0.108817613	15.3333333	0	0	0
6617	0.116072121	15.3333333	0	0	0
6618	0.123326628	15.3333333	0	0	0
6619	0.130581136	15.3333333	0	0	0
6620	0.137835643	15.3333333	0	0	0
6621	0.145090151	15.3333333	0	0	0
6622	0.152344659	15.3333333	0	0	0
6623	0.159599166	15.3333333	0	0	0
6624	0.166853674	15.3333333	0	0	0
6625	0.174108181	15.3333333	0	0	0
6626	0.181362689	15.3333333	0	0	0
6627	0.188617196	15.3333333	0	0	0
6628	0.195871704	15.3333333	0	0	0
6629	0.203126211	15.3333333	0	0	0
6630	0.210380719	15.3333333	0	0	0
6631	0.217635227	15.3333333	0	0	0
6632	0.224889734	15.3333333	0	0	0
6633	0.232144242	15.3333333	0	0	0
6634	0.239398749	15.3333333	0	0	0
6635	0.246653257	15.3333333	0	0	0
6636	0.253907764	15.3333333	0	0	0
6637	0.261162272	15.3333333	0	0	0
6638	0.268416779	15.3333333	0	0	0
6639	0.275671287	15.3333333	0	0	0
6640	0.282925794	15.3333333	0	0	0
6641	0.290180302	15.3333333	0	0	0
6642	0.297434810	15.3333333	0	0	0
6643	0.304689317	15.3333333	0	0	0
6644	0.311943825	15.3333333	0	0	0
6645	0.319198332	15.3333333	0	0	0
6646	0.326452840	15.3333333	0	0	0
6647	0.333707347	15.3333333	0	0	0
6648	0.340961855	15.3333333	0	0	0
6649	0.348216362	15.3333333	0	0	0
6650	0.355470870	15.3333333	0	0	0
6651	0.362725378	15.3333333	0	0	0
6652	0.369979885	15.3333333	0	0	0
6653	0.377234393	15.3333333	0	0	0
6654	0.384488900	15.3333333	0	0	0
6655	0.391743408	15.3333333	0	0	0
6656	0.398997915	15.3333333	0	0	0
6657	0.406252423	15.3333333	0	0	0
6658	0.413506930	15.3333333	0	0	0
6659	0.420761438	15.3333333	0	0	0
6660	0.428015945	15.3333333	0	0	0

6661	0.435270453	15.3333333	0	0	0
6662	0.442524961	15.3333333	0	0	0
6663	0.449779468	15.3333333	0	0	0
6664	0.457033976	15.3333333	0	0	0
6665	0.464288483	15.3333333	0	0	0
6666	0.471542991	15.3333333	0	0	0
6667	0.478797498	15.3333333	0	0	0
6668	0.486052006	15.3333333	0	0	0
6669	0.493306513	15.3333333	0	0	0
6670	0.500561021	15.3333333	0	0	0
6671	0.507815529	15.3333333	0	0	0
6672	0.515070036	15.3333333	0	0	0
6673	0.522324544	15.3333333	0	0	0
6674	0.529579051	15.3333333	0	0	0
6675	0.536833559	15.3333333	0	0	0
6676	0.544088066	15.3333333	0	0	0
6677	0.551342574	15.3333333	0	0	0
6678	0.558597081	15.3333333	0	0	0
6679	0.565851589	15.3333333	0	0	0
6680	0.573106097	15.3333333	0	0	0
6681	0.580360604	15.3333333	0	0	0
6682	0.587615112	15.3333333	0	0	0
6683	0.594869619	15.3333333	0	0	0
6684	0.602124127	15.3333333	0	0	0
6685	0.609378634	15.3333333	0	0	0
6686	0.616633142	15.3333333	0	0	0
6687	0.623887649	15.3333333	0	0	0
6688	0.631142157	15.3333333	0	0	0
6689	0.638396664	15.3333333	0	0	0
6690	0.645651172	15.3333333	0	0	0
6691	0.652905680	15.3333333	0	0	0
6692	0.660160187	15.3333333	0	0	0
6693	0.667414695	15.3333333	0	0	0
6694	0.674669202	15.3333333	0	0	0
6695	0.681923710	15.3333333	0	0	0
6696	0.689178217	15.3333333	0	0	0
6697	0.696432725	15.3333333	0	0	0
6698	0.703687232	15.3333333	0	0	0
6699	0.710941740	15.3333333	0	0	0
6700	0.718196248	15.3333333	0	0	0
6701	0.000000000	15.5656566	0	0	0
6702	0.007254508	15.5656566	0	0	0
6703	0.014509015	15.5656566	0	0	0
6704	0.021763523	15.5656566	0	0	0
6705	0.029018030	15.5656566	0	0	0

6706	0.036272538	15.5656566	0	0	0
6707	0.043527045	15.5656566	0	0	0
6708	0.050781553	15.5656566	0	0	0
6709	0.058036060	15.5656566	0	0	0
6710	0.065290568	15.5656566	0	0	0
6711	0.072545076	15.5656566	0	0	0
6712	0.079799583	15.5656566	0	0	0
6713	0.087054091	15.5656566	0	0	0
6714	0.094308598	15.5656566	0	0	0
6715	0.101563106	15.5656566	0	0	0
6716	0.108817613	15.5656566	0	0	0
6717	0.116072121	15.5656566	0	0	0
6718	0.123326628	15.5656566	0	0	0
6719	0.130581136	15.5656566	0	0	0
6720	0.137835643	15.5656566	0	0	0
6721	0.145090151	15.5656566	0	0	0
6722	0.152344659	15.5656566	0	0	0
6723	0.159599166	15.5656566	0	0	0
6724	0.166853674	15.5656566	0	0	0
6725	0.174108181	15.5656566	0	0	0
6726	0.181362689	15.5656566	0	0	0
6727	0.188617196	15.5656566	0	0	0
6728	0.195871704	15.5656566	0	0	0
6729	0.203126211	15.5656566	0	0	0
6730	0.210380719	15.5656566	0	0	0
6731	0.217635227	15.5656566	0	0	0
6732	0.224889734	15.5656566	0	0	0
6733	0.232144242	15.5656566	0	0	0
6734	0.239398749	15.5656566	0	0	0
6735	0.246653257	15.5656566	0	0	0
6736	0.253907764	15.5656566	0	0	0
6737	0.261162272	15.5656566	0	0	0
6738	0.268416779	15.5656566	0	0	0
6739	0.275671287	15.5656566	0	0	0
6740	0.282925794	15.5656566	0	0	0
6741	0.290180302	15.5656566	0	0	0
6742	0.297434810	15.5656566	0	0	0
6743	0.304689317	15.5656566	0	0	0
6744	0.311943825	15.5656566	0	0	0
6745	0.319198332	15.5656566	0	0	0
6746	0.326452840	15.5656566	0	0	0
6747	0.333707347	15.5656566	0	0	0
6748	0.340961855	15.5656566	0	0	0
6749	0.348216362	15.5656566	0	0	0
6750	0.355470870	15.5656566	0	0	0

6751	0.362725378	15.5656566	0	0	0
6752	0.369979885	15.5656566	0	0	0
6753	0.377234393	15.5656566	0	0	0
6754	0.384488900	15.5656566	0	0	0
6755	0.391743408	15.5656566	0	0	0
6756	0.398997915	15.5656566	0	0	0
6757	0.406252423	15.5656566	0	0	0
6758	0.413506930	15.5656566	0	0	0
6759	0.420761438	15.5656566	0	0	0
6760	0.428015945	15.5656566	0	0	0
6761	0.435270453	15.5656566	0	0	0
6762	0.442524961	15.5656566	0	0	0
6763	0.449779468	15.5656566	0	0	0
6764	0.457033976	15.5656566	0	0	0
6765	0.464288483	15.5656566	0	0	0
6766	0.471542991	15.5656566	0	0	0
6767	0.478797498	15.5656566	0	0	0
6768	0.486052006	15.5656566	0	0	0
6769	0.493306513	15.5656566	0	0	0
6770	0.500561021	15.5656566	0	0	0
6771	0.507815529	15.5656566	0	0	0
6772	0.515070036	15.5656566	0	0	0
6773	0.522324544	15.5656566	0	0	0
6774	0.529579051	15.5656566	0	0	0
6775	0.536833559	15.5656566	0	0	0
6776	0.544088066	15.5656566	0	0	0
6777	0.551342574	15.5656566	0	0	0
6778	0.558597081	15.5656566	0	0	0
6779	0.565851589	15.5656566	0	0	0
6780	0.573106097	15.5656566	0	0	0
6781	0.580360604	15.5656566	0	0	0
6782	0.587615112	15.5656566	0	0	0
6783	0.594869619	15.5656566	0	0	0
6784	0.602124127	15.5656566	0	0	0
6785	0.609378634	15.5656566	0	0	0
6786	0.616633142	15.5656566	0	0	0
6787	0.623887649	15.5656566	0	0	0
6788	0.631142157	15.5656566	0	0	0
6789	0.638396664	15.5656566	0	0	0
6790	0.645651172	15.5656566	0	0	0
6791	0.652905680	15.5656566	0	0	0
6792	0.660160187	15.5656566	0	0	0
6793	0.667414695	15.5656566	0	0	0
6794	0.674669202	15.5656566	0	0	0
6795	0.681923710	15.5656566	0	0	0

6796	0.689178217	15.5656566	0	0	0
6797	0.696432725	15.5656566	0	0	0
6798	0.703687232	15.5656566	0	0	0
6799	0.710941740	15.5656566	0	0	0
6800	0.718196248	15.5656566	0	0	0
6801	0.000000000	15.7979798	0	0	0
6802	0.007254508	15.7979798	0	0	0
6803	0.014509015	15.7979798	0	0	0
6804	0.021763523	15.7979798	0	0	0
6805	0.029018030	15.7979798	0	0	0
6806	0.036272538	15.7979798	0	0	0
6807	0.043527045	15.7979798	0	0	0
6808	0.050781553	15.7979798	0	0	0
6809	0.058036060	15.7979798	0	0	0
6810	0.065290568	15.7979798	0	0	0
6811	0.072545076	15.7979798	0	0	0
6812	0.079799583	15.7979798	0	0	0
6813	0.087054091	15.7979798	0	0	0
6814	0.094308598	15.7979798	0	0	0
6815	0.101563106	15.7979798	0	0	0
6816	0.108817613	15.7979798	0	0	0
6817	0.116072121	15.7979798	0	0	0
6818	0.123326628	15.7979798	0	0	0
6819	0.130581136	15.7979798	0	0	0
6820	0.137835643	15.7979798	0	0	0
6821	0.145090151	15.7979798	0	0	0
6822	0.152344659	15.7979798	0	0	0
6823	0.159599166	15.7979798	0	0	0
6824	0.166853674	15.7979798	0	0	0
6825	0.174108181	15.7979798	0	0	0
6826	0.181362689	15.7979798	0	0	0
6827	0.188617196	15.7979798	0	0	0
6828	0.195871704	15.7979798	0	0	0
6829	0.203126211	15.7979798	0	0	0
6830	0.210380719	15.7979798	0	0	0
6831	0.217635227	15.7979798	0	0	0
6832	0.224889734	15.7979798	0	0	0
6833	0.232144242	15.7979798	0	0	0
6834	0.239398749	15.7979798	0	0	0
6835	0.246653257	15.7979798	0	0	0
6836	0.253907764	15.7979798	0	0	0
6837	0.261162272	15.7979798	0	0	0
6838	0.268416779	15.7979798	0	0	0
6839	0.275671287	15.7979798	0	0	0
6840	0.282925794	15.7979798	0	0	0

6841	0.290180302	15.7979798	0	0	0
6842	0.297434810	15.7979798	0	0	0
6843	0.304689317	15.7979798	0	0	0
6844	0.311943825	15.7979798	0	0	0
6845	0.319198332	15.7979798	0	0	0
6846	0.326452840	15.7979798	0	0	0
6847	0.333707347	15.7979798	0	0	0
6848	0.340961855	15.7979798	0	0	0
6849	0.348216362	15.7979798	0	0	0
6850	0.355470870	15.7979798	0	0	0
6851	0.362725378	15.7979798	0	0	0
6852	0.369979885	15.7979798	0	0	0
6853	0.377234393	15.7979798	0	0	0
6854	0.384488900	15.7979798	0	0	0
6855	0.391743408	15.7979798	0	0	0
6856	0.398997915	15.7979798	0	0	0
6857	0.406252423	15.7979798	0	0	0
6858	0.413506930	15.7979798	0	0	0
6859	0.420761438	15.7979798	0	0	0
6860	0.428015945	15.7979798	0	0	0
6861	0.435270453	15.7979798	0	0	0
6862	0.442524961	15.7979798	0	0	0
6863	0.449779468	15.7979798	0	0	0
6864	0.457033976	15.7979798	0	0	0
6865	0.464288483	15.7979798	0	0	0
6866	0.471542991	15.7979798	0	0	0
6867	0.478797498	15.7979798	0	0	0
6868	0.486052006	15.7979798	0	0	0
6869	0.493306513	15.7979798	0	0	0
6870	0.500561021	15.7979798	0	0	0
6871	0.507815529	15.7979798	0	0	0
6872	0.515070036	15.7979798	0	0	0
6873	0.522324544	15.7979798	0	0	0
6874	0.529579051	15.7979798	0	0	0
6875	0.536833559	15.7979798	0	0	0
6876	0.544088066	15.7979798	0	0	0
6877	0.551342574	15.7979798	0	0	0
6878	0.558597081	15.7979798	0	0	0
6879	0.565851589	15.7979798	0	0	0
6880	0.573106097	15.7979798	0	0	0
6881	0.580360604	15.7979798	0	0	0
6882	0.587615112	15.7979798	0	0	0
6883	0.594869619	15.7979798	0	0	0
6884	0.602124127	15.7979798	0	0	0
6885	0.609378634	15.7979798	0	0	0

6886	0.616633142	15.7979798	0	0	0
6887	0.623887649	15.7979798	0	0	0
6888	0.631142157	15.7979798	0	0	0
6889	0.638396664	15.7979798	0	0	0
6890	0.645651172	15.7979798	0	0	0
6891	0.652905680	15.7979798	0	0	0
6892	0.660160187	15.7979798	0	0	0
6893	0.667414695	15.7979798	0	0	0
6894	0.674669202	15.7979798	0	0	0
6895	0.681923710	15.7979798	0	0	0
6896	0.689178217	15.7979798	0	0	0
6897	0.696432725	15.7979798	0	0	0
6898	0.703687232	15.7979798	0	0	0
6899	0.710941740	15.7979798	0	0	0
6900	0.718196248	15.7979798	0	0	0
6901	0.000000000	16.0303030	0	0	0
6902	0.007254508	16.0303030	0	0	0
6903	0.014509015	16.0303030	0	0	0
6904	0.021763523	16.0303030	0	0	0
6905	0.029018030	16.0303030	0	0	0
6906	0.036272538	16.0303030	0	0	0
6907	0.043527045	16.0303030	0	0	0
6908	0.050781553	16.0303030	0	0	0
6909	0.058036060	16.0303030	0	0	0
6910	0.065290568	16.0303030	0	0	0
6911	0.072545076	16.0303030	0	0	0
6912	0.079799583	16.0303030	0	0	0
6913	0.087054091	16.0303030	0	0	0
6914	0.094308598	16.0303030	0	0	0
6915	0.101563106	16.0303030	0	0	0
6916	0.108817613	16.0303030	0	0	0
6917	0.116072121	16.0303030	0	0	0
6918	0.123326628	16.0303030	0	0	0
6919	0.130581136	16.0303030	0	0	0
6920	0.137835643	16.0303030	0	0	0
6921	0.145090151	16.0303030	0	0	0
6922	0.152344659	16.0303030	0	0	0
6923	0.159599166	16.0303030	0	0	0
6924	0.166853674	16.0303030	0	0	0
6925	0.174108181	16.0303030	0	0	0
6926	0.181362689	16.0303030	0	0	0
6927	0.188617196	16.0303030	0	0	0
6928	0.195871704	16.0303030	0	0	0
6929	0.203126211	16.0303030	0	0	0
6930	0.210380719	16.0303030	0	0	0

6931	0.217635227	16.0303030	0	0	0
6932	0.224889734	16.0303030	0	0	0
6933	0.232144242	16.0303030	0	0	0
6934	0.239398749	16.0303030	0	0	0
6935	0.246653257	16.0303030	0	0	0
6936	0.253907764	16.0303030	0	0	0
6937	0.261162272	16.0303030	0	0	0
6938	0.268416779	16.0303030	0	0	0
6939	0.275671287	16.0303030	0	0	0
6940	0.282925794	16.0303030	0	0	0
6941	0.290180302	16.0303030	0	0	0
6942	0.297434810	16.0303030	0	0	0
6943	0.304689317	16.0303030	0	0	0
6944	0.311943825	16.0303030	0	0	0
6945	0.319198332	16.0303030	0	0	0
6946	0.326452840	16.0303030	0	0	0
6947	0.333707347	16.0303030	0	0	0
6948	0.340961855	16.0303030	0	0	0
6949	0.348216362	16.0303030	0	0	0
6950	0.355470870	16.0303030	0	0	0
6951	0.362725378	16.0303030	0	0	0
6952	0.369979885	16.0303030	0	0	0
6953	0.377234393	16.0303030	0	0	0
6954	0.384488900	16.0303030	0	0	0
6955	0.391743408	16.0303030	0	0	0
6956	0.398997915	16.0303030	0	0	0
6957	0.406252423	16.0303030	0	0	0
6958	0.413506930	16.0303030	0	0	0
6959	0.420761438	16.0303030	0	0	0
6960	0.428015945	16.0303030	0	0	0
6961	0.435270453	16.0303030	0	0	0
6962	0.442524961	16.0303030	0	0	0
6963	0.449779468	16.0303030	0	0	0
6964	0.457033976	16.0303030	0	0	0
6965	0.464288483	16.0303030	0	0	0
6966	0.471542991	16.0303030	0	0	0
6967	0.478797498	16.0303030	0	0	0
6968	0.486052006	16.0303030	0	0	0
6969	0.493306513	16.0303030	0	0	0
6970	0.500561021	16.0303030	0	0	0
6971	0.507815529	16.0303030	0	0	0
6972	0.515070036	16.0303030	0	0	0
6973	0.522324544	16.0303030	0	0	0
6974	0.529579051	16.0303030	0	0	0
6975	0.536833559	16.0303030	0	0	0

6976	0.544088066	16.0303030	0	0	0
6977	0.551342574	16.0303030	0	0	0
6978	0.558597081	16.0303030	0	0	0
6979	0.565851589	16.0303030	0	0	0
6980	0.573106097	16.0303030	0	0	0
6981	0.580360604	16.0303030	0	0	0
6982	0.587615112	16.0303030	0	0	0
6983	0.594869619	16.0303030	0	0	0
6984	0.602124127	16.0303030	0	0	0
6985	0.609378634	16.0303030	0	0	0
6986	0.616633142	16.0303030	0	0	0
6987	0.623887649	16.0303030	0	0	0
6988	0.631142157	16.0303030	0	0	0
6989	0.638396664	16.0303030	0	0	0
6990	0.645651172	16.0303030	0	0	0
6991	0.652905680	16.0303030	0	0	0
6992	0.660160187	16.0303030	0	0	0
6993	0.667414695	16.0303030	0	0	0
6994	0.674669202	16.0303030	0	0	0
6995	0.681923710	16.0303030	0	0	0
6996	0.689178217	16.0303030	0	0	0
6997	0.696432725	16.0303030	0	0	0
6998	0.703687232	16.0303030	0	0	0
6999	0.710941740	16.0303030	0	0	0
7000	0.718196248	16.0303030	0	0	0
7001	0.000000000	16.2626263	0	0	0
7002	0.007254508	16.2626263	0	0	0
7003	0.014509015	16.2626263	0	0	0
7004	0.021763523	16.2626263	0	0	0
7005	0.029018030	16.2626263	0	0	0
7006	0.036272538	16.2626263	0	0	0
7007	0.043527045	16.2626263	0	0	0
7008	0.050781553	16.2626263	0	0	0
7009	0.058036060	16.2626263	0	0	0
7010	0.065290568	16.2626263	0	0	0
7011	0.072545076	16.2626263	0	0	0
7012	0.079799583	16.2626263	0	0	0
7013	0.087054091	16.2626263	0	0	0
7014	0.094308598	16.2626263	0	0	0
7015	0.101563106	16.2626263	0	0	0
7016	0.108817613	16.2626263	0	0	0
7017	0.116072121	16.2626263	0	0	0
7018	0.123326628	16.2626263	0	0	0
7019	0.130581136	16.2626263	0	0	0
7020	0.137835643	16.2626263	0	0	0

7021	0.145090151	16.2626263	0	0	0
7022	0.152344659	16.2626263	0	0	0
7023	0.159599166	16.2626263	0	0	0
7024	0.166853674	16.2626263	0	0	0
7025	0.174108181	16.2626263	0	0	0
7026	0.181362689	16.2626263	0	0	0
7027	0.188617196	16.2626263	0	0	0
7028	0.195871704	16.2626263	0	0	0
7029	0.203126211	16.2626263	0	0	0
7030	0.210380719	16.2626263	0	0	0
7031	0.217635227	16.2626263	0	0	0
7032	0.224889734	16.2626263	0	0	0
7033	0.232144242	16.2626263	0	0	0
7034	0.239398749	16.2626263	0	0	0
7035	0.246653257	16.2626263	0	0	0
7036	0.253907764	16.2626263	0	0	0
7037	0.261162272	16.2626263	0	0	0
7038	0.268416779	16.2626263	0	0	0
7039	0.275671287	16.2626263	0	0	0
7040	0.282925794	16.2626263	0	0	0
7041	0.290180302	16.2626263	0	0	0
7042	0.297434810	16.2626263	0	0	0
7043	0.304689317	16.2626263	0	0	0
7044	0.311943825	16.2626263	0	0	0
7045	0.319198332	16.2626263	0	0	0
7046	0.326452840	16.2626263	0	0	0
7047	0.333707347	16.2626263	0	0	0
7048	0.340961855	16.2626263	0	0	0
7049	0.348216362	16.2626263	0	0	0
7050	0.355470870	16.2626263	0	0	0
7051	0.362725378	16.2626263	0	0	0
7052	0.369979885	16.2626263	0	0	0
7053	0.377234393	16.2626263	0	0	0
7054	0.384488900	16.2626263	0	0	0
7055	0.391743408	16.2626263	0	0	0
7056	0.398997915	16.2626263	0	0	0
7057	0.406252423	16.2626263	0	0	0
7058	0.413506930	16.2626263	0	0	0
7059	0.420761438	16.2626263	0	0	0
7060	0.428015945	16.2626263	0	0	0
7061	0.435270453	16.2626263	0	0	0
7062	0.442524961	16.2626263	0	0	0
7063	0.449779468	16.2626263	0	0	0
7064	0.457033976	16.2626263	0	0	0
7065	0.464288483	16.2626263	0	0	0

7066	0.471542991	16.2626263	0	0	0
7067	0.478797498	16.2626263	0	0	0
7068	0.486052006	16.2626263	0	0	0
7069	0.493306513	16.2626263	0	0	0
7070	0.500561021	16.2626263	0	0	0
7071	0.507815529	16.2626263	0	0	0
7072	0.515070036	16.2626263	0	0	0
7073	0.522324544	16.2626263	0	0	0
7074	0.529579051	16.2626263	0	0	0
7075	0.536833559	16.2626263	0	0	0
7076	0.544088066	16.2626263	0	0	0
7077	0.551342574	16.2626263	0	0	0
7078	0.558597081	16.2626263	0	0	0
7079	0.565851589	16.2626263	0	0	0
7080	0.573106097	16.2626263	0	0	0
7081	0.580360604	16.2626263	0	0	0
7082	0.587615112	16.2626263	0	0	0
7083	0.594869619	16.2626263	0	0	0
7084	0.602124127	16.2626263	0	0	0
7085	0.609378634	16.2626263	0	0	0
7086	0.616633142	16.2626263	0	0	0
7087	0.623887649	16.2626263	0	0	0
7088	0.631142157	16.2626263	0	0	0
7089	0.638396664	16.2626263	0	0	0
7090	0.645651172	16.2626263	0	0	0
7091	0.652905680	16.2626263	0	0	0
7092	0.660160187	16.2626263	0	0	0
7093	0.667414695	16.2626263	0	0	0
7094	0.674669202	16.2626263	0	0	0
7095	0.681923710	16.2626263	0	0	0
7096	0.689178217	16.2626263	0	0	0
7097	0.696432725	16.2626263	0	0	0
7098	0.703687232	16.2626263	0	0	0
7099	0.710941740	16.2626263	0	0	0
7100	0.718196248	16.2626263	0	0	0
7101	0.000000000	16.4949495	0	0	0
7102	0.007254508	16.4949495	0	0	0
7103	0.014509015	16.4949495	0	0	0
7104	0.021763523	16.4949495	0	0	0
7105	0.029018030	16.4949495	0	0	0
7106	0.036272538	16.4949495	0	0	0
7107	0.043527045	16.4949495	0	0	0
7108	0.050781553	16.4949495	0	0	0
7109	0.058036060	16.4949495	0	0	0
7110	0.065290568	16.4949495	0	0	0

7111	0.072545076	16.4949495	0	0	0
7112	0.079799583	16.4949495	0	0	0
7113	0.087054091	16.4949495	0	0	0
7114	0.094308598	16.4949495	0	0	0
7115	0.101563106	16.4949495	0	0	0
7116	0.108817613	16.4949495	0	0	0
7117	0.116072121	16.4949495	0	0	0
7118	0.123326628	16.4949495	0	0	0
7119	0.130581136	16.4949495	0	0	0
7120	0.137835643	16.4949495	0	0	0
7121	0.145090151	16.4949495	0	0	0
7122	0.152344659	16.4949495	0	0	0
7123	0.159599166	16.4949495	0	0	0
7124	0.166853674	16.4949495	0	0	0
7125	0.174108181	16.4949495	0	0	0
7126	0.181362689	16.4949495	0	0	0
7127	0.188617196	16.4949495	0	0	0
7128	0.195871704	16.4949495	0	0	0
7129	0.203126211	16.4949495	0	0	0
7130	0.210380719	16.4949495	0	0	0
7131	0.217635227	16.4949495	0	0	0
7132	0.224889734	16.4949495	0	0	0
7133	0.232144242	16.4949495	0	0	0
7134	0.239398749	16.4949495	0	0	0
7135	0.246653257	16.4949495	0	0	0
7136	0.253907764	16.4949495	0	0	0
7137	0.261162272	16.4949495	0	0	0
7138	0.268416779	16.4949495	0	0	0
7139	0.275671287	16.4949495	0	0	0
7140	0.282925794	16.4949495	0	0	0
7141	0.290180302	16.4949495	0	0	0
7142	0.297434810	16.4949495	0	0	0
7143	0.304689317	16.4949495	0	0	0
7144	0.311943825	16.4949495	0	0	0
7145	0.319198332	16.4949495	0	0	0
7146	0.326452840	16.4949495	0	0	0
7147	0.333707347	16.4949495	0	0	0
7148	0.340961855	16.4949495	0	0	0
7149	0.348216362	16.4949495	0	0	0
7150	0.355470870	16.4949495	0	0	0
7151	0.362725378	16.4949495	0	0	0
7152	0.369979885	16.4949495	0	0	0
7153	0.377234393	16.4949495	0	0	0
7154	0.384488900	16.4949495	0	0	0
7155	0.391743408	16.4949495	0	0	0

7156	0.398997915	16.4949495	0	0	0
7157	0.406252423	16.4949495	0	0	0
7158	0.413506930	16.4949495	0	0	0
7159	0.420761438	16.4949495	0	0	0
7160	0.428015945	16.4949495	0	0	0
7161	0.435270453	16.4949495	0	0	0
7162	0.442524961	16.4949495	0	0	0
7163	0.449779468	16.4949495	0	0	0
7164	0.457033976	16.4949495	0	0	0
7165	0.464288483	16.4949495	0	0	0
7166	0.471542991	16.4949495	0	0	0
7167	0.478797498	16.4949495	0	0	0
7168	0.486052006	16.4949495	0	0	0
7169	0.493306513	16.4949495	0	0	0
7170	0.500561021	16.4949495	0	0	0
7171	0.507815529	16.4949495	0	0	0
7172	0.515070036	16.4949495	0	0	0
7173	0.522324544	16.4949495	0	0	0
7174	0.529579051	16.4949495	0	0	0
7175	0.536833559	16.4949495	0	0	0
7176	0.544088066	16.4949495	0	0	0
7177	0.551342574	16.4949495	0	0	0
7178	0.558597081	16.4949495	0	0	0
7179	0.565851589	16.4949495	0	0	0
7180	0.573106097	16.4949495	0	0	0
7181	0.580360604	16.4949495	0	0	0
7182	0.587615112	16.4949495	0	0	0
7183	0.594869619	16.4949495	0	0	0
7184	0.602124127	16.4949495	0	0	0
7185	0.609378634	16.4949495	0	0	0
7186	0.616633142	16.4949495	0	0	0
7187	0.623887649	16.4949495	0	0	0
7188	0.631142157	16.4949495	0	0	0
7189	0.638396664	16.4949495	0	0	0
7190	0.645651172	16.4949495	0	0	0
7191	0.652905680	16.4949495	0	0	0
7192	0.660160187	16.4949495	0	0	0
7193	0.667414695	16.4949495	0	0	0
7194	0.674669202	16.4949495	0	0	0
7195	0.681923710	16.4949495	0	0	0
7196	0.689178217	16.4949495	0	0	0
7197	0.696432725	16.4949495	0	0	0
7198	0.703687232	16.4949495	0	0	0
7199	0.710941740	16.4949495	0	0	0
7200	0.718196248	16.4949495	0	0	0

7201	0.000000000	16.7272727	0	0	0
7202	0.007254508	16.7272727	0	0	0
7203	0.014509015	16.7272727	0	0	0
7204	0.021763523	16.7272727	0	0	0
7205	0.029018030	16.7272727	0	0	0
7206	0.036272538	16.7272727	0	0	0
7207	0.043527045	16.7272727	0	0	0
7208	0.050781553	16.7272727	0	0	0
7209	0.058036060	16.7272727	0	0	0
7210	0.065290568	16.7272727	0	0	0
7211	0.072545076	16.7272727	0	0	0
7212	0.079799583	16.7272727	0	0	0
7213	0.087054091	16.7272727	0	0	0
7214	0.094308598	16.7272727	0	0	0
7215	0.101563106	16.7272727	0	0	0
7216	0.108817613	16.7272727	0	0	0
7217	0.116072121	16.7272727	0	0	0
7218	0.123326628	16.7272727	0	0	0
7219	0.130581136	16.7272727	0	0	0
7220	0.137835643	16.7272727	0	0	0
7221	0.145090151	16.7272727	0	0	0
7222	0.152344659	16.7272727	0	0	0
7223	0.159599166	16.7272727	0	0	0
7224	0.166853674	16.7272727	0	0	0
7225	0.174108181	16.7272727	0	0	0
7226	0.181362689	16.7272727	0	0	0
7227	0.188617196	16.7272727	0	0	0
7228	0.195871704	16.7272727	0	0	0
7229	0.203126211	16.7272727	0	0	0
7230	0.210380719	16.7272727	0	0	0
7231	0.217635227	16.7272727	0	0	0
7232	0.224889734	16.7272727	0	0	0
7233	0.232144242	16.7272727	0	0	0
7234	0.239398749	16.7272727	0	0	0
7235	0.246653257	16.7272727	0	0	0
7236	0.253907764	16.7272727	0	0	0
7237	0.261162272	16.7272727	0	0	0
7238	0.268416779	16.7272727	0	0	0
7239	0.275671287	16.7272727	0	0	0
7240	0.282925794	16.7272727	0	0	0
7241	0.290180302	16.7272727	0	0	0
7242	0.297434810	16.7272727	0	0	0
7243	0.304689317	16.7272727	0	0	0
7244	0.311943825	16.7272727	0	0	0
7245	0.319198332	16.7272727	0	0	0

7246	0.326452840	16.7272727	0	0	0
7247	0.333707347	16.7272727	0	0	0
7248	0.340961855	16.7272727	0	0	0
7249	0.348216362	16.7272727	0	0	0
7250	0.355470870	16.7272727	0	0	0
7251	0.362725378	16.7272727	0	0	0
7252	0.369979885	16.7272727	0	0	0
7253	0.377234393	16.7272727	0	0	0
7254	0.384488900	16.7272727	0	0	0
7255	0.391743408	16.7272727	0	0	0
7256	0.398997915	16.7272727	0	0	0
7257	0.406252423	16.7272727	0	0	0
7258	0.413506930	16.7272727	0	0	0
7259	0.420761438	16.7272727	0	0	0
7260	0.428015945	16.7272727	0	0	0
7261	0.435270453	16.7272727	0	0	0
7262	0.442524961	16.7272727	0	0	0
7263	0.449779468	16.7272727	0	0	0
7264	0.457033976	16.7272727	0	0	0
7265	0.464288483	16.7272727	0	0	0
7266	0.471542991	16.7272727	0	0	0
7267	0.478797498	16.7272727	0	0	0
7268	0.486052006	16.7272727	0	0	0
7269	0.493306513	16.7272727	0	0	0
7270	0.500561021	16.7272727	0	0	0
7271	0.507815529	16.7272727	0	0	0
7272	0.515070036	16.7272727	0	0	0
7273	0.522324544	16.7272727	0	0	0
7274	0.529579051	16.7272727	0	0	0
7275	0.536833559	16.7272727	0	0	0
7276	0.544088066	16.7272727	0	0	0
7277	0.551342574	16.7272727	0	0	0
7278	0.558597081	16.7272727	0	0	0
7279	0.565851589	16.7272727	0	0	0
7280	0.573106097	16.7272727	0	0	0
7281	0.580360604	16.7272727	0	0	0
7282	0.587615112	16.7272727	0	0	0
7283	0.594869619	16.7272727	0	0	0
7284	0.602124127	16.7272727	0	0	0
7285	0.609378634	16.7272727	0	0	0
7286	0.616633142	16.7272727	0	0	0
7287	0.623887649	16.7272727	0	0	0
7288	0.631142157	16.7272727	0	0	0
7289	0.638396664	16.7272727	0	0	0
7290	0.645651172	16.7272727	0	0	0

7291	0.652905680	16.7272727	0	0	0
7292	0.660160187	16.7272727	0	0	0
7293	0.667414695	16.7272727	0	0	0
7294	0.674669202	16.7272727	0	0	0
7295	0.681923710	16.7272727	0	0	0
7296	0.689178217	16.7272727	0	0	0
7297	0.696432725	16.7272727	0	0	0
7298	0.703687232	16.7272727	0	0	0
7299	0.710941740	16.7272727	0	0	0
7300	0.718196248	16.7272727	0	0	0
7301	0.000000000	16.9595960	0	0	0
7302	0.007254508	16.9595960	0	0	0
7303	0.014509015	16.9595960	0	0	0
7304	0.021763523	16.9595960	0	0	0
7305	0.029018030	16.9595960	0	0	0
7306	0.036272538	16.9595960	0	0	0
7307	0.043527045	16.9595960	0	0	0
7308	0.050781553	16.9595960	0	0	0
7309	0.058036060	16.9595960	0	0	0
7310	0.065290568	16.9595960	0	0	0
7311	0.072545076	16.9595960	0	0	0
7312	0.079799583	16.9595960	0	0	0
7313	0.087054091	16.9595960	0	0	0
7314	0.094308598	16.9595960	0	0	0
7315	0.101563106	16.9595960	0	0	0
7316	0.108817613	16.9595960	0	0	0
7317	0.116072121	16.9595960	0	0	0
7318	0.123326628	16.9595960	0	0	0
7319	0.130581136	16.9595960	0	0	0
7320	0.137835643	16.9595960	0	0	0
7321	0.145090151	16.9595960	0	0	0
7322	0.152344659	16.9595960	0	0	0
7323	0.159599166	16.9595960	0	0	0
7324	0.166853674	16.9595960	0	0	0
7325	0.174108181	16.9595960	0	0	0
7326	0.181362689	16.9595960	0	0	0
7327	0.188617196	16.9595960	0	0	0
7328	0.195871704	16.9595960	0	0	0
7329	0.203126211	16.9595960	0	0	0
7330	0.210380719	16.9595960	0	0	0
7331	0.217635227	16.9595960	0	0	0
7332	0.224889734	16.9595960	0	0	0
7333	0.232144242	16.9595960	0	0	0
7334	0.239398749	16.9595960	0	0	0
7335	0.246653257	16.9595960	0	0	0

7336	0.253907764	16.9595960	0	0	0
7337	0.261162272	16.9595960	0	0	0
7338	0.268416779	16.9595960	0	0	0
7339	0.275671287	16.9595960	0	0	0
7340	0.282925794	16.9595960	0	0	0
7341	0.290180302	16.9595960	0	0	0
7342	0.297434810	16.9595960	0	0	0
7343	0.304689317	16.9595960	0	0	0
7344	0.311943825	16.9595960	0	0	0
7345	0.319198332	16.9595960	0	0	0
7346	0.326452840	16.9595960	0	0	0
7347	0.333707347	16.9595960	0	0	0
7348	0.340961855	16.9595960	0	0	0
7349	0.348216362	16.9595960	0	0	0
7350	0.355470870	16.9595960	0	0	0
7351	0.362725378	16.9595960	0	0	0
7352	0.369979885	16.9595960	0	0	0
7353	0.377234393	16.9595960	0	0	0
7354	0.384488900	16.9595960	0	0	0
7355	0.391743408	16.9595960	0	0	0
7356	0.398997915	16.9595960	0	0	0
7357	0.406252423	16.9595960	0	0	0
7358	0.413506930	16.9595960	0	0	0
7359	0.420761438	16.9595960	0	0	0
7360	0.428015945	16.9595960	0	0	0
7361	0.435270453	16.9595960	0	0	0
7362	0.442524961	16.9595960	0	0	0
7363	0.449779468	16.9595960	0	0	0
7364	0.457033976	16.9595960	0	0	0
7365	0.464288483	16.9595960	0	0	0
7366	0.471542991	16.9595960	0	0	0
7367	0.478797498	16.9595960	0	0	0
7368	0.486052006	16.9595960	0	0	0
7369	0.493306513	16.9595960	0	0	0
7370	0.500561021	16.9595960	0	0	0
7371	0.507815529	16.9595960	0	0	0
7372	0.515070036	16.9595960	0	0	0
7373	0.522324544	16.9595960	0	0	0
7374	0.529579051	16.9595960	0	0	0
7375	0.536833559	16.9595960	0	0	0
7376	0.544088066	16.9595960	0	0	0
7377	0.551342574	16.9595960	0	0	0
7378	0.558597081	16.9595960	0	0	0
7379	0.565851589	16.9595960	0	0	0
7380	0.573106097	16.9595960	0	0	0

7381	0.580360604	16.9595960	0	0	0
7382	0.587615112	16.9595960	0	0	0
7383	0.594869619	16.9595960	0	0	0
7384	0.602124127	16.9595960	0	0	0
7385	0.609378634	16.9595960	0	0	0
7386	0.616633142	16.9595960	0	0	0
7387	0.623887649	16.9595960	0	0	0
7388	0.631142157	16.9595960	0	0	0
7389	0.638396664	16.9595960	0	0	0
7390	0.645651172	16.9595960	0	0	0
7391	0.652905680	16.9595960	0	0	0
7392	0.660160187	16.9595960	0	0	0
7393	0.667414695	16.9595960	0	0	0
7394	0.674669202	16.9595960	0	0	0
7395	0.681923710	16.9595960	0	0	0
7396	0.689178217	16.9595960	0	0	0
7397	0.696432725	16.9595960	0	0	0
7398	0.703687232	16.9595960	0	0	0
7399	0.710941740	16.9595960	0	0	0
7400	0.718196248	16.9595960	0	0	0
7401	0.000000000	17.1919192	0	0	0
7402	0.007254508	17.1919192	0	0	0
7403	0.014509015	17.1919192	0	0	0
7404	0.021763523	17.1919192	0	0	0
7405	0.029018030	17.1919192	0	0	0
7406	0.036272538	17.1919192	0	0	0
7407	0.043527045	17.1919192	0	0	0
7408	0.050781553	17.1919192	0	0	0
7409	0.058036060	17.1919192	0	0	0
7410	0.065290568	17.1919192	0	0	0
7411	0.072545076	17.1919192	0	0	0
7412	0.079799583	17.1919192	0	0	0
7413	0.087054091	17.1919192	0	0	0
7414	0.094308598	17.1919192	0	0	0
7415	0.101563106	17.1919192	0	0	0
7416	0.108817613	17.1919192	0	0	0
7417	0.116072121	17.1919192	0	0	0
7418	0.123326628	17.1919192	0	0	0
7419	0.130581136	17.1919192	0	0	0
7420	0.137835643	17.1919192	0	0	0
7421	0.145090151	17.1919192	0	0	0
7422	0.152344659	17.1919192	0	0	0
7423	0.159599166	17.1919192	0	0	0
7424	0.166853674	17.1919192	0	0	0
7425	0.174108181	17.1919192	0	0	0

7426	0.181362689	17.1919192	0	0	0
7427	0.188617196	17.1919192	0	0	0
7428	0.195871704	17.1919192	0	0	0
7429	0.203126211	17.1919192	0	0	0
7430	0.210380719	17.1919192	0	0	0
7431	0.217635227	17.1919192	0	0	0
7432	0.224889734	17.1919192	0	0	0
7433	0.232144242	17.1919192	0	0	0
7434	0.239398749	17.1919192	0	0	0
7435	0.246653257	17.1919192	0	0	0
7436	0.253907764	17.1919192	0	0	0
7437	0.261162272	17.1919192	0	0	0
7438	0.268416779	17.1919192	0	0	0
7439	0.275671287	17.1919192	0	0	0
7440	0.282925794	17.1919192	0	0	0
7441	0.290180302	17.1919192	0	0	0
7442	0.297434810	17.1919192	0	0	0
7443	0.304689317	17.1919192	0	0	0
7444	0.311943825	17.1919192	0	0	0
7445	0.319198332	17.1919192	0	0	0
7446	0.326452840	17.1919192	0	0	0
7447	0.333707347	17.1919192	0	0	0
7448	0.340961855	17.1919192	0	0	0
7449	0.348216362	17.1919192	0	0	0
7450	0.355470870	17.1919192	0	0	0
7451	0.362725378	17.1919192	0	0	0
7452	0.369979885	17.1919192	0	0	0
7453	0.377234393	17.1919192	0	0	0
7454	0.384488900	17.1919192	0	0	0
7455	0.391743408	17.1919192	0	0	0
7456	0.398997915	17.1919192	0	0	0
7457	0.406252423	17.1919192	0	0	0
7458	0.413506930	17.1919192	0	0	0
7459	0.420761438	17.1919192	0	0	0
7460	0.428015945	17.1919192	0	0	0
7461	0.435270453	17.1919192	0	0	0
7462	0.442524961	17.1919192	0	0	0
7463	0.449779468	17.1919192	0	0	0
7464	0.457033976	17.1919192	0	0	0
7465	0.464288483	17.1919192	0	0	0
7466	0.471542991	17.1919192	0	0	0
7467	0.478797498	17.1919192	0	0	0
7468	0.486052006	17.1919192	0	0	0
7469	0.493306513	17.1919192	0	0	0
7470	0.500561021	17.1919192	0	0	0

7471	0.507815529	17.1919192	0	0	0
7472	0.515070036	17.1919192	0	0	0
7473	0.522324544	17.1919192	0	0	0
7474	0.529579051	17.1919192	0	0	0
7475	0.536833559	17.1919192	0	0	0
7476	0.544088066	17.1919192	0	0	0
7477	0.551342574	17.1919192	0	0	0
7478	0.558597081	17.1919192	0	0	0
7479	0.565851589	17.1919192	0	0	0
7480	0.573106097	17.1919192	0	0	0
7481	0.580360604	17.1919192	0	0	0
7482	0.587615112	17.1919192	0	0	0
7483	0.594869619	17.1919192	0	0	0
7484	0.602124127	17.1919192	0	0	0
7485	0.609378634	17.1919192	0	0	0
7486	0.616633142	17.1919192	0	0	0
7487	0.623887649	17.1919192	0	0	0
7488	0.631142157	17.1919192	0	0	0
7489	0.638396664	17.1919192	0	0	0
7490	0.645651172	17.1919192	0	0	0
7491	0.652905680	17.1919192	0	0	0
7492	0.660160187	17.1919192	0	0	0
7493	0.667414695	17.1919192	0	0	0
7494	0.674669202	17.1919192	0	0	0
7495	0.681923710	17.1919192	0	0	0
7496	0.689178217	17.1919192	0	0	0
7497	0.696432725	17.1919192	0	0	0
7498	0.703687232	17.1919192	0	0	0
7499	0.710941740	17.1919192	0	0	0
7500	0.718196248	17.1919192	0	0	0
7501	0.000000000	17.4242424	0	0	0
7502	0.007254508	17.4242424	0	0	0
7503	0.014509015	17.4242424	0	0	0
7504	0.021763523	17.4242424	0	0	0
7505	0.029018030	17.4242424	0	0	0
7506	0.036272538	17.4242424	0	0	0
7507	0.043527045	17.4242424	0	0	0
7508	0.050781553	17.4242424	0	0	0
7509	0.058036060	17.4242424	0	0	0
7510	0.065290568	17.4242424	0	0	0
7511	0.072545076	17.4242424	0	0	0
7512	0.079799583	17.4242424	0	0	0
7513	0.087054091	17.4242424	0	0	0
7514	0.094308598	17.4242424	0	0	0
7515	0.101563106	17.4242424	0	0	0

7516	0.108817613	17.4242424	0	0	0
7517	0.116072121	17.4242424	0	0	0
7518	0.123326628	17.4242424	0	0	0
7519	0.130581136	17.4242424	0	0	0
7520	0.137835643	17.4242424	0	0	0
7521	0.145090151	17.4242424	0	0	0
7522	0.152344659	17.4242424	0	0	0
7523	0.159599166	17.4242424	0	0	0
7524	0.166853674	17.4242424	0	0	0
7525	0.174108181	17.4242424	0	0	0
7526	0.181362689	17.4242424	0	0	0
7527	0.188617196	17.4242424	0	0	0
7528	0.195871704	17.4242424	0	0	0
7529	0.203126211	17.4242424	0	0	0
7530	0.210380719	17.4242424	0	0	0
7531	0.217635227	17.4242424	0	0	0
7532	0.224889734	17.4242424	0	0	0
7533	0.232144242	17.4242424	0	0	0
7534	0.239398749	17.4242424	0	0	0
7535	0.246653257	17.4242424	0	0	0
7536	0.253907764	17.4242424	0	0	0
7537	0.261162272	17.4242424	0	0	0
7538	0.268416779	17.4242424	0	0	0
7539	0.275671287	17.4242424	0	0	0
7540	0.282925794	17.4242424	0	0	0
7541	0.290180302	17.4242424	0	0	0
7542	0.297434810	17.4242424	0	0	0
7543	0.304689317	17.4242424	0	0	0
7544	0.311943825	17.4242424	0	0	0
7545	0.319198332	17.4242424	0	0	0
7546	0.326452840	17.4242424	0	0	0
7547	0.333707347	17.4242424	0	0	0
7548	0.340961855	17.4242424	0	0	0
7549	0.348216362	17.4242424	0	0	0
7550	0.355470870	17.4242424	0	0	0
7551	0.362725378	17.4242424	0	0	0
7552	0.369979885	17.4242424	0	0	0
7553	0.377234393	17.4242424	0	0	0
7554	0.384488900	17.4242424	0	0	0
7555	0.391743408	17.4242424	0	0	0
7556	0.398997915	17.4242424	0	0	0
7557	0.406252423	17.4242424	0	0	0
7558	0.413506930	17.4242424	0	0	0
7559	0.420761438	17.4242424	0	0	0
7560	0.428015945	17.4242424	0	0	0

7561	0.435270453	17.4242424	0	0	0
7562	0.442524961	17.4242424	0	0	0
7563	0.449779468	17.4242424	0	0	0
7564	0.457033976	17.4242424	0	0	0
7565	0.464288483	17.4242424	0	0	0
7566	0.471542991	17.4242424	0	0	0
7567	0.478797498	17.4242424	0	0	0
7568	0.486052006	17.4242424	0	0	0
7569	0.493306513	17.4242424	0	0	0
7570	0.500561021	17.4242424	0	0	0
7571	0.507815529	17.4242424	0	0	0
7572	0.515070036	17.4242424	0	0	0
7573	0.522324544	17.4242424	0	0	0
7574	0.529579051	17.4242424	0	0	0
7575	0.536833559	17.4242424	0	0	0
7576	0.544088066	17.4242424	0	0	0
7577	0.551342574	17.4242424	0	0	0
7578	0.558597081	17.4242424	0	0	0
7579	0.565851589	17.4242424	0	0	0
7580	0.573106097	17.4242424	0	0	0
7581	0.580360604	17.4242424	0	0	0
7582	0.587615112	17.4242424	0	0	0
7583	0.594869619	17.4242424	0	0	0
7584	0.602124127	17.4242424	0	0	0
7585	0.609378634	17.4242424	0	0	0
7586	0.616633142	17.4242424	0	0	0
7587	0.623887649	17.4242424	0	0	0
7588	0.631142157	17.4242424	0	0	0
7589	0.638396664	17.4242424	0	0	0
7590	0.645651172	17.4242424	0	0	0
7591	0.652905680	17.4242424	0	0	0
7592	0.660160187	17.4242424	0	0	0
7593	0.667414695	17.4242424	0	0	0
7594	0.674669202	17.4242424	0	0	0
7595	0.681923710	17.4242424	0	0	0
7596	0.689178217	17.4242424	0	0	0
7597	0.696432725	17.4242424	0	0	0
7598	0.703687232	17.4242424	0	0	0
7599	0.710941740	17.4242424	0	0	0
7600	0.718196248	17.4242424	0	0	0
7601	0.000000000	17.6565657	0	0	0
7602	0.007254508	17.6565657	0	0	0
7603	0.014509015	17.6565657	0	0	0
7604	0.021763523	17.6565657	0	0	0
7605	0.029018030	17.6565657	0	0	0

7606	0.036272538	17.6565657	0	0	0
7607	0.043527045	17.6565657	0	0	0
7608	0.050781553	17.6565657	0	0	0
7609	0.058036060	17.6565657	0	0	0
7610	0.065290568	17.6565657	0	0	0
7611	0.072545076	17.6565657	0	0	0
7612	0.079799583	17.6565657	0	0	0
7613	0.087054091	17.6565657	0	0	0
7614	0.094308598	17.6565657	0	0	0
7615	0.101563106	17.6565657	0	0	0
7616	0.108817613	17.6565657	0	0	0
7617	0.116072121	17.6565657	0	0	0
7618	0.123326628	17.6565657	0	0	0
7619	0.130581136	17.6565657	0	0	0
7620	0.137835643	17.6565657	0	0	0
7621	0.145090151	17.6565657	0	0	0
7622	0.152344659	17.6565657	0	0	0
7623	0.159599166	17.6565657	0	0	0
7624	0.166853674	17.6565657	0	0	0
7625	0.174108181	17.6565657	0	0	0
7626	0.181362689	17.6565657	0	0	0
7627	0.188617196	17.6565657	0	0	0
7628	0.195871704	17.6565657	0	0	0
7629	0.203126211	17.6565657	0	0	0
7630	0.210380719	17.6565657	0	0	0
7631	0.217635227	17.6565657	0	0	0
7632	0.224889734	17.6565657	0	0	0
7633	0.232144242	17.6565657	0	0	0
7634	0.239398749	17.6565657	0	0	0
7635	0.246653257	17.6565657	0	0	0
7636	0.253907764	17.6565657	0	0	0
7637	0.261162272	17.6565657	0	0	0
7638	0.268416779	17.6565657	0	0	0
7639	0.275671287	17.6565657	0	0	0
7640	0.282925794	17.6565657	0	0	0
7641	0.290180302	17.6565657	0	0	0
7642	0.297434810	17.6565657	0	0	0
7643	0.304689317	17.6565657	0	0	0
7644	0.311943825	17.6565657	0	0	0
7645	0.319198332	17.6565657	0	0	0
7646	0.326452840	17.6565657	0	0	0
7647	0.333707347	17.6565657	0	0	0
7648	0.340961855	17.6565657	0	0	0
7649	0.348216362	17.6565657	0	0	0
7650	0.355470870	17.6565657	0	0	0

7651	0.362725378	17.6565657	0	0	0
7652	0.369979885	17.6565657	0	0	0
7653	0.377234393	17.6565657	0	0	0
7654	0.384488900	17.6565657	0	0	0
7655	0.391743408	17.6565657	0	0	0
7656	0.398997915	17.6565657	0	0	0
7657	0.406252423	17.6565657	0	0	0
7658	0.413506930	17.6565657	0	0	0
7659	0.420761438	17.6565657	0	0	0
7660	0.428015945	17.6565657	0	0	0
7661	0.435270453	17.6565657	0	0	0
7662	0.442524961	17.6565657	0	0	0
7663	0.449779468	17.6565657	0	0	0
7664	0.457033976	17.6565657	0	0	0
7665	0.464288483	17.6565657	0	0	0
7666	0.471542991	17.6565657	0	0	0
7667	0.478797498	17.6565657	0	0	0
7668	0.486052006	17.6565657	0	0	0
7669	0.493306513	17.6565657	0	0	0
7670	0.500561021	17.6565657	0	0	0
7671	0.507815529	17.6565657	0	0	0
7672	0.515070036	17.6565657	0	0	0
7673	0.522324544	17.6565657	0	0	0
7674	0.529579051	17.6565657	0	0	0
7675	0.536833559	17.6565657	0	0	0
7676	0.544088066	17.6565657	0	0	0
7677	0.551342574	17.6565657	0	0	0
7678	0.558597081	17.6565657	0	0	0
7679	0.565851589	17.6565657	0	0	0
7680	0.573106097	17.6565657	0	0	0
7681	0.580360604	17.6565657	0	0	0
7682	0.587615112	17.6565657	0	0	0
7683	0.594869619	17.6565657	0	0	0
7684	0.602124127	17.6565657	0	0	0
7685	0.609378634	17.6565657	0	0	0
7686	0.616633142	17.6565657	0	0	0
7687	0.623887649	17.6565657	0	0	0
7688	0.631142157	17.6565657	0	0	0
7689	0.638396664	17.6565657	0	0	0
7690	0.645651172	17.6565657	0	0	0
7691	0.652905680	17.6565657	0	0	0
7692	0.660160187	17.6565657	0	0	0
7693	0.667414695	17.6565657	0	0	0
7694	0.674669202	17.6565657	0	0	0
7695	0.681923710	17.6565657	0	0	0

7696	0.689178217	17.6565657	0	0	0
7697	0.696432725	17.6565657	0	0	0
7698	0.703687232	17.6565657	0	0	0
7699	0.710941740	17.6565657	0	0	0
7700	0.718196248	17.6565657	0	0	0
7701	0.000000000	17.8888889	0	0	0
7702	0.007254508	17.8888889	0	0	0
7703	0.014509015	17.8888889	0	0	0
7704	0.021763523	17.8888889	0	0	0
7705	0.029018030	17.8888889	0	0	0
7706	0.036272538	17.8888889	0	0	0
7707	0.043527045	17.8888889	0	0	0
7708	0.050781553	17.8888889	0	0	0
7709	0.058036060	17.8888889	0	0	0
7710	0.065290568	17.8888889	0	0	0
7711	0.072545076	17.8888889	0	0	0
7712	0.079799583	17.8888889	0	0	0
7713	0.087054091	17.8888889	0	0	0
7714	0.094308598	17.8888889	0	0	0
7715	0.101563106	17.8888889	0	0	0
7716	0.108817613	17.8888889	0	0	0
7717	0.116072121	17.8888889	0	0	0
7718	0.123326628	17.8888889	0	0	0
7719	0.130581136	17.8888889	0	0	0
7720	0.137835643	17.8888889	0	0	0
7721	0.145090151	17.8888889	0	0	0
7722	0.152344659	17.8888889	0	0	0
7723	0.159599166	17.8888889	0	0	0
7724	0.166853674	17.8888889	0	0	0
7725	0.174108181	17.8888889	0	0	0
7726	0.181362689	17.8888889	0	0	0
7727	0.188617196	17.8888889	0	0	0
7728	0.195871704	17.8888889	0	0	0
7729	0.203126211	17.8888889	0	0	0
7730	0.210380719	17.8888889	0	0	0
7731	0.217635227	17.8888889	0	0	0
7732	0.224889734	17.8888889	0	0	0
7733	0.232144242	17.8888889	0	0	0
7734	0.239398749	17.8888889	0	0	0
7735	0.246653257	17.8888889	0	0	0
7736	0.253907764	17.8888889	0	0	0
7737	0.261162272	17.8888889	0	0	0
7738	0.268416779	17.8888889	0	0	0
7739	0.275671287	17.8888889	0	0	0
7740	0.282925794	17.8888889	0	0	0

7741	0.290180302	17.8888889	0	0	0
7742	0.297434810	17.8888889	0	0	0
7743	0.304689317	17.8888889	0	0	0
7744	0.311943825	17.8888889	0	0	0
7745	0.319198332	17.8888889	0	0	0
7746	0.326452840	17.8888889	0	0	0
7747	0.333707347	17.8888889	0	0	0
7748	0.340961855	17.8888889	0	0	0
7749	0.348216362	17.8888889	0	0	0
7750	0.355470870	17.8888889	0	0	0
7751	0.362725378	17.8888889	0	0	0
7752	0.369979885	17.8888889	0	0	0
7753	0.377234393	17.8888889	0	0	0
7754	0.384488900	17.8888889	0	0	0
7755	0.391743408	17.8888889	0	0	0
7756	0.398997915	17.8888889	0	0	0
7757	0.406252423	17.8888889	0	0	0
7758	0.413506930	17.8888889	0	0	0
7759	0.420761438	17.8888889	0	0	0
7760	0.428015945	17.8888889	0	0	0
7761	0.435270453	17.8888889	0	0	0
7762	0.442524961	17.8888889	0	0	0
7763	0.449779468	17.8888889	0	0	0
7764	0.457033976	17.8888889	0	0	0
7765	0.464288483	17.8888889	0	0	0
7766	0.471542991	17.8888889	0	0	0
7767	0.478797498	17.8888889	0	0	0
7768	0.486052006	17.8888889	0	0	0
7769	0.493306513	17.8888889	0	0	0
7770	0.500561021	17.8888889	0	0	0
7771	0.507815529	17.8888889	0	0	0
7772	0.515070036	17.8888889	0	0	0
7773	0.522324544	17.8888889	0	0	0
7774	0.529579051	17.8888889	0	0	0
7775	0.536833559	17.8888889	0	0	0
7776	0.544088066	17.8888889	0	0	0
7777	0.551342574	17.8888889	0	0	0
7778	0.558597081	17.8888889	0	0	0
7779	0.565851589	17.8888889	0	0	0
7780	0.573106097	17.8888889	0	0	0
7781	0.580360604	17.8888889	0	0	0
7782	0.587615112	17.8888889	0	0	0
7783	0.594869619	17.8888889	0	0	0
7784	0.602124127	17.8888889	0	0	0
7785	0.609378634	17.8888889	0	0	0

7786	0.616633142	17.8888889	0	0	0
7787	0.623887649	17.8888889	0	0	0
7788	0.631142157	17.8888889	0	0	0
7789	0.638396664	17.8888889	0	0	0
7790	0.645651172	17.8888889	0	0	0
7791	0.652905680	17.8888889	0	0	0
7792	0.660160187	17.8888889	0	0	0
7793	0.667414695	17.8888889	0	0	0
7794	0.674669202	17.8888889	0	0	0
7795	0.681923710	17.8888889	0	0	0
7796	0.689178217	17.8888889	0	0	0
7797	0.696432725	17.8888889	0	0	0
7798	0.703687232	17.8888889	0	0	0
7799	0.710941740	17.8888889	0	0	0
7800	0.718196248	17.8888889	0	0	0
7801	0.000000000	18.1212121	0	0	0
7802	0.007254508	18.1212121	0	0	0
7803	0.014509015	18.1212121	0	0	0
7804	0.021763523	18.1212121	0	0	0
7805	0.029018030	18.1212121	0	0	0
7806	0.036272538	18.1212121	0	0	0
7807	0.043527045	18.1212121	0	0	0
7808	0.050781553	18.1212121	0	0	0
7809	0.058036060	18.1212121	0	0	0
7810	0.065290568	18.1212121	0	0	0
7811	0.072545076	18.1212121	0	0	0
7812	0.079799583	18.1212121	0	0	0
7813	0.087054091	18.1212121	0	0	0
7814	0.094308598	18.1212121	0	0	0
7815	0.101563106	18.1212121	0	0	0
7816	0.108817613	18.1212121	0	0	0
7817	0.116072121	18.1212121	0	0	0
7818	0.123326628	18.1212121	0	0	0
7819	0.130581136	18.1212121	0	0	0
7820	0.137835643	18.1212121	0	0	0
7821	0.145090151	18.1212121	0	0	0
7822	0.152344659	18.1212121	0	0	0
7823	0.159599166	18.1212121	0	0	0
7824	0.166853674	18.1212121	0	0	0
7825	0.174108181	18.1212121	0	0	0
7826	0.181362689	18.1212121	0	0	0
7827	0.188617196	18.1212121	0	0	0
7828	0.195871704	18.1212121	0	0	0
7829	0.203126211	18.1212121	0	0	0
7830	0.210380719	18.1212121	0	0	0

7831	0.217635227	18.1212121	0	0	0
7832	0.224889734	18.1212121	0	0	0
7833	0.232144242	18.1212121	0	0	0
7834	0.239398749	18.1212121	0	0	0
7835	0.246653257	18.1212121	0	0	0
7836	0.253907764	18.1212121	0	0	0
7837	0.261162272	18.1212121	0	0	0
7838	0.268416779	18.1212121	0	0	0
7839	0.275671287	18.1212121	0	0	0
7840	0.282925794	18.1212121	0	0	0
7841	0.290180302	18.1212121	0	0	0
7842	0.297434810	18.1212121	0	0	0
7843	0.304689317	18.1212121	0	0	0
7844	0.311943825	18.1212121	0	0	0
7845	0.319198332	18.1212121	0	0	0
7846	0.326452840	18.1212121	0	0	0
7847	0.333707347	18.1212121	0	0	0
7848	0.340961855	18.1212121	0	0	0
7849	0.348216362	18.1212121	0	0	0
7850	0.355470870	18.1212121	0	0	0
7851	0.362725378	18.1212121	0	0	0
7852	0.369979885	18.1212121	0	0	0
7853	0.377234393	18.1212121	0	0	0
7854	0.384488900	18.1212121	0	0	0
7855	0.391743408	18.1212121	0	0	0
7856	0.398997915	18.1212121	0	0	0
7857	0.406252423	18.1212121	0	0	0
7858	0.413506930	18.1212121	0	0	0
7859	0.420761438	18.1212121	0	0	0
7860	0.428015945	18.1212121	0	0	0
7861	0.435270453	18.1212121	0	0	0
7862	0.442524961	18.1212121	0	0	0
7863	0.449779468	18.1212121	0	0	0
7864	0.457033976	18.1212121	0	0	0
7865	0.464288483	18.1212121	0	0	0
7866	0.471542991	18.1212121	0	0	0
7867	0.478797498	18.1212121	0	0	0
7868	0.486052006	18.1212121	0	0	0
7869	0.493306513	18.1212121	0	0	0
7870	0.500561021	18.1212121	0	0	0
7871	0.507815529	18.1212121	0	0	0
7872	0.515070036	18.1212121	0	0	0
7873	0.522324544	18.1212121	0	0	0
7874	0.529579051	18.1212121	0	0	0
7875	0.536833559	18.1212121	0	0	0

7876	0.544088066	18.1212121	0	0	0
7877	0.551342574	18.1212121	0	0	0
7878	0.558597081	18.1212121	0	0	0
7879	0.565851589	18.1212121	0	0	0
7880	0.573106097	18.1212121	0	0	0
7881	0.580360604	18.1212121	0	0	0
7882	0.587615112	18.1212121	0	0	0
7883	0.594869619	18.1212121	0	0	0
7884	0.602124127	18.1212121	0	0	0
7885	0.609378634	18.1212121	0	0	0
7886	0.616633142	18.1212121	0	0	0
7887	0.623887649	18.1212121	0	0	0
7888	0.631142157	18.1212121	0	0	0
7889	0.638396664	18.1212121	0	0	0
7890	0.645651172	18.1212121	0	0	0
7891	0.652905680	18.1212121	0	0	0
7892	0.660160187	18.1212121	0	0	0
7893	0.667414695	18.1212121	0	0	0
7894	0.674669202	18.1212121	0	0	0
7895	0.681923710	18.1212121	0	0	0
7896	0.689178217	18.1212121	0	0	0
7897	0.696432725	18.1212121	0	0	0
7898	0.703687232	18.1212121	0	0	0
7899	0.710941740	18.1212121	0	0	0
7900	0.718196248	18.1212121	0	0	0
7901	0.000000000	18.3535354	0	0	0
7902	0.007254508	18.3535354	0	0	0
7903	0.014509015	18.3535354	0	0	0
7904	0.021763523	18.3535354	0	0	0
7905	0.029018030	18.3535354	0	0	0
7906	0.036272538	18.3535354	0	0	0
7907	0.043527045	18.3535354	0	0	0
7908	0.050781553	18.3535354	0	0	0
7909	0.058036060	18.3535354	0	0	0
7910	0.065290568	18.3535354	0	0	0
7911	0.072545076	18.3535354	0	0	0
7912	0.079799583	18.3535354	0	0	0
7913	0.087054091	18.3535354	0	0	0
7914	0.094308598	18.3535354	0	0	0
7915	0.101563106	18.3535354	0	0	0
7916	0.108817613	18.3535354	0	0	0
7917	0.116072121	18.3535354	0	0	0
7918	0.123326628	18.3535354	0	0	0
7919	0.130581136	18.3535354	0	0	0
7920	0.137835643	18.3535354	0	0	0

7921	0.145090151	18.3535354	0	0	0
7922	0.152344659	18.3535354	0	0	0
7923	0.159599166	18.3535354	0	0	0
7924	0.166853674	18.3535354	0	0	0
7925	0.174108181	18.3535354	0	0	0
7926	0.181362689	18.3535354	0	0	0
7927	0.188617196	18.3535354	0	0	0
7928	0.195871704	18.3535354	0	0	0
7929	0.203126211	18.3535354	0	0	0
7930	0.210380719	18.3535354	0	0	0
7931	0.217635227	18.3535354	0	0	0
7932	0.224889734	18.3535354	0	0	0
7933	0.232144242	18.3535354	0	0	0
7934	0.239398749	18.3535354	0	0	0
7935	0.246653257	18.3535354	0	0	0
7936	0.253907764	18.3535354	0	0	0
7937	0.261162272	18.3535354	0	0	0
7938	0.268416779	18.3535354	0	0	0
7939	0.275671287	18.3535354	0	0	0
7940	0.282925794	18.3535354	0	0	0
7941	0.290180302	18.3535354	0	0	0
7942	0.297434810	18.3535354	0	0	0
7943	0.304689317	18.3535354	0	0	0
7944	0.311943825	18.3535354	0	0	0
7945	0.319198332	18.3535354	0	0	0
7946	0.326452840	18.3535354	0	0	0
7947	0.333707347	18.3535354	0	0	0
7948	0.340961855	18.3535354	0	0	0
7949	0.348216362	18.3535354	0	0	0
7950	0.355470870	18.3535354	0	0	0
7951	0.362725378	18.3535354	0	0	0
7952	0.369979885	18.3535354	0	0	0
7953	0.377234393	18.3535354	0	0	0
7954	0.384488900	18.3535354	0	0	0
7955	0.391743408	18.3535354	0	0	0
7956	0.398997915	18.3535354	0	0	0
7957	0.406252423	18.3535354	0	0	0
7958	0.413506930	18.3535354	0	0	0
7959	0.420761438	18.3535354	0	0	0
7960	0.428015945	18.3535354	0	0	0
7961	0.435270453	18.3535354	0	0	0
7962	0.442524961	18.3535354	0	0	0
7963	0.449779468	18.3535354	0	0	0
7964	0.457033976	18.3535354	0	0	0
7965	0.464288483	18.3535354	0	0	0

7966	0.471542991	18.3535354	0	0	0
7967	0.478797498	18.3535354	0	0	0
7968	0.486052006	18.3535354	0	0	0
7969	0.493306513	18.3535354	0	0	0
7970	0.500561021	18.3535354	0	0	0
7971	0.507815529	18.3535354	0	0	0
7972	0.515070036	18.3535354	0	0	0
7973	0.522324544	18.3535354	0	0	0
7974	0.529579051	18.3535354	0	0	0
7975	0.536833559	18.3535354	0	0	0
7976	0.544088066	18.3535354	0	0	0
7977	0.551342574	18.3535354	0	0	0
7978	0.558597081	18.3535354	0	0	0
7979	0.565851589	18.3535354	0	0	0
7980	0.573106097	18.3535354	0	0	0
7981	0.580360604	18.3535354	0	0	0
7982	0.587615112	18.3535354	0	0	0
7983	0.594869619	18.3535354	0	0	0
7984	0.602124127	18.3535354	0	0	0
7985	0.609378634	18.3535354	0	0	0
7986	0.616633142	18.3535354	0	0	0
7987	0.623887649	18.3535354	0	0	0
7988	0.631142157	18.3535354	0	0	0
7989	0.638396664	18.3535354	0	0	0
7990	0.645651172	18.3535354	0	0	0
7991	0.652905680	18.3535354	0	0	0
7992	0.660160187	18.3535354	0	0	0
7993	0.667414695	18.3535354	0	0	0
7994	0.674669202	18.3535354	0	0	0
7995	0.681923710	18.3535354	0	0	0
7996	0.689178217	18.3535354	0	0	0
7997	0.696432725	18.3535354	0	0	0
7998	0.703687232	18.3535354	0	0	0
7999	0.710941740	18.3535354	0	0	0
8000	0.718196248	18.3535354	0	0	0
8001	0.000000000	18.5858586	0	0	0
8002	0.007254508	18.5858586	0	0	0
8003	0.014509015	18.5858586	0	0	0
8004	0.021763523	18.5858586	0	0	0
8005	0.029018030	18.5858586	0	0	0
8006	0.036272538	18.5858586	0	0	0
8007	0.043527045	18.5858586	0	0	0
8008	0.050781553	18.5858586	0	0	0
8009	0.058036060	18.5858586	0	0	0
8010	0.065290568	18.5858586	0	0	0

8011	0.072545076	18.5858586	0	0	0
8012	0.079799583	18.5858586	0	0	0
8013	0.087054091	18.5858586	0	0	0
8014	0.094308598	18.5858586	0	0	0
8015	0.101563106	18.5858586	0	0	0
8016	0.108817613	18.5858586	0	0	0
8017	0.116072121	18.5858586	0	0	0
8018	0.123326628	18.5858586	0	0	0
8019	0.130581136	18.5858586	0	0	0
8020	0.137835643	18.5858586	0	0	0
8021	0.145090151	18.5858586	0	0	0
8022	0.152344659	18.5858586	0	0	0
8023	0.159599166	18.5858586	0	0	0
8024	0.166853674	18.5858586	0	0	0
8025	0.174108181	18.5858586	0	0	0
8026	0.181362689	18.5858586	0	0	0
8027	0.188617196	18.5858586	0	0	0
8028	0.195871704	18.5858586	0	0	0
8029	0.203126211	18.5858586	0	0	0
8030	0.210380719	18.5858586	0	0	0
8031	0.217635227	18.5858586	0	0	0
8032	0.224889734	18.5858586	0	0	0
8033	0.232144242	18.5858586	0	0	0
8034	0.239398749	18.5858586	0	0	0
8035	0.246653257	18.5858586	0	0	0
8036	0.253907764	18.5858586	0	0	0
8037	0.261162272	18.5858586	0	0	0
8038	0.268416779	18.5858586	0	0	0
8039	0.275671287	18.5858586	0	0	0
8040	0.282925794	18.5858586	0	0	0
8041	0.290180302	18.5858586	0	0	0
8042	0.297434810	18.5858586	0	0	0
8043	0.304689317	18.5858586	0	0	0
8044	0.311943825	18.5858586	0	0	0
8045	0.319198332	18.5858586	0	0	0
8046	0.326452840	18.5858586	0	0	0
8047	0.333707347	18.5858586	0	0	0
8048	0.340961855	18.5858586	0	0	0
8049	0.348216362	18.5858586	0	0	0
8050	0.355470870	18.5858586	0	0	0
8051	0.362725378	18.5858586	0	0	0
8052	0.369979885	18.5858586	0	0	0
8053	0.377234393	18.5858586	0	0	0
8054	0.384488900	18.5858586	0	0	0
8055	0.391743408	18.5858586	0	0	0

8056	0.398997915	18.5858586	0	0	0
8057	0.406252423	18.5858586	0	0	0
8058	0.413506930	18.5858586	0	0	0
8059	0.420761438	18.5858586	0	0	0
8060	0.428015945	18.5858586	0	0	0
8061	0.435270453	18.5858586	0	0	0
8062	0.442524961	18.5858586	0	0	0
8063	0.449779468	18.5858586	0	0	0
8064	0.457033976	18.5858586	0	0	0
8065	0.464288483	18.5858586	0	0	0
8066	0.471542991	18.5858586	0	0	0
8067	0.478797498	18.5858586	0	0	0
8068	0.486052006	18.5858586	0	0	0
8069	0.493306513	18.5858586	0	0	0
8070	0.500561021	18.5858586	0	0	0
8071	0.507815529	18.5858586	0	0	0
8072	0.515070036	18.5858586	0	0	0
8073	0.522324544	18.5858586	0	0	0
8074	0.529579051	18.5858586	0	0	0
8075	0.536833559	18.5858586	0	0	0
8076	0.544088066	18.5858586	0	0	0
8077	0.551342574	18.5858586	0	0	0
8078	0.558597081	18.5858586	0	0	0
8079	0.565851589	18.5858586	0	0	0
8080	0.573106097	18.5858586	0	0	0
8081	0.580360604	18.5858586	0	0	0
8082	0.587615112	18.5858586	0	0	0
8083	0.594869619	18.5858586	0	0	0
8084	0.602124127	18.5858586	0	0	0
8085	0.609378634	18.5858586	0	0	0
8086	0.616633142	18.5858586	0	0	0
8087	0.623887649	18.5858586	0	0	0
8088	0.631142157	18.5858586	0	0	0
8089	0.638396664	18.5858586	0	0	0
8090	0.645651172	18.5858586	0	0	0
8091	0.652905680	18.5858586	0	0	0
8092	0.660160187	18.5858586	0	0	0
8093	0.667414695	18.5858586	0	0	0
8094	0.674669202	18.5858586	0	0	0
8095	0.681923710	18.5858586	0	0	0
8096	0.689178217	18.5858586	0	0	0
8097	0.696432725	18.5858586	0	0	0
8098	0.703687232	18.5858586	0	0	0
8099	0.710941740	18.5858586	0	0	0
8100	0.718196248	18.5858586	0	0	0

8101	0.000000000	18.8181818	0	0	0
8102	0.007254508	18.8181818	0	0	0
8103	0.014509015	18.8181818	0	0	0
8104	0.021763523	18.8181818	0	0	0
8105	0.029018030	18.8181818	0	0	0
8106	0.036272538	18.8181818	0	0	0
8107	0.043527045	18.8181818	0	0	0
8108	0.050781553	18.8181818	0	0	0
8109	0.058036060	18.8181818	0	0	0
8110	0.065290568	18.8181818	0	0	0
8111	0.072545076	18.8181818	0	0	0
8112	0.079799583	18.8181818	0	0	0
8113	0.087054091	18.8181818	0	0	0
8114	0.094308598	18.8181818	0	0	0
8115	0.101563106	18.8181818	0	0	0
8116	0.108817613	18.8181818	0	0	0
8117	0.116072121	18.8181818	0	0	0
8118	0.123326628	18.8181818	0	0	0
8119	0.130581136	18.8181818	0	0	0
8120	0.137835643	18.8181818	0	0	0
8121	0.145090151	18.8181818	0	0	0
8122	0.152344659	18.8181818	0	0	0
8123	0.159599166	18.8181818	0	0	0
8124	0.166853674	18.8181818	0	0	0
8125	0.174108181	18.8181818	0	0	0
8126	0.181362689	18.8181818	0	0	0
8127	0.188617196	18.8181818	0	0	0
8128	0.195871704	18.8181818	0	0	0
8129	0.203126211	18.8181818	0	0	0
8130	0.210380719	18.8181818	0	0	0
8131	0.217635227	18.8181818	0	0	0
8132	0.224889734	18.8181818	0	0	0
8133	0.232144242	18.8181818	0	0	0
8134	0.239398749	18.8181818	0	0	0
8135	0.246653257	18.8181818	0	0	0
8136	0.253907764	18.8181818	0	0	0
8137	0.261162272	18.8181818	0	0	0
8138	0.268416779	18.8181818	0	0	0
8139	0.275671287	18.8181818	0	0	0
8140	0.282925794	18.8181818	0	0	0
8141	0.290180302	18.8181818	0	0	0
8142	0.297434810	18.8181818	0	0	0
8143	0.304689317	18.8181818	0	0	0
8144	0.311943825	18.8181818	0	0	0
8145	0.319198332	18.8181818	0	0	0

8146	0.326452840	18.8181818	0	0	0
8147	0.333707347	18.8181818	0	0	0
8148	0.340961855	18.8181818	0	0	0
8149	0.348216362	18.8181818	0	0	0
8150	0.355470870	18.8181818	0	0	0
8151	0.362725378	18.8181818	0	0	0
8152	0.369979885	18.8181818	0	0	0
8153	0.377234393	18.8181818	0	0	0
8154	0.384488900	18.8181818	0	0	0
8155	0.391743408	18.8181818	0	0	0
8156	0.398997915	18.8181818	0	0	0
8157	0.406252423	18.8181818	0	0	0
8158	0.413506930	18.8181818	0	0	0
8159	0.420761438	18.8181818	0	0	0
8160	0.428015945	18.8181818	0	0	0
8161	0.435270453	18.8181818	0	0	0
8162	0.442524961	18.8181818	0	0	0
8163	0.449779468	18.8181818	0	0	0
8164	0.457033976	18.8181818	0	0	0
8165	0.464288483	18.8181818	0	0	0
8166	0.471542991	18.8181818	0	0	0
8167	0.478797498	18.8181818	0	0	0
8168	0.486052006	18.8181818	0	0	0
8169	0.493306513	18.8181818	0	0	0
8170	0.500561021	18.8181818	0	0	0
8171	0.507815529	18.8181818	0	0	0
8172	0.515070036	18.8181818	0	0	0
8173	0.522324544	18.8181818	0	0	0
8174	0.529579051	18.8181818	0	0	0
8175	0.536833559	18.8181818	0	0	0
8176	0.544088066	18.8181818	0	0	0
8177	0.551342574	18.8181818	0	0	0
8178	0.558597081	18.8181818	0	0	0
8179	0.565851589	18.8181818	0	0	0
8180	0.573106097	18.8181818	0	0	0
8181	0.580360604	18.8181818	0	0	0
8182	0.587615112	18.8181818	0	0	0
8183	0.594869619	18.8181818	0	0	0
8184	0.602124127	18.8181818	0	0	0
8185	0.609378634	18.8181818	0	0	0
8186	0.616633142	18.8181818	0	0	0
8187	0.623887649	18.8181818	0	0	0
8188	0.631142157	18.8181818	0	0	0
8189	0.638396664	18.8181818	0	0	0
8190	0.645651172	18.8181818	0	0	0

8191	0.652905680	18.8181818	0	0	0
8192	0.660160187	18.8181818	0	0	0
8193	0.667414695	18.8181818	0	0	0
8194	0.674669202	18.8181818	0	0	0
8195	0.681923710	18.8181818	0	0	0
8196	0.689178217	18.8181818	0	0	0
8197	0.696432725	18.8181818	0	0	0
8198	0.703687232	18.8181818	0	0	0
8199	0.710941740	18.8181818	0	0	0
8200	0.718196248	18.8181818	0	0	0
8201	0.000000000	19.0505051	0	0	0
8202	0.007254508	19.0505051	0	0	0
8203	0.014509015	19.0505051	0	0	0
8204	0.021763523	19.0505051	0	0	0
8205	0.029018030	19.0505051	0	0	0
8206	0.036272538	19.0505051	0	0	0
8207	0.043527045	19.0505051	0	0	0
8208	0.050781553	19.0505051	0	0	0
8209	0.058036060	19.0505051	0	0	0
8210	0.065290568	19.0505051	0	0	0
8211	0.072545076	19.0505051	0	0	0
8212	0.079799583	19.0505051	0	0	0
8213	0.087054091	19.0505051	0	0	0
8214	0.094308598	19.0505051	0	0	0
8215	0.101563106	19.0505051	0	0	0
8216	0.108817613	19.0505051	0	0	0
8217	0.116072121	19.0505051	0	0	0
8218	0.123326628	19.0505051	0	0	0
8219	0.130581136	19.0505051	0	0	0
8220	0.137835643	19.0505051	0	0	0
8221	0.145090151	19.0505051	0	0	0
8222	0.152344659	19.0505051	0	0	0
8223	0.159599166	19.0505051	0	0	0
8224	0.166853674	19.0505051	0	0	0
8225	0.174108181	19.0505051	0	0	0
8226	0.181362689	19.0505051	0	0	0
8227	0.188617196	19.0505051	0	0	0
8228	0.195871704	19.0505051	0	0	0
8229	0.203126211	19.0505051	0	0	0
8230	0.210380719	19.0505051	0	0	0
8231	0.217635227	19.0505051	0	0	0
8232	0.224889734	19.0505051	0	0	0
8233	0.232144242	19.0505051	0	0	0
8234	0.239398749	19.0505051	0	0	0
8235	0.246653257	19.0505051	0	0	0

8236	0.253907764	19.0505051	0	0	0
8237	0.261162272	19.0505051	0	0	0
8238	0.268416779	19.0505051	0	0	0
8239	0.275671287	19.0505051	0	0	0
8240	0.282925794	19.0505051	0	0	0
8241	0.290180302	19.0505051	0	0	0
8242	0.297434810	19.0505051	0	0	0
8243	0.304689317	19.0505051	0	0	0
8244	0.311943825	19.0505051	0	0	0
8245	0.319198332	19.0505051	0	0	0
8246	0.326452840	19.0505051	0	0	0
8247	0.333707347	19.0505051	0	0	0
8248	0.340961855	19.0505051	0	0	0
8249	0.348216362	19.0505051	0	0	0
8250	0.355470870	19.0505051	0	0	0
8251	0.362725378	19.0505051	0	0	0
8252	0.369979885	19.0505051	0	0	0
8253	0.377234393	19.0505051	0	0	0
8254	0.384488900	19.0505051	0	0	0
8255	0.391743408	19.0505051	0	0	0
8256	0.398997915	19.0505051	0	0	0
8257	0.406252423	19.0505051	0	0	0
8258	0.413506930	19.0505051	0	0	0
8259	0.420761438	19.0505051	0	0	0
8260	0.428015945	19.0505051	0	0	0
8261	0.435270453	19.0505051	0	0	0
8262	0.442524961	19.0505051	0	0	0
8263	0.449779468	19.0505051	0	0	0
8264	0.457033976	19.0505051	0	0	0
8265	0.464288483	19.0505051	0	0	0
8266	0.471542991	19.0505051	0	0	0
8267	0.478797498	19.0505051	0	0	0
8268	0.486052006	19.0505051	0	0	0
8269	0.493306513	19.0505051	0	0	0
8270	0.500561021	19.0505051	0	0	0
8271	0.507815529	19.0505051	0	0	0
8272	0.515070036	19.0505051	0	0	0
8273	0.522324544	19.0505051	0	0	0
8274	0.529579051	19.0505051	0	0	0
8275	0.536833559	19.0505051	0	0	0
8276	0.544088066	19.0505051	0	0	0
8277	0.551342574	19.0505051	0	0	0
8278	0.558597081	19.0505051	0	0	0
8279	0.565851589	19.0505051	0	0	0
8280	0.573106097	19.0505051	0	0	0

8281	0.580360604	19.0505051	0	0	0
8282	0.587615112	19.0505051	0	0	0
8283	0.594869619	19.0505051	0	0	0
8284	0.602124127	19.0505051	0	0	0
8285	0.609378634	19.0505051	0	0	0
8286	0.616633142	19.0505051	0	0	0
8287	0.623887649	19.0505051	0	0	0
8288	0.631142157	19.0505051	0	0	0
8289	0.638396664	19.0505051	0	0	0
8290	0.645651172	19.0505051	0	0	0
8291	0.652905680	19.0505051	0	0	0
8292	0.660160187	19.0505051	0	0	0
8293	0.667414695	19.0505051	0	0	0
8294	0.674669202	19.0505051	0	0	0
8295	0.681923710	19.0505051	0	0	0
8296	0.689178217	19.0505051	0	0	0
8297	0.696432725	19.0505051	0	0	0
8298	0.703687232	19.0505051	0	0	0
8299	0.710941740	19.0505051	0	0	0
8300	0.718196248	19.0505051	0	0	0
8301	0.000000000	19.2828283	0	0	0
8302	0.007254508	19.2828283	0	0	0
8303	0.014509015	19.2828283	0	0	0
8304	0.021763523	19.2828283	0	0	0
8305	0.029018030	19.2828283	0	0	0
8306	0.036272538	19.2828283	0	0	0
8307	0.043527045	19.2828283	0	0	0
8308	0.050781553	19.2828283	0	0	0
8309	0.058036060	19.2828283	0	0	0
8310	0.065290568	19.2828283	0	0	0
8311	0.072545076	19.2828283	0	0	0
8312	0.079799583	19.2828283	0	0	0
8313	0.087054091	19.2828283	0	0	0
8314	0.094308598	19.2828283	0	0	0
8315	0.101563106	19.2828283	0	0	0
8316	0.108817613	19.2828283	0	0	0
8317	0.116072121	19.2828283	0	0	0
8318	0.123326628	19.2828283	0	0	0
8319	0.130581136	19.2828283	0	0	0
8320	0.137835643	19.2828283	0	0	0
8321	0.145090151	19.2828283	0	0	0
8322	0.152344659	19.2828283	0	0	0
8323	0.159599166	19.2828283	0	0	0
8324	0.166853674	19.2828283	0	0	0
8325	0.174108181	19.2828283	0	0	0

8326	0.181362689	19.2828283	0	0	0
8327	0.188617196	19.2828283	0	0	0
8328	0.195871704	19.2828283	0	0	0
8329	0.203126211	19.2828283	0	0	0
8330	0.210380719	19.2828283	0	0	0
8331	0.217635227	19.2828283	0	0	0
8332	0.224889734	19.2828283	0	0	0
8333	0.232144242	19.2828283	0	0	0
8334	0.239398749	19.2828283	0	0	0
8335	0.246653257	19.2828283	0	0	0
8336	0.253907764	19.2828283	0	0	0
8337	0.261162272	19.2828283	0	0	0
8338	0.268416779	19.2828283	0	0	0
8339	0.275671287	19.2828283	0	0	0
8340	0.282925794	19.2828283	0	0	0
8341	0.290180302	19.2828283	0	0	0
8342	0.297434810	19.2828283	0	0	0
8343	0.304689317	19.2828283	0	0	0
8344	0.311943825	19.2828283	0	0	0
8345	0.319198332	19.2828283	0	0	0
8346	0.326452840	19.2828283	0	0	0
8347	0.333707347	19.2828283	0	0	0
8348	0.340961855	19.2828283	0	0	0
8349	0.348216362	19.2828283	0	0	0
8350	0.355470870	19.2828283	0	0	0
8351	0.362725378	19.2828283	0	0	0
8352	0.369979885	19.2828283	0	0	0
8353	0.377234393	19.2828283	0	0	0
8354	0.384488900	19.2828283	0	0	0
8355	0.391743408	19.2828283	0	0	0
8356	0.398997915	19.2828283	0	0	0
8357	0.406252423	19.2828283	0	0	0
8358	0.413506930	19.2828283	0	0	0
8359	0.420761438	19.2828283	0	0	0
8360	0.428015945	19.2828283	0	0	0
8361	0.435270453	19.2828283	0	0	0
8362	0.442524961	19.2828283	0	0	0
8363	0.449779468	19.2828283	0	0	0
8364	0.457033976	19.2828283	0	0	0
8365	0.464288483	19.2828283	0	0	0
8366	0.471542991	19.2828283	0	0	0
8367	0.478797498	19.2828283	0	0	0
8368	0.486052006	19.2828283	0	0	0
8369	0.493306513	19.2828283	0	0	0
8370	0.500561021	19.2828283	0	0	0

8371	0.507815529	19.2828283	0	0	0
8372	0.515070036	19.2828283	0	0	0
8373	0.522324544	19.2828283	0	0	0
8374	0.529579051	19.2828283	0	0	0
8375	0.536833559	19.2828283	0	0	0
8376	0.544088066	19.2828283	0	0	0
8377	0.551342574	19.2828283	0	0	0
8378	0.558597081	19.2828283	0	0	0
8379	0.565851589	19.2828283	0	0	0
8380	0.573106097	19.2828283	0	0	0
8381	0.580360604	19.2828283	0	0	0
8382	0.587615112	19.2828283	0	0	0
8383	0.594869619	19.2828283	0	0	0
8384	0.602124127	19.2828283	0	0	0
8385	0.609378634	19.2828283	0	0	0
8386	0.616633142	19.2828283	0	0	0
8387	0.623887649	19.2828283	0	0	0
8388	0.631142157	19.2828283	0	0	0
8389	0.638396664	19.2828283	0	0	0
8390	0.645651172	19.2828283	0	0	0
8391	0.652905680	19.2828283	0	0	0
8392	0.660160187	19.2828283	0	0	0
8393	0.667414695	19.2828283	0	0	0
8394	0.674669202	19.2828283	0	0	0
8395	0.681923710	19.2828283	0	0	0
8396	0.689178217	19.2828283	0	0	0
8397	0.696432725	19.2828283	0	0	0
8398	0.703687232	19.2828283	0	0	0
8399	0.710941740	19.2828283	0	0	0
8400	0.718196248	19.2828283	0	0	0
8401	0.000000000	19.5151515	0	0	0
8402	0.007254508	19.5151515	0	0	0
8403	0.014509015	19.5151515	0	0	0
8404	0.021763523	19.5151515	0	0	0
8405	0.029018030	19.5151515	0	0	0
8406	0.036272538	19.5151515	0	0	0
8407	0.043527045	19.5151515	0	0	0
8408	0.050781553	19.5151515	0	0	0
8409	0.058036060	19.5151515	0	0	0
8410	0.065290568	19.5151515	0	0	0
8411	0.072545076	19.5151515	0	0	0
8412	0.079799583	19.5151515	0	0	0
8413	0.087054091	19.5151515	0	0	0
8414	0.094308598	19.5151515	0	0	0
8415	0.101563106	19.5151515	0	0	0

8416	0.108817613	19.5151515	0	0	0
8417	0.116072121	19.5151515	0	0	0
8418	0.123326628	19.5151515	0	0	0
8419	0.130581136	19.5151515	0	0	0
8420	0.137835643	19.5151515	0	0	0
8421	0.145090151	19.5151515	0	0	0
8422	0.152344659	19.5151515	0	0	0
8423	0.159599166	19.5151515	0	0	0
8424	0.166853674	19.5151515	0	0	0
8425	0.174108181	19.5151515	0	0	0
8426	0.181362689	19.5151515	0	0	0
8427	0.188617196	19.5151515	0	0	0
8428	0.195871704	19.5151515	0	0	0
8429	0.203126211	19.5151515	0	0	0
8430	0.210380719	19.5151515	0	0	0
8431	0.217635227	19.5151515	0	0	0
8432	0.224889734	19.5151515	0	0	0
8433	0.232144242	19.5151515	0	0	0
8434	0.239398749	19.5151515	0	0	0
8435	0.246653257	19.5151515	0	0	0
8436	0.253907764	19.5151515	0	0	0
8437	0.261162272	19.5151515	0	0	0
8438	0.268416779	19.5151515	0	0	0
8439	0.275671287	19.5151515	0	0	0
8440	0.282925794	19.5151515	0	0	0
8441	0.290180302	19.5151515	0	0	0
8442	0.297434810	19.5151515	0	0	0
8443	0.304689317	19.5151515	0	0	0
8444	0.311943825	19.5151515	0	0	0
8445	0.319198332	19.5151515	0	0	0
8446	0.326452840	19.5151515	0	0	0
8447	0.333707347	19.5151515	0	0	0
8448	0.340961855	19.5151515	0	0	0
8449	0.348216362	19.5151515	0	0	0
8450	0.355470870	19.5151515	0	0	0
8451	0.362725378	19.5151515	0	0	0
8452	0.369979885	19.5151515	0	0	0
8453	0.377234393	19.5151515	0	0	0
8454	0.384488900	19.5151515	0	0	0
8455	0.391743408	19.5151515	0	0	0
8456	0.398997915	19.5151515	0	0	0
8457	0.406252423	19.5151515	0	0	0
8458	0.413506930	19.5151515	0	0	0
8459	0.420761438	19.5151515	0	0	0
8460	0.428015945	19.5151515	0	0	0

8461	0.435270453	19.5151515	0	0	0
8462	0.442524961	19.5151515	0	0	0
8463	0.449779468	19.5151515	0	0	0
8464	0.457033976	19.5151515	0	0	0
8465	0.464288483	19.5151515	0	0	0
8466	0.471542991	19.5151515	0	0	0
8467	0.478797498	19.5151515	0	0	0
8468	0.486052006	19.5151515	0	0	0
8469	0.493306513	19.5151515	0	0	0
8470	0.500561021	19.5151515	0	0	0
8471	0.507815529	19.5151515	0	0	0
8472	0.515070036	19.5151515	0	0	0
8473	0.522324544	19.5151515	0	0	0
8474	0.529579051	19.5151515	0	0	0
8475	0.536833559	19.5151515	0	0	0
8476	0.544088066	19.5151515	0	0	0
8477	0.551342574	19.5151515	0	0	0
8478	0.558597081	19.5151515	0	0	0
8479	0.565851589	19.5151515	0	0	0
8480	0.573106097	19.5151515	0	0	0
8481	0.580360604	19.5151515	0	0	0
8482	0.587615112	19.5151515	0	0	0
8483	0.594869619	19.5151515	0	0	0
8484	0.602124127	19.5151515	0	0	0
8485	0.609378634	19.5151515	0	0	0
8486	0.616633142	19.5151515	0	0	0
8487	0.623887649	19.5151515	0	0	0
8488	0.631142157	19.5151515	0	0	0
8489	0.638396664	19.5151515	0	0	0
8490	0.645651172	19.5151515	0	0	0
8491	0.652905680	19.5151515	0	0	0
8492	0.660160187	19.5151515	0	0	0
8493	0.667414695	19.5151515	0	0	0
8494	0.674669202	19.5151515	0	0	0
8495	0.681923710	19.5151515	0	0	0
8496	0.689178217	19.5151515	0	0	0
8497	0.696432725	19.5151515	0	0	0
8498	0.703687232	19.5151515	0	0	0
8499	0.710941740	19.5151515	0	0	0
8500	0.718196248	19.5151515	0	0	0
8501	0.000000000	19.7474747	0	0	0
8502	0.007254508	19.7474747	0	0	0
8503	0.014509015	19.7474747	0	0	0
8504	0.021763523	19.7474747	0	0	0
8505	0.029018030	19.7474747	0	0	0

8506	0.036272538	19.7474747	0	0	0
8507	0.043527045	19.7474747	0	0	0
8508	0.050781553	19.7474747	0	0	0
8509	0.058036060	19.7474747	0	0	0
8510	0.065290568	19.7474747	0	0	0
8511	0.072545076	19.7474747	0	0	0
8512	0.079799583	19.7474747	0	0	0
8513	0.087054091	19.7474747	0	0	0
8514	0.094308598	19.7474747	0	0	0
8515	0.101563106	19.7474747	0	0	0
8516	0.108817613	19.7474747	0	0	0
8517	0.116072121	19.7474747	0	0	0
8518	0.123326628	19.7474747	0	0	0
8519	0.130581136	19.7474747	0	0	0
8520	0.137835643	19.7474747	0	0	0
8521	0.145090151	19.7474747	0	0	0
8522	0.152344659	19.7474747	0	0	0
8523	0.159599166	19.7474747	0	0	0
8524	0.166853674	19.7474747	0	0	0
8525	0.174108181	19.7474747	0	0	0
8526	0.181362689	19.7474747	0	0	0
8527	0.188617196	19.7474747	0	0	0
8528	0.195871704	19.7474747	0	0	0
8529	0.203126211	19.7474747	0	0	0
8530	0.210380719	19.7474747	0	0	0
8531	0.217635227	19.7474747	0	0	0
8532	0.224889734	19.7474747	0	0	0
8533	0.232144242	19.7474747	0	0	0
8534	0.239398749	19.7474747	0	0	0
8535	0.246653257	19.7474747	0	0	0
8536	0.253907764	19.7474747	0	0	0
8537	0.261162272	19.7474747	0	0	0
8538	0.268416779	19.7474747	0	0	0
8539	0.275671287	19.7474747	0	0	0
8540	0.282925794	19.7474747	0	0	0
8541	0.290180302	19.7474747	0	0	0
8542	0.297434810	19.7474747	0	0	0
8543	0.304689317	19.7474747	0	0	0
8544	0.311943825	19.7474747	0	0	0
8545	0.319198332	19.7474747	0	0	0
8546	0.326452840	19.7474747	0	0	0
8547	0.333707347	19.7474747	0	0	0
8548	0.340961855	19.7474747	0	0	0
8549	0.348216362	19.7474747	0	0	0
8550	0.355470870	19.7474747	0	0	0

8551	0.362725378	19.7474747	0	0	0
8552	0.369979885	19.7474747	0	0	0
8553	0.377234393	19.7474747	0	0	0
8554	0.384488900	19.7474747	0	0	0
8555	0.391743408	19.7474747	0	0	0
8556	0.398997915	19.7474747	0	0	0
8557	0.406252423	19.7474747	0	0	0
8558	0.413506930	19.7474747	0	0	0
8559	0.420761438	19.7474747	0	0	0
8560	0.428015945	19.7474747	0	0	0
8561	0.435270453	19.7474747	0	0	0
8562	0.442524961	19.7474747	0	0	0
8563	0.449779468	19.7474747	0	0	0
8564	0.457033976	19.7474747	0	0	0
8565	0.464288483	19.7474747	0	0	0
8566	0.471542991	19.7474747	0	0	0
8567	0.478797498	19.7474747	0	0	0
8568	0.486052006	19.7474747	0	0	0
8569	0.493306513	19.7474747	0	0	0
8570	0.500561021	19.7474747	0	0	0
8571	0.507815529	19.7474747	0	0	0
8572	0.515070036	19.7474747	0	0	0
8573	0.522324544	19.7474747	0	0	0
8574	0.529579051	19.7474747	0	0	0
8575	0.536833559	19.7474747	0	0	0
8576	0.544088066	19.7474747	0	0	0
8577	0.551342574	19.7474747	0	0	0
8578	0.558597081	19.7474747	0	0	0
8579	0.565851589	19.7474747	0	0	0
8580	0.573106097	19.7474747	0	0	0
8581	0.580360604	19.7474747	0	0	0
8582	0.587615112	19.7474747	0	0	0
8583	0.594869619	19.7474747	0	0	0
8584	0.602124127	19.7474747	0	0	0
8585	0.609378634	19.7474747	0	0	0
8586	0.616633142	19.7474747	0	0	0
8587	0.623887649	19.7474747	0	0	0
8588	0.631142157	19.7474747	0	0	0
8589	0.638396664	19.7474747	0	0	0
8590	0.645651172	19.7474747	0	0	0
8591	0.652905680	19.7474747	0	0	0
8592	0.660160187	19.7474747	0	0	0
8593	0.667414695	19.7474747	0	0	0
8594	0.674669202	19.7474747	0	0	0
8595	0.681923710	19.7474747	0	0	0

8596	0.689178217	19.7474747	0	0	0
8597	0.696432725	19.7474747	0	0	0
8598	0.703687232	19.7474747	0	0	0
8599	0.710941740	19.7474747	0	0	0
8600	0.718196248	19.7474747	0	0	0
8601	0.000000000	19.9797980	0	0	0
8602	0.007254508	19.9797980	0	0	0
8603	0.014509015	19.9797980	0	0	0
8604	0.021763523	19.9797980	0	0	0
8605	0.029018030	19.9797980	0	0	0
8606	0.036272538	19.9797980	0	0	0
8607	0.043527045	19.9797980	0	0	0
8608	0.050781553	19.9797980	0	0	0
8609	0.058036060	19.9797980	0	0	0
8610	0.065290568	19.9797980	0	0	0
8611	0.072545076	19.9797980	0	0	0
8612	0.079799583	19.9797980	0	0	0
8613	0.087054091	19.9797980	0	0	0
8614	0.094308598	19.9797980	0	0	0
8615	0.101563106	19.9797980	0	0	0
8616	0.108817613	19.9797980	0	0	0
8617	0.116072121	19.9797980	0	0	0
8618	0.123326628	19.9797980	0	0	0
8619	0.130581136	19.9797980	0	0	0
8620	0.137835643	19.9797980	0	0	0
8621	0.145090151	19.9797980	0	0	0
8622	0.152344659	19.9797980	0	0	0
8623	0.159599166	19.9797980	0	0	0
8624	0.166853674	19.9797980	0	0	0
8625	0.174108181	19.9797980	0	0	0
8626	0.181362689	19.9797980	0	0	0
8627	0.188617196	19.9797980	0	0	0
8628	0.195871704	19.9797980	0	0	0
8629	0.203126211	19.9797980	0	0	0
8630	0.210380719	19.9797980	0	0	0
8631	0.217635227	19.9797980	0	0	0
8632	0.224889734	19.9797980	0	0	0
8633	0.232144242	19.9797980	0	0	0
8634	0.239398749	19.9797980	0	0	0
8635	0.246653257	19.9797980	0	0	0
8636	0.253907764	19.9797980	0	0	0
8637	0.261162272	19.9797980	0	0	0
8638	0.268416779	19.9797980	0	0	0
8639	0.275671287	19.9797980	0	0	0
8640	0.282925794	19.9797980	0	0	0

8641	0.290180302	19.9797980	0	0	0
8642	0.297434810	19.9797980	0	0	0
8643	0.304689317	19.9797980	0	0	0
8644	0.311943825	19.9797980	0	0	0
8645	0.319198332	19.9797980	0	0	0
8646	0.326452840	19.9797980	0	0	0
8647	0.333707347	19.9797980	0	0	0
8648	0.340961855	19.9797980	0	0	0
8649	0.348216362	19.9797980	0	0	0
8650	0.355470870	19.9797980	0	0	0
8651	0.362725378	19.9797980	0	0	0
8652	0.369979885	19.9797980	0	0	0
8653	0.377234393	19.9797980	0	0	0
8654	0.384488900	19.9797980	0	0	0
8655	0.391743408	19.9797980	0	0	0
8656	0.398997915	19.9797980	0	0	0
8657	0.406252423	19.9797980	0	0	0
8658	0.413506930	19.9797980	0	0	0
8659	0.420761438	19.9797980	0	0	0
8660	0.428015945	19.9797980	0	0	0
8661	0.435270453	19.9797980	0	0	0
8662	0.442524961	19.9797980	0	0	0
8663	0.449779468	19.9797980	0	0	0
8664	0.457033976	19.9797980	0	0	0
8665	0.464288483	19.9797980	0	0	0
8666	0.471542991	19.9797980	0	0	0
8667	0.478797498	19.9797980	0	0	0
8668	0.486052006	19.9797980	0	0	0
8669	0.493306513	19.9797980	0	0	0
8670	0.500561021	19.9797980	0	0	0
8671	0.507815529	19.9797980	0	0	0
8672	0.515070036	19.9797980	0	0	0
8673	0.522324544	19.9797980	0	0	0
8674	0.529579051	19.9797980	0	0	0
8675	0.536833559	19.9797980	0	0	0
8676	0.544088066	19.9797980	0	0	0
8677	0.551342574	19.9797980	0	0	0
8678	0.558597081	19.9797980	0	0	0
8679	0.565851589	19.9797980	0	0	0
8680	0.573106097	19.9797980	0	0	0
8681	0.580360604	19.9797980	0	0	0
8682	0.587615112	19.9797980	0	0	0
8683	0.594869619	19.9797980	0	0	0
8684	0.602124127	19.9797980	0	0	0
8685	0.609378634	19.9797980	0	0	0

8686	0.616633142	19.9797980	0	0	0
8687	0.623887649	19.9797980	0	0	0
8688	0.631142157	19.9797980	0	0	0
8689	0.638396664	19.9797980	0	0	0
8690	0.645651172	19.9797980	0	0	0
8691	0.652905680	19.9797980	0	0	0
8692	0.660160187	19.9797980	0	0	0
8693	0.667414695	19.9797980	0	0	0
8694	0.674669202	19.9797980	0	0	0
8695	0.681923710	19.9797980	0	0	0
8696	0.689178217	19.9797980	0	0	0
8697	0.696432725	19.9797980	0	0	0
8698	0.703687232	19.9797980	0	0	0
8699	0.710941740	19.9797980	0	0	0
8700	0.718196248	19.9797980	0	0	0
8701	0.000000000	20.2121212	0	0	0
8702	0.007254508	20.2121212	0	0	0
8703	0.014509015	20.2121212	0	0	0
8704	0.021763523	20.2121212	0	0	0
8705	0.029018030	20.2121212	0	0	0
8706	0.036272538	20.2121212	0	0	0
8707	0.043527045	20.2121212	0	0	0
8708	0.050781553	20.2121212	0	0	0
8709	0.058036060	20.2121212	0	0	0
8710	0.065290568	20.2121212	0	0	0
8711	0.072545076	20.2121212	0	0	0
8712	0.079799583	20.2121212	0	0	0
8713	0.087054091	20.2121212	0	0	0
8714	0.094308598	20.2121212	0	0	0
8715	0.101563106	20.2121212	0	0	0
8716	0.108817613	20.2121212	0	0	0
8717	0.116072121	20.2121212	0	0	0
8718	0.123326628	20.2121212	0	0	0
8719	0.130581136	20.2121212	0	0	0
8720	0.137835643	20.2121212	0	0	0
8721	0.145090151	20.2121212	0	0	0
8722	0.152344659	20.2121212	0	0	0
8723	0.159599166	20.2121212	0	0	0
8724	0.166853674	20.2121212	0	0	0
8725	0.174108181	20.2121212	0	0	0
8726	0.181362689	20.2121212	0	0	0
8727	0.188617196	20.2121212	0	0	0
8728	0.195871704	20.2121212	0	0	0
8729	0.203126211	20.2121212	0	0	0
8730	0.210380719	20.2121212	0	0	0

8731	0.217635227	20.2121212	0	0	0
8732	0.224889734	20.2121212	0	0	0
8733	0.232144242	20.2121212	0	0	0
8734	0.239398749	20.2121212	0	0	0
8735	0.246653257	20.2121212	0	0	0
8736	0.253907764	20.2121212	0	0	0
8737	0.261162272	20.2121212	0	0	0
8738	0.268416779	20.2121212	0	0	0
8739	0.275671287	20.2121212	0	0	0
8740	0.282925794	20.2121212	0	0	0
8741	0.290180302	20.2121212	0	0	0
8742	0.297434810	20.2121212	0	0	0
8743	0.304689317	20.2121212	0	0	0
8744	0.311943825	20.2121212	0	0	0
8745	0.319198332	20.2121212	0	0	0
8746	0.326452840	20.2121212	0	0	0
8747	0.333707347	20.2121212	0	0	0
8748	0.340961855	20.2121212	0	0	0
8749	0.348216362	20.2121212	0	0	0
8750	0.355470870	20.2121212	0	0	0
8751	0.362725378	20.2121212	0	0	0
8752	0.369979885	20.2121212	0	0	0
8753	0.377234393	20.2121212	0	0	0
8754	0.384488900	20.2121212	0	0	0
8755	0.391743408	20.2121212	0	0	0
8756	0.398997915	20.2121212	0	0	0
8757	0.406252423	20.2121212	0	0	0
8758	0.413506930	20.2121212	0	0	0
8759	0.420761438	20.2121212	0	0	0
8760	0.428015945	20.2121212	0	0	0
8761	0.435270453	20.2121212	0	0	0
8762	0.442524961	20.2121212	0	0	0
8763	0.449779468	20.2121212	0	0	0
8764	0.457033976	20.2121212	0	0	0
8765	0.464288483	20.2121212	0	0	0
8766	0.471542991	20.2121212	0	0	0
8767	0.478797498	20.2121212	0	0	0
8768	0.486052006	20.2121212	0	0	0
8769	0.493306513	20.2121212	0	0	0
8770	0.500561021	20.2121212	0	0	0
8771	0.507815529	20.2121212	0	0	0
8772	0.515070036	20.2121212	0	0	0
8773	0.522324544	20.2121212	0	0	0
8774	0.529579051	20.2121212	0	0	0
8775	0.536833559	20.2121212	0	0	0

8776	0.544088066	20.2121212	0	0	0
8777	0.551342574	20.2121212	0	0	0
8778	0.558597081	20.2121212	0	0	0
8779	0.565851589	20.2121212	0	0	0
8780	0.573106097	20.2121212	0	0	0
8781	0.580360604	20.2121212	0	0	0
8782	0.587615112	20.2121212	0	0	0
8783	0.594869619	20.2121212	0	0	0
8784	0.602124127	20.2121212	0	0	0
8785	0.609378634	20.2121212	0	0	0
8786	0.616633142	20.2121212	0	0	0
8787	0.623887649	20.2121212	0	0	0
8788	0.631142157	20.2121212	0	0	0
8789	0.638396664	20.2121212	0	0	0
8790	0.645651172	20.2121212	0	0	0
8791	0.652905680	20.2121212	0	0	0
8792	0.660160187	20.2121212	0	0	0
8793	0.667414695	20.2121212	0	0	0
8794	0.674669202	20.2121212	0	0	0
8795	0.681923710	20.2121212	0	0	0
8796	0.689178217	20.2121212	0	0	0
8797	0.696432725	20.2121212	0	0	0
8798	0.703687232	20.2121212	0	0	0
8799	0.710941740	20.2121212	0	0	0
8800	0.718196248	20.2121212	0	0	0
8801	0.000000000	20.4444444	0	0	0
8802	0.007254508	20.4444444	0	0	0
8803	0.014509015	20.4444444	0	0	0
8804	0.021763523	20.4444444	0	0	0
8805	0.029018030	20.4444444	0	0	0
8806	0.036272538	20.4444444	0	0	0
8807	0.043527045	20.4444444	0	0	0
8808	0.050781553	20.4444444	0	0	0
8809	0.058036060	20.4444444	0	0	0
8810	0.065290568	20.4444444	0	0	0
8811	0.072545076	20.4444444	0	0	0
8812	0.079799583	20.4444444	0	0	0
8813	0.087054091	20.4444444	0	0	0
8814	0.094308598	20.4444444	0	0	0
8815	0.101563106	20.4444444	0	0	0
8816	0.108817613	20.4444444	0	0	0
8817	0.116072121	20.4444444	0	0	0
8818	0.123326628	20.4444444	0	0	0
8819	0.130581136	20.4444444	0	0	0
8820	0.137835643	20.4444444	0	0	0

8821	0.145090151	20.44444444	0	0	0
8822	0.152344659	20.44444444	0	0	0
8823	0.159599166	20.44444444	0	0	0
8824	0.166853674	20.44444444	0	0	0
8825	0.174108181	20.44444444	0	0	0
8826	0.181362689	20.44444444	0	0	0
8827	0.188617196	20.44444444	0	0	0
8828	0.195871704	20.44444444	0	0	0
8829	0.203126211	20.44444444	0	0	0
8830	0.210380719	20.44444444	0	0	0
8831	0.217635227	20.44444444	0	0	0
8832	0.224889734	20.44444444	0	0	0
8833	0.232144242	20.44444444	0	0	0
8834	0.239398749	20.44444444	0	0	0
8835	0.246653257	20.44444444	0	0	0
8836	0.253907764	20.44444444	0	0	0
8837	0.261162272	20.44444444	0	0	0
8838	0.268416779	20.44444444	0	0	0
8839	0.275671287	20.44444444	0	0	0
8840	0.282925794	20.44444444	0	0	0
8841	0.290180302	20.44444444	0	0	0
8842	0.297434810	20.44444444	0	0	0
8843	0.304689317	20.44444444	0	0	0
8844	0.311943825	20.44444444	0	0	0
8845	0.319198332	20.44444444	0	0	0
8846	0.326452840	20.44444444	0	0	0
8847	0.333707347	20.44444444	0	0	0
8848	0.340961855	20.44444444	0	0	0
8849	0.348216362	20.44444444	0	0	0
8850	0.355470870	20.44444444	0	0	0
8851	0.362725378	20.44444444	0	0	0
8852	0.369979885	20.44444444	0	0	0
8853	0.377234393	20.44444444	0	0	0
8854	0.384488900	20.44444444	0	0	0
8855	0.391743408	20.44444444	0	0	0
8856	0.398997915	20.44444444	0	0	0
8857	0.406252423	20.44444444	0	0	0
8858	0.413506930	20.44444444	0	0	0
8859	0.420761438	20.44444444	0	0	0
8860	0.428015945	20.44444444	0	0	0
8861	0.435270453	20.44444444	0	0	0
8862	0.442524961	20.44444444	0	0	0
8863	0.449779468	20.44444444	0	0	0
8864	0.457033976	20.44444444	0	0	0
8865	0.464288483	20.44444444	0	0	0

8866	0.471542991	20.44444444	0	0	0
8867	0.478797498	20.44444444	0	0	0
8868	0.486052006	20.44444444	0	0	0
8869	0.493306513	20.44444444	0	0	0
8870	0.500561021	20.44444444	0	0	0
8871	0.507815529	20.44444444	0	0	0
8872	0.515070036	20.44444444	0	0	0
8873	0.522324544	20.44444444	0	0	0
8874	0.529579051	20.44444444	0	0	0
8875	0.536833559	20.44444444	0	0	0
8876	0.544088066	20.44444444	0	0	0
8877	0.551342574	20.44444444	0	0	0
8878	0.558597081	20.44444444	0	0	0
8879	0.565851589	20.44444444	0	0	0
8880	0.573106097	20.44444444	0	0	0
8881	0.580360604	20.44444444	0	0	0
8882	0.587615112	20.44444444	0	0	0
8883	0.594869619	20.44444444	0	0	0
8884	0.602124127	20.44444444	0	0	0
8885	0.609378634	20.44444444	0	0	0
8886	0.616633142	20.44444444	0	0	0
8887	0.623887649	20.44444444	0	0	0
8888	0.631142157	20.44444444	0	0	0
8889	0.638396664	20.44444444	0	0	0
8890	0.645651172	20.44444444	0	0	0
8891	0.652905680	20.44444444	0	0	0
8892	0.660160187	20.44444444	0	0	0
8893	0.667414695	20.44444444	0	0	0
8894	0.674669202	20.44444444	0	0	0
8895	0.681923710	20.44444444	0	0	0
8896	0.689178217	20.44444444	0	0	0
8897	0.696432725	20.44444444	0	0	0
8898	0.703687232	20.44444444	0	0	0
8899	0.710941740	20.44444444	0	0	0
8900	0.718196248	20.44444444	0	0	0
8901	0.000000000	20.6767677	0	0	0
8902	0.007254508	20.6767677	0	0	0
8903	0.014509015	20.6767677	0	0	0
8904	0.021763523	20.6767677	0	0	0
8905	0.029018030	20.6767677	0	0	0
8906	0.036272538	20.6767677	0	0	0
8907	0.043527045	20.6767677	0	0	0
8908	0.050781553	20.6767677	0	0	0
8909	0.058036060	20.6767677	0	0	0
8910	0.065290568	20.6767677	0	0	0

8911	0.072545076	20.6767677	0	0	0
8912	0.079799583	20.6767677	0	0	0
8913	0.087054091	20.6767677	0	0	0
8914	0.094308598	20.6767677	0	0	0
8915	0.101563106	20.6767677	0	0	0
8916	0.108817613	20.6767677	0	0	0
8917	0.116072121	20.6767677	0	0	0
8918	0.123326628	20.6767677	0	0	0
8919	0.130581136	20.6767677	0	0	0
8920	0.137835643	20.6767677	0	0	0
8921	0.145090151	20.6767677	0	0	0
8922	0.152344659	20.6767677	0	0	0
8923	0.159599166	20.6767677	0	0	0
8924	0.166853674	20.6767677	0	0	0
8925	0.174108181	20.6767677	0	0	0
8926	0.181362689	20.6767677	0	0	0
8927	0.188617196	20.6767677	0	0	0
8928	0.195871704	20.6767677	0	0	0
8929	0.203126211	20.6767677	0	0	0
8930	0.210380719	20.6767677	0	0	0
8931	0.217635227	20.6767677	0	0	0
8932	0.224889734	20.6767677	0	0	0
8933	0.232144242	20.6767677	0	0	0
8934	0.239398749	20.6767677	0	0	0
8935	0.246653257	20.6767677	0	0	0
8936	0.253907764	20.6767677	0	0	0
8937	0.261162272	20.6767677	0	0	0
8938	0.268416779	20.6767677	0	0	0
8939	0.275671287	20.6767677	0	0	0
8940	0.282925794	20.6767677	0	0	0
8941	0.290180302	20.6767677	0	0	0
8942	0.297434810	20.6767677	0	0	0
8943	0.304689317	20.6767677	0	0	0
8944	0.311943825	20.6767677	0	0	0
8945	0.319198332	20.6767677	0	0	0
8946	0.326452840	20.6767677	0	0	0
8947	0.333707347	20.6767677	0	0	0
8948	0.340961855	20.6767677	0	0	0
8949	0.348216362	20.6767677	0	0	0
8950	0.355470870	20.6767677	0	0	0
8951	0.362725378	20.6767677	0	0	0
8952	0.369979885	20.6767677	0	0	0
8953	0.377234393	20.6767677	0	0	0
8954	0.384488900	20.6767677	0	0	0
8955	0.391743408	20.6767677	0	0	0

8956	0.398997915	20.6767677	0	0	0
8957	0.406252423	20.6767677	0	0	0
8958	0.413506930	20.6767677	0	0	0
8959	0.420761438	20.6767677	0	0	0
8960	0.428015945	20.6767677	0	0	0
8961	0.435270453	20.6767677	0	0	0
8962	0.442524961	20.6767677	0	0	0
8963	0.449779468	20.6767677	0	0	0
8964	0.457033976	20.6767677	0	0	0
8965	0.464288483	20.6767677	0	0	0
8966	0.471542991	20.6767677	0	0	0
8967	0.478797498	20.6767677	0	0	0
8968	0.486052006	20.6767677	0	0	0
8969	0.493306513	20.6767677	0	0	0
8970	0.500561021	20.6767677	0	0	0
8971	0.507815529	20.6767677	0	0	0
8972	0.515070036	20.6767677	0	0	0
8973	0.522324544	20.6767677	0	0	0
8974	0.529579051	20.6767677	0	0	0
8975	0.536833559	20.6767677	0	0	0
8976	0.544088066	20.6767677	0	0	0
8977	0.551342574	20.6767677	0	0	0
8978	0.558597081	20.6767677	0	0	0
8979	0.565851589	20.6767677	0	0	0
8980	0.573106097	20.6767677	0	0	0
8981	0.580360604	20.6767677	0	0	0
8982	0.587615112	20.6767677	0	0	0
8983	0.594869619	20.6767677	0	0	0
8984	0.602124127	20.6767677	0	0	0
8985	0.609378634	20.6767677	0	0	0
8986	0.616633142	20.6767677	0	0	0
8987	0.623887649	20.6767677	0	0	0
8988	0.631142157	20.6767677	0	0	0
8989	0.638396664	20.6767677	0	0	0
8990	0.645651172	20.6767677	0	0	0
8991	0.652905680	20.6767677	0	0	0
8992	0.660160187	20.6767677	0	0	0
8993	0.667414695	20.6767677	0	0	0
8994	0.674669202	20.6767677	0	0	0
8995	0.681923710	20.6767677	0	0	0
8996	0.689178217	20.6767677	0	0	0
8997	0.696432725	20.6767677	0	0	0
8998	0.703687232	20.6767677	0	0	0
8999	0.710941740	20.6767677	0	0	0
9000	0.718196248	20.6767677	0	0	0

9001	0.000000000	20.9090909	0	0	0
9002	0.007254508	20.9090909	0	0	0
9003	0.014509015	20.9090909	0	0	0
9004	0.021763523	20.9090909	0	0	0
9005	0.029018030	20.9090909	0	0	0
9006	0.036272538	20.9090909	0	0	0
9007	0.043527045	20.9090909	0	0	0
9008	0.050781553	20.9090909	0	0	0
9009	0.058036060	20.9090909	0	0	0
9010	0.065290568	20.9090909	0	0	0
9011	0.072545076	20.9090909	0	0	0
9012	0.079799583	20.9090909	0	0	0
9013	0.087054091	20.9090909	0	0	0
9014	0.094308598	20.9090909	0	0	0
9015	0.101563106	20.9090909	0	0	0
9016	0.108817613	20.9090909	0	0	0
9017	0.116072121	20.9090909	0	0	0
9018	0.123326628	20.9090909	0	0	0
9019	0.130581136	20.9090909	0	0	0
9020	0.137835643	20.9090909	0	0	0
9021	0.145090151	20.9090909	0	0	0
9022	0.152344659	20.9090909	0	0	0
9023	0.159599166	20.9090909	0	0	0
9024	0.166853674	20.9090909	0	0	0
9025	0.174108181	20.9090909	0	0	0
9026	0.181362689	20.9090909	0	0	0
9027	0.188617196	20.9090909	0	0	0
9028	0.195871704	20.9090909	0	0	0
9029	0.203126211	20.9090909	0	0	0
9030	0.210380719	20.9090909	0	0	0
9031	0.217635227	20.9090909	0	0	0
9032	0.224889734	20.9090909	0	0	0
9033	0.232144242	20.9090909	0	0	0
9034	0.239398749	20.9090909	0	0	0
9035	0.246653257	20.9090909	0	0	0
9036	0.253907764	20.9090909	0	0	0
9037	0.261162272	20.9090909	0	0	0
9038	0.268416779	20.9090909	0	0	0
9039	0.275671287	20.9090909	0	0	0
9040	0.282925794	20.9090909	0	0	0
9041	0.290180302	20.9090909	0	0	0
9042	0.297434810	20.9090909	0	0	0
9043	0.304689317	20.9090909	0	0	0
9044	0.311943825	20.9090909	0	0	0
9045	0.319198332	20.9090909	0	0	0

9046	0.326452840	20.9090909	0	0	0
9047	0.333707347	20.9090909	0	0	0
9048	0.340961855	20.9090909	0	0	0
9049	0.348216362	20.9090909	0	0	0
9050	0.355470870	20.9090909	0	0	0
9051	0.362725378	20.9090909	0	0	0
9052	0.369979885	20.9090909	0	0	0
9053	0.377234393	20.9090909	0	0	0
9054	0.384488900	20.9090909	0	0	0
9055	0.391743408	20.9090909	0	0	0
9056	0.398997915	20.9090909	0	0	0
9057	0.406252423	20.9090909	0	0	0
9058	0.413506930	20.9090909	0	0	0
9059	0.420761438	20.9090909	0	0	0
9060	0.428015945	20.9090909	0	0	0
9061	0.435270453	20.9090909	0	0	0
9062	0.442524961	20.9090909	0	0	0
9063	0.449779468	20.9090909	0	0	0
9064	0.457033976	20.9090909	0	0	0
9065	0.464288483	20.9090909	0	0	0
9066	0.471542991	20.9090909	0	0	0
9067	0.478797498	20.9090909	0	0	0
9068	0.486052006	20.9090909	0	0	0
9069	0.493306513	20.9090909	0	0	0
9070	0.500561021	20.9090909	0	0	0
9071	0.507815529	20.9090909	0	0	0
9072	0.515070036	20.9090909	0	0	0
9073	0.522324544	20.9090909	0	0	0
9074	0.529579051	20.9090909	0	0	0
9075	0.536833559	20.9090909	0	0	0
9076	0.544088066	20.9090909	0	0	0
9077	0.551342574	20.9090909	0	0	0
9078	0.558597081	20.9090909	0	0	0
9079	0.565851589	20.9090909	0	0	0
9080	0.573106097	20.9090909	0	0	0
9081	0.580360604	20.9090909	0	0	0
9082	0.587615112	20.9090909	0	0	0
9083	0.594869619	20.9090909	0	0	0
9084	0.602124127	20.9090909	0	0	0
9085	0.609378634	20.9090909	0	0	0
9086	0.616633142	20.9090909	0	0	0
9087	0.623887649	20.9090909	0	0	0
9088	0.631142157	20.9090909	0	0	0
9089	0.638396664	20.9090909	0	0	0
9090	0.645651172	20.9090909	0	0	0

9091	0.652905680	20.9090909	0	0	0
9092	0.660160187	20.9090909	0	0	0
9093	0.667414695	20.9090909	0	0	0
9094	0.674669202	20.9090909	0	0	0
9095	0.681923710	20.9090909	0	0	0
9096	0.689178217	20.9090909	0	0	0
9097	0.696432725	20.9090909	0	0	0
9098	0.703687232	20.9090909	0	0	0
9099	0.710941740	20.9090909	0	0	0
9100	0.718196248	20.9090909	0	0	0
9101	0.000000000	21.1414141	0	0	0
9102	0.007254508	21.1414141	0	0	0
9103	0.014509015	21.1414141	0	0	0
9104	0.021763523	21.1414141	0	0	0
9105	0.029018030	21.1414141	0	0	0
9106	0.036272538	21.1414141	0	0	0
9107	0.043527045	21.1414141	0	0	0
9108	0.050781553	21.1414141	0	0	0
9109	0.058036060	21.1414141	0	0	0
9110	0.065290568	21.1414141	0	0	0
9111	0.072545076	21.1414141	0	0	0
9112	0.079799583	21.1414141	0	0	0
9113	0.087054091	21.1414141	0	0	0
9114	0.094308598	21.1414141	0	0	0
9115	0.101563106	21.1414141	0	0	0
9116	0.108817613	21.1414141	0	0	0
9117	0.116072121	21.1414141	0	0	0
9118	0.123326628	21.1414141	0	0	0
9119	0.130581136	21.1414141	0	0	0
9120	0.137835643	21.1414141	0	0	0
9121	0.145090151	21.1414141	0	0	0
9122	0.152344659	21.1414141	0	0	0
9123	0.159599166	21.1414141	0	0	0
9124	0.166853674	21.1414141	0	0	0
9125	0.174108181	21.1414141	0	0	0
9126	0.181362689	21.1414141	0	0	0
9127	0.188617196	21.1414141	0	0	0
9128	0.195871704	21.1414141	0	0	0
9129	0.203126211	21.1414141	0	0	0
9130	0.210380719	21.1414141	0	0	0
9131	0.217635227	21.1414141	0	0	0
9132	0.224889734	21.1414141	0	0	0
9133	0.232144242	21.1414141	0	0	0
9134	0.239398749	21.1414141	0	0	0
9135	0.246653257	21.1414141	0	0	0

9136	0.253907764	21.1414141	0	0	0
9137	0.261162272	21.1414141	0	0	0
9138	0.268416779	21.1414141	0	0	0
9139	0.275671287	21.1414141	0	0	0
9140	0.282925794	21.1414141	0	0	0
9141	0.290180302	21.1414141	0	0	0
9142	0.297434810	21.1414141	0	0	0
9143	0.304689317	21.1414141	0	0	0
9144	0.311943825	21.1414141	0	0	0
9145	0.319198332	21.1414141	0	0	0
9146	0.326452840	21.1414141	0	0	0
9147	0.333707347	21.1414141	0	0	0
9148	0.340961855	21.1414141	0	0	0
9149	0.348216362	21.1414141	0	0	0
9150	0.355470870	21.1414141	0	0	0
9151	0.362725378	21.1414141	0	0	0
9152	0.369979885	21.1414141	0	0	0
9153	0.377234393	21.1414141	0	0	0
9154	0.384488900	21.1414141	0	0	0
9155	0.391743408	21.1414141	0	0	0
9156	0.398997915	21.1414141	0	0	0
9157	0.406252423	21.1414141	0	0	0
9158	0.413506930	21.1414141	0	0	0
9159	0.420761438	21.1414141	0	0	0
9160	0.428015945	21.1414141	0	0	0
9161	0.435270453	21.1414141	0	0	0
9162	0.442524961	21.1414141	0	0	0
9163	0.449779468	21.1414141	0	0	0
9164	0.457033976	21.1414141	0	0	0
9165	0.464288483	21.1414141	0	0	0
9166	0.471542991	21.1414141	0	0	0
9167	0.478797498	21.1414141	0	0	0
9168	0.486052006	21.1414141	0	0	0
9169	0.493306513	21.1414141	0	0	0
9170	0.500561021	21.1414141	0	0	0
9171	0.507815529	21.1414141	0	0	0
9172	0.515070036	21.1414141	0	0	0
9173	0.522324544	21.1414141	0	0	0
9174	0.529579051	21.1414141	0	0	0
9175	0.536833559	21.1414141	0	0	0
9176	0.544088066	21.1414141	0	0	0
9177	0.551342574	21.1414141	0	0	0
9178	0.558597081	21.1414141	0	0	0
9179	0.565851589	21.1414141	0	0	0
9180	0.573106097	21.1414141	0	0	0

9181	0.580360604	21.1414141	0	0	0
9182	0.587615112	21.1414141	0	0	0
9183	0.594869619	21.1414141	0	0	0
9184	0.602124127	21.1414141	0	0	0
9185	0.609378634	21.1414141	0	0	0
9186	0.616633142	21.1414141	0	0	0
9187	0.623887649	21.1414141	0	0	0
9188	0.631142157	21.1414141	0	0	0
9189	0.638396664	21.1414141	0	0	0
9190	0.645651172	21.1414141	0	0	0
9191	0.652905680	21.1414141	0	0	0
9192	0.660160187	21.1414141	0	0	0
9193	0.667414695	21.1414141	0	0	0
9194	0.674669202	21.1414141	0	0	0
9195	0.681923710	21.1414141	0	0	0
9196	0.689178217	21.1414141	0	0	0
9197	0.696432725	21.1414141	0	0	0
9198	0.703687232	21.1414141	0	0	0
9199	0.710941740	21.1414141	0	0	0
9200	0.718196248	21.1414141	0	0	0
9201	0.000000000	21.3737374	0	0	0
9202	0.007254508	21.3737374	0	0	0
9203	0.014509015	21.3737374	0	0	0
9204	0.021763523	21.3737374	0	0	0
9205	0.029018030	21.3737374	0	0	0
9206	0.036272538	21.3737374	0	0	0
9207	0.043527045	21.3737374	0	0	0
9208	0.050781553	21.3737374	0	0	0
9209	0.058036060	21.3737374	0	0	0
9210	0.065290568	21.3737374	0	0	0
9211	0.072545076	21.3737374	0	0	0
9212	0.079799583	21.3737374	0	0	0
9213	0.087054091	21.3737374	0	0	0
9214	0.094308598	21.3737374	0	0	0
9215	0.101563106	21.3737374	0	0	0
9216	0.108817613	21.3737374	0	0	0
9217	0.116072121	21.3737374	0	0	0
9218	0.123326628	21.3737374	0	0	0
9219	0.130581136	21.3737374	0	0	0
9220	0.137835643	21.3737374	0	0	0
9221	0.145090151	21.3737374	0	0	0
9222	0.152344659	21.3737374	0	0	0
9223	0.159599166	21.3737374	0	0	0
9224	0.166853674	21.3737374	0	0	0
9225	0.174108181	21.3737374	0	0	0

9226	0.181362689	21.3737374	0	0	0
9227	0.188617196	21.3737374	0	0	0
9228	0.195871704	21.3737374	0	0	0
9229	0.203126211	21.3737374	0	0	0
9230	0.210380719	21.3737374	0	0	0
9231	0.217635227	21.3737374	0	0	0
9232	0.224889734	21.3737374	0	0	0
9233	0.232144242	21.3737374	0	0	0
9234	0.239398749	21.3737374	0	0	0
9235	0.246653257	21.3737374	0	0	0
9236	0.253907764	21.3737374	0	0	0
9237	0.261162272	21.3737374	0	0	0
9238	0.268416779	21.3737374	0	0	0
9239	0.275671287	21.3737374	0	0	0
9240	0.282925794	21.3737374	0	0	0
9241	0.290180302	21.3737374	0	0	0
9242	0.297434810	21.3737374	0	0	0
9243	0.304689317	21.3737374	0	0	0
9244	0.311943825	21.3737374	0	0	0
9245	0.319198332	21.3737374	0	0	0
9246	0.326452840	21.3737374	0	0	0
9247	0.333707347	21.3737374	0	0	0
9248	0.340961855	21.3737374	0	0	0
9249	0.348216362	21.3737374	0	0	0
9250	0.355470870	21.3737374	0	0	0
9251	0.362725378	21.3737374	0	0	0
9252	0.369979885	21.3737374	0	0	0
9253	0.377234393	21.3737374	0	0	0
9254	0.384488900	21.3737374	0	0	0
9255	0.391743408	21.3737374	0	0	0
9256	0.398997915	21.3737374	0	0	0
9257	0.406252423	21.3737374	0	0	0
9258	0.413506930	21.3737374	0	0	0
9259	0.420761438	21.3737374	0	0	0
9260	0.428015945	21.3737374	0	0	0
9261	0.435270453	21.3737374	0	0	0
9262	0.442524961	21.3737374	0	0	0
9263	0.449779468	21.3737374	0	0	0
9264	0.457033976	21.3737374	0	0	0
9265	0.464288483	21.3737374	0	0	0
9266	0.471542991	21.3737374	0	0	0
9267	0.478797498	21.3737374	0	0	0
9268	0.486052006	21.3737374	0	0	0
9269	0.493306513	21.3737374	0	0	0
9270	0.500561021	21.3737374	0	0	0

9271	0.507815529	21.3737374	0	0	0
9272	0.515070036	21.3737374	0	0	0
9273	0.522324544	21.3737374	0	0	0
9274	0.529579051	21.3737374	0	0	0
9275	0.536833559	21.3737374	0	0	0
9276	0.544088066	21.3737374	0	0	0
9277	0.551342574	21.3737374	0	0	0
9278	0.558597081	21.3737374	0	0	0
9279	0.565851589	21.3737374	0	0	0
9280	0.573106097	21.3737374	0	0	0
9281	0.580360604	21.3737374	0	0	0
9282	0.587615112	21.3737374	0	0	0
9283	0.594869619	21.3737374	0	0	0
9284	0.602124127	21.3737374	0	0	0
9285	0.609378634	21.3737374	0	0	0
9286	0.616633142	21.3737374	0	0	0
9287	0.623887649	21.3737374	0	0	0
9288	0.631142157	21.3737374	0	0	0
9289	0.638396664	21.3737374	0	0	0
9290	0.645651172	21.3737374	0	0	0
9291	0.652905680	21.3737374	0	0	0
9292	0.660160187	21.3737374	0	0	0
9293	0.667414695	21.3737374	0	0	0
9294	0.674669202	21.3737374	0	0	0
9295	0.681923710	21.3737374	0	0	0
9296	0.689178217	21.3737374	0	0	0
9297	0.696432725	21.3737374	0	0	0
9298	0.703687232	21.3737374	0	0	0
9299	0.710941740	21.3737374	0	0	0
9300	0.718196248	21.3737374	0	0	0
9301	0.000000000	21.6060606	0	0	0
9302	0.007254508	21.6060606	0	0	0
9303	0.014509015	21.6060606	0	0	0
9304	0.021763523	21.6060606	0	0	0
9305	0.029018030	21.6060606	0	0	0
9306	0.036272538	21.6060606	0	0	0
9307	0.043527045	21.6060606	0	0	0
9308	0.050781553	21.6060606	0	0	0
9309	0.058036060	21.6060606	0	0	0
9310	0.065290568	21.6060606	0	0	0
9311	0.072545076	21.6060606	0	0	0
9312	0.079799583	21.6060606	0	0	0
9313	0.087054091	21.6060606	0	0	0
9314	0.094308598	21.6060606	0	0	0
9315	0.101563106	21.6060606	0	0	0

9316	0.108817613	21.6060606	0	0	0
9317	0.116072121	21.6060606	0	0	0
9318	0.123326628	21.6060606	0	0	0
9319	0.130581136	21.6060606	0	0	0
9320	0.137835643	21.6060606	0	0	0
9321	0.145090151	21.6060606	0	0	0
9322	0.152344659	21.6060606	0	0	0
9323	0.159599166	21.6060606	0	0	0
9324	0.166853674	21.6060606	0	0	0
9325	0.174108181	21.6060606	0	0	0
9326	0.181362689	21.6060606	0	0	0
9327	0.188617196	21.6060606	0	0	0
9328	0.195871704	21.6060606	0	0	0
9329	0.203126211	21.6060606	0	0	0
9330	0.210380719	21.6060606	0	0	0
9331	0.217635227	21.6060606	0	0	0
9332	0.224889734	21.6060606	0	0	0
9333	0.232144242	21.6060606	0	0	0
9334	0.239398749	21.6060606	0	0	0
9335	0.246653257	21.6060606	0	0	0
9336	0.253907764	21.6060606	0	0	0
9337	0.261162272	21.6060606	0	0	0
9338	0.268416779	21.6060606	0	0	0
9339	0.275671287	21.6060606	0	0	0
9340	0.282925794	21.6060606	0	0	0
9341	0.290180302	21.6060606	0	0	0
9342	0.297434810	21.6060606	0	0	0
9343	0.304689317	21.6060606	0	0	0
9344	0.311943825	21.6060606	0	0	0
9345	0.319198332	21.6060606	0	0	0
9346	0.326452840	21.6060606	0	0	0
9347	0.333707347	21.6060606	0	0	0
9348	0.340961855	21.6060606	0	0	0
9349	0.348216362	21.6060606	0	0	0
9350	0.355470870	21.6060606	0	0	0
9351	0.362725378	21.6060606	0	0	0
9352	0.369979885	21.6060606	0	0	0
9353	0.377234393	21.6060606	0	0	0
9354	0.384488900	21.6060606	0	0	0
9355	0.391743408	21.6060606	0	0	0
9356	0.398997915	21.6060606	0	0	0
9357	0.406252423	21.6060606	0	0	0
9358	0.413506930	21.6060606	0	0	0
9359	0.420761438	21.6060606	0	0	0
9360	0.428015945	21.6060606	0	0	0

9361	0.435270453	21.6060606	0	0	0
9362	0.442524961	21.6060606	0	0	0
9363	0.449779468	21.6060606	0	0	0
9364	0.457033976	21.6060606	0	0	0
9365	0.464288483	21.6060606	0	0	0
9366	0.471542991	21.6060606	0	0	0
9367	0.478797498	21.6060606	0	0	0
9368	0.486052006	21.6060606	0	0	0
9369	0.493306513	21.6060606	0	0	0
9370	0.500561021	21.6060606	0	0	0
9371	0.507815529	21.6060606	0	0	0
9372	0.515070036	21.6060606	0	0	0
9373	0.522324544	21.6060606	0	0	0
9374	0.529579051	21.6060606	0	0	0
9375	0.536833559	21.6060606	0	0	0
9376	0.544088066	21.6060606	0	0	0
9377	0.551342574	21.6060606	0	0	0
9378	0.558597081	21.6060606	0	0	0
9379	0.565851589	21.6060606	0	0	0
9380	0.573106097	21.6060606	0	0	0
9381	0.580360604	21.6060606	0	0	0
9382	0.587615112	21.6060606	0	0	0
9383	0.594869619	21.6060606	0	0	0
9384	0.602124127	21.6060606	0	0	0
9385	0.609378634	21.6060606	0	0	0
9386	0.616633142	21.6060606	0	0	0
9387	0.623887649	21.6060606	0	0	0
9388	0.631142157	21.6060606	0	0	0
9389	0.638396664	21.6060606	0	0	0
9390	0.645651172	21.6060606	0	0	0
9391	0.652905680	21.6060606	0	0	0
9392	0.660160187	21.6060606	0	0	0
9393	0.667414695	21.6060606	0	0	0
9394	0.674669202	21.6060606	0	0	0
9395	0.681923710	21.6060606	0	0	0
9396	0.689178217	21.6060606	0	0	0
9397	0.696432725	21.6060606	0	0	0
9398	0.703687232	21.6060606	0	0	0
9399	0.710941740	21.6060606	0	0	0
9400	0.718196248	21.6060606	0	0	0
9401	0.000000000	21.8383838	0	0	0
9402	0.007254508	21.8383838	0	0	0
9403	0.014509015	21.8383838	0	0	0
9404	0.021763523	21.8383838	0	0	0
9405	0.029018030	21.8383838	0	0	0

9406	0.036272538	21.8383838	0	0	0
9407	0.043527045	21.8383838	0	0	0
9408	0.050781553	21.8383838	0	0	0
9409	0.058036060	21.8383838	0	0	0
9410	0.065290568	21.8383838	0	0	0
9411	0.072545076	21.8383838	0	0	0
9412	0.079799583	21.8383838	0	0	0
9413	0.087054091	21.8383838	0	0	0
9414	0.094308598	21.8383838	0	0	0
9415	0.101563106	21.8383838	0	0	0
9416	0.108817613	21.8383838	0	0	0
9417	0.116072121	21.8383838	0	0	0
9418	0.123326628	21.8383838	0	0	0
9419	0.130581136	21.8383838	0	0	0
9420	0.137835643	21.8383838	0	0	0
9421	0.145090151	21.8383838	0	0	0
9422	0.152344659	21.8383838	0	0	0
9423	0.159599166	21.8383838	0	0	0
9424	0.166853674	21.8383838	0	0	0
9425	0.174108181	21.8383838	0	0	0
9426	0.181362689	21.8383838	0	0	0
9427	0.188617196	21.8383838	0	0	0
9428	0.195871704	21.8383838	0	0	0
9429	0.203126211	21.8383838	0	0	0
9430	0.210380719	21.8383838	0	0	0
9431	0.217635227	21.8383838	0	0	0
9432	0.224889734	21.8383838	0	0	0
9433	0.232144242	21.8383838	0	0	0
9434	0.239398749	21.8383838	0	0	0
9435	0.246653257	21.8383838	0	0	0
9436	0.253907764	21.8383838	0	0	0
9437	0.261162272	21.8383838	0	0	0
9438	0.268416779	21.8383838	0	0	0
9439	0.275671287	21.8383838	0	0	0
9440	0.282925794	21.8383838	0	0	0
9441	0.290180302	21.8383838	0	0	0
9442	0.297434810	21.8383838	0	0	0
9443	0.304689317	21.8383838	0	0	0
9444	0.311943825	21.8383838	0	0	0
9445	0.319198332	21.8383838	0	0	0
9446	0.326452840	21.8383838	0	0	0
9447	0.333707347	21.8383838	0	0	0
9448	0.340961855	21.8383838	0	0	0
9449	0.348216362	21.8383838	0	0	0
9450	0.355470870	21.8383838	0	0	0

9451	0.362725378	21.8383838	0	0	0
9452	0.369979885	21.8383838	0	0	0
9453	0.377234393	21.8383838	0	0	0
9454	0.384488900	21.8383838	0	0	0
9455	0.391743408	21.8383838	0	0	0
9456	0.398997915	21.8383838	0	0	0
9457	0.406252423	21.8383838	0	0	0
9458	0.413506930	21.8383838	0	0	0
9459	0.420761438	21.8383838	0	0	0
9460	0.428015945	21.8383838	0	0	0
9461	0.435270453	21.8383838	0	0	0
9462	0.442524961	21.8383838	0	0	0
9463	0.449779468	21.8383838	0	0	0
9464	0.457033976	21.8383838	0	0	0
9465	0.464288483	21.8383838	0	0	0
9466	0.471542991	21.8383838	0	0	0
9467	0.478797498	21.8383838	0	0	0
9468	0.486052006	21.8383838	0	0	0
9469	0.493306513	21.8383838	0	0	0
9470	0.500561021	21.8383838	0	0	0
9471	0.507815529	21.8383838	0	0	0
9472	0.515070036	21.8383838	0	0	0
9473	0.522324544	21.8383838	0	0	0
9474	0.529579051	21.8383838	0	0	0
9475	0.536833559	21.8383838	0	0	0
9476	0.544088066	21.8383838	0	0	0
9477	0.551342574	21.8383838	0	0	0
9478	0.558597081	21.8383838	0	0	0
9479	0.565851589	21.8383838	0	0	0
9480	0.573106097	21.8383838	0	0	0
9481	0.580360604	21.8383838	0	0	0
9482	0.587615112	21.8383838	0	0	0
9483	0.594869619	21.8383838	0	0	0
9484	0.602124127	21.8383838	0	0	0
9485	0.609378634	21.8383838	0	0	0
9486	0.616633142	21.8383838	0	0	0
9487	0.623887649	21.8383838	0	0	0
9488	0.631142157	21.8383838	0	0	0
9489	0.638396664	21.8383838	0	0	0
9490	0.645651172	21.8383838	0	0	0
9491	0.652905680	21.8383838	0	0	0
9492	0.660160187	21.8383838	0	0	0
9493	0.667414695	21.8383838	0	0	0
9494	0.674669202	21.8383838	0	0	0
9495	0.681923710	21.8383838	0	0	0

9496	0.689178217	21.8383838	0	0	0
9497	0.696432725	21.8383838	0	0	0
9498	0.703687232	21.8383838	0	0	0
9499	0.710941740	21.8383838	0	0	0
9500	0.718196248	21.8383838	0	0	0
9501	0.000000000	22.0707071	0	0	0
9502	0.007254508	22.0707071	0	0	0
9503	0.014509015	22.0707071	0	0	0
9504	0.021763523	22.0707071	0	0	0
9505	0.029018030	22.0707071	0	0	0
9506	0.036272538	22.0707071	0	0	0
9507	0.043527045	22.0707071	0	0	0
9508	0.050781553	22.0707071	0	0	0
9509	0.058036060	22.0707071	0	0	0
9510	0.065290568	22.0707071	0	0	0
9511	0.072545076	22.0707071	0	0	0
9512	0.079799583	22.0707071	0	0	0
9513	0.087054091	22.0707071	0	0	0
9514	0.094308598	22.0707071	0	0	0
9515	0.101563106	22.0707071	0	0	0
9516	0.108817613	22.0707071	0	0	0
9517	0.116072121	22.0707071	0	0	0
9518	0.123326628	22.0707071	0	0	0
9519	0.130581136	22.0707071	0	0	0
9520	0.137835643	22.0707071	0	0	0
9521	0.145090151	22.0707071	0	0	0
9522	0.152344659	22.0707071	0	0	0
9523	0.159599166	22.0707071	0	0	0
9524	0.166853674	22.0707071	0	0	0
9525	0.174108181	22.0707071	0	0	0
9526	0.181362689	22.0707071	0	0	0
9527	0.188617196	22.0707071	0	0	0
9528	0.195871704	22.0707071	0	0	0
9529	0.203126211	22.0707071	0	0	0
9530	0.210380719	22.0707071	0	0	0
9531	0.217635227	22.0707071	0	0	0
9532	0.224889734	22.0707071	0	0	0
9533	0.232144242	22.0707071	0	0	0
9534	0.239398749	22.0707071	0	0	0
9535	0.246653257	22.0707071	0	0	0
9536	0.253907764	22.0707071	0	0	0
9537	0.261162272	22.0707071	0	0	0
9538	0.268416779	22.0707071	0	0	0
9539	0.275671287	22.0707071	0	0	0
9540	0.282925794	22.0707071	0	0	0

9541	0.290180302	22.0707071	0	0	0
9542	0.297434810	22.0707071	0	0	0
9543	0.304689317	22.0707071	0	0	0
9544	0.311943825	22.0707071	0	0	0
9545	0.319198332	22.0707071	0	0	0
9546	0.326452840	22.0707071	0	0	0
9547	0.333707347	22.0707071	0	0	0
9548	0.340961855	22.0707071	0	0	0
9549	0.348216362	22.0707071	0	0	0
9550	0.355470870	22.0707071	0	0	0
9551	0.362725378	22.0707071	0	0	0
9552	0.369979885	22.0707071	0	0	0
9553	0.377234393	22.0707071	0	0	0
9554	0.384488900	22.0707071	0	0	0
9555	0.391743408	22.0707071	0	0	0
9556	0.398997915	22.0707071	0	0	0
9557	0.406252423	22.0707071	0	0	0
9558	0.413506930	22.0707071	0	0	0
9559	0.420761438	22.0707071	0	0	0
9560	0.428015945	22.0707071	0	0	0
9561	0.435270453	22.0707071	0	0	0
9562	0.442524961	22.0707071	0	0	0
9563	0.449779468	22.0707071	0	0	0
9564	0.457033976	22.0707071	0	0	0
9565	0.464288483	22.0707071	0	0	0
9566	0.471542991	22.0707071	0	0	0
9567	0.478797498	22.0707071	0	0	0
9568	0.486052006	22.0707071	0	0	0
9569	0.493306513	22.0707071	0	0	0
9570	0.500561021	22.0707071	0	0	0
9571	0.507815529	22.0707071	0	0	0
9572	0.515070036	22.0707071	0	0	0
9573	0.522324544	22.0707071	0	0	0
9574	0.529579051	22.0707071	0	0	0
9575	0.536833559	22.0707071	0	0	0
9576	0.544088066	22.0707071	0	0	0
9577	0.551342574	22.0707071	0	0	0
9578	0.558597081	22.0707071	0	0	0
9579	0.565851589	22.0707071	0	0	0
9580	0.573106097	22.0707071	0	0	0
9581	0.580360604	22.0707071	0	0	0
9582	0.587615112	22.0707071	0	0	0
9583	0.594869619	22.0707071	0	0	0
9584	0.602124127	22.0707071	0	0	0
9585	0.609378634	22.0707071	0	0	0

9586	0.616633142	22.0707071	0	0	0
9587	0.623887649	22.0707071	0	0	0
9588	0.631142157	22.0707071	0	0	0
9589	0.638396664	22.0707071	0	0	0
9590	0.645651172	22.0707071	0	0	0
9591	0.652905680	22.0707071	0	0	0
9592	0.660160187	22.0707071	0	0	0
9593	0.667414695	22.0707071	0	0	0
9594	0.674669202	22.0707071	0	0	0
9595	0.681923710	22.0707071	0	0	0
9596	0.689178217	22.0707071	0	0	0
9597	0.696432725	22.0707071	0	0	0
9598	0.703687232	22.0707071	0	0	0
9599	0.710941740	22.0707071	0	0	0
9600	0.718196248	22.0707071	0	0	0
9601	0.000000000	22.3030303	0	0	0
9602	0.007254508	22.3030303	0	0	0
9603	0.014509015	22.3030303	0	0	0
9604	0.021763523	22.3030303	0	0	0
9605	0.029018030	22.3030303	0	0	0
9606	0.036272538	22.3030303	0	0	0
9607	0.043527045	22.3030303	0	0	0
9608	0.050781553	22.3030303	0	0	0
9609	0.058036060	22.3030303	0	0	0
9610	0.065290568	22.3030303	0	0	0
9611	0.072545076	22.3030303	0	0	0
9612	0.079799583	22.3030303	0	0	0
9613	0.087054091	22.3030303	0	0	0
9614	0.094308598	22.3030303	0	0	0
9615	0.101563106	22.3030303	0	0	0
9616	0.108817613	22.3030303	0	0	0
9617	0.116072121	22.3030303	0	0	0
9618	0.123326628	22.3030303	0	0	0
9619	0.130581136	22.3030303	0	0	0
9620	0.137835643	22.3030303	0	0	0
9621	0.145090151	22.3030303	0	0	0
9622	0.152344659	22.3030303	0	0	0
9623	0.159599166	22.3030303	0	0	0
9624	0.166853674	22.3030303	0	0	0
9625	0.174108181	22.3030303	0	0	0
9626	0.181362689	22.3030303	0	0	0
9627	0.188617196	22.3030303	0	0	0
9628	0.195871704	22.3030303	0	0	0
9629	0.203126211	22.3030303	0	0	0
9630	0.210380719	22.3030303	0	0	0

9631	0.217635227	22.3030303	0	0	0
9632	0.224889734	22.3030303	0	0	0
9633	0.232144242	22.3030303	0	0	0
9634	0.239398749	22.3030303	0	0	0
9635	0.246653257	22.3030303	0	0	0
9636	0.253907764	22.3030303	0	0	0
9637	0.261162272	22.3030303	0	0	0
9638	0.268416779	22.3030303	0	0	0
9639	0.275671287	22.3030303	0	0	0
9640	0.282925794	22.3030303	0	0	0
9641	0.290180302	22.3030303	0	0	0
9642	0.297434810	22.3030303	0	0	0
9643	0.304689317	22.3030303	0	0	0
9644	0.311943825	22.3030303	0	0	0
9645	0.319198332	22.3030303	0	0	0
9646	0.326452840	22.3030303	0	0	0
9647	0.333707347	22.3030303	0	0	0
9648	0.340961855	22.3030303	0	0	0
9649	0.348216362	22.3030303	0	0	0
9650	0.355470870	22.3030303	0	0	0
9651	0.362725378	22.3030303	0	0	0
9652	0.369979885	22.3030303	0	0	0
9653	0.377234393	22.3030303	0	0	0
9654	0.384488900	22.3030303	0	0	0
9655	0.391743408	22.3030303	0	0	0
9656	0.398997915	22.3030303	0	0	0
9657	0.406252423	22.3030303	0	0	0
9658	0.413506930	22.3030303	0	0	0
9659	0.420761438	22.3030303	0	0	0
9660	0.428015945	22.3030303	0	0	0
9661	0.435270453	22.3030303	0	0	0
9662	0.442524961	22.3030303	0	0	0
9663	0.449779468	22.3030303	0	0	0
9664	0.457033976	22.3030303	0	0	0
9665	0.464288483	22.3030303	0	0	0
9666	0.471542991	22.3030303	0	0	0
9667	0.478797498	22.3030303	0	0	0
9668	0.486052006	22.3030303	0	0	0
9669	0.493306513	22.3030303	0	0	0
9670	0.500561021	22.3030303	0	0	0
9671	0.507815529	22.3030303	0	0	0
9672	0.515070036	22.3030303	0	0	0
9673	0.522324544	22.3030303	0	0	0
9674	0.529579051	22.3030303	0	0	0
9675	0.536833559	22.3030303	0	0	0

9676	0.544088066	22.3030303	0	0	0
9677	0.551342574	22.3030303	0	0	0
9678	0.558597081	22.3030303	0	0	0
9679	0.565851589	22.3030303	0	0	0
9680	0.573106097	22.3030303	0	0	0
9681	0.580360604	22.3030303	0	0	0
9682	0.587615112	22.3030303	0	0	0
9683	0.594869619	22.3030303	0	0	0
9684	0.602124127	22.3030303	0	0	0
9685	0.609378634	22.3030303	0	0	0
9686	0.616633142	22.3030303	0	0	0
9687	0.623887649	22.3030303	0	0	0
9688	0.631142157	22.3030303	0	0	0
9689	0.638396664	22.3030303	0	0	0
9690	0.645651172	22.3030303	0	0	0
9691	0.652905680	22.3030303	0	0	0
9692	0.660160187	22.3030303	0	0	0
9693	0.667414695	22.3030303	0	0	0
9694	0.674669202	22.3030303	0	0	0
9695	0.681923710	22.3030303	0	0	0
9696	0.689178217	22.3030303	0	0	0
9697	0.696432725	22.3030303	0	0	0
9698	0.703687232	22.3030303	0	0	0
9699	0.710941740	22.3030303	0	0	0
9700	0.718196248	22.3030303	0	0	0
9701	0.000000000	22.5353535	0	0	0
9702	0.007254508	22.5353535	0	0	0
9703	0.014509015	22.5353535	0	0	0
9704	0.021763523	22.5353535	0	0	0
9705	0.029018030	22.5353535	0	0	0
9706	0.036272538	22.5353535	0	0	0
9707	0.043527045	22.5353535	0	0	0
9708	0.050781553	22.5353535	0	0	0
9709	0.058036060	22.5353535	0	0	0
9710	0.065290568	22.5353535	0	0	0
9711	0.072545076	22.5353535	0	0	0
9712	0.079799583	22.5353535	0	0	0
9713	0.087054091	22.5353535	0	0	0
9714	0.094308598	22.5353535	0	0	0
9715	0.101563106	22.5353535	0	0	0
9716	0.108817613	22.5353535	0	0	0
9717	0.116072121	22.5353535	0	0	0
9718	0.123326628	22.5353535	0	0	0
9719	0.130581136	22.5353535	0	0	0
9720	0.137835643	22.5353535	0	0	0

9721	0.145090151	22.5353535	0	0	0
9722	0.152344659	22.5353535	0	0	0
9723	0.159599166	22.5353535	0	0	0
9724	0.166853674	22.5353535	0	0	0
9725	0.174108181	22.5353535	0	0	0
9726	0.181362689	22.5353535	0	0	0
9727	0.188617196	22.5353535	0	0	0
9728	0.195871704	22.5353535	0	0	0
9729	0.203126211	22.5353535	0	0	0
9730	0.210380719	22.5353535	0	0	0
9731	0.217635227	22.5353535	0	0	0
9732	0.224889734	22.5353535	0	0	0
9733	0.232144242	22.5353535	0	0	0
9734	0.239398749	22.5353535	0	0	0
9735	0.246653257	22.5353535	0	0	0
9736	0.253907764	22.5353535	0	0	0
9737	0.261162272	22.5353535	0	0	0
9738	0.268416779	22.5353535	0	0	0
9739	0.275671287	22.5353535	0	0	0
9740	0.282925794	22.5353535	0	0	0
9741	0.290180302	22.5353535	0	0	0
9742	0.297434810	22.5353535	0	0	0
9743	0.304689317	22.5353535	0	0	0
9744	0.311943825	22.5353535	0	0	0
9745	0.319198332	22.5353535	0	0	0
9746	0.326452840	22.5353535	0	0	0
9747	0.333707347	22.5353535	0	0	0
9748	0.340961855	22.5353535	0	0	0
9749	0.348216362	22.5353535	0	0	0
9750	0.355470870	22.5353535	0	0	0
9751	0.362725378	22.5353535	0	0	0
9752	0.369979885	22.5353535	0	0	0
9753	0.377234393	22.5353535	0	0	0
9754	0.384488900	22.5353535	0	0	0
9755	0.391743408	22.5353535	0	0	0
9756	0.398997915	22.5353535	0	0	0
9757	0.406252423	22.5353535	0	0	0
9758	0.413506930	22.5353535	0	0	0
9759	0.420761438	22.5353535	0	0	0
9760	0.428015945	22.5353535	0	0	0
9761	0.435270453	22.5353535	0	0	0
9762	0.442524961	22.5353535	0	0	0
9763	0.449779468	22.5353535	0	0	0
9764	0.457033976	22.5353535	0	0	0
9765	0.464288483	22.5353535	0	0	0

9766	0.471542991	22.5353535	0	0	0
9767	0.478797498	22.5353535	0	0	0
9768	0.486052006	22.5353535	0	0	0
9769	0.493306513	22.5353535	0	0	0
9770	0.500561021	22.5353535	0	0	0
9771	0.507815529	22.5353535	0	0	0
9772	0.515070036	22.5353535	0	0	0
9773	0.522324544	22.5353535	0	0	0
9774	0.529579051	22.5353535	0	0	0
9775	0.536833559	22.5353535	0	0	0
9776	0.544088066	22.5353535	0	0	0
9777	0.551342574	22.5353535	0	0	0
9778	0.558597081	22.5353535	0	0	0
9779	0.565851589	22.5353535	0	0	0
9780	0.573106097	22.5353535	0	0	0
9781	0.580360604	22.5353535	0	0	0
9782	0.587615112	22.5353535	0	0	0
9783	0.594869619	22.5353535	0	0	0
9784	0.602124127	22.5353535	0	0	0
9785	0.609378634	22.5353535	0	0	0
9786	0.616633142	22.5353535	0	0	0
9787	0.623887649	22.5353535	0	0	0
9788	0.631142157	22.5353535	0	0	0
9789	0.638396664	22.5353535	0	0	0
9790	0.645651172	22.5353535	0	0	0
9791	0.652905680	22.5353535	0	0	0
9792	0.660160187	22.5353535	0	0	0
9793	0.667414695	22.5353535	0	0	0
9794	0.674669202	22.5353535	0	0	0
9795	0.681923710	22.5353535	0	0	0
9796	0.689178217	22.5353535	0	0	0
9797	0.696432725	22.5353535	0	0	0
9798	0.703687232	22.5353535	0	0	0
9799	0.710941740	22.5353535	0	0	0
9800	0.718196248	22.5353535	0	0	0
9801	0.000000000	22.7676768	0	0	0
9802	0.007254508	22.7676768	0	0	0
9803	0.014509015	22.7676768	0	0	0
9804	0.021763523	22.7676768	0	0	0
9805	0.029018030	22.7676768	0	0	0
9806	0.036272538	22.7676768	0	0	0
9807	0.043527045	22.7676768	0	0	0
9808	0.050781553	22.7676768	0	0	0
9809	0.058036060	22.7676768	0	0	0
9810	0.065290568	22.7676768	0	0	0

9811	0.072545076	22.7676768	0	0	0
9812	0.079799583	22.7676768	0	0	0
9813	0.087054091	22.7676768	0	0	0
9814	0.094308598	22.7676768	0	0	0
9815	0.101563106	22.7676768	0	0	0
9816	0.108817613	22.7676768	0	0	0
9817	0.116072121	22.7676768	0	0	0
9818	0.123326628	22.7676768	0	0	0
9819	0.130581136	22.7676768	0	0	0
9820	0.137835643	22.7676768	0	0	0
9821	0.145090151	22.7676768	0	0	0
9822	0.152344659	22.7676768	0	0	0
9823	0.159599166	22.7676768	0	0	0
9824	0.166853674	22.7676768	0	0	0
9825	0.174108181	22.7676768	0	0	0
9826	0.181362689	22.7676768	0	0	0
9827	0.188617196	22.7676768	0	0	0
9828	0.195871704	22.7676768	0	0	0
9829	0.203126211	22.7676768	0	0	0
9830	0.210380719	22.7676768	0	0	0
9831	0.217635227	22.7676768	0	0	0
9832	0.224889734	22.7676768	0	0	0
9833	0.232144242	22.7676768	0	0	0
9834	0.239398749	22.7676768	0	0	0
9835	0.246653257	22.7676768	0	0	0
9836	0.253907764	22.7676768	0	0	0
9837	0.261162272	22.7676768	0	0	0
9838	0.268416779	22.7676768	0	0	0
9839	0.275671287	22.7676768	0	0	0
9840	0.282925794	22.7676768	0	0	0
9841	0.290180302	22.7676768	0	0	0
9842	0.297434810	22.7676768	0	0	0
9843	0.304689317	22.7676768	0	0	0
9844	0.311943825	22.7676768	0	0	0
9845	0.319198332	22.7676768	0	0	0
9846	0.326452840	22.7676768	0	0	0
9847	0.333707347	22.7676768	0	0	0
9848	0.340961855	22.7676768	0	0	0
9849	0.348216362	22.7676768	0	0	0
9850	0.355470870	22.7676768	0	0	0
9851	0.362725378	22.7676768	0	0	0
9852	0.369979885	22.7676768	0	0	0
9853	0.377234393	22.7676768	0	0	0
9854	0.384488900	22.7676768	0	0	0
9855	0.391743408	22.7676768	0	0	0

9856	0.398997915	22.7676768	0	0	0
9857	0.406252423	22.7676768	0	0	0
9858	0.413506930	22.7676768	0	0	0
9859	0.420761438	22.7676768	0	0	0
9860	0.428015945	22.7676768	0	0	0
9861	0.435270453	22.7676768	0	0	0
9862	0.442524961	22.7676768	0	0	0
9863	0.449779468	22.7676768	0	0	0
9864	0.457033976	22.7676768	0	0	0
9865	0.464288483	22.7676768	0	0	0
9866	0.471542991	22.7676768	0	0	0
9867	0.478797498	22.7676768	0	0	0
9868	0.486052006	22.7676768	0	0	0
9869	0.493306513	22.7676768	0	0	0
9870	0.500561021	22.7676768	0	0	0
9871	0.507815529	22.7676768	0	0	0
9872	0.515070036	22.7676768	0	0	0
9873	0.522324544	22.7676768	0	0	0
9874	0.529579051	22.7676768	0	0	0
9875	0.536833559	22.7676768	0	0	0
9876	0.544088066	22.7676768	0	0	0
9877	0.551342574	22.7676768	0	0	0
9878	0.558597081	22.7676768	0	0	0
9879	0.565851589	22.7676768	0	0	0
9880	0.573106097	22.7676768	0	0	0
9881	0.580360604	22.7676768	0	0	0
9882	0.587615112	22.7676768	0	0	0
9883	0.594869619	22.7676768	0	0	0
9884	0.602124127	22.7676768	0	0	0
9885	0.609378634	22.7676768	0	0	0
9886	0.616633142	22.7676768	0	0	0
9887	0.623887649	22.7676768	0	0	0
9888	0.631142157	22.7676768	0	0	0
9889	0.638396664	22.7676768	0	0	0
9890	0.645651172	22.7676768	0	0	0
9891	0.652905680	22.7676768	0	0	0
9892	0.660160187	22.7676768	0	0	0
9893	0.667414695	22.7676768	0	0	0
9894	0.674669202	22.7676768	0	0	0
9895	0.681923710	22.7676768	0	0	0
9896	0.689178217	22.7676768	0	0	0
9897	0.696432725	22.7676768	0	0	0
9898	0.703687232	22.7676768	0	0	0
9899	0.710941740	22.7676768	0	0	0
9900	0.718196248	22.7676768	0	0	0

9901	0.000000000	23.0000000	0	0	0
9902	0.007254508	23.0000000	0	0	0
9903	0.014509015	23.0000000	0	0	0
9904	0.021763523	23.0000000	0	0	0
9905	0.029018030	23.0000000	0	0	0
9906	0.036272538	23.0000000	0	0	0
9907	0.043527045	23.0000000	0	0	0
9908	0.050781553	23.0000000	0	0	0
9909	0.058036060	23.0000000	0	0	0
9910	0.065290568	23.0000000	0	0	0
9911	0.072545076	23.0000000	0	0	0
9912	0.079799583	23.0000000	0	0	0
9913	0.087054091	23.0000000	0	0	0
9914	0.094308598	23.0000000	0	0	0
9915	0.101563106	23.0000000	0	0	0
9916	0.108817613	23.0000000	0	0	0
9917	0.116072121	23.0000000	0	0	0
9918	0.123326628	23.0000000	0	0	0
9919	0.130581136	23.0000000	0	0	0
9920	0.137835643	23.0000000	0	0	0
9921	0.145090151	23.0000000	0	0	0
9922	0.152344659	23.0000000	0	0	0
9923	0.159599166	23.0000000	0	0	0
9924	0.166853674	23.0000000	0	0	0
9925	0.174108181	23.0000000	0	0	0
9926	0.181362689	23.0000000	0	0	0
9927	0.188617196	23.0000000	0	0	0
9928	0.195871704	23.0000000	0	0	0
9929	0.203126211	23.0000000	0	0	0
9930	0.210380719	23.0000000	0	0	0
9931	0.217635227	23.0000000	0	0	0
9932	0.224889734	23.0000000	0	0	0
9933	0.232144242	23.0000000	0	0	0
9934	0.239398749	23.0000000	0	0	0
9935	0.246653257	23.0000000	0	0	0
9936	0.253907764	23.0000000	0	0	0
9937	0.261162272	23.0000000	0	0	0
9938	0.268416779	23.0000000	0	0	0
9939	0.275671287	23.0000000	0	0	0
9940	0.282925794	23.0000000	0	0	0
9941	0.290180302	23.0000000	0	0	0
9942	0.297434810	23.0000000	0	0	0
9943	0.304689317	23.0000000	0	0	0
9944	0.311943825	23.0000000	0	0	0
9945	0.319198332	23.0000000	0	0	0

9946	0.326452840	23.0000000	0	0	0
9947	0.333707347	23.0000000	0	0	0
9948	0.340961855	23.0000000	0	0	0
9949	0.348216362	23.0000000	0	0	0
9950	0.355470870	23.0000000	0	0	0
9951	0.362725378	23.0000000	0	0	0
9952	0.369979885	23.0000000	0	0	0
9953	0.377234393	23.0000000	0	0	0
9954	0.384488900	23.0000000	0	0	0
9955	0.391743408	23.0000000	0	0	0
9956	0.398997915	23.0000000	0	0	0
9957	0.406252423	23.0000000	0	0	0
9958	0.413506930	23.0000000	0	0	0
9959	0.420761438	23.0000000	0	0	0
9960	0.428015945	23.0000000	0	0	0
9961	0.435270453	23.0000000	0	0	0
9962	0.442524961	23.0000000	0	0	0
9963	0.449779468	23.0000000	0	0	0
9964	0.457033976	23.0000000	0	0	0
9965	0.464288483	23.0000000	0	0	0
9966	0.471542991	23.0000000	0	0	0
9967	0.478797498	23.0000000	0	0	0
9968	0.486052006	23.0000000	0	0	0
9969	0.493306513	23.0000000	0	0	0
9970	0.500561021	23.0000000	0	0	0
9971	0.507815529	23.0000000	0	0	0
9972	0.515070036	23.0000000	0	0	0
9973	0.522324544	23.0000000	0	0	0
9974	0.529579051	23.0000000	0	0	0
9975	0.536833559	23.0000000	0	0	0
9976	0.544088066	23.0000000	0	0	0
9977	0.551342574	23.0000000	0	0	0
9978	0.558597081	23.0000000	0	0	0
9979	0.565851589	23.0000000	0	0	0
9980	0.573106097	23.0000000	0	0	0
9981	0.580360604	23.0000000	0	0	0
9982	0.587615112	23.0000000	0	0	0
9983	0.594869619	23.0000000	0	0	0
9984	0.602124127	23.0000000	0	0	0
9985	0.609378634	23.0000000	0	0	0
9986	0.616633142	23.0000000	0	0	0
9987	0.623887649	23.0000000	0	0	0
9988	0.631142157	23.0000000	0	0	0
9989	0.638396664	23.0000000	0	0	0
9990	0.645651172	23.0000000	0	0	0

9991	0.652905680	23.0000000	0	0	0
9992	0.660160187	23.0000000	0	0	0
9993	0.667414695	23.0000000	0	0	0
9994	0.674669202	23.0000000	0	0	0
9995	0.681923710	23.0000000	0	0	0
9996	0.689178217	23.0000000	0	0	0
9997	0.696432725	23.0000000	0	0	0
9998	0.703687232	23.0000000	0	0	0
9999	0.710941740	23.0000000	0	0	0
10000	0.718196248	23.0000000	0	0	0

```
# Get predictions on log-scale with standard errors
predictions <- predict(ssf_gam_temporal,
                      newdata = ndvi_hour_values,
                      type = "link", # log-scale
                      se.fit = TRUE)

# Extract log RSS and standard errors
gam_log_rss <- predictions$fit
gam_se_log_rss <- predictions$se.fit

# Calculate confidence intervals on log-scale
gam_log_rss_lower <- gam_log_rss - 1.96 * gam_se_log_rss
gam_log_rss_upper <- gam_log_rss + 1.96 * gam_se_log_rss

# Transform to RSS scale
rss <- exp(gam_log_rss)
rss_lower <- exp(gam_log_rss_lower)
rss_upper <- exp(gam_log_rss_upper)

ndvi_hour_values <- ndvi_hour_values %>% mutate(
  gam_log_rss = gam_log_rss,
  gam_rss = exp(gam_log_rss),
  gam_log_rss_lower = as.numeric(gam_log_rss_lower),
  gam_log_rss_upper = as.numeric(gam_log_rss_upper)
)

max_abs_colour <- max(abs(range(ndvi_hour_values$gam_log_rss)))

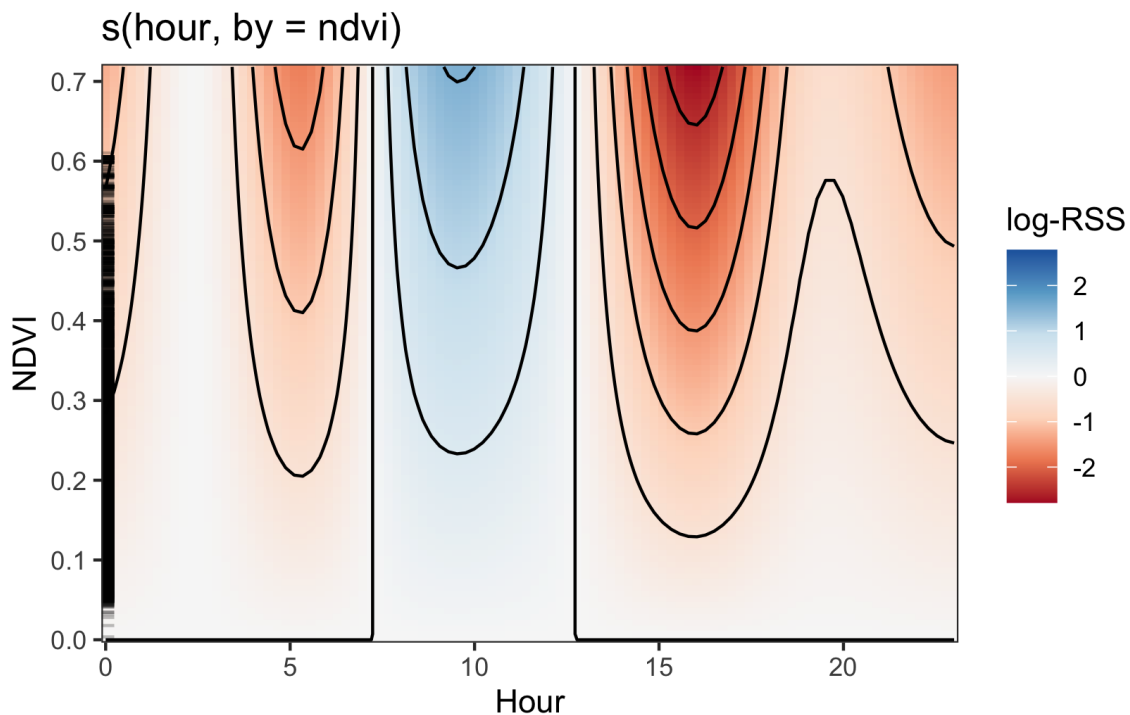
# plot the selection surface
ggplot() +
  geom_raster(data = ndvi_hour_values,
             aes(x = hour, y = ndvi, fill = gam_log_rss)) +
  labs(title = "s(hour, by = ndvi)", x = "Hour", y = "NDVI", fill = "log-RSS") +
  geom_contour(data = ndvi_hour_values,
```



```

aes(x = hour, y = ndvi, z = gam_log_rss), colour = "black") +
scale_fill_distiller(palette = "RdBu", type = "div", direction = 1,
                      limits = c(-max_abs_colour, max_abs_colour)) +
coord_cartesian(expand = FALSE) +
geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
         aes(y = ndvi), alpha = 0.25, length = unit(0.015, "npc"), sides="l") +
scale_y_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
theme_bw()

```



```

ggsave(paste0(plot_save_path, "/ndvi_linear_gam_response_id", which_buffalo, ".png"),
        width = 150, height = 80, units = "mm", dpi = 600)

```

Fit 2D GAM smooth

```

# fit an ssf gam model
# As above, set the cyclic knots for the hour margin explicitly at 0 and 24
ssf_gam2d_temporal2 <- mgcv::gam(cbind(times, step_id_) ~
                                te(hour, ndvi, bs = c("cc", "cr")) + #, k = c(5, 5)
                                sl_ + log_sl + cos_ta,
                                knots = list(hour = c(0, 24)),
                                data = buffalo_ssf_data,

```

```

        family = cox.ph,
        weight = case_)

summary(ssf_gam2d_temporal2)

```

Family: Cox PH

Link function: identity

Formula:

```

cbind(times, step_id_) ~ te(hour, ndvi, bs = c("cc", "cr")) +
    sl_ + log_sl + cos_ta

```

Parametric coefficients:

	Estimate	Std. Error	z value	Pr(> z)
sl_	-2.009e-05	3.194e-05	-0.629	0.529
log_sl	6.985e-03	5.756e-03	1.214	0.225
cos_ta	-3.941e-03	1.505e-02	-0.262	0.793

Approximate significance of smooth terms:

	edf	Ref.df	Chi.sq	p-value
te(hour,ndvi)	12.25	13.83	346.1	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Deviance explained = 0.0413%

-REML = 23103 Scale est. = 1 n = 106678

```

AIC(ssf_gam2d_temporal2)

```

```

[1] 46156.49

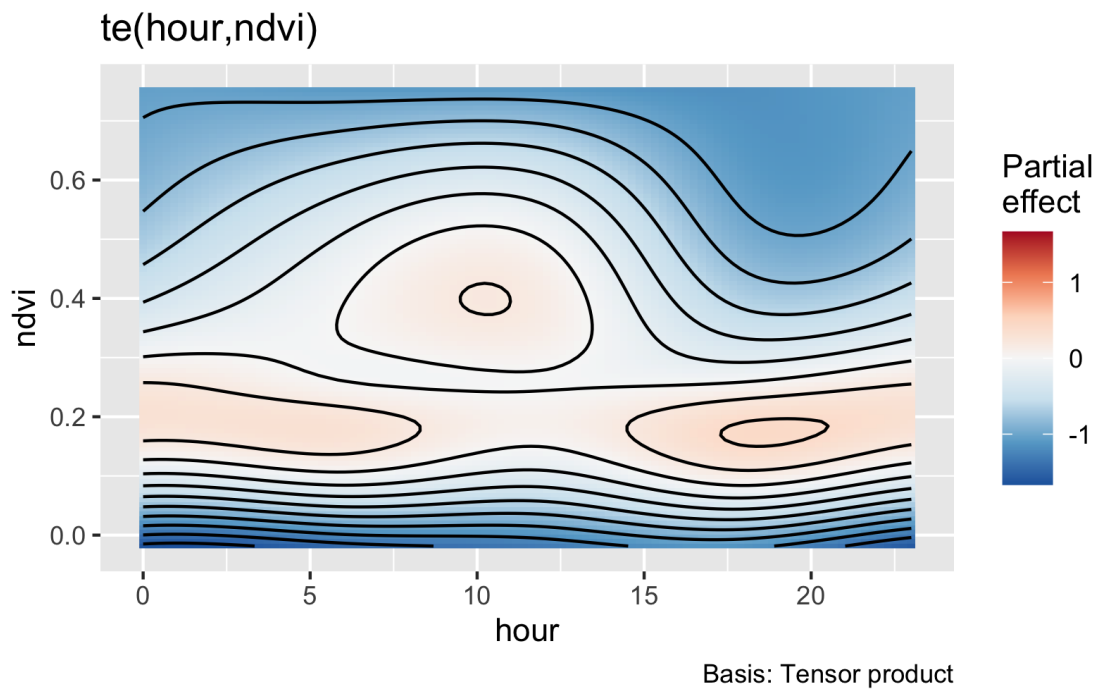
```

Plot the smooth selection surface

```

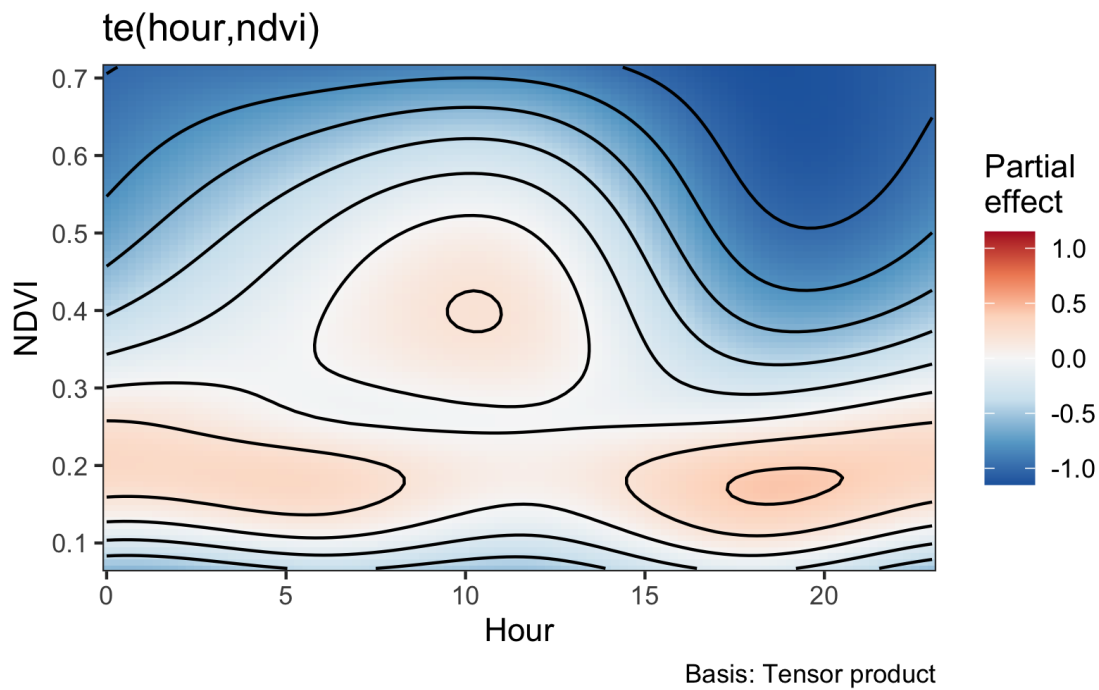
gratia::draw(ssf_gam2d_temporal2, rug = FALSE)

```



```
gam_smooths <- smooth_estimates(ssf_gam2d_temporal2)
filtered_gam_smooths <- gam_smooths %>% filter(ndvi > ndvi_quantiles[1] & ndvi < ndvi_quantiles[2])

gratia::draw(filtered_gam_smooths) +
  coord_cartesian(expand = FALSE) +
  scale_y_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  labs(x = "Hour", y = "NDVI") +
  theme_bw()
```



```
ndvi_hour_values <- expand.grid(ndvi = ndvi_values,
                               hour = seq(0, 23, length.out = length(ndvi_values)))

ndvi_hour_values <- ndvi_hour_values %>% mutate(
  sl_ = 0,
  log_sl = 0,
  cos_ta = 0
)

predictions <- predict(ssf_gam2d_temporal2,
                       newdata = ndvi_hour_values,
                       type = "link", # log-scale
                       se.fit = TRUE)

# Extract log RSS and standard errors
gam_log_rss <- predictions$fit
gam_se_log_rss <- predictions$se.fit

# Calculate confidence intervals on log-scale
gam_log_rss_lower <- gam_log_rss - 1.96 * gam_se_log_rss
gam_log_rss_upper <- gam_log_rss + 1.96 * gam_se_log_rss

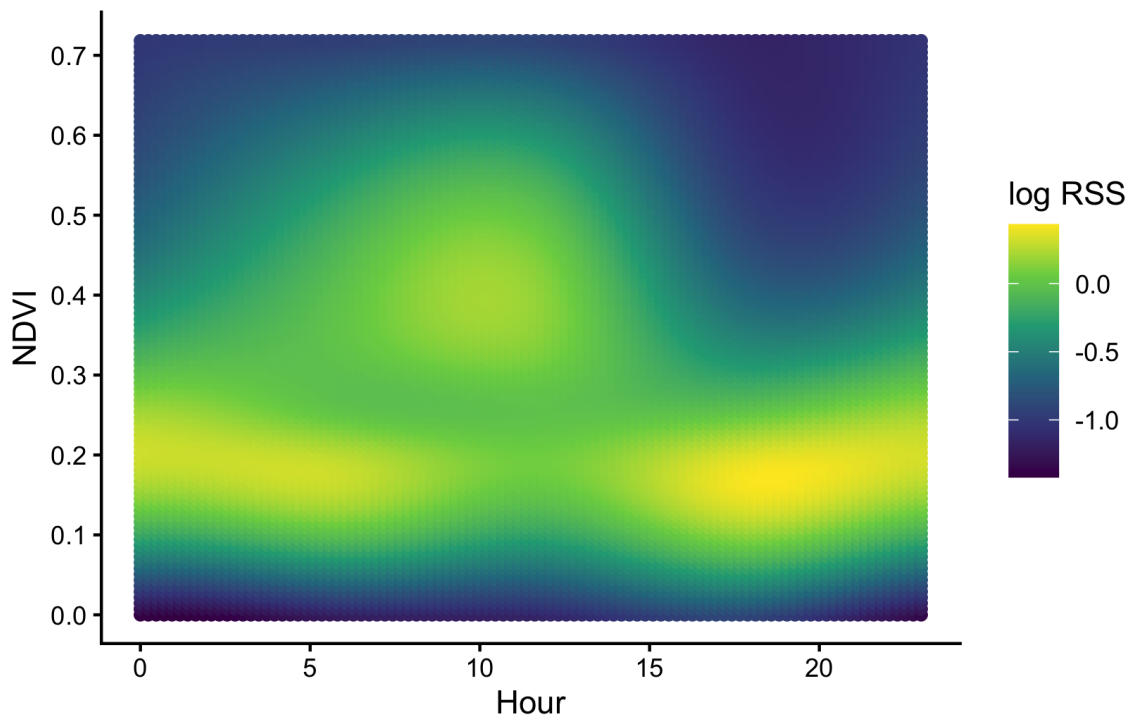
ndvi_hour_values <- ndvi_hour_values %>% mutate(
```

```

gam_log_rss = gam_log_rss,
gam_rss = exp(gam_log_rss),
gam_log_rss_lower = as.numeric(gam_log_rss_lower),
gam_log_rss_upper = as.numeric(gam_log_rss_upper)
)

# plot the selection surface
ggplot(data = ndvi_hour_values) +
  geom_point(aes(x = hour, y = ndvi, colour = gam_log_rss)) +
  labs(x = "Hour", y = "NDVI", colour = "log RSS") +
  scale_colour_viridis_c() +
  scale_y_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  theme_classic()

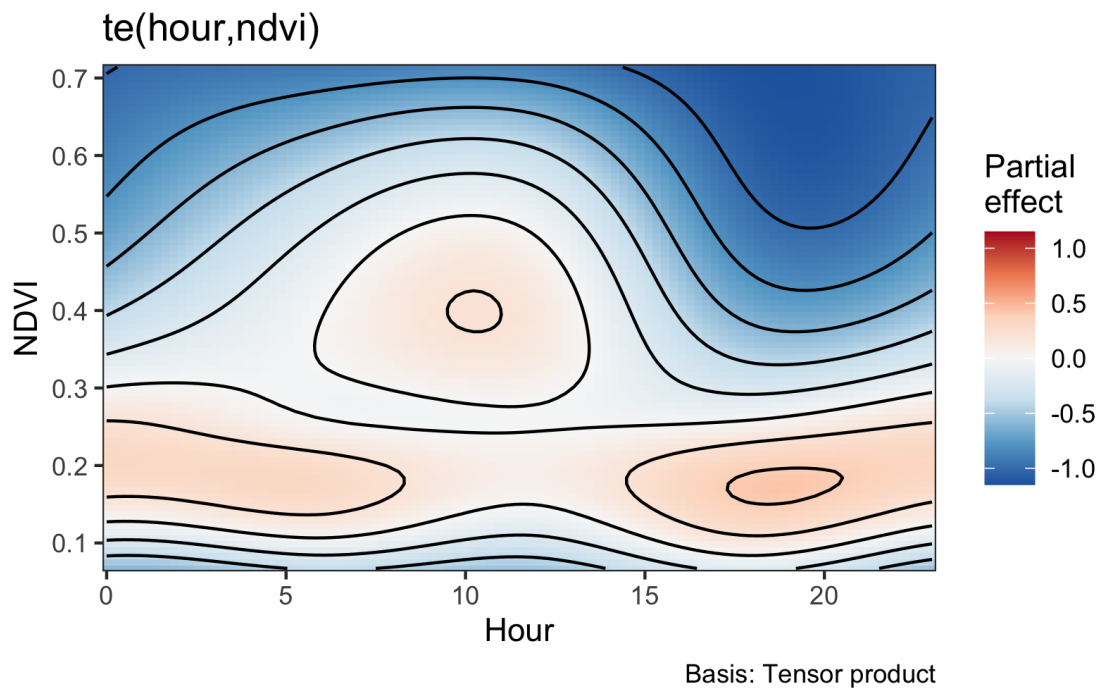
```



```

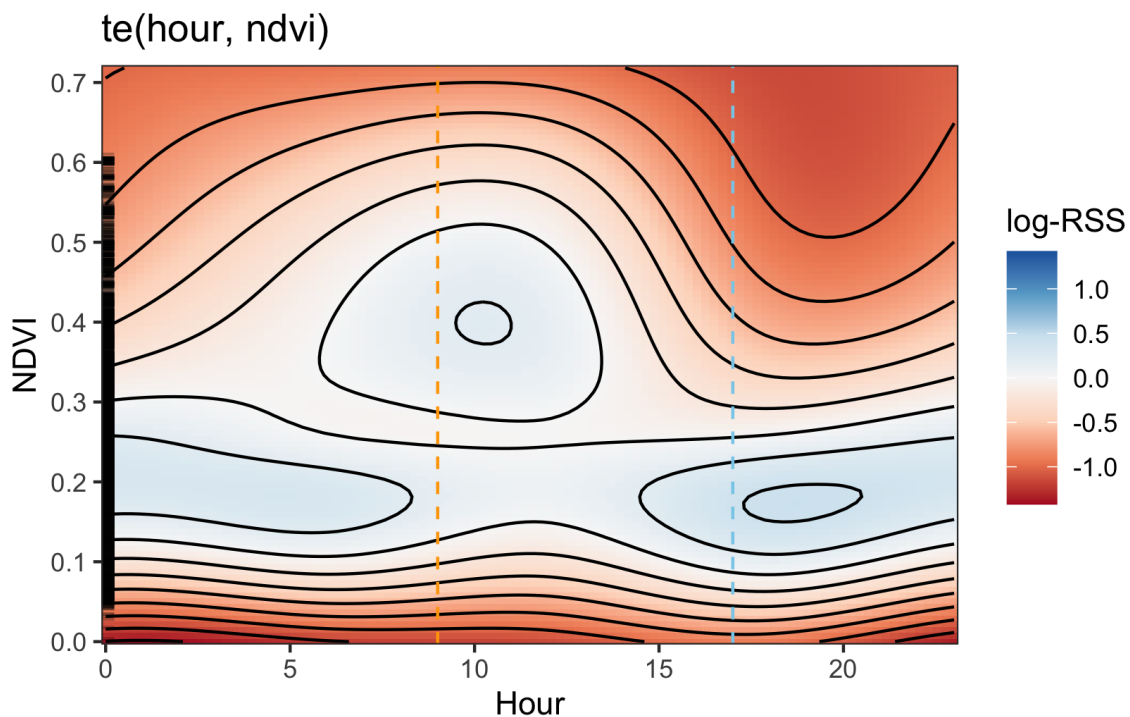
gratia::draw(filtered_gam_smooths) +
  coord_cartesian(expand = FALSE) +
  scale_y_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  labs(x = "Hour", y = "NDVI") +
  theme_bw()

```



```
max_abs_colour <- max(abs(range(ndvi_hour_values$gam_log_rss)))

# plot the selection surface
ggplot() +
  geom_raster(data = ndvi_hour_values,
    aes(x = hour, y = ndvi, fill = gam_log_rss)) +
  labs(title = "te(hour, ndvi)", x = "Hour", y = "NDVI", fill = "log-RSS") +
  geom_contour(data = ndvi_hour_values,
    aes(x = hour, y = ndvi, z = gam_log_rss), colour = "black") +
  geom_vline(xintercept = 9, linetype = "dashed", colour = "orange") +
  geom_vline(xintercept = 17, linetype = "dashed", colour = "skyblue") +
  scale_fill_distiller(palette = "RdBu", type = "div", direction = 1,
    limits = c(-max_abs_colour, max_abs_colour)) +
  coord_cartesian(expand = FALSE) +
  geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
    aes(y = ndvi), alpha = 0.25, length = unit(0.015, "npc"), sides="l") +
  scale_y_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  theme_bw()
```



```
ggsave(paste0(plot_save_path, "/ndvi_tensor_gam_response_id", which_buffalo, ".png"),
       width = 150, height = 80, units = "mm", dpi = 600)
```

Pull out a response curve from one of the hours

```
# Get predictions on log-scale with standard errors
predictions <- predict(ssf_gam2d_temporal2,
                      newdata = pred_data_hour9_17,
                      type = "link", # log-scale
                      se.fit = TRUE)

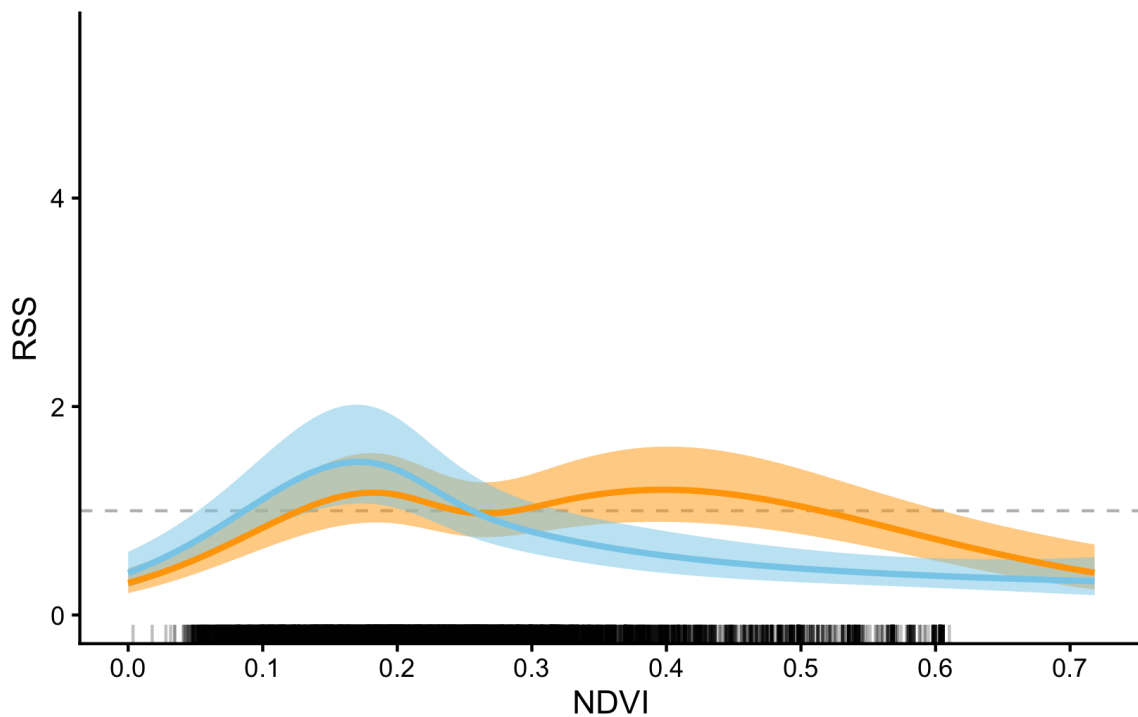
pred_data_hour9_17 <- pred_data_hour9_17 %>%
  mutate(
    # Extract log RSS and standard errors
    log_rss = predictions$fit,
    se_log_rss = predictions$se.fit,
    # Calculate confidence intervals on log-scale
    log_rss_lower = log_rss - 1.96 * se_log_rss,
    log_rss_upper = log_rss + 1.96 * se_log_rss,
    # Transform to RSS scale
    rss = exp(log_rss),
```

```

    rss_lower = exp(log_rss_lower),
    rss_upper = exp(log_rss_upper)
  )

# plot the response curve
ggplot(data = pred_data_hour9_17) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey") +
  geom_ribbon(aes(x = ndvi,
                 ymin = rss_lower, ymax = rss_upper,
                 fill = as.factor(hour)),
            alpha = 0.5) +
  geom_line(aes(x = ndvi, y = rss, colour = as.factor(hour)),
            linewidth = 1) +
  geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
           aes(x = ndvi), alpha = 0.25, sides="b") +
  scale_colour_manual(values = c("orange", "skyblue")) +
  scale_fill_manual(values = c("orange", "skyblue")) +
  scale_x_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  scale_y_continuous(limits = c(0, 5.5)) +
  labs(x = "NDVI", y = "RSS") +
  theme_classic() +
  theme(legend.position = "none")

```




```
ggsave(paste0(plot_save_path, "/ndvi_gam2d_response_sample_hour_id", which_buffalo, ".png"),  
       width = 70, height = 80, units = "mm", dpi = 600)
```

References

Klappstein, Natasha J, Théo Michelot, John Fieberg, Eric J Pedersen, and Joanna Mills Flemming. 2024. "Step selection functions with non-linear and random effects." *Methods in Ecology and Evolution*, ahead of print, June 24. <https://doi.org/10.1111/2041-210x.14367>.